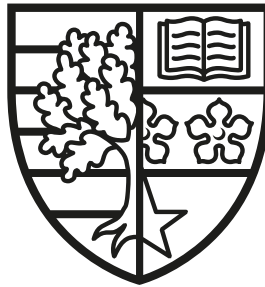


Relating Geometries: An Exploration of the Landscape of Courant Algebroid Relations

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Abstract

This thesis explores the framework of Courant algebroid relations as a tool for connecting geometric structures on seemingly unrelated manifolds. Such relations extend the notion of morphisms in the category of Courant algebroids and provide a natural setting for studying the morphisms of additional structures, including generalised metrics, divergence operators, spinors, Dirac structures, and generalised complex/Kähler geometries. Applications range from T-duality and mirror symmetry in string theory to quasi-Hamiltonian reduction in symplectic geometry.

A particular focus is given to the case of T-duality, where we show that it should be described by a Courant algebroid relation given by the composition of two reduction relations. Under suitable invariance assumptions, we demonstrate how a generalised metric, divergence operator, spinor module, and generalised complex/Kähler structure on a manifold $\mathcal{Q}_1$ can be transported to corresponding T-dual structures on a dual manifold $\mathcal{Q}_2$. The resulting transformation laws are shown to recover classical results such as the Buscher rules and the dilaton shift. In addition, we recover the Fourier–Mukai transform and show how one can extend it to the case of non-compact fibres. Finally, we establish the transformation law for the classical Ricci tensor, yielding the Ricci–Buscher rules.

Current-borne, wave-flung, tugged hugely by the whole might of ocean, the jellyfish drifts in the tidal abyss. Borne, flung, tugged from anywhere to anywhere, for in the deep sea there is no compass but nearer and farther, higher and lower. . . Hanging, swaying, pulsing, the most vulnerable and insubstantial creature, it has for its defense the violence and power of the whole ocean, to which it has entrusted its being, its going, and its will.

Ursula K. Le Guin, *The Lathe of Heaven*

This wise old whiskery fish swims up to three young fish and goes, 'Morning, boys, how's the water?' and swims away; and the three young fish watch him swim away and look at each other and go, 'What the fuck is water?' and swim away.

David Foster Wallace, *Infinite Jest*

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Chapter 1

Introduction

The aim of this thesis is to develop a rigorous geometric framework for understanding T-duality, a fundamental concept in string theory that relates physical theories defined on distinct spaces. These physical theories arise from certain two-dimensional variational problems known as sigma models. T-duality asserts that two seemingly different sigma models, defined on separate manifolds, can nonetheless describe equivalent physics.

A central challenge in formulating this geometric framework for T-duality lies in constructing a concrete geometric structure, similar to a map, that can transfer the data of one sigma model to another, even when the underlying manifolds are not diffeomorphic. The mathematical object encoding the sigma model is a certain type of vector bundle called a Courant algebroid. However, since traditional bundle maps require a diffeomorphism between the base spaces, a more general notion is needed.

This thesis explores the concept of a *Courant algebroid relation*, a geometric object that generalizes the idea of a vector bundle map between two Courant algebroids to the setting where the base manifolds may differ. Courant algebroid relations thus serve as the geometric structure enabling a rigorous definition for T-duality.

The notion of Courant algebroid relations in this thesis was introduced by Vysoký in [1], generalising Courant morphisms introduced in [2, 3], and inspired by Weinstein's symplectic category [4], which encodes symplectic geometry through Lagrangian submanifolds relations rather than maps. The definition of a Courant algebroid relation is general, allowing it to be adapted depending on the require-

ments of the theory. A Courant algebroid is a vector bundle with various additional operations, such as a bracket and a pairing, and a Courant algebroid relation is a structure between Courant algebroids such that these operations are “preserved” in an appropriate sense. When further structures, such as generalised Riemannian metrics, are imposed on the Courant algebroid, the notion of a Courant algebroid relation can be refined in order ensure compatibility. The focus of this thesis is thus to describe the interplay between these added structures and Courant algebroid relations, particularly in the case when the Courant algebroid relation describes T-duality.

The material presented in this thesis is based in part on the ongoing collaborations with Vincenzo Emilio Marotta and Richard J. Szabo:

- [5] T. C. De Fraja, V. E. Marotta, and R. J. Szabo, *T-Dualities and Courant Algebroid Relations*, Communications in Mathematical Physics **406** (2025) 21 [2308.15147 [math.DG]]
- [6] T. C. De Fraja, V. E. Marotta, and R. J. Szabo, *Courant Algebroid Relations, T-Dualities and Generalised Ricci Flow*, arXiv preprint arXiv:2502.02318(2025)
- [7] T. C. De Fraja, V. E. Marotta, and R. J. Szabo, *Generalised Complex and Spinor Relations*, in preparation(2025)

Chapter 2 primarily serves as a review of foundational material and contains minimal original content. Chapter 3 follows closely the exposition of [1], completing the necessary background for the thesis. Section 3.2.3 and Chapter 4 contain original results developed in [5]. Chapter 5 presents new contributions based on [6], while Chapter 6 includes novel results from ongoing joint work with Vincenzo Emilio Marotta and Richard J. Szabo.

1.1 Courant algebroids

Courant algebroids, as introduced in [8–11], have proved to be an indispensable geometric tool for understanding many aspects of supergravity and string theory, see

e.g. [12, 13] and references therein, as well as aiding the understanding of Hamiltonian reduction and its various generalisations, [2, 14–16]. An account of the history of Courant algebroids is given in [17], which I summarise here.

Courant algebroids were first described in [8, 14], where Ted Courant and Alan Weinstein sought integrability conditions for certain subbundles, called Dirac structures, of $TM \oplus T^*M$, the doubled tangent bundle of a manifold M . This bundle provides a natural framework for unifying Poisson and (pre)symplectic structures¹. Studying independently at the same time, Irene Dorfman was also studying Dirac structures from a more algebraic perspective [18], seeking the same integrability condition. The integrability condition they sought is given by closure under the Courant bracket — analogous to the integrability of distributions of the tangent bundle TM equipped with the Lie bracket — and describes the integrability of both Poisson and symplectic structures.

After the emergence of Lie algebroids as the infinitesimal counterpart of Lie groupoids, MacKenzie and Xu [19] defined the Lie bialgebroid as the infinitesimal counterpart of Poisson groupoids. Since the double of a Lie bialgebra is a Drinfel’d double [20], this naturally led Liu, Weinstein and Xu consider how this correspondence generalises to the algebroid setting, i.e. they sought to describe the double of a Lie bialgebroid. In [21] they found this to be a Courant algebroid, and the first axiomatic definition of a Courant algebroid was given therein: A Courant algebroid is a vector bundle E over a manifold M , equipped with a bracket on its space of sections, a symmetric non-degenerate pairing, and an *anchor map* taking sections of E to vector fields on M , satisfying a certain set of compatibility conditions.

The definition of Courant algebroids subsequently evolved, with Roytenberg’s [22] introduction of the non-skew-symmetric Dorfman bracket replacing the skew-symmetric Courant bracket, where skew-symmetry is sacrificed in order to satisfy a Jacobi identity. The Dorfman bracket became the defining bracket for Courant algebroids, and the number of defining axioms was slowly whittled down from five to three [23, 24].

As these developments occurred, other manifestations of Courant algebroids

¹Observe that the graphs of Poisson bivectors and presymplectic forms define isotropic subbundles of $TM \oplus T^*M$, with respect to a canonical pairing, making them key examples of the Dirac structures.

emerged, including their being in one-to-one correspondence with a certain type of dg symplectic manifolds [25].

More relevant to the subject of this thesis, in his now infamous letters to Weinstein [10], Ševera explains how Courant algebroids appear in the context of two-dimensional variational problems, more specifically two-dimensional sigma models. Also relevant to this work, Kosmann-Schwarzberg and Roytenberg [25, 26] showed how one can recover the Courant bracket as a “derived bracket”.

We will now briefly describe how Courant algebroids appear in both these last two situations, i.e. how they appear in the context of sigma models, and how they appear from derived brackets.

Courant Algebroids from Sigma-Models

The background data for the bosonic part of the worldsheet theory of closed oriented strings consists of a closed oriented Riemann surface (Σ, h) (the string), a Riemannian manifold (M, g) (the ambient space) and a closed three-form $H \in \Omega_{\text{cl}}^3(M)$, called an *H-flux*, which represents an integer cohomology class $[H] \in H^3(M, \mathbb{Z})$. The string sigma-model is a field theory of smooth maps $\mathbb{X} \in C^\infty(\Sigma, M)$ mapping the string into the ambient space, whose action functional is a sum of two terms, a kinetic term and a topological term.

The kinetic term is given by a Dirichlet-type functional, called the *Polyakov functional*. It is defined by endowing the space of maps $d\mathbb{X}: T\Sigma \rightarrow \mathbb{X}^*TM$ with a metric induced by g , regarded as a metric on the vector space of sections of the pullback $\mathbb{X}^*TM$ of the tangent bundle TM to Σ by $\mathbb{X}$, and the cometric h^{-1} on $T^*\Sigma$. This gives a well-defined metric $h^{-1} \otimes \mathbb{X}^*g$ on the vector bundle $T^*\Sigma \otimes \mathbb{X}^*TM$ over Σ which allows one to write the Polyakov functional as

$$S_g[\mathbb{X}] = \frac{1}{2} \int_{\Sigma} h^{-1}(\mathbb{X}^*g(d\mathbb{X}, d\mathbb{X})) \, d\mu_h, \quad (1.1)$$

where $d\mathbb{X} \in \Gamma(T^*\Sigma \otimes \mathbb{X}^*TM)$ and μ_h is the area measure on Σ induced by h .

More immediately relevant to us is the topological term, called the *Wess-Zumino functional*. It is the functional

$$S_H: C^\infty(\Sigma, M) \longrightarrow \mathbb{R}/\mathbb{Z}$$

defined by

$$S_H[\mathbb{X}] := \int_V \mathbb{X}_V^* H , \quad (1.2)$$

where V is any three-manifold with boundary Σ , and $\mathbb{X}_V: V \rightarrow M$ is any smooth extension of $\mathbb{X} \in C^\infty(\Sigma, M)$ to V . The space of Lagrangian densities of the Wess-Zumino functional S_H is $\Omega_{\text{cl}}^3(M)$.

Consider the variational problem for the topological term. A variation of the Wess-Zumino functional (1.2) is generated by a vector field $X \in \Gamma(TM)$ acting via the Lie derivative. Since H is closed, by Stokes' Theorem this is given by²

$$\delta_X S_H[\mathbb{X}] = \int_\Sigma \mathbb{X}^* (\iota_X H) . \quad (1.3)$$

The solutions of the variational problem $\delta_X S_H[\mathbb{X}] = 0$ (see e.g. [10, 27, 28]) are given by the maps $\mathbb{X} \in C^\infty(\Sigma, M)$ such that, for all $X \in \Gamma(TM)$, there exists $\bar{\alpha} \in \Gamma(T^*\Sigma)$ satisfying

$$\mathbb{X}^* (\iota_X H) = d\bar{\alpha} . \quad (1.4)$$

Let us introduce the vector bundle

$$\mathbb{T}M := TM \oplus T^*M ,$$

called the *generalised tangent bundle* of M . Given a section $X + \alpha \in \Gamma(\mathbb{T}M)$, known as a *generalised tangent vector*, Equation (1.4) is then satisfied if

$$\iota_X H = d\alpha , \quad (1.5)$$

where Equation (1.4) is obtained by pulling back Equation (1.5) by $\mathbb{X}$. Equation (1.5) can be interpreted as giving the tangent directions on $C^\infty(\Sigma, M)$ along which S_H is constant. We shall now show that these flat directions are related to symmetries of the Wess-Zumino functional (1.2).

²Notation: we denote the contraction of a section of a vector bundle with a tensor of the dual vector bundle by $\iota_{(\cdot)}$.

Following [28], we define the action of $\text{Diff}(M)$ on $C^\infty(\Sigma, M)$ by $(\phi, \mathbb{X}) \rightarrow \phi^{-1} \circ \mathbb{X}$. This group action maps the functional (1.2) to $S_H(\phi^{-1} \circ \mathbb{X}) = S_{(\phi^{-1})^* H}(\mathbb{X})$, hence the action by ϕ is a symmetry of the functional (1.2) if and only if $\phi^* H = H$. However, this is not the full group of symmetries of $S_H(\mathbb{X})$. The Lie group

$$\mathbf{G} := \text{Diff}(M) \ltimes \Omega^2(M)$$

acts on the space of Lagrangian densities $\Omega_{\text{cl}}^3(M)$ by

$$(\phi, B) \cdot H = (\phi^{-1})^*(H - \text{d}B) =: H' , \quad (1.6)$$

for all $(\phi, B) \in \mathbf{G}$ and $H \in \Omega_{\text{cl}}^3(M)$. Under the group action (1.6), the Wess-Zumino functional S_H transforms as

$$S_{H'}[\mathbb{X}] = S_H[\phi^{-1} \circ \mathbb{X}] - \int_{\Sigma} \mathbb{X}^*((\phi^{-1})^* B) , \quad (1.7)$$

where $H' = (\phi^{-1})^*(H - \text{d}B)$. Thus, the functional (1.2) is invariant if and only if $\phi^* H = H - \text{d}B$. Such pairs (ϕ, B) define the isotropy subgroup $\mathbf{G}_H \subset \mathbf{G}$ of the H -flux, see [28, Proposition 2.23].

Let $\text{Lie}(\mathbf{G}_H) \subset \Gamma(TM \oplus \wedge^2 T^*M)$ be the Lie algebra of the isotropy group $\mathbf{G}_H$. A representation of $\mathbf{G}$ is given by the normal Lie subalgebra of $\text{Lie}(\mathbf{G}_H)$

$$\text{Lie}(\mathbf{G}_H)^{\mathfrak{n}} = \{ X + \text{d}\alpha - \iota_X H \mid X \in \Gamma(TM) , \alpha \in \Gamma(T^*M) \} , \quad (1.8)$$

which is parametrised by elements in $\Gamma(\mathbb{T}M)$. See [28, Proposition 2.24] for the proof.

The Lie group $\mathbf{G}$ acts on $\Gamma(\mathbb{T}M)$ by

$$(\phi, B) \cdot (X + \alpha) = \phi_* X + (\phi^{-1})^*(\alpha + \iota_X B) , \quad (1.9)$$

for all $(\phi, B) \in \mathbf{G}$ and $X + \alpha \in \Gamma(\mathbb{T}M)$. Let $\Psi: \Gamma(\mathbb{T}M) \rightarrow \text{Lie}(\mathbf{G}_H)^{\mathfrak{n}}$ be the $\mathbf{G}_H$ -equivariant map defined by

$$\Psi(X + \alpha) = X + \text{d}\alpha - \iota_X H , \quad (1.10)$$

for all $X + \alpha \in \Gamma(\mathbb{T}M)$. The Dorfman bracket $\llbracket \cdot, \cdot \rrbracket_H$ is obtained from the Lie algebra action induced by the restriction of the representation of $\mathbf{G}$ on $\Gamma(\mathbb{T}M)$ to $\text{im}(\Psi) \subset \text{Lie}(\mathbf{G})$:

$$\Psi(X + \alpha) \cdot (Y + \beta) = [X, Y] + \mathcal{L}_X \beta - \iota_Y d\alpha + \iota_Y \iota_X H =: \llbracket X + \alpha, Y + \beta \rrbracket_H, \quad (1.11)$$

for all $X + \alpha, Y + \beta \in \Gamma(\mathbb{T}M)$.³ The Jacobi identity for $\llbracket \cdot, \cdot \rrbracket_H$ is a consequence of $\mathbf{G}_H$ -equivariance of Ψ .

The vector bundle $\mathbb{T}M$ endowed with the bracket $\llbracket \cdot, \cdot \rrbracket_H$ (and a suitable pairing) on its sections is our first example of a (exact) Courant algebroid. The cohomology class $[H] \in H^3(M)$ is enough to determine the isomorphism class of the Courant algebroid. $[H]$ is known as the *Ševera class*. We will see later that the isomorphism class is the same as the isotropy group $\mathbf{G}_H \subset \mathbf{G}$ of H for the action (1.6) on $\Omega_{\text{cl}}^3(M)$, i.e. the symmetries leaving S_H invariant (see Corollary 2.34). Thus the Ševera class determines the isotropy group of S_H , and vice versa. Incorporating the kinetic term (1.1) amounts to introducing what is called a *generalised metric* on $\mathbb{T}M$, which is the appropriate generalisation of a Riemannian metric to $\mathbb{T}M$. We conclude that, from the worldsheet perspective, the data of a string background is encoded in an exact Courant algebroid endowed with a generalised metric.

Courant Algebroids from Derived Brackets on Clifford modules

Recall that for a vector space V with a symmetric, bilinear pairing $\langle \cdot, \cdot \rangle$, one can form the Clifford algebra $\text{Cl}(V) = \bigotimes_{n=0}^{\infty} V/I(V)$, where $I(V)$ is the two sided ideal generated by the element $v \otimes v - \langle v, v \rangle 1$. The Clifford algebra $\text{Cl}(V)$ has a $\mathbb{Z}_2$ grading, into elements of even and odd degree.

Let M be a n -manifold. The notion of a Clifford module can be extended to vector bundles over a manifold. Any rank $2n$ pseudo-Euclidean vector bundle E with split signature pairing admits a $\text{Spin}(n, n)$ -structure [29]. For instance, consider again the generalised tangent bundle $\mathbb{T}M$, now equipped with the canonical split

³Notation: $[X, Y]$ denotes the Lie bracket of vector fields $X, Y \in \Gamma(TM)$, and $\mathcal{L}_X$ denotes the Lie derivative along X .

signature metric $\langle \cdot, \cdot \rangle$ induced by the duality pairing:

$$\langle X + \alpha, Y + \beta \rangle := \iota_X \beta + \iota_Y \alpha, \quad (1.12)$$

for all $X + \alpha, Y + \beta \in \Gamma(\mathbb{T}M)$. An action $\cdot : \Gamma(\mathbb{T}M) \times \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$ of generalised tangent vectors on the space of forms is given by

$$(X + \xi) \cdot \omega := \iota_X \omega + \xi \wedge \omega, \quad (1.13)$$

for any $X + \xi \in \Gamma(\mathbb{T}M)$, and $\omega \in \Omega^\bullet(M)$. This extends to the tensor product by

$$((X + \xi) \otimes (Y + \eta)) \cdot \omega = (X + \xi) \cdot ((Y + \eta) \cdot \omega) \quad (1.14)$$

and satisfies

$$((X + \xi) \otimes (X + \xi)) \cdot \omega = \langle X + \xi, X + \xi \rangle \omega. \quad (1.15)$$

Thus the action descends to $\text{Cl}(\mathbb{T}M)$, and makes $\bigwedge^\bullet T^*M$ a Clifford module, so we have

$$\text{Cl}(\mathbb{T}M) \cong \text{End}(\bigwedge^\bullet T^*M). \quad (1.16)$$

The space of forms $\Omega^\bullet(M)$ is also called the space of *spinors*.

The $\mathbb{Z}_2$ -grading of $\text{Cl}(\mathbb{T}M)$ allows one to define the super commutator of homogenous elements $A, B \in \Gamma(\text{Cl}(\mathbb{T}M))$, acting on $\omega \in \Omega^\bullet(M)$, given by

$$[A, B] \cdot \omega = A \cdot B \cdot \omega - (-1)^{\deg A \deg B} B \cdot A \cdot \omega. \quad (1.17)$$

We may introduce an odd differential operator on $\Omega^\bullet(M)$, the *H-twisted differential* $d_H : \Omega^\bullet(M) \rightarrow \Omega^\bullet(M)$, given by

$$d_H \omega := d\omega + H \wedge \omega, \quad (1.18)$$

where $H \in \Omega_{\text{cl}}^3(M)$, for any $\omega \in \Omega^\bullet(M)$, and d is the usual de Rham differential of

forms. This is an example of a *Dirac generating operator*, as introduced by [30], see also [31]. This similarly satisfies a super commutation law as in Equation (1.17), with $\deg d_H = 1$. We may define the twisted Dorfman bracket $\llbracket \cdot, \cdot \rrbracket_H$ as a derived bracket of d_H , extending the usual formula for the Lie bracket in terms of the exterior derivative and interior product to the Clifford action (1.13), i.e.

$$\llbracket e_1, e_2 \rrbracket_H \cdot \omega := [[d_H, e_1 \cdot], e_2 \cdot] \omega, \quad (1.19)$$

for all $e_1, e_2 \in \Gamma(TM)$, $\omega \in \Omega^\bullet(M)$. This is in agreement with (1.11).

The role of spinors in generalised geometry are two fold. On the String Theory side, the spinor fields define the fermionic sector of Type II superstring theory. The twisted differential defines the integrability condition for these structures. This, among other structures placed on the Courant algebroid, will allow for the complete geometric description of T-duality of Type II supergravity.

On the mathematical side, they give an equivalent description of generalised complex geometry, as introduced by Gualtieri in his thesis [29]. Generalised complex geometry is the generalisation of complex structures on TM to the bundle TM , and allows for the unification of the symplectic and complex structures of ordinary geometry. These are the two structures appearing in Mirror Symmetry, now a widely studied field in Mathematics, and which has a very similar flavour to the T-duality of String Theory. For instance, in [32], it was argued that mirror symmetry can be described by T-duality. This is expanded on in [33] using the language of generalised complex geometry, or more precisely the language of generalised Kähler geometry [34] and generalised Calabi-Yau geometry [11]. Though we do not discuss Mirror Symmetry in detail in this thesis, generalised geometry is an important tool for studying it, and thus a concrete mathematical tool for relating generalised complex structures is an important step. This will be elaborated on in Chapter 6.

1.2 Courant Algebroid Relations

As previously mentioned, Courant algebroid relations are inspired by Weinstein's symplectic category, where morphisms are given by Lagrangian correspondences. That is, symplectic manifolds (M_i, ω_i) with symplectic 2-forms ω_i for $i = 1, 2$ are in

Lagrangian correspondence if there is a Lagrangian submanifold L of the symplectic manifold $(M_1 \times M_2, p_1^*\omega_1 - p_2^*\omega_2)$, i.e.

$$L \subset (M_1 \times M_2, p_1^*\omega_1 - p_2^*\omega_2) , \quad \text{and} \quad (p_1^*\omega_1 - p_2^*\omega_2)|_L = 0 . \quad (1.20)$$

Note that the graph $\text{gr}(\phi) \subset M_1 \times M_2$ of a symplectomorphism $\phi: (M_1, \omega_1) \rightarrow (M_2, \omega_2)$ always gives a Lagrangian correspondence. If a Lie group G has a Hamiltonian action on a symplectic manifold (M, ω) with moment map $\mu: M \rightarrow \mathfrak{g}^*$, then $\mu^{-1}(0)$ gives a Lagrangian correspondence between (M, ω) and its symplectic reduction $(\mu^{-1}(0)/G, \underline{\omega})$. Symplectic reduction can be generalised to Poisson manifolds P by considering a Lie group-valued moment map $\mu: P \rightarrow G^*$, where G is now a Poisson-Lie group, and G^* integrates the dual space $\mathfrak{g}^*$ [35]. This can be generalised further to quasi-Poisson manifolds, as introduced in [36], where the Schouten-Nijenhuis bracket of the quasi-Poisson bivector is given by a trivector, rather than vanishing as in the Poisson case. When the action of G is so-called quasi-Hamiltonian, the moment map is then a map $\mu: P \rightarrow D/G$, where $(\mathfrak{d}, \mathfrak{g})$ is a Manin pair⁴ and D integrates $\mathfrak{d}$ [2, 37]. As previously mentioned, the theory of Manin pairs (or equivalently Lie bialgebras) can be encapsulated by study of pairs (E, L) , where E is a Courant algebroid equipped with a Dirac structures $L \subset E$.⁵ Thus an adequate theory of quasi-Poisson reduction should resemble the Lagrangian correspondence of symplectic reduction, replacing the symplectic reduction by Courant algebroid reduction. In [3, 15], this reduction was described as a *Courant morphism*, defined as a Dirac structure in $E \times \overline{E}$ supported on the graph of a function between the underlying manifolds. Here, $\overline{E}$ is the reduced Courant algebroid and $\overline{E}$ flips the sign of the pairing on $\underline{E}$, similar to flipping the sign of ω_2 in Lagrangian correspondence in (1.20). Courant algebroid relations are the generalisation of Courant morphisms to when there is no map between the spaces on which the Courant algebroids are defined over. Thus we see that the theory of quasi-Hamiltonian reduction is encapsulated by Courant algebroid relations. In [38]⁶, it is argued that T-duality should be an “almost isomorphism” between two quasi-Hamiltonian spaces, the

⁴i.e. $\mathfrak{d}$ is a Lie algebra with symmetric pairing of split signature, and $\mathfrak{g}$ is a maximally isotropic subalgebra.

⁵The usual notion of a Manin pair is taken by considering Courant algebroids over a point.

⁶See also <http://thphys.irb.hr/dualities2017/files/Jun06Severa.pdf>

quasi-Hamiltonian spaces being obtained through quasi-Hamiltonian reduction, and the “almost isomorphism” described by a Courant algebroid relation. In this thesis, as well as seeing how Courant algebroid relations give a robust notion of T-duality, we will explore how Courant algebroid relations describe this reduction procedure for Courant algebroids. In Chapter 6 in particular, we consider the reduction of Dirac structures under this procedure. Courant algebroid reduction is also crucial aspect of our formulation of T-duality, which we now discuss.

As we have seen from Equation (1.6) and the proceeding discussion, the Weiss-Zumino functional S_H on $C^\infty(\Sigma, M)$ is invariant under the isotropy subgroup $\mathbf{G}_H \subset \mathbf{G}$. $\mathbf{G}_H$ consists of pairs (ϕ, B) , where ϕ is a diffeomorphism of M , and B is a two-form on M , and such pairs must satisfy $\phi^*H = H - \mathrm{d}B$. Such pairs act on the bundle $\mathbb{T}M$ through (1.9), and the condition that the Courant bracket equipped on $\mathbb{T}M$ is preserved under this action is precisely that $\phi^*H = H - \mathrm{d}B$. That is, isomorphisms of the Courant algebroids are in one-to-one correspondence with the symmetries of sigma models.

In string theory, the notion of T-duality for sigma-models [39–41] may go beyond this concept of symmetry, with the underlying manifolds between T-dual spaces not required to be diffeomorphic [42–44]. Finalising a fully geometric description of T-duality and its various generalisations, such as non-abelian T-duality, Poisson-Lie T-duality, and non-isometric T-duality, has proven to be an elusive task. In this thesis, we explore how Courant algebroid relations can provide such a geometric description of T-duality. We will be interested in their role in capturing the topological, geometric, and physical properties of spaces related by T-duality symmetries of string theory, see e.g. [1, 10, 27, 45–49]. Similar to [10], in [12] Coimbra et al. show how each layer of data of a sigma model has a geometric counterpart exhibited as structures on an exact Courant algebroid. The aim of this thesis is to describe how Courant algebroid relations behave when constrained to this geometric data. Roughly speaking, the data of the string sigma-model is as follows: the H -flux of the Weiss-Zumino term corresponds to the bracket on $\mathbb{T}M$; the dynamical term (1.1) of the sigma model corresponds to a *generalised metric* on $\mathbb{T}M$, that is, a (positive definite) subbundle $V^+ \subset \mathbb{T}M$ of whose rank is $\dim M$. Courant algebroid relations relate T-dual H -fluxes, whose topology is encoded in the Dorfman bracket, and

these relations are constrained to be *generalised isometries* in order to describe the generalised metric under T-duality. These objects are described in Chapter 3, based on the approach taken in [1]. The dilaton of the string sigma model corresponds to a divergence on the Courant algebroid; the full bosonic equation of motion involves the generalised Ricci and generalised scalar curvature, which are defined by the generalised Riemannian geometry of a generalised metric on the Courant algebroid. In Chapter 5, we describe how divergences and generalised Ricci tensors behave under a Courant algebroid relation. Finally, RR field strengths correspond to spinors in the spinor bundle of the Courant algebroid. In T-duality, spinors are transported via a Fourier-Mukai transform. In Chapter 6, we describe this process via spin representation theory of the Courant algebroid relation, which recovers the Fourier-Mukai transform.

A Courant algebroid relation mimics a Courant algebroid isomorphism by imitating the properties of the graph of said isomorphism in the following way. If E_1 and E_2 are Courant algebroids, the graph $\text{gr}(\Phi)$ of an isomorphism $\Phi: E_1 \rightarrow E_2$ covering $\phi: M_1 \rightarrow M_2$ is a subbundle of the product Courant algebroid $\text{gr}(\Phi) \subset E_1 \times E_2$, supported on $\text{gr}(\phi) \subset M_1 \times M_2$. Thus a Courant algebroid relation $R: E_1 \dashrightarrow E_2$ is a subbundle $R \subset E_1 \times E_2$ supported on a submanifold $C \subset M_1 \times M_2$. If the isomorphism Φ preserves the bracket from E_1 to E_2 , then $\text{gr}(\Phi)$ will be an involutive subbundle of $E_1 \times E_2$, and thus one demands a Courant algebroid relation be an involutive subbundle of $E_1 \times E_2$. When the Courant algebroids are equipped with a non-degenerate pairing (such as (1.12)), the isomorphism Φ must preserve this structure. This can be expressed as $\text{gr}(\Phi)$ being an isotropic subbundle of $E_1 \times \overline{E}_2$, where again $\overline{E}_2$ denotes the flipping the sign of the pairing on E_2 . Thus R should be an isotropic subbundle of $E_1 \times \overline{E}_2$. One proceeds in this fashion, adding structure to the Courant algebroid and mimicking the situation for isomorphisms. A priori, this definition is extremely general and suffers from behaving generally quite badly. However, when layers of structure are added, such as the pairing on the Courant algebroid and generalised metrics, the definition of Courant algebroid relations is strengthened, allowing them to become more manageable objects.

In this sense, Courant algebroid relations allow one to naïvely extend the set of arrows in the category of Courant algebroids. Courant algebroid relations, while less

constraining than isomorphisms, present subtle and more complex features that before the work of Vysoký [1] were not fully understood. For instance, the composition of two Courant algebroid relations can only be performed under certain conditions, and so within this extended set of arrows not all morphisms compose.

The Courant algebroid relation R involved in T-duality is constructed in the following way. Suppose there is an exact Courant algebroid E over a manifold M . M is assumed to be foliated by two foliations $\mathcal{F}_1, \mathcal{F}_2$, with smooth leaf spaces $\mathcal{Q}_1, \mathcal{Q}_2$ respectively

$$\begin{array}{ccc} & M & \\ /_{\mathcal{F}_1} & & \searrow_{\mathcal{F}_2} \\ \mathcal{Q}_1 & & \mathcal{Q}_2. \end{array} \quad (1.21)$$

Under the correct conditions, the Courant algebroid E can undergo *reduction* by the foliations $\mathcal{F}_i$, a process developed in [50], building on the works of [10, 51]. This will be discussed in detail in Section 2.1.3. The reduction yields exact Courant algebroids $\underline{E}_i$ over $\mathcal{Q}_i$. Such a reduction process can be described by a Courant algebroid relation $Q(\mathcal{F}_i): E \dashrightarrow \underline{E}_i$. The *T-duality relation* is then the composition $Q(\mathcal{F}_2) \circ Q(\mathcal{F}_1)^\top$

$$\begin{array}{ccc} & E & \\ Q(\mathcal{F}_1)^\top \dashrightarrow & & \searrow_{Q(\mathcal{F}_2)} \\ \underline{E}_1 & \dashrightarrow^R & \underline{E}_2 \end{array} \quad (1.22)$$

where $Q(\mathcal{F}_1)^\top$ is the *transpose* of $Q(\mathcal{F}_1)$.

This construction is inspired by different approaches to T-duality, including the correspondence space picture for principal torus bundles of [46, 52], and Poisson-Lie T-duality, as in [1, 49]. Indeed, Courant algebroid relations pave the way for a complete formulation of T-duality as a generalised isometry. As already mentioned, this idea was first proposed by Ševera⁷ in order to describe Poisson-Lie T-duality as an “almost isomorphism” of Hamiltonian systems [38]. The almost isomorphism arises from a Courant algebroid relation that is required to be both Lagrangian – i.e. a Dirac structure for the product Courant algebroid – and a generalised isometry. This is explored in detail in [1], where Poisson-Lie T-duality is shown to be a generalised isometry between two Courant algebroids over principal bundles with the same base. However, there is no general geometric framework encompassing

⁷See <http://thphys.irb.hr/dualities2017/files/Jun06Severa.pdf>.

T-duality in all of its flavours. In this thesis we will provide such a framework for T-dualities arising between general fiber bundles equipped with Courant algebroids, and not necessarily over the same base. A particularly relevant instance that we are interested in is the T-duality for torus bundles with H -flux discussed in [46, 52–54]. This has also recently been discussed in [55] from a more general geometric perspective which is closer in spirit to our treatment; however, our framework also covers more general cases of affine torus bundles [56].

Poisson-Lie T-duality, generalised metrics and divergences

Let us discuss Poisson-Lie T-duality in more detail, following the approach in [49], which originates as a non-isometric generalisation of non-abelian T-duality in the non-linear sigma-model [57]. Let $E \rightarrow M$ be a Courant algebroid endowed with a generalised metric $V^+ \subset E$. Let div be a divergence operator on E , i.e. a map from $\Gamma(E)$ to $C^\infty(M)$ satisfying a Leibniz-like rule. Let $\varphi: M' \rightarrow M$ be a smooth map. Then there is the pullback $E' := \varphi^*E$, which is a Courant algebroid $E' \rightarrow M'$

$$\begin{array}{ccc} & & E' \\ & \nearrow & \\ E & & \end{array} \quad (1.23)$$

In order to define equivariant Poisson-Lie T-duality, also known as Poisson-Lie T-duality of dressing cosets in the gauged non-linear sigma-model [58], assume that the Courant algebroid $E \rightarrow M$ admits an action $\psi: \mathfrak{g} \rightarrow \Gamma(E)$ induced by the action of a Lie group G with Lie algebra $\mathfrak{g}$. Further assume that this action pulls back by φ to a $\mathfrak{g}$ -action $\psi': \mathfrak{g} \rightarrow \Gamma(E')$ turning E' into a G -equivariant Courant algebroid. It then follows that G acts⁸ freely and properly on M' .

The equivariant Courant algebroid can then undergo reduction by the G action. Denote by $\underline{E}'$ the reduced Courant algebroid over the quotient manifold $\mathcal{Q}' = M'/G$. Suppose that the pullback generalised metric $V' := \varphi^*V^+$ and divergence $\text{div}' := \varphi^*\text{div}$ are reducible by the $\mathfrak{g}$ -action $\psi': \mathfrak{g} \rightarrow \Gamma(E')$, with reduced generalised metric $\underline{V}'$ and divergence $\underline{\text{div}}'$ on $\underline{E}'$. Then Poisson-Lie T-duality is the statement that the physically relevant properties of $(\underline{V}', \underline{\text{div}}')$, such as the non-linear sigma-model and

⁸through the use of the anchor $E' \rightarrow TM'$, to be defined in Chapter 2

the generalised string background equations, can be completely formulated in terms of (V^+, div, ψ) , without any reference to $\mathcal{Q}'$ and $\underline{E}'$ [49, 57–59].

$$\begin{array}{ccc} & E' & \\ \nearrow & & \searrow \\ E & & \underline{E}' \end{array} \quad (1.24)$$

A formulation of Poisson-Lie T-duality in terms of Courant algebroid relations, in a slightly different setting to the one presented above, can be found in the work of Vysoký [1]. We will come back to Vysoký’s formulation in Chapter 7.

In the above setup, Ševera and Valach go as far as showing that the renormalisation group flow and string background equations are “compatible” [49]. The role of the divergence in this formulation is two-fold: firstly, as already mentioned, a divergence corresponds to a dilaton field in two-dimensional string sigma models. Secondly, the divergence is essential for defining such string background equations and the renormalisation group flows through generalised Riemannian geometry, which we will now describe.

Generalised Riemannian Geometry and Flows

One of the main goals of generalised geometry is to define notions of differential geometry when the tangent bundle is replaced by a Courant algebroid. For instance, one might wish to find the counterpart of Riemannian geometry in generalised geometry. The counterpart of Riemannian metrics is given by generalised metrics on Courant algebroids, i.e. maximally positive-definite subbundles (with respect to the pairing). Other structures associated with a metric can then be defined. For instance, connections on Courant algebroids have been extensively studied, see e.g. [28, 60, 61], and have even been explored in the context of Courant algebroid relations [1]. Their caveat is the non-uniqueness of a generalised Levi-Civita connection which, in order to define further geometric structures such as Ricci tensors and scalar curvature, necessitates the introduction of divergence operators [62].

Thus, given a pair of a generalised metric $V^+ \subset E$ and a divergence operator div on the sections of E , one can pick out a single Levi-Civita connection, and define the generalised Riemann tensor as in the classical case. The generalised

Ricci tensor $\text{GRic}_{V^+, \text{div}}$ and generalised scalar curvature $\text{Gr}_{V^+, \text{div}}$ are subsequently defined. One can define directly the generalised Ricci tensor and scalar curvature from the divergence and the metric. Indeed there are many approaches to defining the generalised Ricci tensor available in the literature, see e.g. [13, 60–62], however it has been recently shown in [63] that all of them are equivalent.

As previously mentioned, the physical motivation for generalised Riemannian geometry comes from considerations of the low-energy effective theory and renormalisation group flow for strings. Such theories occur when studying the quantization of the bosonic string, which involves adding the term

$$S_{R_h}[\mathbb{X}] = \int_{\Sigma} \mathbb{X}^* \phi R_h d\mu_h \quad (1.25)$$

to (1.1) and (1.2), where μ_h is the volume form associated to h and R_h its scalar curvature, and $\phi \in C^\infty(M)$ is the dilaton. In terms of the defining background data (g, H, ϕ) , the quantization of this action is conformally invariant at the quantum level if this data satisfies the vanishing of the 1-loop beta functions (i.e. the generalised Ricci flow). This condition can also be recovered from the equations of motion of the following *low energy string effective action* of the background data (g, H, ϕ)

$$S[g, H, \phi] = \int_M \left(R_g - \frac{1}{12} \|H\|_g^2 + \|\nabla_g \phi\|^2 \right) e^{-\phi} d\mu_g. \quad (1.26)$$

This action also arises from considering a generalised geometric version of the Einstein-Hilbert functional of general relativity on the space of string backgrounds (g, H, ϕ) . Recall that such a string backgrounds are in correspondence with generalised metrics V^+ and divergence operators⁹ div_ϕ on an exact Courant algebroid E with Ševera class $[H]$. Let us assume that M is a compact manifold. Denote by $\mathcal{M}_E$ the space of generalised metrics on E , and by $\mathcal{H}^\times$ the space of nowhere-vanishing half-densities on M , i.e. the invertible sections of the line bundle of half-densities on M . Following [61], the *generalized Einstein-Hilbert functional* is the function

⁹given ϕ , a divergence operator on E is given by $\text{div}_\phi e = \frac{1}{e^{-\phi} \mu_g} \mathcal{L}_{\rho(e)}(e^{-\phi} \mu_g)$, for $e \in \Gamma(E)$, where ρ is the anchor on E and μ_g is the density form associated to g . This will be expanded on in Section 5.1.

$\mathcal{S}: \mathcal{M}_E \times \mathcal{H}^\times \rightarrow \mathbb{R}$ given by

$$\mathcal{S}(V^+, \varkappa) = \int_M \text{GR}_{V^+, \text{div}_\phi} \varkappa^2, \quad (1.27)$$

where ϕ is such that $\varkappa^2 = e^{-\phi} \mu_g$, for μ_g the density associated to the Riemannian metric g . We will see later that a generalised metric is equivalent to the specification of a Riemannian metric g and a specific representative H of the Ševera class $[H]$. As we shall see from the formula (5.78), (1.27) is equivalent (1.26). Thus it is useful to study these generalised quantities, and how they are transformed under T-duality.

The *generalised Ricci flow* is the gradient flow of $\frac{1}{2} \text{grad}(\mathcal{S})$, where the gradient on $\mathcal{M}_E \times \mathcal{H}^\times$ is defined by the metric induced by integration over the measure $\varkappa^2$ and the generalised metric (see [61] for more details). The governing equation is similar to Hamilton's Ricci flow, with the classical structures g and Ric_g replaced by their generalised geometric counterparts. A slightly more modern approach incorporates a flow equation for the dilaton, involving the generalised scalar curvature. The generalised Ricci flow is the geometric recasting of the one-loop renormalisation group flow of string non-linear sigma models.

All these structures are expected, in a certain sense, to be compatible under T-duality. For Poisson-Lie T-duality, this was proved in [49]. The geometric approach adopted therein allows for a simple proof of this compatibility. Indeed in the present thesis, with the more general setup as in (1.22), one need only consider certain invariance conditions on the generalised metric $\underline{V}_1^+$ and divergence $\underline{\text{div}}_1$ on $\underline{E}_1$. One can then transport these to $\underline{E}_2$ via E in a way that is compatible with the relation R . From there, the generalised curvature tensors can be defined on both sides, and the respective generalised Ricci flows can subsequently be performed. The Courant algebroid relation will guarantee that these structures are compatible.

At this point, the full picture for type II supergravity is almost complete. The final piece of data is the Ramond–Ramond potentials. In generalised geometry, these arise as spinors on the Courant algebroid.

T-duality and the Fourier-Mukai Transform

We have already seen the space of spinors for the exact Courant algebroid $\mathbb{T}M$, namely the $\text{Cl}(\mathbb{T}M)$ -Clifford module $\bigwedge^\bullet T^*M$. Consider the string background with H -flux given by the closed three-form $H \in \Omega^3(M)$. The Ramond–Ramond fields are spinors $\omega \in \Omega^\bullet(M)$, satisfying a certain Bianchi identity

$$d\omega + H \wedge \omega = 0. \quad (1.28)$$

As before, the H -twisted exterior derivative is $d_H = d + H \wedge$, for which $d_H^2 = 0$ follows as H is closed. The spinors are thus closed under d_H , and hence of interest is their (twisted) cohomology class in the cohomology group of the differential complex $(\Omega^\bullet(M), d_H)$.

In the picture proposed by [53] and later expanded in [46], T-duality is accompanied by an isomorphism of twisted cohomology of torus bundles with a common base. It arises from a (smooth version of the) Fourier-Mukai transform through a correspondence between these bundles. We will see in Chapter 6 how the graph $\text{gr}(\varrho)$ of the Fourier-Mukai transform $\varrho: \Omega^\bullet(\mathcal{Q}_1) \rightarrow \Omega^\bullet(\mathcal{Q}_2)$ can be expressed as an irreducible spin representation of the Clifford algebra $\text{Cl}(R)$ generated by the T-duality Courant algebroid relation $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$ of (1.22). This thus allows for a reinterpretation of the Fourier-Mukai transform not necessitating fiber integration. Let us discuss the Fourier-Mukai transform in this approach to T-duality in more depth.

The setting of [46, 53] involves two principal $\mathbb{T}^k$ -bundles $\mathcal{Q}_1$ and $\mathcal{Q}_2$ over a common base manifold $\mathcal{B}$. Let $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ be the fibred product, with respective projections ϖ_1 and ϖ_2 to $\mathcal{Q}_1$ and $\mathcal{Q}_2$. This is a principal $\mathbb{T}^{2k}$ -bundle over $\mathcal{B}$, called a correspondence space or a doubled torus bundle, which sits in the commutative diagram

$$\begin{array}{ccc} & M & \\ \varpi_1 \swarrow & & \searrow \varpi_2 \\ \mathcal{Q}_1 & & \mathcal{Q}_2 \\ & \searrow & \swarrow \\ & \mathcal{B} & \end{array} \quad (1.29)$$

Suppose that both $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are endowed with closed $\mathbb{T}^k$ -invariant three-forms

$\underline{H}_1$ and $\underline{H}_2$, respectively. Then $(\mathcal{Q}_1, \underline{H}_1)$ and $(\mathcal{Q}_2, \underline{H}_2)$ are said to be T-dual if

$$\varpi_1^* \underline{H}_1 - \varpi_2^* \underline{H}_2 = dB \quad (1.30)$$

on the correspondence space M , for some $\mathbb{T}^{2k}$ -invariant two-form $B \in \Omega_{\mathbb{T}^{2k}}^2(\mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2)$ whose restriction to $\ker(\varpi_{1*}) \otimes \ker(\varpi_{2*})$ is non-degenerate. Gauging the string sigma-model for the doubled torus bundle M then relates the sigma-models for the quotient spaces $\mathcal{Q}_1$ and $\mathcal{Q}_2$ via reduction along the projections ϖ_1 and ϖ_2 , respectively [64].

In this case, since the fibers are compact, the Fourier-Mukai transform is well-defined and given by

$$\varrho(\omega) := \int_{\mathbb{T}^k} e^B \wedge \varpi_1^* \omega \quad (1.31)$$

for any $\omega \in \Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_1)$, where the fibrewise integration is the pushforward of forms by the projection $\varpi_2 : M \rightarrow \mathcal{Q}_2$. It is shown in [53] that ϱ defines a degree-shifting isomorphism between the twisted differential complexes $(\Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_1), d_{\underline{H}_1})$ and $(\Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_2), d_{\underline{H}_2})$. It describes the transformation of Ramond–Ramond fields in type II string theory under T-duality.

The Fourier-Mukai transform ϱ becomes an isomorphism of irreducible Clifford modules once an isomorphism

$$\mathcal{R} : \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_1) \longrightarrow \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_2) \quad (1.32)$$

of $C^\infty(\mathcal{B})$ -modules is chosen such that

$$\varrho(e \cdot \omega) = \mathcal{R}(e) \cdot \varrho(\omega) , \quad (1.33)$$

for any $e \in \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_1)$ and $\omega \in \Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_1)$, where $e \cdot \omega$ is the action defined in (1.13). In [46] this isomorphism was shown to exist and to be unique.

Applying the construction of the T-duality relation outlined earlier, where the foliations $\mathcal{F}_1, \mathcal{F}_2$ are generated by the two $\mathbb{T}^k$ torus actions on $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_1$, we obtain a Courant algebroid relation $R : \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$. In terms of invariant sections of $\mathbb{T}\mathcal{Q}_i$, this recovers $\mathcal{R}$. From there, we can consider the Clifford algebra $\text{Cl}(R)$

generated¹⁰ by R . Sections of $\text{Cl}(R)$ act on $\Omega^\bullet(\mathcal{Q}_1) \times \Omega^\bullet(\mathcal{Q}_2)$. Observe that, if $(e_1, e_2) \in R$, then, by (1.33)

$$(e_1, e_2) \cdot (\omega, \varrho(\omega)) = (e_1 \cdot \omega, e_2 \cdot \varrho(\omega)) = (e_1 \cdot \omega, \varrho(e_1 \cdot \omega)) \in \text{gr}(\varrho). \quad (1.34)$$

Thus $\text{gr}(\varrho)$ is invariant under this action, and hence defines a spinor representation. Moreover, one can show that this is an irreducible representation.

This motivates us to define what we call an *R-Clifford relation*, for an arbitrary Courant algebroid relation $R: E_1 \rightarrow E_2$. Here, E_i are Courant algebroids with spinor bundles S_i for $\text{Cl}(E_i)$, for $i = 1, 2$. An *R-Clifford relation* is then a subbundle $S_R \subset S_1 \times S_2$ providing an irreducible representation for $\text{Cl}(R)$, i.e. S_R is a spinor bundle for $\text{Cl}(R)$.

While the construction of the isomorphism of (twisted) differential complexes outlined above is restricted to the possibility of integrating along the fibres, our novel approach provides an alternative construction without introducing the Fourier-Mukai transform ϱ , based on the spin representation theory of Courant algebroid relations. Our approach is thus powerful enough to cover T-dualities based on correspondence spaces which involve non-compact fibres, such as those which arise in Poisson-Lie T-duality [65, 66].

Outline of the Thesis

The main goal of this work is to propose a rigorous geometric description of T-duality in its various forms. We do this using Courant algebroid relations, a recently developed tool designed to supplement Courant algebroid isomorphisms, which we argue makes them suitable for the application of T-duality. Their power comes from their malleability, allowing additional structure to be added as required. The later Chapters show how this is done. The thesis is thus outlined as follows.

In Chapter 2 we introduce Courant algebroids and their isomorphisms. As discussed, Courant algebroid relations are closely related to Courant algebroid isomorphisms, thus a detailed discussion of isomorphisms is given, leading to the classification of Courant algebroids of Ševera. The reduction of Courant algebroids is then

¹⁰since $R \subset \mathbb{T}\mathcal{Q}_1 \times \overline{\mathbb{T}\mathcal{Q}_2}$ is isotropic, the Clifford algebra is defined by considering $R \subset \mathbb{T}\mathcal{Q}_1 \times \mathbb{T}\mathcal{Q}_2$. The isotropic condition ensures that the Clifford action is well defined. See Chapter 6 for details.

discussed, which will be important for the proceeding chapters.

Chapter 3 introduces the main idea of the thesis: Courant algebroid relations. First examples are given, including how the aforementioned reduction of Courant algebroids can be interpreted via Courant algebroid relation. The subtle subject of their composition is then discussed. We then introduce the first extra structure to the Courant algebroid, generalised metrics, and explain how the notion of generalised isometries allows Courant algebroid relations to preserve generalised metrics in some sense. This will be critical for the formulation of the main results of the next chapter. We also discuss transverse generalised metrics and isometries, necessary for transporting the metric data through the reduction relation.

In Chapter 4, T-duality is formulated using Courant algebroid relations. We first discuss topological T-duality, where we explain in detail the conditions under which the T-duality relation (1.22) can be constructed. The generalised metric is then introduced, and we define geometric T-duality in terms of Courant algebroid relations. The main result of this section shows when a generalised metric can be transported across the T-duality relation. We also give the local picture of T-duality, giving the local version of the main result.

In Chapter 5, we continue adding geometric data to the picture. This begins with defining what it means for relations to preserve divergences between Courant algebroids. We show, for the particular case of the T-duality relation, how and when one can transport a divergence across the relation. Divergences allow one to define generalised Riemannian geometry on Courant algebroids, which we describe therein. The constructions of this chapter for divergences, and the power of Courant algebroid relations, allows one to describe how the generalised curvature tensors and the generalised Ricci flow behaves under the T-duality relation with little additional work. In particular, this yields the Ricci-Buscher rules, which gives a formula for the Ricci tensor of the T-dual Riemannian metric, which can often be a computationally expensive task.

Important to T-duality and more broadly in mathematics is the Fourier-Mukai transform of spinors involved in generalised complex geometry, and Chapter 6 discusses how this is described in the language of Courant algebroid relations. We describe a novel technique for relating spinors on Courant algebroids between a

Courant algebroid relation. We also define relations for Dirac structures, and show how the correspondence between Dirac structures and spinors gives rise to a correspondence between related Dirac structures and related spinors. This allows for the reduction of Dirac structures and generalised complex structures to be described in terms of spinor relations. In the T-duality setting, we give the construction of a spinor relation for the T-duality relation, which recovers the Fourier-Mukai transform in the case of torus bundles. Using the notion of generalised isometry introduced in Chapter 3, we discuss the theory of related generalised Kähler and generalised Calabi-Yau structures.

Finally, in Chapter 7, examples and applications are explored. In particular, we look at how Poisson-Lie T-duality can be described as a T-duality relation, and that this gives a relation between the generalised Riemannian structures and spinors of the T-dual spaces. We then provide explicit examples, including T-duality for affine torus bundles (in particular the Klein bottle and its three dimensional cousins), examples of T-dual solutions of the generalised Ricci flow, and T-dual generalised Kähler structures not coming from ordinary Kähler structures, relating harmonic functions and manifolds with two symplectic forms.

Chapter 2

Courant algebroids

This chapter focuses on introducing Courant algebroids, along with some of the relevant theory and constructions associated to them. This includes their isomorphisms and isomorphism classes, up to the classification by Ševera [10], and the reduction procedure by foliations [50, 51].

2.1 Exact Courant Algebroids: Isomorphism and Reduction

We briefly recall the main notions and results concerning Courant algebroids. We refer to [13, 28, 29, 67] and references therein for more extensive treatments.

2.1.1 Courant Algebroids

The structure of the vector bundle $\mathbb{T}M = TM \oplus T^*M$ endowed with the Dorfman bracket $\llbracket \cdot, \cdot \rrbracket_H$ introduced in Section 1.1 suggests that the fundamental notion to introduce is that of a Courant algebroid. In the following we provide a brief account of Courant algebroids.

Definition 2.1. A *Courant algebroid* is a quadruple $(E, \llbracket \cdot, \cdot \rrbracket, \langle \cdot, \cdot \rangle, \rho)$, where E is a vector bundle over a manifold M with a fibrewise non-degenerate pairing $\langle \cdot, \cdot \rangle \in \Gamma(\odot^2 E^*)$, a vector bundle morphism $\rho: E \rightarrow TM$ called the *anchor*, and a bracket operation

$$\llbracket \cdot, \cdot \rrbracket: \Gamma(E) \times \Gamma(E) \longrightarrow \Gamma(E) \quad (2.2)$$

called the *Dorfman bracket*, which together satisfy

- (i) $\rho(e) \cdot \langle e_1, e_2 \rangle = \langle \llbracket e, e_1 \rrbracket, e_2 \rangle + \langle e_1, \llbracket e, e_2 \rrbracket \rangle$,
- (ii) $\langle \llbracket e, e \rrbracket, e_1 \rangle = \frac{1}{2} \rho(e_1) \cdot \langle e, e \rangle$,
- (iii) $\llbracket e, \llbracket e_1, e_2 \rrbracket \rrbracket = \llbracket \llbracket e, e_1 \rrbracket, e_2 \rrbracket + \llbracket e_1, \llbracket e, e_2 \rrbracket \rrbracket$,

for all $e, e_1, e_2 \in \Gamma(E)$.¹

Remark 2.3. The anchored Leibniz rule for the Dorfman bracket

$$\llbracket e_1, f e_2 \rrbracket = f \llbracket e_1, e_2 \rrbracket + (\rho(e_1) \cdot f) e_2 , \quad (2.4)$$

for all $e_1, e_2 \in \Gamma(E)$ and $f \in C^\infty(M)$, follows from item (i). The Jacobi identity (iii) and the anchored Leibniz rule (2.4) imply that the anchor ρ is a bracket homomorphism:

$$\rho(\llbracket e_1, e_2 \rrbracket) = [\rho(e_1), \rho(e_2)] , \quad (2.5)$$

for all $e_1, e_2 \in \Gamma(E)$. We refer to [67, Section 2] for a complete account of all the main properties of Courant algebroids.

A Courant algebroid E is called *regular* if its anchor map $\rho: E \rightarrow TM$ has constant rank, and *transitive* if ρ is surjective.

Remark 2.6. For any Courant algebroid E over M there is a map $\rho^*: T^*M \rightarrow E$ given by

$$\langle \rho^*(\alpha), e \rangle := \iota_e \rho^\dagger(\alpha) , \quad (2.7)$$

for all $\alpha \in \Omega^1(M)$ and $e \in \Gamma(E)$, where $\rho^\dagger: T^*M \rightarrow E^*$ is the transpose of ρ . The map ρ^* induces a map $\mathcal{D}: C^\infty(M) \rightarrow \Gamma(E)$ defined by

$$\mathcal{D}f = \rho^* df , \quad (2.8)$$

¹Notation: throughout this thesis we will denote the Courant algebroid $(E, \llbracket \cdot, \cdot \rrbracket, \langle \cdot, \cdot \rangle, \rho)$ simply by E when there is no ambiguity in the structure maps. If there is more than one Courant algebroid E involved in the discussion, we label its operations with a subscript E . If it is the case that the Courant algebroids are labelled by subscripts i , we may adopt the notation of labelling their operations by the same subscript i , for example the bracket of a Courant algebroid E_i may be denoted $\llbracket \cdot, \cdot \rrbracket_i$.

for all $f \in C^\infty(M)$, which obeys a derivation-like rule and is the natural generalisation of the exterior derivative on functions into the algebroid E . Then item (ii) of Definition 2.1 is equivalent to

$$\llbracket e, e \rrbracket = \frac{1}{2} \mathcal{D}\langle e, e \rangle , \quad (2.9)$$

which together with Equation (2.4) imply the additional Leibniz rule

$$\llbracket f e_1, e_2 \rrbracket = f \llbracket e_1, e_2 \rrbracket - (\rho(e_2) \cdot f) e_1 + \langle e_1, e_2 \rangle \mathcal{D}f . \quad (2.10)$$

Applying ρ to (2.9), one has

$$\rho \circ \rho^* = 0 , \quad (2.11)$$

which motivates

Definition 2.12. A Courant algebroid E over M is *exact* if the chain complex

$$0 \longrightarrow T^*M \xrightarrow{\rho^*} E \xrightarrow{\rho} TM \longrightarrow 0 \quad (2.13)$$

is a short exact sequence of vector bundles.

Example 2.14 (Standard Courant Algebroids). Let M be a manifold, and let $H \in \Omega_{\text{cl}}^3(M)$. The (*H-twisted*) *standard Courant algebroid* over M is the quadruple $(\mathbb{T}M, \llbracket \cdot, \cdot \rrbracket_H, \langle \cdot, \cdot \rangle_{\mathbb{T}M, \text{pr}_1})$, where

$$\mathbb{T}M := TM \oplus T^*M \quad (2.15)$$

is the generalised tangent bundle, the Dorfman bracket is

$$\llbracket X + \alpha, Y + \beta \rrbracket_H := [X, Y] + \mathcal{L}_X \beta - \iota_Y d\alpha + \iota_Y \iota_X H , \quad (2.16)$$

as in (1.11), the pairing is given by

$$\langle X + \alpha, Y + \beta \rangle = \iota_Y \alpha + \iota_X \beta , \quad (2.17)$$

as in (1.12), and the anchor pr_1 is the projection to the first summand of $\mathbb{T}M = TM \oplus T^*M$. Then pr_1^* is the inclusion $i: T^*M \rightarrow \mathbb{T}M$, from which it follows that the sequence (2.13) is exact. From now on, we will denote the H -twisted standard Courant algebroid over M by $(\mathbb{T}M, H)$.

2.1.2 Isomorphisms of Courant Algebroids

Here we recall the main properties of Courant algebroid isomorphisms, together with some of the standard examples, closely following [28, 29].

Definition 2.18. Let E_1 and E_2 be Courant algebroids over manifolds M_1 and M_2 , respectively. A vector bundle morphism $\Phi: E_1 \rightarrow E_2$ covering a diffeomorphism $\phi: M_1 \rightarrow M_2$ is a *Courant algebroid morphism* if it satisfies

- (i) $\langle \Phi(\cdot), \Phi(\cdot) \rangle_{E_2} \circ \phi = \langle \cdot, \cdot \rangle_{E_1}$, (isometry)
- (ii) $\llbracket \Phi(\cdot), \Phi(\cdot) \rrbracket_{E_2} = \Phi(\llbracket \cdot, \cdot \rrbracket_{E_1})$. (bracket homomorphism)

Here and in the following the word *isomorphism* in the context of Courant algebroids refers to vector bundle isomorphisms covering diffeomorphisms between different Courant algebroid structures, even if defined on the same vector bundle.

Remark 2.19. Definition 2.18 together with the anchored Leibniz rule (2.4) implies a compatibility condition with the anchor maps:

$$\rho_{E_2} \circ \Phi = \phi_* \circ \rho_{E_1} . \quad (2.20)$$

Example 2.21. Let M_1, M_2 be manifolds, and consider the H -twisted standard Courant algebroid $(\mathbb{T}M_1, H)$ for $H \in \Omega_{\text{cl}}(M_1)$. For any $\phi \in \text{Diff}(M_1, M_2)$, consider the induced vector bundle isomorphism

$$\bar{\phi} := \phi_* + (\phi^{-1})^* : \mathbb{T}M_1 \longrightarrow \mathbb{T}M_2 \quad (2.22)$$

which covers ϕ . One has that

$$\langle \bar{\phi}(X + \alpha), \bar{\phi}(Y + \beta) \rangle = \phi^* \langle X + \alpha, Y + \beta \rangle . \quad (2.23)$$

The Dorfman bracket on $\mathbb{T}M_1$ transforms to

$$\bar{\phi}(\llbracket \bar{\phi}^{-1}(X + \alpha), \bar{\phi}^{-1}(Y + \beta) \rrbracket_{H_1}) = \llbracket X + \alpha, Y + \beta \rrbracket_{(\phi^{-1})^*H_1}, \quad (2.24)$$

for all $X + \alpha, Y + \beta \in \Gamma(\mathbb{T}M_2)$. Thus $\bar{\phi}$ is an isomorphism $(\mathbb{T}M_1, H) \rightarrow (\mathbb{T}M_2, (\phi^{-1})^*H)$ between the H_1 -twisted standard Courant algebroid $\mathbb{T}M_1$ and the $(\phi^{-1})^*H_1$ -twisted standard Courant algebroid $\mathbb{T}M_2$.

Example 2.25. Let $(\mathbb{T}M, H)$ be the H -twisted standard Courant algebroid over M , where H is the three-form on M characterising its Dorfman bracket. Let $B \in \Omega^2(M)$ be any two-form on M ; it induces the vector bundle morphism $B^\flat: TM \rightarrow T^*M$ given by $B^\flat(X) = \iota_X B$, for all $X \in \Gamma(TM)$. Define the vector bundle isomorphism $e^B: \mathbb{T}M \rightarrow \mathbb{T}M$ by

$$e^B(X + \alpha) = X + B^\flat(X) + \alpha, \quad (2.26)$$

for all $X \in \Gamma(TM)$ and $\alpha \in \Gamma(T^*M)$. Then

$$\begin{aligned} \langle e^B(X + \alpha), e^B(Y + \beta) \rangle &= B(Y, X) + B(X, Y) + \iota_X \beta + \iota_Y \alpha \\ &= \langle X + \alpha, Y + \beta \rangle. \end{aligned} \quad (2.27)$$

Moreover, the Dorfman bracket satisfies,

$$\llbracket e^B(X + \alpha), e^B(Y + \beta) \rrbracket_H = e^B(\llbracket X + \alpha, Y + \beta \rrbracket_H) + \iota_Y \iota_X dB, \quad (2.28)$$

and hence

$$\llbracket e^B(X + \alpha), e^B(Y + \beta) \rrbracket_{H-dB} = e^B(\llbracket X + \alpha, Y + \beta \rrbracket_H). \quad (2.29)$$

Thus e^B is an isomorphism $e^B: (\mathbb{T}M, H) \rightarrow (\mathbb{T}M, H - dB)$, see e.g. [29, Proposition 3.23].

Definition 2.30. The vector bundle isomorphism $e^B: (\mathbb{T}M, H) \rightarrow (\mathbb{T}M, H - dB)$ of Example 2.25 is a *B-field transformation*.

We adapt the arguments used in [29, Section 3.2] to characterise isomorphisms

of exact Courant algebroids as

Proposition 2.31. Let $\Phi: \mathbb{T}M_1 \rightarrow \mathbb{T}M_2$ be an isomorphism of the twisted standard Courant algebroids $\mathbb{T}M_1$ and $\mathbb{T}M_2$, covering $\phi \in \text{Diff}(M_1, M_2)$, with twisting three-forms $H_1 \in \Omega_{\text{cl}}^3(M_1)$ and $H_2 \in \Omega_{\text{cl}}^3(M_2)$. Then Φ can be expressed as the composition $\Phi = \bar{\phi} \circ e^B$ of a B -field transformation with two-form $B \in \Omega^2(M_1)$, and $\phi^*H_2 = H_1 - dB$.

Proof. Set $F := \bar{\phi}^{-1} \circ \Phi$, which covers the identity 1_{M_1} . By Remark 2.19 it follows that

$$\rho_{E_1} \circ F = \rho_{E_1} . \quad (2.32)$$

Equation (2.32) gives

$$\langle e, F^{-1}(df) \rangle_{E_1} = \langle F(e), df \rangle_{E_1} = \iota_{\rho_{E_1}(F(e))} df = \iota_{\rho_{E_1}(e)} df = \langle e, df \rangle_{E_1} \quad (2.33)$$

for all $e \in \Gamma(\mathbb{T}M_1)$ and $f \in C^\infty(M_1)$. This shows that F^{-1} acts as the identity on T^*M_1 . Therefore $F = e^B$ for some $B \in \Omega^2(M_1)$, because F is an isomorphism. Thus we obtain, from the calculation of the Dorfman bracket that $\phi^*H_2 = H_1 - dB$. $\square$

When $M_1 = M_2 = M$, we have

Corollary 2.34. The automorphism group of the H -twisted standard Courant algebroid $(\mathbb{T}M, H)$ consists of those $\Phi = \bar{\phi} \circ e^B$ such that $\phi^*H = H - dB$.

Note that the action of $\text{Diff}(M) \times \Omega^2(M)$ defined by Proposition 2.31 is precisely the group action (1.6) transforming the Wess-Zumino functional S_H of the sigma-model discussed in Section 1.1. In other words, $\text{Aut}(\mathbb{T}M, H) = \mathbf{G}_H$, where $\mathbf{G}_H$ is the isotropy subgroup of the symmetries of S_H .

Remark 2.35 (Ševera's Classification). The classification of exact Courant algebroids E , due to Ševera [10], arises from the choice of any isotropic splitting $\sigma: TM \rightarrow E$ of the short exact sequence (2.13), that is, splittings such that $\text{Im}(\sigma)$ is isotropic with respect to $\langle \cdot, \cdot \rangle$. This yields the Courant algebroid isomorphism $\sigma + \frac{1}{2}\rho^*$ which identifies E with $\mathbb{T}M$ and defines a closed three-form $H_\sigma \in \Omega_{\text{cl}}^3(M)$

given by

$$H_\sigma(X, Y, Z) = \langle \llbracket \sigma(X), \sigma(Y) \rrbracket, \sigma(Z) \rangle \quad (2.36)$$

for $X, Y, Z \in \Gamma(TM)$.

The equivalence classes of Courant algebroids (up to small isomorphisms²) are in one-to-one correspondence with cohomology classes in $H^3(M, \mathbb{R})$. Since the difference between two isotropic splittings $\sigma - \sigma'$ defines a two-form $B \in \Omega^2(M)$ via $(\sigma - \sigma')(X) = \rho_E^*(\iota_X B)$, H_σ is shifted by dB under a change of splitting. Hence there is a well-defined cohomology class $[H_\sigma] \in H^3(M, \mathbb{R})$ associated to the exact sequence (2.13) which completely determines the Courant algebroid structure. This is called the Ševera class of the exact Courant algebroid. A general discussion can be found in [28, Section 2.2].

As a consequence of the Ševera classification and Proposition 2.31, any isomorphism between exact Courant algebroids E_1 and E_2 with representatives of their Ševera class H_1 and H_2 , respectively, covering a diffeomorphism $\phi: M_1 \rightarrow M_2$ is such that $\phi^*H_2 = H_1 - dB$, for some $B \in \Omega^2(M_1)$.

2.1.3 Reduction of Exact Courant Algebroids

The reduction of Courant algebroids will play an important role in our formulation of T-duality. The process refers to obtaining a new Courant algebroid on the leaf space of a manifold with a foliation from a Courant algebroid on the original manifold. Reduction by a group action was developed by Bursztyn-Cavalcanti-Gualtieri [51] and generalised by Zambon to foliations that are not necessarily generated by a group action [50]. We specialise Zambon's discussion to subbundles of a Courant algebroid over M which are supported on all of M , rather than on some submanifold of M . As discussed in [10, 27], this case naturally arises in the description of sigma-models related to equivariant exact Courant algebroids. In particular, the infinitesimal symmetries of the Wess-Zumino functional (1.2) generate directions in which S_H is constant.

This motivates a “gauging-like” reduction of the string background defining the

²A small isomorphism is a Courant algebroid isomorphism covering a diffeomorphism lying in the identity component of $\text{Diff}(M)$, see [28, Definition 2.18]

sigma-model: the condition (1.5) naturally arises from the definition of the map Ψ in Equation (1.10) by looking at the sections $X + \alpha \in \Gamma(TM)$ for which $\Psi(X + \alpha) = X$, i.e. $\iota_X H = d\alpha$. Therefore the infinitesimal action associated with the variation generated by $X \in \Gamma(TM)$ is given by the differential operator $\llbracket X + \alpha, \cdot \rrbracket_H$, for any $X + \alpha \in \Gamma(TM)$ for which $\Psi(X + \alpha) = X$, which are the elements associated with the flat directions for S_H . This shows how reduction of exact Courant algebroids naturally arises from the symmetries of the topological term of a string sigma-model. The construction which follows can be motivated by the observation [68] that the constraints for gauging the full sigma-model by isometries of M are equivalent to requiring that the sections $X + \alpha$ span an involutive isotropic subbundle $K \subset TM$.

Let E be an exact Courant algebroid over M , and K an isotropic subbundle of E supported on M such that $\rho(K^\perp) = TM$.

Definition 2.37. The space of sections of $K^\perp$ which are *basic with respect to K* is

$$\Gamma_{\text{bas}}(K^\perp) := \{ e \in \Gamma(K^\perp) \mid \llbracket \Gamma(K), e \rrbracket \subset \Gamma(K) \} .$$

When there are enough basic sections, i.e. $\Gamma_{\text{bas}}(K^\perp)$ spans $K^\perp$ pointwise, then K and $K^\perp$ satisfy the main properties [50, Lemma 3.5]

$$\llbracket \Gamma(K), \Gamma(K^\perp) \rrbracket \subset \Gamma(K^\perp) , \quad (2.38)$$

$$\llbracket \Gamma(K), \Gamma(K) \rrbracket \subset \Gamma(K) . \quad (2.39)$$

In particular, the distribution $\rho(K) \subset TM$ induces a smooth integrating foliation $\mathcal{F}$ of M .

These properties imply the following central result for reduction of exact Courant algebroids over foliated manifolds, as stated by Zambon in [50, Theorem 3.7], generalising the analogous statement of Bursztyn-Cavalcanti-Gualtieri in [51, Theorem 3.3] for the case of reduction by group actions.

Theorem 2.40. Let E be an exact Courant algebroid over M , and K an isotropic subbundle of E supported on M such that $\rho(K^\perp) = TM$. Assume that the space of basic sections $\Gamma_{\text{bas}}(K^\perp)$ spans $K^\perp$ pointwise, and that the quotient $\mathcal{Q}$ of M by the foliation $\mathcal{F}$ integrating $\rho(K)$ is a smooth manifold, so that the quotient map

$\varpi: M \rightarrow \mathcal{Q}$ is a surjective submersion. Then there is an exact Courant algebroid $\underline{E}$ over $\mathcal{Q}$ which fits in the pullback diagram

$$\begin{array}{ccc} K^\perp/K & \longrightarrow & \underline{E} \\ \downarrow & & \downarrow \\ M & \xrightarrow{\varpi} & \mathcal{Q} \end{array}$$

of vector bundles.

Sketch of Proof. We sketch the first part of the proof given in [50] in order to discuss how elements in $K^\perp/K$ at two points in a leaf of $\mathcal{F}$ are identified and to introduce our notation. Recall that since $K^\perp$ has enough basic sections, Equation (2.39) holds, i.e., K is involutive. Let $\mathcal{P}: K^\perp \rightarrow K^\perp/K$ be the quotient map and denote by $N_q = \varpi^{-1}(q)$, for any $q \in \mathcal{Q}$, a leaf of $\mathcal{F}$. Then for any $m, m' \in N_q$, we say that $e \in (K^\perp/K)_m$ and $e' \in (K^\perp/K)_{m'}$ are identified if and only if there exists a basic section $\hat{e} \in \Gamma_{\text{bas}}(K^\perp)$ such that

$$\mathcal{P}(\hat{e}(m)) = e \quad \text{and} \quad \mathcal{P}(\hat{e}(m')) = e' . \quad (2.41)$$

This means that there exists a trivialisation of $K^\perp/K|_{N_q}$ obtained by projecting onto it basic sections giving a frame for $K^\perp$. By assumption there are enough basic sections, so that they induce a frame for $K^\perp/K|_{N_q}$.

It is shown in [50] that this identification is well-defined, i.e. for any $\hat{e}, \check{e} \in \Gamma_{\text{bas}}(K^\perp)$ such that $\mathcal{P}(\hat{e}(m)) = \mathcal{P}(\check{e}(m))$, then $\mathcal{P}(\hat{e}(m')) = \mathcal{P}(\check{e}(m'))$. This is achieved by considering a finite sequence $\{k_i\}_{i=1, \dots, n}$ of (local) sections $k_i \in \Gamma(K)$ such that the integral curves of their corresponding vector fields $\rho(k_i)$ can be linked so as to join m and m' . Let us denote by t_i the parameter for each integral curve. Denote by

$$\exp(\text{ad}_{k_i}) \in \text{Aut}(E) \quad (2.42)$$

the automorphism of E obtained by integrating the K -action $\text{ad}_{k_i} = \llbracket k_i, \cdot \rrbracket$.

Then we can define the automorphism

$$\mathbf{Ad}_k := \exp(t_1 \mathbf{ad}_{k_1}) \circ \cdots \circ \exp(t_n \mathbf{ad}_{k_n}) . \quad (2.43)$$

Since $\hat{e}$ and $\check{e}$ are basic and the property (2.38) holds, it follows that $\mathbf{Ad}_k(\hat{e}(m)) - \hat{e}(m') \in K_{m'}$ and $\mathbf{Ad}_k(\check{e}(m)) - \check{e}(m') \in K_{m'}$. We assumed that $\hat{e}(m)$ and $\check{e}(m)$ reduce to the same element in $(K^\perp/K)_m$, hence $\hat{e}(m) - \check{e}(m) \in K_m$. Thus $\mathbf{Ad}_k(\hat{e}(m) - \check{e}(m)) \in K_{m'}$ by Equation (2.39). Therefore $\hat{e}(m') - \check{e}(m') \in K_{m'}$, which means that $\hat{e}(m')$ and $\check{e}(m')$ project to the same element in $(K^\perp/K)_{m'}$.

The identification (2.41) yields a pointwise isomorphism

$$\mathcal{J}_{q,m} : \underline{E}_q \longrightarrow (K^\perp/K)_m \quad (2.44)$$

for any $m \in N_q$, which together with the surjective submersion ϖ show that $\underline{E}$ is a vector bundle. The map $\mathcal{P}$, together with the pointwise isomorphisms $\mathcal{J}_{q,m}$, imply that

$$\Gamma(\underline{E}) \simeq \Gamma_{\text{bas}}(K^\perp) / \Gamma(K) \quad (2.45)$$

as $C^\infty(\mathcal{Q})$ -modules, where the $C^\infty(\mathcal{Q})$ -module structure on $\Gamma_{\text{bas}}(K^\perp)/\Gamma(K)$ is given by pulling back functions to M with ϖ .³

The completion of the proof can be found in [50, 51], wherein $\underline{E}$ is shown to be an exact Courant algebroid with the bracket, anchor and pairing inherited from E . $\square$

In the notation of [50, 51], we write

$$\underline{E} = \frac{K^\perp}{K} \Big/ \mathcal{F} \quad (2.46)$$

for the above construction of $\underline{E}$, and we denote by

$$\natural : K^\perp \longrightarrow \underline{E} \quad (2.47)$$

³This is the same construction given in [69] for the quotient of $\mathbf{G}$ -equivariant vector bundles by the action of a group $\mathbf{G}$. In that case the invariance condition is dictated by the involutive vector subbundle K inducing the foliation of the base manifold, which in turn corresponds to the orbits of the $\mathbf{G}$ -action.

the vector bundle morphism covering ϖ given by the two quotients by K and $\mathcal{F}$.

When Theorem 2.40 holds, we can also establish that basic sections have the property of

Lemma 2.48. Let E be an exact Courant algebroid endowed with an involutive isotropic subbundle K satisfying the assumptions of Theorem 2.40. Consider the short exact sequence of vector bundles

$$0 \longrightarrow K \longrightarrow K^\perp \longrightarrow K^\perp/K \longrightarrow 0 . \quad (2.49)$$

Then any splitting $s: K^\perp/K \rightarrow K^\perp$ of this sequence satisfies $s([e]) \in \Gamma_{\text{bas}}(K^\perp)$, for any $[e] \in \Gamma_{\text{bas}}(K^\perp)/\Gamma(K)$ corresponding to $\underline{e} \in \Gamma(\underline{E})$.

Proof. By the isomorphism $\Gamma_{\text{bas}}(K^\perp)/\Gamma(K) \simeq \Gamma(\underline{E})$ of $C^\infty(\mathcal{Q})$ -modules, to any section $\underline{e}$ of $\underline{E}$ there corresponds an element $[e] \in \Gamma_{\text{bas}}(K^\perp)/\Gamma(K)$. Thus for any splitting s of the sequence we get

$$s([e]) = e + k_s , \quad (2.50)$$

for some $k_s \in \Gamma(K)$, where $e \in \Gamma_{\text{bas}}(K^\perp)$. Hence

$$[[k, s([e])]] = [[k, e]] + [[k, k_s]] \in \Gamma(K) , \quad (2.51)$$

for any section k of K , by Definition 2.37 and involutivity of K . □

Example 2.52 (Reduction of Standard Courant Algebroids). Of importance to us is the case of a foliated manifold $(M, \mathcal{F})$ such that $\mathcal{Q} = M/\mathcal{F}$ is smooth. Take

$$K = T\mathcal{F} \oplus \{0\} \quad (2.53)$$

as a subbundle of the untwisted standard Courant algebroid $(TM, 0)$. Then, for $X \in T\mathcal{F}$ and $Y + \alpha \in TM$,

$$\langle X, Y + \alpha \rangle = \iota_X \alpha. \quad (2.54)$$

Thus $K^\perp = TM \oplus \text{Ann}(T\mathcal{F})$, and basic sections are given by projectable vector

fields and one-forms of the type $d(\varpi^*f)$, where $f \in C^\infty(\mathcal{Q})$, which respectively span TM and $\text{Ann}(T\mathcal{F})$ pointwise. Hence the conditions of Theorem 2.40 are satisfied, and our reduced Courant algebroid becomes

$$\underline{E} = (TM/T\mathcal{F} \oplus \text{Ann}(T\mathcal{F})) / \mathcal{F} = T\mathcal{Q} \oplus T^*\mathcal{Q} = \mathbb{T}\mathcal{Q} . \quad (2.55)$$

Remark 2.71 below gives the reduction of the twisted standard Courant algebroid $(\mathbb{T}M, H)$, after introducing adapted splittings.

Reduction of Subbundles of Exact Courant Algebroids

Following [50, Proposition 4.1] which deals with the reduction of Dirac structures, a useful result concerning the reduction of subbundles of E is

Proposition 2.56. Let E be an exact Courant algebroid over M endowed with an isotropic subbundle K satisfying the assumptions of Theorem 2.40, and let W be a subbundle of E such that $W \cap K^\perp$ has constant rank. If

$$\llbracket \Gamma(K), \Gamma(W \cap K^\perp) \rrbracket \subset \Gamma(W + K) , \quad (2.57)$$

then W descends to a subbundle $\underline{W}$ of the reduced Courant algebroid $\underline{E}$ over $\mathcal{Q}$.

Proof. To show that elements of $W \cap K^\perp$ have a well-defined notion of identification under the map $\mathcal{P}: K^\perp \rightarrow K^\perp/K$ when restricted to a leaf $N_q \subset \mathcal{F}$, for some $q \in \mathcal{Q}$, recall the definition of the automorphism Ad_k from Equation (2.43), together with the identification of elements in $K^\perp$ given in the sketch of the proof of Theorem 2.40. If the condition (2.57) holds, then

$$\text{Ad}_k(W \cap K^\perp) \subseteq (W + K) \cap K^\perp = (W \cap K^\perp) + K , \quad (2.58)$$

where we also use Equation (2.38). Therefore $\underline{W} := \mathfrak{d}(W \cap K^\perp)$ is a subbundle of $\underline{E}$, since $W \cap K^\perp$ has constant rank. $\square$

Remark 2.59. If $K \subset W \subset K^\perp$, then Equation (2.57) becomes

$$\llbracket \Gamma(K), \Gamma(W) \rrbracket \subset \Gamma(W) . \quad (2.60)$$

We will be particularly interested in this case in the following sections. Any subbundle W satisfying these properties is pointwise the span of $\Gamma_{\text{bas}}(W)$: there are $w_i \in \Gamma(W)$ such that $[\Gamma(K), w_i] \subset \Gamma(K)$, and $W = \text{Span}\{w_i\}$. The proof can be found in [70, Theorem 4.1].⁴

Adapted Splittings of Exact Courant Algebroids

We give a brief summary of Zambon's notion [50, Section 5] of adapted splittings and the reduced Ševera class.

Let E be an exact Courant algebroid over M , and let K be an isotropic subbundle of E such that $\rho(K^\perp) = TM$ and $\rho(K)$ is an integrable distribution with smooth leaf space $\mathcal{Q}$.

Definition 2.61. A splitting $\sigma: TM \rightarrow E$ of the short exact sequence (2.13) is *adapted to K* if

- (a) the image of σ is isotropic,
- (b) $\sigma(TM) \subset K^\perp$,
- (c) $\sigma(X) \in \Gamma_{\text{bas}}(K^\perp)$, for any vector field X on M which is projectable to $\mathcal{Q}$.

Remark 2.62 (Properties of Adapted Splittings). Adapted splittings σ are in one-to-one correspondence with maximally isotropic subbundles $L_\sigma \subset E$ satisfying

- (i) $\rho(L_\sigma \cap K^\perp) = TM$,
- (ii) $[\Gamma(K), \Gamma(L_\sigma \cap K^\perp)] \subset \Gamma(L_\sigma + K)$,

via the prescription $L_\sigma = \sigma(TM)$.

If σ is an adapted splitting, then

$$\sigma(\rho(K)) \subseteq K, \quad (2.63)$$

see [50, Remark 5.2]. Using this construction, we can establish properties of the anchor through

Lemma 2.64. Let E be an exact Courant algebroid with an involutive isotropic subbundle K such that $\rho(K^\perp) = TM$, endowed with a splitting σ adapted to K . Then $\rho|_K$ is injective and $\sigma \circ \rho|_K = \mathbb{1}_K$.

⁴In their notation, $S = K$, $D = D^S = W$, and “canonical” means existence of basic sections. Note that we do not assume that W is a Dirac structure, and hence W is not necessarily involutive, thus we only have the implication c) $\implies$ a) of [70, Theorem 4.1], rather than the full equivalence.

Proof. Since $K^\perp \cap \rho^*(T^*M) = \rho^*(\text{Ann}(\rho(K)))$, it follows that

$$K^\perp = \sigma(TM) \oplus \rho^*(\text{Ann}(\rho(K))) , \quad (2.65)$$

where we used $\sigma(TM) \cap \rho^*(T^*M) = \{0\}$. Equation (2.65) and the Rank-Nullity Theorem imply

$$\text{rk}(K^\perp) = 2\text{rk}(TM) - \text{rk}(K) + \text{rk}(\ker(\rho|_K)) . \quad (2.66)$$

Since $\text{rk}(K^\perp) = \text{rk}(E) - \text{rk}(K)$, we get $\text{rk}(\ker(\rho|_K)) = 0$, because E is exact (note that ρ^* is injective). Thus $\rho|_K$ is injective.

Since $\rho|_K$ is injective, it follows that $\sigma \circ \rho|_K = \mathbb{1}_K$, because

$$0 = \rho(k) - \rho(k) = \rho(\sigma(\rho(k))) - \rho(k) \implies \sigma(\rho(k)) - k = 0 , \quad (2.67)$$

for all $k \in K$. □

Remark 2.68 (Existence of Adapted Splittings). The condition of having enough basic sections is closely related to the existence of adapted splittings: by [50, Proposition 5.5], splittings adapted to K exist if and only if $\Gamma_{\text{bas}}(K^\perp)$ spans $K^\perp$ pointwise. This result will be very useful in subsequent chapters.

Remark 2.69 (Reduction of Adapted Splittings). By [50, Proposition 5.7], a splitting σ of an exact Courant algebroid E adapted to K , where K satisfies the assumptions of Theorem 2.40, induces a splitting $\underline{\sigma}$ of the reduced Courant algebroid $\underline{E}$ over $\mathcal{Q}$. This follows from property (ii) of Remark 2.62 applied to the maximally isotropic subbundle $L_\sigma = \sigma(TM)$, then applying Proposition 2.56. Thus L_σ reduces to a maximally isotropic subbundle $\underline{L}_{\underline{\sigma}}$ of $\underline{E}$ which is the image of the splitting $\underline{\sigma}$ of $\underline{E}$.

This shows how the three-form H_σ from Equation (2.36) descends to a closed three-form $\underline{H}_{\underline{\sigma}}$ representing the Ševera class of $\underline{E}$: by applying Lemma 2.48 to any $\underline{X}_i \in \Gamma(T\mathcal{Q})$ for $i = 1, 2, 3$ that are the images of projectable vector fields

$X_i \in \Gamma(TM)$, we can lift $\underline{\sigma}(X_i) \in \Gamma(\underline{E})$ to $\sigma(X_i) \in \Gamma_{\text{bas}}(K^\perp)$. Therefore

$$\begin{aligned} H_\sigma(X_1, X_2, X_3) &= \langle \llbracket \sigma(X_1), \sigma(X_2) \rrbracket_E, \sigma(X_3) \rangle_E \\ &= \langle \llbracket \underline{\sigma}(X_1), \underline{\sigma}(X_2) \rrbracket_{\underline{E}}, \underline{\sigma}(X_3) \rangle_{\underline{E}} =: \underline{H}_\sigma(X_1, X_2, X_3), \end{aligned} \quad (2.70)$$

showing that H_σ descends to the three-form $\underline{H}_\sigma$ associated with the reduced splitting $\underline{\sigma}$.

Example 2.71 (Reduction of Twisted Standard Courant Algebroids). Let $(M, \mathcal{F})$ be a foliated manifold with smooth leaf space $\mathcal{Q} = M/\mathcal{F}$. Consider the twisted standard Courant algebroid $(\mathbb{T}M, H)$ for some $H \in \Omega_{\text{cl}}^3(M)$. We take as our maximally isotropic subbundle L_σ in Remark 2.69 to be the tangent bundle

$$L_\sigma = TM \oplus \{0\}. \quad (2.72)$$

Setting $K = T\mathcal{F} \oplus \{0\}$, we have

$$\llbracket X, Y \rrbracket_H = [X, Y] + \iota_Y \iota_X H, \quad (2.73)$$

for every $X \in \Gamma(T\mathcal{F}), Y \in \Gamma(TM)$, hence $\llbracket \Gamma(K), \Gamma(L_\sigma) \rrbracket \subset \Gamma(L_\sigma)$ if and only if $\iota_X H = 0$ for every $X \in \Gamma(T\mathcal{F})$. This suffices to show how H reduces, since $K \subset L_\sigma \subset K^\perp$ in this case, hence Remark 2.59 applies. Thus $\mathcal{L}_X H = 0$, for all $X \in \Gamma(T\mathcal{F})$, hence H is the pullback of a three-form $\underline{H} \in \Omega_{\text{cl}}^3(\mathcal{Q})$ by the quotient map $\varpi : M \rightarrow \mathcal{Q}$, and the reduced Courant algebroid is $\underline{E} = (\mathbb{T}\mathcal{Q}, \underline{H})$. Note that by Remark 2.68, $K^\perp = TM \oplus \text{Ann}(T\mathcal{F})$ has enough basic sections, which, as in Example 2.52, are given by projectable vector fields and one-forms of the form $d\varpi^* f$ for $f \in C^\infty(\mathcal{Q})$.

Chapter 3

Courant Algebroid Relations

In this chapter, we introduce the main concept of this thesis. Courant algebroid relations, as defined in [1], found applications in the theory of quasi-Poisson geometry and quasi-Hamiltonian reduction. We will discuss the closely related concept of Courant algebroid reduction previously discussed in Section 2.1.3. In [1] and in this thesis, we argue how T-duality can be interpreted as a certain Courant algebroid relation.

3.1 Courant Algebroid Relations: Reduction and Composition

In this section we introduce the notion of Courant algebroid relations, following the work of Vysoký [1].

3.1.1 Courant Algebroid Relations

To define Courant algebroid relations, we first need the notion of involutive structures on a Courant algebroid [71].

Definition 3.1. Let E be a Courant algebroid over M . A subbundle L of E supported on a submanifold $C \subset M$ is an *almost involutive structure supported on C* if

- (i) L is isotropic,
- (ii) $L^\perp$ is compatible with the anchor: $\rho(L^\perp) \subset TC$.

Denoting by $\Gamma(E; L)$ the $C^\infty(M)$ -submodule of sections of E which take values in L when restricted to C , then L is *involutive* if

$$[\Gamma(E; L), \Gamma(E; L)] \subseteq \Gamma(E; L) . \quad (3.2)$$

In this case, we call L an *involutive structure supported on C* . If in addition $L = L^\perp$, i.e. L is maximally isotropic, then L is a *Dirac structure supported on C* .

For two Courant algebroids E_1 and E_2 over M_1 and M_2 , respectively, the product $(E_1 \times E_2, [\cdot, \cdot], \langle \cdot, \cdot \rangle, \rho)$ is a Courant algebroid over $M_1 \times M_2$, with the Courant algebroid structures defined by

$$\rho(e_1, e_2) := (\rho_{E_1}(e_1), \rho_{E_2}(e_2)) , \quad (3.3)$$

$$\langle (e_1, e_2), (e'_1, e'_2) \rangle := \langle e_1, e'_1 \rangle_{E_1} \circ \text{pr}_1 + \langle e_2, e'_2 \rangle_{E_2} \circ \text{pr}_2 , \quad (3.4)$$

$$[(e_1, e_2), (e'_1, e'_2)] := ([e_1, e'_1]_{E_1}, [e_2, e'_2]_{E_2}) , \quad (3.5)$$

where $\text{pr}_i: M_1 \times M_2 \rightarrow M_i$ are the projection maps for $i = 1, 2$; for the sake of brevity, we will usually not write the projections explicitly in the following. Denote by $\overline{E}$ the Courant algebroid $(E, [\cdot, \cdot]_E, -\langle \cdot, \cdot \rangle_E, \rho_E)$. From [1, Definition 3.1], we state the following

Definition 3.6. Let E_1 and E_2 be Courant algebroids over M_1 and M_2 respectively.

A *Courant algebroid relation from E_1 to E_2* is a Dirac structure $R \subseteq E_1 \times \overline{E}_2$ supported on a submanifold $C \subseteq M_1 \times M_2$, denoted $R: E_1 \dashrightarrow E_2$.

If C is the graph of a smooth map $\varphi: M_1 \rightarrow M_2$, then R is a *Courant algebroid morphism from E_1 to E_2 over φ* , denoted $R: E_1 \rightarrowtail E_2$.

If R is further the graph of some vector bundle map $\Phi: E_1 \rightarrow E_2$ covering $\varphi: M_1 \rightarrow M_2$, then Φ is a *classical Courant algebroid morphism over φ* .

By viewing a Courant algebroid relation R as a subset of $E_2 \times \overline{E}_1$, we obtain the *transpose relation* $R^\top: E_2 \dashrightarrow E_1$, whose support is denoted $C^\top \subseteq M_2 \times M_1$, and similarly the *transpose morphism* $R^\top: E_2 \rightarrowtail E_1$ when φ is a diffeomorphism.

Note that in [1], the above definition is given in terms of involutive structures, i.e. isotropic subbundles, rather than only Dirac structures. We do not need this degree

of generality in this work and defer to [1, Example 4.33] for motivations justifying this generalisation.

Example 3.7. Let $\Phi: E_1 \rightarrow E_2$ be a vector bundle map which covers a diffeomorphism $\phi: M_1 \rightarrow M_2$. To see why the notion of Courant algebroid relation is well-defined, suppose that its graph $\text{gr}(\Phi) \subseteq E_1 \times \overline{E}_2$ defines a classical Courant algebroid morphism. Then for any $e, e' \in \Gamma(E_1)$, isotropy of $\text{gr}(\Phi)$ gives

$$0 = \langle (e, \Phi(e)), (e', \Phi(e')) \rangle = \langle e, e' \rangle_{E_1} - \langle \Phi(e), \Phi(e') \rangle_{E_2} , \quad (3.8)$$

so Φ is an isometry. Note the minus sign, which is why we consider subbundles of $E_1 \times \overline{E}_2$, rather than of $E_1 \times E_2$. Since $\text{gr}(\Phi)$ is involutive, Φ is a bracket morphism, and hence a Courant algebroid morphism. That $\text{gr}(\Phi)^\perp$ is compatible with the anchor (item (ii) of Definition 3.1) gives Equation (2.20).

Conversely, if Φ is a Courant algebroid isomorphism covering a diffeomorphism ϕ , then $\text{gr}(\Phi)$ is an isotropic and involutive subbundle of $E_1 \times \overline{E}_2$. Since the diagram

$$\begin{array}{ccc} T^*M_1 & \xrightarrow{(\phi^{-1})^*} & T^*M_2 \\ \rho_{E_1}^* \downarrow & & \downarrow \rho_{E_2}^* \\ E_1 & \xrightarrow{\Phi} & E_2 \end{array} \quad (3.9)$$

commutes, where $\rho_{E_i}^*$ are defined by Equation (2.7), $\text{gr}(\Phi)^\perp$ is compatible with the anchor ρ , and hence Φ defines a classical Courant algebroid morphism. This last condition is satisfied when $E_i = \mathbb{T}M_i$ and ρ_{E_i} is the projection to TM_i .

We will also need a notion of sections and functions related by a Courant algebroid relation and its support.

Definition 3.10. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation. Two sections $e_1 \in \Gamma(E_1)$ and $e_2 \in \Gamma(E_2)$ are *R-related*, denoted $e_1 \sim_R e_2$, if $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$.

Two functions $f_1 \in C^\infty(M_1)$ and $f_2 \in C^\infty(M_2)$ are *C-related*, denoted $f_1 \sim_C f_2$, if $f_1(c_1) = f_2(c_2)$ for all $(c_1, c_2) \in C$, that is,

$$\text{pr}_1^* f_1|_C = \text{pr}_2^* f_2|_C . \quad (3.11)$$

For brevity, we will sometimes loosely write $f_1 = f_2$ for C -related functions when no confusion can arise.

A useful result we will make use of is

Lemma 3.12. If $e_1 \sim_R e_2$ and $f_1 \sim_C f_2$, then $\mathcal{L}_{\rho_1(e_1)}f_1 \sim_C \mathcal{L}_{\rho_2(e_2)}f_2$.

Proof. Let $X_i = \rho_i(e_i)$ for $i = 1, 2$. Since R is compatible with the anchor, $(X_1, X_2)_c \in T_c C$ for every $c = (m_1, m_2) \in C$. Let (Φ_1^t, Φ_2^t) be the flow of (X_1, X_2) and $\epsilon > 0$ such that $(\Phi_1^t, \Phi_2^t)(c) \in C$ for $t \in (-\epsilon, \epsilon)$. Then $f_1(\Phi_1^t(m_1)) = f_2(\Phi_2^t(m_2))$ since $f_1 \sim_C f_2$, and the result follows by differentiating this equality with respect to t at $t = 0$. $\square$

3.1.2 Relational Approach to Reduction

In the reduction procedure of Theorem 2.40, there can be no vector bundle morphism between the exact Courant algebroids E and $\underline{E}$. Using the language of Courant algebroid relations, we can describe the reduction of exact Courant algebroids over foliated manifolds, analogously to the approach given in [1, Section 4.3] for equivariant exact Courant algebroids.

Let E be an exact Courant algebroid as in Theorem 2.40, i.e. E is endowed with an isotropic subbundle K such that $\rho(K^\perp) = TM$ and $\Gamma_{\text{bas}}(K^\perp)$ spans $K^\perp$ pointwise, inducing a foliation $\mathcal{F}$ of the base manifold M with smooth leaf space $\mathcal{Q}$ such that the quotient map $\varpi: M \rightarrow \mathcal{Q}$ is a surjective submersion. Thus $\natural: K^\perp \rightarrow \underline{E}$ is a vector bundle map covering ϖ . Consider the vector subbundle, supported on $\text{gr}(\varpi)$, defined fibrewise as

$$Q(K)_{(m, \varpi(m))} = \{ (e, \natural(e)) \mid e \in K_m^\perp \} \subset E_m \times \overline{E}_{\varpi(m)}, \quad (3.13)$$

for any $m \in M$. We show that $Q(K)$ defines an involutive structure on $E \times \overline{E}$, in the sense of Definition 3.1, following the approach used in [1].

Lemma 3.14. $Q(K)$ is a maximally isotropic almost involutive structure.

Proof. Since the pairing on $\underline{E}$ is induced by the pairing on E , it follows that $Q(K)$ is isotropic, proving item (i) of Definition 3.1.

Take $(e, \underline{e}) \in Q(K)^\perp$. Note there is some $\hat{e} \in K^\perp$ such that $\natural(\hat{e}) = \underline{e}$. Then for all $(e', \natural(e')) \in Q(K)$ we have

$$0 = \langle e, e' \rangle_E - \langle \underline{e}, \natural(e') \rangle_{\underline{E}} = \langle e - \hat{e}, e' \rangle_E, \quad (3.15)$$

and hence $e - \hat{e} \in K$. Since $K \subset K^\perp$, it follows that $e \in K^\perp$ and has $\natural(e) = \underline{e}$. Hence $Q(K)^\perp \subseteq Q(K)$, so that $Q(K)$ is Lagrangian, i.e. maximally isotropic.

Since the anchor on $\underline{E}$ descends from the anchor on E , it follows that $Q(K)$ is compatible with the anchor ρ . $\square$

Proposition 3.16. $Q(K)$ is a Dirac structure supported on $\text{gr}(\varpi)$, hence it defines a Courant algebroid morphism $Q(K): E \rightarrow \underline{E}$.

Proof. Suppose e and e' are basic sections for $K^\perp$, and consider the corresponding elements $(e, \natural(e))$ and $(e', \natural(e'))$ of $\Gamma(Q(K))$. Since such sections span $Q(K)$ pointwise, by [1, Proposition 2.23] it is enough to show that $\llbracket (e, \natural(e)), (e', \natural(e')) \rrbracket \in \Gamma(Q(K))$. For every $k \in \Gamma(K)$, using property (i) from Definition 2.1 we compute

$$\langle k, \llbracket e, e' \rrbracket_E \rangle_E = \rho_E(e) \cdot \langle k, e' \rangle_E - \langle \llbracket e, k \rrbracket_E, e' \rangle_E. \quad (3.17)$$

The first term vanishes as $e' \in \Gamma(K^\perp)$. In the second term, by Equation (2.9), we have $\llbracket e, k \rrbracket = -\llbracket k, e \rrbracket$, and since e is basic, the second term vanishes. Hence $\llbracket e, e' \rrbracket_E \in \Gamma(K^\perp)$. Further, because of the Jacobi identity (iii),

$$\llbracket k, \llbracket e, e' \rrbracket_E \rrbracket_E = \llbracket \llbracket k, e \rrbracket_E, e' \rrbracket_E + \llbracket e, \llbracket k, e' \rrbracket_E \rrbracket_E \quad (3.18)$$

and thus by Equation (2.9), the above is a section of K , and so it follows that $\llbracket e, e' \rrbracket_E \in \Gamma_{\text{bas}}(K^\perp)$.

Thus, by the way the bracket on $\underline{E}$ is constructed,

$$\llbracket (e, \natural(e)), (e', \natural(e')) \rrbracket = (\llbracket e, e' \rrbracket_E, \natural(\llbracket e, e' \rrbracket_E)) \in \Gamma(Q(K)), \quad (3.19)$$

hence involutivity follows. $\square$

Following on from Example 2.71, we then have

Corollary 3.20. If $E = (\mathbb{T}M, H)$ is a twisted standard Courant algebroid, let $K = T\mathcal{F} \oplus \{0\}$ where $\mathcal{F}$ is a foliation of M with smooth leaf space $\mathcal{Q} = M/\mathcal{F}$. Then $Q(K)$ is a Courant algebroid relation if and only if $\iota_X H = 0$ for every $X \in \Gamma(T\mathcal{F})$.

In the setting of Corollary 3.20, we denote $Q(K)$ by $Q(\mathcal{F})$.

In light of the above construction of $Q(K)$, we make the following definitions

Definition 3.21. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on C . The *kernel* and *cokernel* are given by

$$K_R := (E_1 \times 0_{E_2})|_C \cap R, \quad \text{and} \quad CK_R := (0_{E_1} \times E_2)|_C \cap R, \quad (3.22)$$

respectively, where 0_{E_i} is the zero section of E_i , for $i = 1, 2$. Note that both K_R and CK_R are possibly singular subbundles of R . We also denote $\text{Ker}(R) = p_1(K_R)$ and $\text{CoKer}(R) = p_2(CK_R)$

Example 3.23. The kernel and cokernel of $Q(K)$ are given by

$$\text{Ker}(Q(K))_m = K_m, \quad \text{CoKer}(Q(K))_{\varpi(m)} = \{0_{\frac{E}{\varpi(m)}}\},$$

for all $(m, \varpi(m)) \in \text{gr}(\varpi)$, where $0_{\frac{E}{\varpi(m)}}: \mathcal{Q} \rightarrow \frac{E}{\varpi(m)}$ is the zero section. This mimics a quotienting behaviour by K for the morphism $Q(K)$.

3.1.3 Composition of Courant Algebroid Relations

It will be particularly relevant in the rest of this paper to discuss the circumstances under which the composition of Courant algebroid relations gives another Courant algebroid relation. In this brief summary we will follow [1, 71].

Definition 3.24. Let $R: E_1 \dashrightarrow E_2$ and $R': E_2 \dashrightarrow E_3$ be Courant algebroid relations. The *composition* $R' \circ R$ is the subset of $E_1 \times \overline{E_3}$ given by

$$R' \circ R = \{(e_1, e_3) \in E_1 \times \overline{E_3} \mid (e_1, e_2) \in R, (e_2, e_3) \in R' \text{ for some } e_2 \in E_2\}. \quad (3.25)$$

The requirements for the set (3.25) to be a Courant algebroid relation are two-fold, given in Propositions 3.30 and 3.32 below, and can be equivalently formulated

at the level of either the Courant algebroid or the base manifold, but both boil down to checking if the set defined by Equation (3.25) is a smooth subbundle of $E_1 \times \overline{E}_3$.

We recall the important definitions and results from [1]. There are two manifolds that are important: $R \times R'$ and $E_1 \times \Delta(E_2) \times \overline{E}_3$, where $\Delta(E)$ is the diagonal embedding of E into $\overline{E} \times E$. Their intersection, which we denote by

$$R' \diamond R = (R \times R') \cap (E_1 \times \Delta(E_2) \times \overline{E}_3) , \quad (3.26)$$

projects to $R' \circ R$. They are the key ingredients to apply the Grabowski-Rotkiewicz Theorem [72, Theorem 2.3] to $R' \diamond R$. The set $R' \diamond R$ is closed under scalar multiplication, but in general it fails to be a submanifold of $E_1 \times \overline{E}_2 \times E_2 \times \overline{E}_3$. If the latter holds as well, it follows immediately that $R' \diamond R$ is a subbundle of $E_1 \times \overline{E}_2 \times E_2 \times \overline{E}_3$ over

$$C' \diamond C := (C \times C') \cap (M_1 \times \Delta(M_2) \times M_3) \quad (3.27)$$

by the Grabowski-Rotkiewicz Theorem. Thus let us briefly discuss the conditions under which $R' \diamond R$ is a submanifold.

Definition 3.28. Two submanifolds C and C' of a manifold M *intersect cleanly in M* if $C \cap C'$ is a submanifold of M , and

$$T_c(C \cap C') = T_c C \cap T_c C' , \quad (3.29)$$

for each $c \in C \cap C'$.

The inclusion $\subseteq$ is always true. The importance of this condition is that if C and C' intersect cleanly, then C and C' look locally like a pair of intersecting vector subspaces; see e.g. [73, Proposition C.3.1].

Following Vysoký [1], we state two propositions that are essential for the characterisation of composition of relations and will be needed in Chapter 4.

Proposition 3.30. Let $R: E_1 \dashrightarrow E_2$ and $R': E_2 \dashrightarrow E_3$ be Courant algebroid relations over $C \subseteq M_1 \times M_2$ and $C' \subseteq M_2 \times M_3$ respectively. The following conditions are equivalent:

- (i) $R \times R'$ and $E_1 \times \Delta(E_2) \times \overline{E}_3$ intersect cleanly.
- (ii) $C \times C'$ and $M_1 \times \Delta(M_2) \times M_3$ intersect cleanly, and the dimension of $(R' \diamond R)_c$ is independent of $c \in C' \diamond C$.

Both these conditions ensure that $R' \diamond R$ is a subbundle of $E_1 \times \overline{E}_2 \times E_2 \times \overline{E}_3$ over $C' \diamond C$.

Similar conditions can be stated for $R' \circ R$ in order to make it a subbundle of $E_1 \times \overline{E}_3$ over

$$C' \circ C := \{(m_1, m_3) \in M_1 \times M_3 \mid (m_1, m_2) \in C, (m_2, m_3) \in C' \text{ for some } m_2 \in M_2\} \quad (3.31)$$

since $R' \circ R$ is an $\mathbb{R}$ -module.

Proposition 3.32. Let $R: E_1 \dashrightarrow E_2$ and $R': E_2 \dashrightarrow E_3$ be Courant algebroid relations over $C \subseteq M_1 \times M_2$ and $C' \subseteq M_2 \times M_3$ respectively. Suppose at least one of the conditions of Proposition 3.30 are satisfied. The following conditions are equivalent:

- (i) $R' \circ R$ is a submanifold of $E_1 \times \overline{E}_3$ such that the induced map¹ $p: R' \diamond R \rightarrow R' \circ R$ is a smooth surjective submersion.
- (ii) $C' \circ C$ is a submanifold of $M_1 \times M_3$ such that the induced map $\pi: C' \diamond C \rightarrow C' \circ C$ is a smooth surjective submersion and the rank of the linear map $p: (R' \diamond R)_c \rightarrow (R' \circ R)_{\pi(c)}$ is independent of $c \in C' \diamond C$.

Both these conditions ensure that $R' \circ R$ is a subbundle of E' supported on $C' \circ C$ and $p: R' \diamond R \rightarrow R' \circ R$ is a fibrewise surjective vector bundle map over $\pi: C' \diamond C \rightarrow C' \circ C$.

If any of the two equivalent conditions (i) or (ii) occurs, we say that R and R' *compose cleanly*. Vysoký shows in [1] that Propositions 3.30 and 3.32 together give

Theorem 3.33. Let $R: E_1 \dashrightarrow E_2$ and $R': E_2 \dashrightarrow E_3$ be Courant algebroid relations over C and C' , respectively, which compose cleanly. Then $R' \circ R$ is an involutive structure supported on $C' \circ C$, hence it defines a Courant algebroid relation $R' \circ R: E_1 \dashrightarrow E_3$.

¹Here $p: E_1 \times \overline{E}_2 \times E_2 \times \overline{E}_3 \rightarrow E_1 \times \overline{E}_3$ is the projection to the first and the fourth factor of the Cartesian product, and similarly for $\pi: M_1 \times M_2 \times M_2 \times M_3 \rightarrow M_1 \times M_3$ below.

Example 3.34 (Relations between Reduced Courant Algebroids). Let K be an isotropic subbundle of an exact Courant algebroid E over M such that $\rho(K^\perp) = TM$. Suppose there exists a subbundle $K_1 \subset K$ such that $K_1^\perp$ has enough basic sections which are also basic with respect to K . Then since $K^\perp \subset K_1^\perp$, both K and K_1 are involutive subbundles, hence they induce foliations $\mathcal{F}$ and $\mathcal{F}_1$ of M , respectively, such that $\mathcal{F}_1 \subset \mathcal{F}$. We also assume that they have smooth leaf spaces $\mathcal{Q}_1 = M/\mathcal{F}_1$ and $\mathcal{Q} = M/\mathcal{F}$, so that we can complete the diagram

$$\begin{array}{ccc} & M & \\ \varpi_1 \swarrow & & \searrow \varpi \\ \mathcal{Q}_1 & \xrightarrow{\pi} & \mathcal{Q} \end{array} \quad (3.35)$$

of surjective submersions.

Since $K^\perp$ and $K_1^\perp$ have enough basic sections, we may form the Courant algebroid relations $Q(K)$ and $Q(K_1)$, which fit into the diagram

$$\begin{array}{ccc} & E & \\ Q(K_1) \swarrow & & \searrow Q(K) \\ \underline{E}_1 & & \underline{E} \end{array} \quad (3.36)$$

We consider the composition of $Q(K)$ with $Q(K_1)^\top$, and check the conditions in Propositions 3.30 and 3.32.

It can be shown that the submanifolds $\text{gr}(\varpi_1)^\top \times \text{gr}(\varpi)$ and $\mathcal{Q}_1 \times \Delta(M) \times \mathcal{Q}$ intersect cleanly, and the projection $\text{gr}(\varpi) \diamond \text{gr}(\varpi_1)^\top \rightarrow \text{gr}(\pi)$ is a smooth surjective submersion. For $(q_1, \pi(q_1)) \in \text{gr}(\pi)$, the space

$$(Q(K) \diamond Q(K_1)^\top)_{(q_1, \pi(q_1))} = \{ (\natural_{E_1}(e), e, e, \natural(e)) \mid e \in K_m^\perp \} \quad (3.37)$$

is isomorphic to $K_m^\perp$, for any $m \in M$ such that $q_1 = \varpi_1(m)$, and hence has constant dimension. Finally, the projection $Q(K) \diamond Q(K_1)^\top \rightarrow Q(K) \circ Q(K_1)^\top$ has kernel given by $K \cap K_1 = K_1$, and hence has constant rank.

Thus the conditions (i) of both Propositions 3.30 and 3.32 are satisfied, and the composition is clean, giving the relation

$$R(\mathcal{F}_1) := Q(K) \circ Q(K_1)^\top = \{ (\natural_{E_1}(e), \natural(e)) \mid e \in K^\perp \} \subset \underline{E}_1 \times \overline{\underline{E}}, \quad (3.38)$$

supported on the graph $\text{gr}(\pi)$.

Example 3.34 may be seen as the foliated reduction counterpart of the well-known case of reduction of a $\mathbf{G}$ -equivariant exact Courant algebroid by a closed Lie subgroup² $\mathbf{H} \subset \mathbf{G}$ that represents one of the building blocks of Poisson-Lie T-duality (see e.g. [49, Section 5.1]).

3.2 Courant Algebroid Relations as Isometries

In this section we review important ideas of [1, Section 5]. We then generalise these to cases of interest to us, proceeding to define generalised isometries for transverse generalised metrics. We begin with some preliminaries on generalised metrics, the generalised geometric counterpart to Riemannian metrics.

3.2.1 Generalised Metrics

We wish to know when a given Courant algebroid relation is able to carry geometric structure between Courant algebroids. In particular, we are interested in relations between generalised metrics as a building block for our approach to T-duality.

Definition 3.39. A *generalised metric* on a Courant algebroid E is an automorphism $\tau: E \rightarrow E$ covering the identity with $\tau^2 = 1_E$ which, together with the pairing $\langle \cdot, \cdot \rangle$, defines a positive-definite fibrewise metric

$$\mathcal{G}(e, e') = \langle e, \tau(e') \rangle \quad (3.40)$$

on E , for $e, e' \in E$. We denote the Courant algebroid E endowed with a generalised metric τ by (E, τ) .

Generalised metrics on exact Courant algebroids are characterised as follows, see e.g. [13, Proposition 3.5] for details.

Proposition 3.41. Let E be an exact Courant algebroid over M . A generalised metric $\mathcal{G}$, as in Equation (3.40), determines a pair $(g, b) \in \Gamma(\odot^2 T^*M) \times \Omega^2(M)$

²This is usually chosen to be Lagrangian with respect to the split-signature pairing.

where g is non-degenerate. Conversely, a Riemannian metric g and a two-form b on M define a generalised metric $\mathcal{G}$ given by

$$\mathcal{G} = \begin{pmatrix} g - b g^{-1} b & b g^{-1} \\ -g^{-1} b & g^{-1} \end{pmatrix} \quad (3.42)$$

on the isomorphic twisted standard Courant algebroid $\mathbb{T}M = TM \oplus T^*M$.

An alternative characterisation of generalised metrics is

Definition 3.43. A *generalised metric* is a subbundle $V^+ \subseteq E$ which is a maximal positive-definite subbundle of E with respect to $\langle \cdot, \cdot \rangle$. We denote the Courant algebroid E endowed with a generalised metric V^+ by (E, V^+) .

Remark 3.44 (Generalised Metrics as Subbundles). Definitions 3.39 and 3.43 are equivalent, see e.g. [13, Proposition 3.3]: the maximal positive-definite subbundle V^+ is the $+1$ -eigenbundle for the generalised metric τ . Indeed, if τ is given by g and b as in Proposition 3.41, then one can show

$$V^+ = \text{gr}(g + b) = \{ v + \iota_v(g + b) \in \mathbb{T}M \mid v \in TM \} . \quad (3.45)$$

The -1 -eigenbundle of τ is given by $V^- = \text{gr}(-g + b) = (V^+)^\perp$.

Since Definitions 3.39 and 3.43 are equivalent, we can switch between the notations (E, τ) and (E, V^+) .

3.2.2 Generalised Isometries

We now have enough machinery to talk about a notion of isometries of generalised metrics for Courant algebroid relations, following [1].

Definition 3.46. Suppose $R: E_1 \dashrightarrow E_2$ is a Courant algebroid relation supported on a submanifold $C \subseteq M_1 \times M_2$. Let τ_1 and τ_2 be generalised metrics for E_1 and E_2 respectively, and set $\tau := \tau_1 \times \tau_2$. Then R is a *generalised isometry between τ_1 and τ_2* if $\tau(R) = R$.

It follows from the definitions that $\tau = \tau_1 \times \tau_2$ is a generalised metric for $E_1 \times E_2$. It is not, however, a generalised metric for $E_1 \times \overline{E}_2$: the ± 1 -eigenbundles are not

necessarily maximally positive-definite with respect to the pairing on $E_1 \times \overline{E}_2$. See [1, Remark 5.2] for details.

Remark 3.47 (Classical Generalised Isometries). If R is a classical Courant algebroid morphism, then $R = \text{gr}(\Phi)$ for some vector bundle map $\Phi : E_1 \rightarrow E_2$ covering a smooth map $\phi : M_1 \rightarrow M_2$. For $\text{gr}(\Phi)$ to be a generalised isometry, the equation

$$\tau_2 \circ \Phi = \Phi \circ \tau_1 \quad (3.48)$$

must hold, or equivalently

$$\mathcal{G}_2(\Phi(e_1), \Phi(e'_1))_{\phi(m_1)} = \mathcal{G}_1(e_1, e'_1)_{m_1} , \quad (3.49)$$

for each $e_1, e'_1 \in (E_1)_{m_1}$ with $m_1 \in M_1$, for the metrics $\mathcal{G}_1$ and $\mathcal{G}_2$ induced by τ_1 and τ_2 respectively. This justifies the terminology ‘generalised isometry’ [1, Example 5.3].

For later purposes, let us stress that a further equivalent condition for a classical Courant algebroid morphism Φ to be a generalised isometry is

$$\Phi(V_1^+) = V_2^+ , \quad (3.50)$$

where $V_i^+ = \ker(\mathbb{1}_{E_i} - \tau_i)$ is the $+1$ -eigenbundle of τ_i for $i = 1, 2$. From this, setting $V_i^- := (V_i^+)^\perp$, it follows that

$$\text{gr}(\Phi) = \text{gr}(\Phi|_{V_1^+}) \oplus \text{gr}(\Phi|_{V_2^+}). \quad (3.51)$$

When Φ is a Courant algebroid isomorphism, generalised isometries are characterised by

Proposition 3.52. Let E_1 and E_2 be exact Courant algebroids over M_1 and M_2 , respectively, and suppose they are endowed with generalised metrics τ_1 and τ_2 , respectively. A Courant algebroid isomorphism $\Phi : E_1 \rightarrow E_2$, covering $\phi \in \text{Diff}(M_1, M_2)$,

is a generalised isometry if and only if

$$\phi^* g_2 = g_1 \quad \text{and} \quad \phi^* b_2 = b_1 + B, \quad (3.53)$$

where, by using the canonical isomorphism $E_i \simeq TM_i \oplus TM_i^*$ given by any isotropic splitting, τ_i is determined by the pair (g_i, b_i) for $i = 1, 2$, and (with a slight abuse of notation) $\Phi = \bar{\phi} \circ e^B$ for some $B \in \Omega^2(M_1)$.

Proof. By choosing a splitting for E_1 , its generalised metric is $V_1^+ \simeq \text{gr}(g_1 + b_1)$. It then follows that

$$\Phi(v_1 + \iota_{v_1}(g_1 + b_1)) = \phi_*(v_1) + (\phi^{-1})^*(\iota_{v_1}(g_1 + b_1 + B)), \quad (3.54)$$

for all $v_1 \in TM_1$, is an element in V_2^+ if and only if $\phi^* g_2 = g_1$ and $\phi^* b_2 = b_1 + B$, since $V_2^+ \simeq \text{gr}(g_2 + b_2)$ in the chosen splitting for E_2 . $\square$

Remark 3.55. Proposition 3.52 reduces to [28, Proposition 2.41] when $M_1 = M_2 = M$ and $\tau_1 = \tau_2 = \tau$, i.e. in this case Φ is a generalised isometry if and only if $\phi^* g = g$ and $B = 0$.

Following the equivalence between Definitions 3.39 and 3.43, we can formulate an alternative definition of generalised isometry in terms of the eigenbundles of the automorphisms τ_1 and τ_2 .

Definition 3.56. Suppose $R: E_1 \dashrightarrow E_2$ is a Courant algebroid relation supported on a submanifold $C \subseteq M_1 \times M_2$. Let V_1^+ and V_2^+ be generalised metrics for E_1 and E_2 respectively. Then R is a *generalised isometry between V_1^+ and V_2^+* if

$$R_c = R_c^+ \oplus R_c^-, \quad (3.57)$$

for each $c \in C$, where $R_c^\pm = (V_1^\pm \times V_2^\pm)_c \cap R_c$.

We briefly comment on the pointwise nature of this definition, which will be a theme throughout this section. Due to the nature of relations, we only consider their pointwise description, and do not demand that our definition holds globally over the submanifold C that R is supported on. In particular, in Equation (3.57) we do not

demand that $R^\pm$ are both subbundles supported on C , since the intersection might not have constant rank. We could distinguish from cases where global descriptions (on C) are possible, however this is not necessary for the description of generalised isometries, as made clear through

Proposition 3.58. Definition 3.46 and Definition 3.56 are equivalent.

Proof. Suppose $\tau(R) = R$. Let $\mathcal{V}^\pm = V_1^\pm \times V_2^\pm$. The bundles $\mathcal{V}^\pm$ are the ± 1 -eigenbundles for $\tau = \tau_1 \times \tau_2$. Take $c \in C$ and let $r \in R_c$. Since $E_1 \times \overline{E}_2 = \mathcal{V}^+ \oplus \mathcal{V}^-$, we can write $r = r^+ + r^-$, for $r^\pm \in \mathcal{V}_c^\pm$. Thus

$$\tau(r^+ + r^-) = r^+ - r^- \in R_c. \quad (3.59)$$

Hence both $r^+ + r^-$ and $r^+ - r^-$ are in R_c , so that $r^+, r^- \in R_c$, and it follows that $R_c \subseteq R_c^+ \oplus R_c^-$, hence Equation (3.57) holds.

Conversely, if R_c decomposes as in Equation (3.57) for every $c \in C$, then each $r \in R_c$ can be written as $r = r^+ + r^-$ where $r^\pm \in R_c^\pm$. Hence $\tau(r) = r^+ - r^- \in R_c$, so $\tau(R_c) = R_c$. Since this holds for every $c \in C$, it follows that $\tau(R) = R$. $\square$

Composition of generalised isometries works well: the composition $R' \circ R$ of two Courant algebroid relations that are generalised isometries and that compose cleanly is also a generalised isometry, see [1, Proposition 5.7].

Example 3.60 ($Q(K)$ is not a Generalised Isometry). Let us look at the relation $Q(K)$ of Section 3.1.2, where K is an isotropic subbundle³ inducing a regular foliation $\mathcal{F}$ of M given by the integral manifolds of $\rho(K)$ with smooth leaf space $\mathcal{Q} = M/\mathcal{F}$ and projection map $\varpi : M \rightarrow \mathcal{Q}$. Take generalised metrics $\mathcal{V}^+ \subset E$ and $V^+ \subset \underline{E}$: a generalised metric V^+ may be constructed from $\mathcal{V}^+$ whenever $W = \mathcal{V}^+$ satisfies Equation (2.57) of Proposition 2.56. This is similar to [1, Example 5.6], where Equation (2.57) is tantamount to $\mathbf{G}$ -equivariance; therein Vysoký concludes that $Q(K)$ is not a generalised isometry.

In our case we argue as follows. If $Q(K)$ is a generalised isometry between $\mathcal{V}^+$ and V^+ , at a point $(m, \varpi(m)) \in \text{gr}(\varpi)$ we could write

$$Q(K)_{(m, \varpi(m))} = Q(K)_{(m, \varpi(m))}^+ \oplus Q(K)_{(m, \varpi(m))}^-, \quad (3.61)$$

³We always assume that K has non-zero rank.

where $Q(K)^\pm = (\mathcal{V}^\pm \times V^\pm) \cap Q(K)$. Since $(k, 0_{\varpi(m)}^E) \in Q(K)_{(m, \varpi(m))}$, for $k \in K_m$, this implies

$$(k, 0_{\varpi(m)}^E) = (k^+, 0_{\varpi(m)}^E) + (k^-, 0_{\varpi(m)}^E) , \quad (3.62)$$

where $(k^\pm, 0_{\varpi(m)}^E) \in Q(K)_{(m, \varpi(m))}^\pm$. Since these live inside $Q(K)$, and the kernel of $\natural$ is K , it follows that $k^\pm \in K_m$. Thus $k^\pm \in K_m \cap \mathcal{V}_m^\pm$. In particular

$$0 < \langle k^+, k^+ \rangle_E = 0 , \quad (3.63)$$

which is a contradiction. Hence $Q(K)$ cannot be a generalised isometry.

Notice that the isotropy of K here plays a crucial role. It is shown in [1, Example 5.6] that $Q(K)$ may be a generalised isometry for non-isotropic $\mathbf{G}$ -actions only if $K \cap K^\perp = \{0\}$.

Example 3.60 motivates a characterising property of generalised isometries given by

Proposition 3.64. Let $R: E_1 \dashrightarrow E_2$ be a generalised isometry between $\tau_1 \in \text{Aut}(E_1)$ and $\tau_2 \in \text{Aut}(E_2)$ supported on $C \subseteq M_1 \times M_2$. Then $\text{Ker}(R)_{p_1(c)} = \{0_{p_1(c)}^{E_1}\}$, and $\text{CoKer}(R)_{p_2(c)} = \{0_{p_2(c)}^{E_2}\}$ for every $c \in C$.

Proof. Let $c = (m_1, m_2) \in C$ and $k_1 \in \text{Ker}(R)_{m_1}$. Then

$$(\tau_1 \times \tau_2)(k_1, 0) = (\tau_1(k_1), 0) \in \text{Ker}(R)_{m_1} \times \{0_{m_2}^{E_2}\} , \quad (3.65)$$

hence $\text{Ker}(R)_{m_1}$ is invariant under τ_1 . Since R is isotropic, it follows that $\text{Ker}(R)_{m_1}$ is also isotropic, as $\langle k_1, k'_1 \rangle_{E_1} = \langle k_1, k'_1 \rangle_{E_1} - \langle 0, 0 \rangle_{E_2} = \langle (k_1, 0), (k'_1, 0) \rangle = 0$ for all $k_1, k'_1 \in \text{Ker}(R)_{m_1}$, hence

$$\mathcal{G}_1(k_1, k_1) = \langle k_1, \tau_1(k_1) \rangle_{E_1} = 0 . \quad (3.66)$$

Therefore $\text{Ker}(R)_{m_1} = \{0_{m_1}^{E_1}\}$ for every $m_1 \in \text{pr}_1(C)$.

Similarly $\text{CoKer}(R)_{m_2} = \{0_{m_2}^{E_2}\}$ for every $m_2 \in \text{pr}_2(C)$. $\square$

A useful uniqueness result for generalised metrics now follows from

Corollary 3.67. Let $R: E_1 \dashrightarrow E_2$ be a generalised isometry between $\tau_1 \in \text{Aut}(E_1)$ and $\tau_2 \in \text{Aut}(E_2)$ which is also a generalised isometry between $\tau_1 \in \text{Aut}(E_1)$ and $\tau'_2 \in \text{Aut}(E_2)$. Then $\tau_2|_{\text{pr}_2(R)} = \tau'_2|_{\text{pr}_2(R)}$, where $\text{pr}_2: E_1 \times \overline{E}_2 \rightarrow \overline{E}_2$ is the projection.

Proof. Let $(e_1, e_2) \in R$. Then $(\tau_1(e_1), \tau_2(e_2)) \in R$ and $(\tau_1(e_1), \tau'_2(e_2)) \in R$, therefore $(0, \tau_2(e_2) - \tau'_2(e_2)) \in R$. Hence $\tau_2(e_2) = \tau'_2(e_2)$ for all $e_2 \in \text{pr}_2(R)$ by Proposition 3.64. $\square$

3.2.3 Transverse Generalised Isometries

Example 3.60 and Proposition 3.64 highlight the need to go beyond the notion of generalised isometry. In particular, we wish to extend the definition of generalised isometry to the case when the kernel and cokernel of R are non-trivial. For this, we assume from now on that the subsets $\text{Ker}(R)$ and $\text{CoKer}(R)$ are subbundles supported on M_1, M_2 , respectively. Hence $\text{pr}_i(C) = M_i$, where $\text{pr}_i: M_1 \times M_2 \rightarrow M_i$ are the Cartesian projections for $i = 1, 2$.

Transverse Generalised Metrics

We recall the notion of transverse generalised metrics, as first introduced in [74], which will aid in understanding the properties of our extension of generalised isometry.

Definition 3.68. Let E be an exact Courant algebroid over M and K an involutive isotropic subbundle of E . A subbundle $W \subset E$ of rank $\text{rk}(W) = \dim(M)$ is a *pre- K -transverse generalised metric* if $K \subset W \subset K^\perp$ and

$$\langle w, w \rangle_E > 0, \quad (3.69)$$

for all $w \in W$ with $w \notin K$. A pre- K -transverse generalised metric W is a *K -transverse generalised metric* if it is invariant with respect to K , in the sense that

$$[\Gamma(K), \Gamma(W)]_E \subseteq \Gamma(W). \quad (3.70)$$

An exact Courant algebroid E endowed with a transverse generalised metric W is denoted by (E, W) .

Note that this is the same condition appearing in the context of reducible subbundles of exact Courant algebroids, see Remark 2.59.

Remark 3.71 (Transverse Generalised Metrics as Graphs). Similarly to Remark 3.44 (see Equation (3.45)), suppose that the restriction of the anchor $\rho_E|_K: K \rightarrow TM$ to K is injective, and that W is a pre- K -transverse generalised metric. Then in a given splitting, there is a (degenerate) symmetric bilinear pairing g and a (degenerate) two-form b , with $\ker(b) \supset \ker(g)$, such that

$$W \simeq \text{gr}(g + b) = \{ v + \iota_v(g + b) \in TM \mid v \in TM \} . \quad (3.72)$$

In this splitting, we have the isomorphism $K \simeq \ker(g)$.

If W is a K -transverse generalised metric, then

$$\mathcal{L}_X g = \mathcal{L}_X b = 0 \quad \text{and} \quad \iota_X H = 0 , \quad (3.73)$$

for every $X \in \Gamma(\rho_E(K))$, where $H \in \Omega_{\text{cl}}^3(M)$ represents the Ševera class of E . The converse is also true.

The result of Remark 3.71 allows an interpretation in terms of Riemannian submersions and Riemannian foliations, as shown in [74].

Example 3.74 (Riemannian Foliations). This construction extends to any manifold (M, g) endowed with a foliation $\mathcal{F}$ such that the degenerate symmetric bilinear tensor g satisfies $\ker(g) = T\mathcal{F}$ and is leaf-invariant:

$$\mathcal{L}_X g = 0 , \quad (3.75)$$

for all $X \in \Gamma(T\mathcal{F})$. In other words, $(M, g, \mathcal{F})$ is a Riemannian foliation. It is easy to see that

$$W = \text{gr}(g) \quad (3.76)$$

is a K -transverse generalised metric, where $K \simeq T\mathcal{F}^\perp$ is an involutive isotropic subbundle of TM and $K^\perp = TM \oplus \text{Ann}(T\mathcal{F})$.

Relations for Transverse Generalised Metrics

The main goal of this section may be rephrased as giving a definition of isometry of transverse generalised metrics. In order to provide more insight into what our definition should look like, let us first explore the properties of the most natural way to extend generalised isometries: by adapting Equation (3.50) to (pre-)transverse generalised metrics.

Remark 3.77 (Isomorphisms and Transverse Generalised Metrics). Suppose E_1 and E_2 are exact Courant algebroids over M_1 and M_2 , respectively, which are isomorphic via a map $\Phi: E_1 \rightarrow E_2$ covering a diffeomorphism $\phi: M_1 \rightarrow M_2$. Suppose K_1 and K_2 are involutive isotropic subbundles of E_1 and E_2 , respectively, and that W_i are pre- K_i -transverse generalised metrics for $i = 1, 2$. The natural definition would be to call Φ an isometry of these metrics if $\Phi(W_1) = W_2$. This turns out to be too restrictive.

Let us discuss the conditions arising from this choice. Upon choosing splittings for E_i , we have $\Phi = \bar{\phi} \circ e^B$ for some $B \in \Omega^2(M_1)$, and $W_i \simeq \text{gr}(g_i + b_i)$ for some $g_i \in \Gamma(\odot^2 T^*M_i)$ and $b_i \in \Omega^2(M_i)$. As in Proposition 3.52, $\Phi(W_1) = W_2$ if and only if $\phi^*g_2 = g_1$ and $\phi^*b_2 = b_1 + B$. However, the condition that ϕ should map the degenerate symmetric tensors g_i into each other implies that

$$\phi_*(\ker(g_1)) = \ker(g_2) , \quad (3.78)$$

which is equivalent to requiring $\Phi(K_1) = K_2$. In particular, if M_i are endowed with Riemannian foliations, by Example 3.74 this is tantamount to requiring that ϕ is a foliation-preserving diffeomorphism.

We would like to introduce a notion of generalised isometry for transverse generalised metrics that is not constrained to being foliation-preserving. This choice is motivated by the possibility to formalise the description of T-duality where T-dual manifolds may not be diffeomorphic, and hence the foliations defining the reductions to these backgrounds cannot be bijectively mapped into each other (see Section 4.1 for more details). Instead we consider $W_i^+ := W_i/K_i$. We then require $\Phi(\widetilde{W}_1^+) = \widetilde{W}_2^+$ for some lifts $\widetilde{W}_i^+$ of W_i^+ to E_i . In order to extend this condition to a general Courant algebroid relation, we mimic Definition 3.56 using this idea.

Definition 3.79. Let E_1 and E_2 be Courant algebroids over M_1 and M_2 , respectively, and let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on $C \subset M_1 \times M_2$. Suppose K_i are involutive isotropic subbundles of E_i , and that W_i are pre- K_i -transverse generalised metrics for $i = 1, 2$. Set $W_i^+ := W_i/K_i$ and $W_i^- := W_i^\perp/K_i$.

Then R is a *transverse generalised isometry*⁴ between W_1 and W_2 if there are pointwise lifts $\widetilde{W}_i^\pm \subseteq W_i$ of $W_i^\pm$ to E_i such that

$$R_c \cap (\mathcal{W}_c^+ \oplus \mathcal{W}_c^-) = (R_c \cap \mathcal{W}_c^+) \oplus (R_c \cap \mathcal{W}_c^-) , \quad (3.80)$$

for all $c \in C$, where $\mathcal{W}^\pm = \widetilde{W}_1^\pm \times \widetilde{W}_2^\pm$. A transverse generalised isometry R is *regular* if the lifts $\widetilde{W}_i^\pm$ are subbundles of E_i for $i = 1, 2$.

Note that the inclusion $\supseteq$ in Equation (3.80) always holds.

Definition 3.79 may be better understood by looking at the case of a classical Courant algebroid morphism.

Remark 3.81 (Lifts for Transverse Generalised Isometries). The lift $\widetilde{W}_i^+$ is a choice of subbundle of W_i such that

$$\langle w_i, w_i \rangle_{E_i} > 0 , \quad (3.82)$$

for every $w_i \in \widetilde{W}_i^+$. In other words, it is a choice of splitting $s_{m_i}^+: (W_i^+)_{m_i} \rightarrow (W_i)_{m_i}$, at each point $m_i \in M_i$, of the short exact sequence

$$0 \longrightarrow (K_i)_{m_i} \longrightarrow (W_i)_{m_i} \longrightarrow (W_i^+)_{m_i} \longrightarrow 0 , \quad (3.83)$$

with $(\widetilde{W}_i^+)_{m_i} = s_{m_i}^+((W_i^+)_{m_i})$. Similarly, the lift $\widetilde{W}_i^-$ is a choice of negative definite subbundle with respect to the pairing $\langle \cdot, \cdot \rangle_{E_i}$.

We may think of $\widetilde{W}_i^+$ as the counterpart of V_i^+ for transverse generalised metrics, whereas K can be interpreted as the degenerate part of W_i . Definition 3.79 is thus the analogue of Definition 3.56 such that only the positive definite components of

⁴We are aware that the terminology ‘transverse generalised isometry’ might be a misnomer, since these relations do not necessarily preserve the subbundles K_i defining the transversality conditions. We chose this name for the sake of simplicity, since the K_i -preserving case is included in our definition.

W_i are related. Note that we recover Definition 3.56 for generalised metrics when K_i is the zero subbundle.

If the splitting s^+ varies smoothly, then we may consider the short exact sequence (3.83) to be a short exact sequence of vector bundles and the splitting is a bundle map, thus R is a regular transverse generalised isometry. This will be the case in Section 4.1.

The main case in which regular transverse generalised isometries exist is given by

Theorem 3.84. Consider the reduction relation $Q(K): E \rightarrow \underline{E}$ over $\varpi: M \rightarrow \mathcal{Q}$ of Section 3.1.2 and let $V^+ \subset \underline{E}$ be a generalised metric on $\underline{E}$. There exists a unique K -transverse generalised metric W on E such that $Q(K)$ is a regular transverse generalised isometry between W and V^+ .

Conversely, if there is a K -transverse generalised metric W on E , then $V^+ := \mathfrak{h}(W)$ is a generalised metric on $\underline{E}$ and $Q(K)$ is a regular transverse generalised isometry between W and V^+ .

Proof. Take a generalised metric V^+ on $\underline{E}$. Recall that for every $q \in \mathcal{Q}$ there is an isomorphism $\mathcal{J}_{q,m}: \underline{E}_q \rightarrow K_m^\perp/K_m$ given by Equation (2.44). Define $W_m^+ = \mathcal{J}_{q,m}(V_q^+)$. For two points $m, m' \in \mathcal{F}_q$, take $w^+ \in W_m^+$ and $w'^+ \in W_{m'}^+$ such that $\mathcal{J}_{q,m}^{-1}(w^+) = \mathcal{J}_{q,m'}^{-1}(w'^+)$. By the construction in the sketched proof of Theorem 2.40, there is a basic section w such that, under the projection $K^\perp \rightarrow K^\perp/K$, w maps to w^+ at m and w'^+ at m' . Take W to be the span of such basic sections w . Then $K \subset W \subset K^\perp$ and $\langle w, w \rangle_E > 0$ for every $w \in W$ such that $w \notin K$. Hence W is a pre- K -transverse generalised metric on E .

By Remark 2.59, for $w \in W$ we can write $w = \sum_i f_i w_i$, for $f_i \in C^\infty(M)$ (so that the sum is locally finite) and $w_i \in \Gamma_{\text{bas}}(W)$. Then

$$\llbracket k, w \rrbracket_E = \sum_i f_i \llbracket k, w_i \rrbracket_E + \sum_i (\rho_E(k) \cdot f_i) w_i, \quad (3.85)$$

for each $k \in K$. Since w_i are basic, it follows that $\llbracket \Gamma(K), \Gamma(W) \rrbracket_E \subseteq \Gamma(W)$, hence W is a K -transverse generalised metric and $\mathfrak{h}(W) = V^+$.

Uniqueness is seen as follows: Suppose there exists another K -transverse generalised metric W' such that $\mathfrak{h}(W') = V^+$. By Remark 2.59, it follows that W'

is spanned pointwise by $\{w'_i\} \subset \Gamma_{\text{bas}}(W')$. Thus $v_i = \natural(w'_i)$ give a set of sections spanning V^+ pointwise. Now we apply the previous construction to v_i to construct W . Hence we get W spanned by w_i with $\natural(w_i) = \natural(w'_i)$. Therefore w_i and w'_i differ by an element in K , so $W = W'$.

Conversely, starting with a K -transverse generalised metric W on E , then $V^+ := \natural(W)$ defines a generalised metric on $\underline{E}$, since the K -transversality condition makes V_+ a well-defined subbundle of $\underline{E}$ over $\mathcal{Q}$ by Proposition 2.56 and

$$\langle v, v \rangle_{\underline{E}} = \langle w, w \rangle_E \geq 0, \quad (3.86)$$

for every $v \in V^+$, with equality if and only if $w \in K$, i.e. $v = 0$. It also follows that $\text{rk}(V^+) = \dim(\mathcal{Q})$.

We now show, in both cases, that $Q(K)$ is a regular transverse generalised isometry between W and V^+ . Fix $p = (m, \varpi(m)) \in \text{gr}(\varpi) \subset M \times \mathcal{Q}$. Take arbitrary smooth splittings $\widetilde{W}^\pm$ of the sequence (3.83), i.e. a splitting of the short exact sequence of vector bundles

$$0 \longrightarrow K \longrightarrow W \longrightarrow W^+ \longrightarrow 0 \quad (3.87)$$

over M , and set $\mathcal{W}^\pm = \widetilde{W}^\pm \times V^\pm$. Then

$$Q(K)_p \cap (\mathcal{W}^+ \oplus \mathcal{W}^-)_p = \{ (e, \natural(e)) \mid e \in K_m^\perp \cap (\widetilde{W}_m^+ \oplus \widetilde{W}_m^-) \}. \quad (3.88)$$

Thus if $(e, \natural(e)) \in Q(K)_p \cap (\mathcal{W}^+ \oplus \mathcal{W}^-)_p$, then

$$(e, \natural(e)) = (e_+ + e_-, \natural(e_+ + e_-)), \quad (3.89)$$

with $e_\pm \in \widetilde{W}_m^\pm$. By construction $\natural(\widetilde{W}^\pm) = V^\pm$. Hence

$$\begin{aligned} (e, \natural(e)) &= (e_+ + e_-, \underline{e}_+ + \underline{e}_-) \\ &= (e_+, \underline{e}_+) + (e_-, \underline{e}_-) \in (Q(K)_p \cap \mathcal{W}_p^+) \oplus (Q(K)_p \cap \mathcal{W}_p^-), \end{aligned} \quad (3.90)$$

where $\underline{e}_\pm = \natural(e_\pm) = \natural(e)_\pm \in V^\pm$. Thus Equation (3.80) follows, and $Q(K)$ is a

regular transverse generalised isometry between W and V^+ . $\square$

Example 3.91. For a twisted standard Courant algebroid $E = (\mathbb{T}M, H)$, a generalised metric on $\underline{E} = (\mathbb{T}\mathcal{Q}, \underline{H})$ is given by a metric $\underline{g} \in \Gamma(\odot^2 T^*\mathcal{Q})$ and a two-form $\underline{b} \in \Omega^2(\mathcal{Q})$. Then the pullbacks $\varpi^*\underline{g}$ and $\varpi^*\underline{b}$ define a K -transverse generalised metric W such that $\mathfrak{h}(W) = V^+$, and $Q(\mathcal{F})$ is a regular transverse generalised isometry between W and V^+ .

3.2.4 Composition of Transverse Generalised Isometries

Composition of transverse generalised isometries presents a multi-faceted problem by nature of its less restrictive construction compared to generalised isometries. To elucidate why this is so, suppose $R: E_1 \dashrightarrow E_2$ and $R': E_2 \dashrightarrow E_3$ are Courant algebroid relations composing cleanly, and let W_i be transverse generalised metrics on E_i for $i = 1, 2, 3$. Suppose further that R is a transverse generalised isometry between W_1 and W_2 , and R' is a transverse generalised isometry between W_2 and W_3 . The problems encountered are two-fold.

Firstly, since the lifts $\widetilde{W}_i^\pm$ of $W_i^\pm$ are not unique (as would be the case for a generalised metric) it may not be the case that the lift $\widetilde{W}_2^\pm$ making R a transverse generalised metric is the same as the lift $\widetilde{W}_2'^\pm$ making R' a transverse generalised metric.

Secondly, even if we had $\widetilde{W}_2^\pm = \widetilde{W}_2'^\pm$, this may still not be enough. For instance, suppose $(e_1, e_3) \in (R' \circ R) \cap (\widetilde{W}_1^+ \times \widetilde{W}_3^+)$. By definition, there exists e_2 such that $(e_1, e_2) \in R$ and $(e_2, e_3) \in R'$. But it is in no way guaranteed that $e_2 \in \widetilde{W}_2^+$.

The first problem is discussed in Appendix A. For now we assume that $\widetilde{W}_2^\pm = \widetilde{W}_2'^\pm$ and address the second problem through

Proposition 3.92. Suppose that $R_1: E_1 \dashrightarrow E_2$ and $R_2: E_2 \dashrightarrow E_3$ are Courant algebroid relations supported on C_1 and C_2 , respectively, that compose cleanly. For $i = 1, 2, 3$ let K_i be isotropic involutive subbundles of E_i , and let W_i be pre- K_i -transverse generalised metrics such that R_i is a transverse generalised isometry between W_i and W_{i+1} . Suppose there are splittings $\widetilde{W}_1^\pm$, $\widetilde{W}_2^\pm$ and $\widetilde{W}_3^\pm$ giving the

decomposition (3.80) for R_1 and R_2 . Let $\mathcal{A}^\pm = \widetilde{W}_1^\pm \times \widetilde{W}_2^\pm \times \widetilde{W}_2^\pm \times \widetilde{W}_3^\pm$. If

$$\text{rk}((R_2 \diamond R_1)_{(c_1, c_2)} \cap (\mathcal{A}^+ \oplus \mathcal{A}^-)_{(c_1, c_2)}) > 0, \quad (3.93)$$

for every $(c_1, c_2) \in C_2 \diamond C_1$, then $R_2 \circ R_1$ is a transverse generalised isometry between W_1 and W_3 .

Proof. Take $(e_1, e_3) \in (R_2 \circ R_1) \cap (\mathcal{W}^+ \oplus \mathcal{W}^-)$, where $\mathcal{W}^\pm = \widetilde{W}_1^\pm \times \widetilde{W}_3^\pm$. Then there exists $e_2 \in E_2$ such that $(e_i, e_{i+1}) \in R_i$ for $i = 1, 2$. That is, $(e_1, e_2, e_2, e_3) \in R_2 \diamond R_1$. By Equation (3.93) we may assume $e_2 = e_2^+ + e_2^-$ where $e_2^\pm \in \widetilde{W}_2^\pm$. Then

$$\begin{aligned} (e_i, e_{i+1}) &= (e_i^+ + e_i^-, e_{i+1}^+ + e_{i+1}^-) \\ &= (e_i^+, e_{i+1}^+) + (e_i^-, e_{i+1}^-) \in R_i \cap ((\widetilde{W}_i^+ \times \widetilde{W}_{i+1}^+) \oplus (\widetilde{W}_i^- \times \widetilde{W}_{i+1}^-)), \end{aligned} \quad (3.94)$$

for $i = 1, 2$. Since R_i is a transverse generalised isometry between W_i and W_{i+1} , the decomposition (3.80) holds for R_1 and R_2 , hence $(e_i^\pm, e_{i+1}^\pm) \in R_i$. It then follows by definition that $(e_1^\pm, e_3^\pm) \in R_2 \circ R_1$, and so the decomposition (3.80) also holds for $R_2 \circ R_1$. $\square$

Corollary 3.95. If either R_1 or R_2 is the graph of a classical Courant algebroid isomorphism, then $R_2 \circ R_1$ is always a transverse generalised isometry between W_1 and W_3 .

Proof. Suppose $R_1 = \text{gr}(\Phi)$, where $\Phi: E_1 \rightarrow E_2$ is a Courant algebroid isomorphism over a diffeomorphism $\phi: M_1 \rightarrow M_2$ (hence $C_1 = \text{gr}(\phi)$). Since Φ is an isomorphism, we have already seen that $\widetilde{W}_2^\pm = \Phi(\widetilde{W}_1^\pm)$. To show that Equation (3.93) holds, fix $(c_1, c_2) \in C_2 \diamond C_1$. Then at this point

$$R_2 \diamond \text{gr}(\Phi) = \{ (e_1, \Phi(e_1), \Phi(e_1), e_3) \mid e_1 \in E_1, (\Phi(e_1), e_3) \in R_2 \}. \quad (3.96)$$

But if $e_1 \in \widetilde{W}_1^+ \oplus \widetilde{W}_1^-$ then $\Phi(e_1) \in \widetilde{W}_2^+ \oplus \widetilde{W}_2^-$, and hence Equation (3.93) holds.

If R_2 is the graph of a classical Courant algebroid isomorphism Φ , then this argument can be applied to $(R_1 \circ \text{gr}(\Phi))^\top = \text{gr}(\Phi^{-1}) \circ R_1^\top$, as the transpose of a transverse generalised isometry is a transverse generalised isometry. $\square$

Chapter 4

The T-duality relation

Having studied in detail the properties of Courant algebroid relations and (transverse) generalised isometries, we now move on to defining T-duality as a Courant algebroid relation. The first half of this chapter is devoted to the construction of the T-duality relation, by composing a pair of reduction relations. Then, in Section 4.1.2, the problem of transporting a generalised metric across the T-duality relation is discussed, for which we show that the reduction relations compose as transverse generalised isometries, using the machinery developed in Chapter 3.

4.1 T-duality as a Courant Algebroid Relation

In this section we formalise the idea of T-duality as a Courant algebroid relation, where the Courant algebroid relation may be viewed as the relation induced by two reductions of an exact Courant algebroid. That is, we consider an exact Courant algebroid E over a manifold M and take isotropic subbundles $K_1 \subset E$ and $K_2 \subset E$, inducing regular foliations by $\rho_E(K_1)$ and $\rho_E(K_2)$ of M , and apply the reduction procedure of Theorem 2.40. Under the right conditions, the reduced Courant algebroids inherit a Courant algebroid relation from the reductions. In other words, we will explore circumstances under which the Wess-Zumino functionals of our sigma-models are related.

4.1.1 Topological T-duality

Let E be an exact Courant algebroid over M endowed with isotropic subbundles K_1 and K_2 . Here and throughout the rest of the paper we will assume that

$$\mathrm{rk}(K_1) = \mathrm{rk}(K_2) . \quad (4.1)$$

Assume as well that $K_1^\perp$ and $K_2^\perp$ have enough basic sections and denote by $\mathcal{F}_i$ the foliation of M given by the integral manifolds of the distribution $\rho_E(K_i)$. Suppose that the leaf spaces $\mathcal{Q}_i = M/\mathcal{F}_i$ have a smooth structure, hence there are unique surjective submersions $\varpi_i: M \rightarrow \mathcal{Q}_i$ for $i = 1, 2$.

From Section 3.1.2 we can then form the Courant algebroid morphisms $Q(K_i): E \rightarrow \underline{E}_i$ supported on $\mathrm{gr}(\varpi_i)$. This gives the diagram

$$\begin{array}{ccc} & E & \\ Q(K_1) \swarrow & & \searrow Q(K_2) \\ \underline{E}_1 & & \underline{E}_2 \end{array} \quad (4.2)$$

which prepares for our first definition of T-duality.

Definition 4.3. Let E be an exact Courant algebroid fitting into the diagram (4.2). Then $\underline{E}_1$ and $\underline{E}_2$ are *T-duality related* over $\mathcal{Q}_1$ and $\mathcal{Q}_2$ if the composition

$$R = Q(K_2) \circ Q(K_1)^\top: \underline{E}_1 \dashrightarrow \underline{E}_2 \quad (4.4)$$

is a Courant algebroid relation supported on

$$C = \{ (\varpi_1(m), \varpi_2(m)) \mid m \in M \} \subset \mathcal{Q}_1 \times \mathcal{Q}_2 . \quad (4.5)$$

The Courant algebroid relation R is the *T-duality relation* between $\underline{E}_1$ and $\underline{E}_2$.

Remark 4.6. We first discuss the smoothness of the supporting submanifold C , as in general it may not be smooth. The most interesting case is when $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are fibered manifolds over a common base manifold $\mathcal{B}$. In this case there is the

commutative diagram

$$\begin{array}{ccc}
 & M & \\
 \varpi_1 \swarrow & & \searrow \varpi_2 \\
 \mathcal{Q}_1 & & \mathcal{Q}_2 \\
 \pi_1 \searrow & & \swarrow \pi_2 \\
 & \mathcal{B} &
 \end{array} \tag{4.7}$$

Since the fibred product $\mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ also fits into the diagram (4.7), there is a smooth map $M \rightarrow \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ given by the quotient of M by the foliation induced by the distribution $T\mathcal{F}_1 \cap T\mathcal{F}_2$. Thus in this case the rank of $T\mathcal{F}_1 \cap T\mathcal{F}_2$ must be constant.¹

Lemma 4.8. If M fits into the commutative diagram (4.7), then $C = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ and hence is smooth.

Proof. Note that $C \subseteq \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$. For the opposite inclusion, consider a point (q_1, q_2) in $\mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$. Then there exist $m, m' \in M$ such that $q_1 = \varpi_1(m)$ and $q_2 = \varpi_2(m')$, and $b := \pi_1(\varpi_1(m)) = \pi_2(\varpi_2(m')) \in \mathcal{B}$. Since also $\pi_2(\varpi_2(m)) = b$, with $q_0 := \varpi_2(m)$ the points q_2 and q_0 belong to the same fibre of π_2 . Hence there is a path $\gamma : [0, 1] \rightarrow \mathcal{Q}_2$ such that $\gamma(0) = q_0$ and $\gamma(1) = q_2$. By the homotopy lifting property we get a path $\tilde{\gamma}$ in $(\mathcal{F}_1)_{q_1}$ such that $\tilde{\gamma}(0) = m$, where $(\mathcal{F}_1)_{q_1}$ is the leaf of the foliation $\mathcal{F}_1$ given by $\varpi_1^{-1}(q_1)$. We then define $m'' := \tilde{\gamma}(1)$. It follows that $\varpi_1(m'') = q_1$ and $\varpi_2(m'') = q_2$, hence $\mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2 \subseteq C$. $\square$

Assuming C is smooth, we can now state the conditions under which the T-duality relation R can be formed as

Theorem 4.9. Let E be an exact Courant algebroid over M endowed with isotropic subbundles K_1 and K_2 satisfying the conditions for the diagram (4.2) discussed above. Let us also assume that $T\mathcal{F}_1 \cap T\mathcal{F}_2$ has constant rank and that C is a smooth manifold. Then $\underline{E}_1$ and $\underline{E}_2$ are T-duality related if and only if $K_1 \cap K_2$ has constant rank.

Proof. To compose $Q(K_2)$ with $Q(K_1)^\top$, we first check the conditions on the base manifolds. Setting $C_1 = \text{gr}(\varpi_1)$ and $C_2 = \text{gr}(\varpi_2)$, the intersection of $C_1^\top \times C_2$ with $\mathcal{Q}_1 \times \Delta(M) \times \mathcal{Q}_2$ is clean: one can show that they intersect transversally² in

¹We will see below that this must also be true in the general case without a common base $\mathcal{B}$.

²We say that submanifolds S and S' of M intersect transversally if $T_s S + T_s S' = T_s M$ for each $s \in S \cap S'$. Submanifolds that intersect transversally also intersect cleanly [73, Appendix C.3].

$\mathcal{Q}_1 \times M \times M \times \mathcal{Q}_2$. For if $v \in T_{\hat{m}}(\mathcal{Q}_1 \times M \times M \times \mathcal{Q}_2)$, for any $\hat{m} \in (C_1^\top \times C_2) \cap (\mathcal{Q}_1 \times \Delta(M) \times \mathcal{Q}_2)$, then

$$\begin{aligned} v = (\underline{v}_1, v_1, v'_1, v_2) &= ((\varpi_1)_*(v_1), v_1, v'_1, (\varpi_2)_*(v'_1)) \\ &+ (\underline{v}_1 - (\varpi_1)_*(v_1), 0, 0, \underline{v}_2 - (\varpi_2)_*(v'_1)) , \end{aligned} \quad (4.10)$$

where the first term is an element of $T_{\hat{m}}(C_1^\top \times C_2)$ and the second term is an element of $T_{\hat{m}}(\mathcal{Q}_1 \times \Delta(M) \times \mathcal{Q}_2)$. This implies the condition (ii) on the base manifolds of Proposition 3.30. Next since $C = C_2 \circ C_1^\top$ is a smooth manifold and $T\mathcal{F}_1 \cap T\mathcal{F}_2$ has constant rank, the map $C_2 \diamond C_1^\top \rightarrow C_2 \circ C_1^\top$ is a smooth surjective submersion, thereby satisfying the condition (ii) on the base manifolds of Proposition 3.32.

To check the constant rank criteria of conditions (ii) of Propositions 3.30 and 3.32, we note that since $Q(K_1) = \{ (e, \natural_1(e)) \mid e \in K_1^\perp \}$,

$$Q(K_2) \diamond Q(K_1)^\top = \{ (\natural_1(e), e, e, \natural_2(e)) \mid e \in K_1^\perp \cap K_2^\perp \} . \quad (4.11)$$

This is pointwise isomorphic to $K_1^\perp \cap K_2^\perp$, and for $m \in M$ we find

$$\begin{aligned} \dim((K_2^\perp)_m \cap (K_1^\perp)_m) &= \text{rk}(K_1^\perp) + \text{rk}(K_2^\perp) - \dim((K_1^\perp)_m + (K_2^\perp)_m) \\ &= \text{rk}(K_1^\perp) + \text{rk}(K_2^\perp) - \text{rk}(E_2) + \dim((K_1^\perp)_m^\perp \cap (K_2^\perp)_m^\perp) \\ &= \text{rk}(K_1^\perp) + \text{rk}(K_2^\perp) - \text{rk}(E_2) + \dim((K_1)_m \cap (K_2)_m) . \end{aligned} \quad (4.12)$$

It follows that condition (ii) of Proposition 3.30 is satisfied if and only if the dimension of $(K_1)_m \cap (K_2)_m$ is independent of $m \in M$.

Finally, the kernel of the projection of Equation (4.11) to $Q(K_2) \circ Q(K_1)^\top$ is $K_1 \cap K_2$. Hence by item (ii) of Proposition 3.32, the composition is clean if and only if $K_1 \cap K_2$ has constant rank. Thus by Theorem 3.33, we get the Courant algebroid relation (4.4). $\square$

The diagram (4.2) now becomes

$$\begin{array}{ccc}
 & E & \\
 Q(K_1)^\top \nearrow & & \nwarrow Q(K_2) \\
 \underline{E}_1 & \xrightarrow{\quad R \quad} & \underline{E}_2
 \end{array} \tag{4.13}$$

Explicitly, if $\mathfrak{h}_i: K_i^\perp \rightarrow (K_i^\perp/K_i)/\mathcal{F}_i = \underline{E}_i$ denotes the quotient map for $i = 1, 2$, then

$$R = \{ (\mathfrak{h}_1(e), \mathfrak{h}_2(e)) \mid e \in K_1^\perp \cap K_2^\perp \} . \tag{4.14}$$

As mentioned in Section 1, there is a notion that the T-duality relation should be Lagrangian. Our relation is consistent with this, through

Proposition 4.15. R is maximally isotropic, hence a Dirac structure in $\underline{E}_1 \times \overline{\underline{E}}_2$.

Proof. Since R is involutive by construction, we only need to show that it is maximally isotropic in $\underline{E}_1 \times \overline{\underline{E}}_2$. Note that, from the standpoint of graded symplectic geometry, this is a consequence of the fact that the composition of two Lagrangian relations is again Lagrangian.

Using Equation (4.1), we have

$$\mathrm{rk}(\underline{E}_1 \times \overline{\underline{E}}_2) = 2 \left(\mathrm{rk}(E) - 2 \mathrm{rk}(K_1) \right) . \tag{4.16}$$

We have already seen that $Q(K_2) \diamond Q(K_1)^\top$ is isomorphic to $K_1^\perp \cap K_2^\perp$ pointwise, and that its projection to R has kernel $K_1 \cap K_2$. Thus

$$\begin{aligned}
 \mathrm{rk}(R) &= \mathrm{rk}(K_1^\perp \cap K_2^\perp) - \mathrm{rk}(K_1 \cap K_2) \\
 &= \mathrm{rk}(K_1^\perp) + \mathrm{rk}(K_2^\perp) - \mathrm{rk}(E) \\
 &= \mathrm{rk}(E) - 2 \mathrm{rk}(K_1)
 \end{aligned} \tag{4.17}$$

as required. □

Remark 4.18 (Bisubmersions). In the picture of T-duality related Courant algebroids of Definition 4.3, at the level of base manifolds there are surjective submer-

sions

$$\begin{array}{ccc} & M & \\ \varpi_1 \swarrow & & \searrow \varpi_2 \\ \mathcal{Q}_1 & & \mathcal{Q}_2 \end{array}$$

with the fibres of ϖ_1 given by the leaves of the foliation $\mathcal{F}_1$ and the fibres of ϖ_2 given by the leaves of the foliation $\mathcal{F}_2$.

If both surjective submersions have connected fibres and the Lie bracket of vector fields satisfies

$$[\Gamma(\ker(\varpi_{1*})), \Gamma(\ker(\varpi_{2*}))] \subset \Gamma(\ker(\varpi_{1*})) + \Gamma(\ker(\varpi_{2*})) , \quad (4.19)$$

then there exist unique (possibly singular) foliations $\underline{\mathcal{F}}_1$ on $\mathcal{Q}_1$ and $\underline{\mathcal{F}}_2$ on $\mathcal{Q}_2$ such that [75, Corollary 2.16]

$$\varpi_{1*}^{-1}(T\underline{\mathcal{F}}_1) = (\varpi_2)_*^{-1}(T\underline{\mathcal{F}}_2) = \ker(\varpi_{1*}) + \ker(\varpi_{2*}) . \quad (4.20)$$

This means that there is a *bisubmersion* between $(\mathcal{Q}_1, \underline{\mathcal{F}}_1)$ and $(\mathcal{Q}_2, \underline{\mathcal{F}}_2)$; see e.g. [75] and references therein for the general definition of bisubmersion.

This construction is an example of *Hausdorff-Morita equivalent foliations* in the sense of [75, Definition 2.1]. Equation (4.19) is an involutivity condition for the subbundle

$$\ker(\varpi_{1*}) + \ker(\varpi_{2*}) = T\mathcal{F}_1 + T\mathcal{F}_2 , \quad (4.21)$$

hence M is endowed with a (possibly singular) foliation $\mathcal{F}$ that induces the Hausdorff-Morita equivalent foliations. This is a regular foliation if $\ker(\varpi_{1*}) \cap \ker(\varpi_{2*})$ has constant rank. This notion will be crucial in the description of the symmetries of the background fields defined on $\mathcal{Q}_1$ and $\mathcal{Q}_2$, and we will see how it naturally arises in the construction of T-duality relations.

Example 4.22. Our main source of examples will be the following set up: let $E_1 = (\mathbb{T}M_1, H_1)$ and $E_2 = (\mathbb{T}M_2, H_2)$ be isomorphic twisted standard Courant algebroids, isomorphic via $\bar{\phi} \circ e^B$, where $\bar{\phi} = \phi_* + (\phi^{-1})^*$ and $B \in \Omega^2(M_1)$ with $dB = H_1 - \phi^*H_2$, as described in Proposition 2.31. Suppose that M_1 and M_2 are

foliated by $\mathcal{F}_1$ and $\mathcal{F}'_2$, respectively, such that $\mathcal{Q}_1 = M_1/\mathcal{F}_1$ with quotient map ϖ_1 .³ Suppose further that $\iota_{X_1}H_1 = 0$, for all $X_1 \in \Gamma(T\mathcal{F}_1)$ and $\iota_{X_2}H_2 = 0$, for all $X_2 \in \Gamma(T\mathcal{F}'_2)$. Finally, let $\mathcal{Q}_2 = M_1/\mathcal{F}_2$ with $\mathcal{F}_2 = \phi^{-1}(\mathcal{F}'_2)$ and quotient map ϖ_2 , and let $\underline{H}_1$ and $\underline{H}_2$ be such that $\varpi_1^*\underline{H}_1 = H_1$, $\varpi_2^*\underline{H}_2 = \phi^*H_2$. Then

Proposition 4.23. If $T\mathcal{F}_1 \cap T\mathcal{F}_2$ and $\ker(B|_{T\mathcal{F}_1 \cap T\mathcal{F}_2})$ have constant rank, $(\mathbb{T}\mathcal{Q}_1, \underline{H}_1)$ and $(\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$ are T-duality related.

Proof. Take $K_1 = T\mathcal{F}_1$, and $K_2 = e^{-B}(T\mathcal{F}_2)$. Then

$$K_1 \cap K_2 = \{v_1 \in T\mathcal{F}_1 \cap T\mathcal{F}_2 \mid \iota_{v_1}B = 0\} = \ker(B|_{T\mathcal{F}_1 \cap T\mathcal{F}_2}). \quad (4.24)$$

By Corollary 3.20, there are Courant algebroid relations $Q(K_1): (\mathbb{T}M_1, H_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_1, \underline{H}_1)$ and $Q(K_2): (\mathbb{T}M_1, H_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$.

By Theorem 4.9, $R = Q(K_2) \circ Q(K_1)^\top$ is a Courant algebroid relation. $\square$

4.1.2 Geometric T-duality

We now introduce geometric data into our picture of T-duality, which amounts to incorporating the Polyakov functionals (1.1) into our string sigma-models; the symmetries of S_g are generated by Killing vectors of the metric g . We will make the following assumptions. Let E be an exact Courant algebroid over M endowed with involutive isotropic subbundles K_1 and K_2 that have the same rank. Suppose they satisfy the conditions to fit the diagram (4.2) and suppose that the reduced Courant algebroids $\underline{E}_1$ and $\underline{E}_2$ are T-duality related, i.e. the constant rank assumption of Theorem 4.9 holds. Denote the resulting T-duality relation by R . We will work under these assumptions for the remainder of this section.

We begin by endowing the reduced Courant algebroids with generalised metrics.

Definition 4.25. The reduced Courant algebroids $(\underline{E}_1, V_1^+)$ and $(\underline{E}_2, V_2^+)$ are *geometrically T-dual* if R is a generalised isometry between the generalised metrics V_1^+ and V_2^+ .

The starting point of any T-duality transformation is the identification of the symmetries of the original background.

³In general we do not require that $\phi_*T\mathcal{F}_1 = T\mathcal{F}'_2$.

Definition 4.26. Let $\underline{E}_1$ and $\underline{E}_2$ be T-duality related Courant algebroids in the sense of Definition 4.3 and suppose that the unique surjective submersions $\varpi_i: M \rightarrow \mathcal{Q}_i$ satisfy the involutivity condition (4.19) discussed in Remark 4.18, for $i = 1, 2$. The subbundle $D_1 := T\underline{\mathcal{F}}_1 = \varpi_{1*}(T\underline{\mathcal{F}}_2)$ of $T\mathcal{Q}_1$ is the *distribution of T-duality directions*.⁴

Since we may form the relations $Q(K_i)$ for $i = 1, 2$, by Remark 2.68 there exists splittings $\sigma_i: TM \rightarrow E$ of the short exact sequences

$$0 \longrightarrow T^*M \xrightarrow{\rho^*} E \xrightarrow{\rho} TM \longrightarrow 0 \quad (4.27)$$

$\xleftarrow{\sigma_i^*} \quad \xleftarrow{\sigma_i}$

which are adapted to K_i , where $\sigma_i^* := \sigma_i^t \circ \flat_E$ is the induced left splitting and $\flat_E: E \rightarrow E^*$ is the isomorphism induced by the pairing. By Remark 2.62, σ_i induces a splitting $\underline{\sigma}_i$ of $\underline{E}_i$, giving also a left splitting $\underline{\sigma}_i^*$.

We are now ready to relate the distribution D_1 to the symmetries of our initial background and begin to justify our terminology ‘T-duality directions’.

Remark 4.28. Let us discuss how we can state an invariance condition of a generalised metric on $\underline{E}_1$. Let $\underline{\sigma}_1: T\mathcal{Q}_1 \rightarrow \underline{E}_1$ be a splitting of $\underline{E}_1$ descending from an adapted splitting σ_1 of E , then V_1^+ corresponds to a pair $(\underline{g}_1, \underline{b}_1)$ under the isomorphism $(\underline{\sigma}_1 + \frac{1}{2}\rho_{\underline{E}_1}^*)^{-1}: \underline{E}_1 \rightarrow T\mathcal{Q}_1$, i.e. any element $\underline{w}_1 \in \Gamma(V_1^+)$ can be written as

$$\underline{w}_1 = \underline{Y} + \iota_{\underline{Y}}(\underline{g}_1 + \underline{b}_1) , \quad (4.29)$$

for some $\underline{Y} \in \Gamma(T\mathcal{Q}_1)$. Let $\underline{X} \in \Gamma(D_1)$. Then the generalised metric $(\underline{g}_1, \underline{b}_1)$ is invariant with respect to $\underline{X}$ if

$$\mathcal{L}_{\underline{X}}(\underline{g}_1 + \underline{b}_1) = 0 . \quad (4.30)$$

⁴ This definition can be adapted to the case where $\underline{\mathcal{F}}_1$ is a singular foliation. Then we consider the T-duality directions to be defined by the locally finitely generated $C_c^\infty(\mathcal{Q}_1)$ -module $\Gamma_c(D_1)$ of compactly supported vector fields integrating to $\underline{\mathcal{F}}_1$. We do not explicitly discuss this in the following in order to avoid further technicalities. Invariance with respect to a singular foliation relates to the work of Kotov-Strobl [76].

Note that we can write

$$\llbracket \underline{X}, \underline{w}_1 \rrbracket_{H_1} = [\underline{X}, \underline{Y}] + \iota_{\underline{Y}}(\mathcal{L}_{\underline{X}}(\underline{g}_1 + \underline{b}_1)) + \iota_{[\underline{X}, \underline{Y}]}(\underline{g}_1 + \underline{b}_1) + \text{pr}_2(\llbracket \underline{X}, \rho(\underline{w}_1) \rrbracket_{H_1}) , \quad (4.31)$$

where $\text{pr}_2(\llbracket \underline{X}, \underline{Y} \rrbracket_{H_1}) = \iota_{\underline{X}} \iota_{\underline{Y}} \underline{H}$. Hence by imposing the condition (4.30), we obtain

$$\llbracket \underline{X}, \underline{w}_1 \rrbracket_{H_1} - \text{pr}_2(\llbracket \underline{X}, \rho(\underline{w}_1) \rrbracket_{H_1}) \in \Gamma(\text{gr}(\underline{g}_1 + \underline{b}_1)) , \quad (4.32)$$

and vice versa.

In light of this remark, we introduce a new action

$$\text{ad}^\sigma : \Gamma(E) \times \Gamma(E) \longrightarrow \Gamma(E) \quad (4.33)$$

on an exact Courant algebroid E for a given splitting $\sigma : TM \rightarrow E$, called the σ -adjoint action. It is defined by

$$\text{ad}_e^\sigma e' := \llbracket e, e' \rrbracket - \rho^* \sigma^* \llbracket e, \sigma(\rho(e')) \rrbracket , \quad (4.34)$$

for all $e, e' \in \Gamma(E)$. If $X \in \Gamma(TM)$, we denote $\text{ad}_X^\sigma := \text{ad}_{\sigma(X)}^\sigma$. See Appendix B for the properties of the σ -adjoint action ad^σ .

Definition 4.35. Let $\underline{\sigma}_1 : T\mathcal{Q}_1 \rightarrow \underline{E}_1$ be the splitting of the exact Courant algebroid $\underline{E}_1$ over $\mathcal{Q}_1$ induced by an isotropic splitting $\sigma_1 : TM \rightarrow E$ of E which is adapted to $K_1 \subset E$. A generalised metric V_1^+ on $\underline{E}_1$ is *invariant with respect to D_1* , or simply *D_1 -invariant*, if there is a locally finitely generating⁵ Lie subalgebra $\mathfrak{k}_{\tau_1}$ of the $C^\infty(\mathcal{Q}_1)$ -submodule $\Gamma(D_1)$ such that

$$\text{ad}_{\underline{X}}^{\underline{\sigma}_1}(\underline{w}_1) \in \Gamma(V_1^+) , \quad (4.36)$$

for all $\underline{X} \in \mathfrak{k}_{\tau_1}$, $\underline{w}_1 \in \Gamma(V_1^+)$. The Lie subalgebra $\mathfrak{k}_{\tau_1}$ of $\Gamma(T\mathcal{Q}_1)$ is the *isometry algebra of D_1* or the *Lie algebra of Killing vector fields for V_1^+* .

Remark 4.37. On an exact Courant algebroid E over M with Ševera class $[H]$ and

⁵i.e. at every point of $\mathcal{Q}_1$ there is an open neighborhood such that D_1 is generated by only finitely many elements of $\mathfrak{k}_{\tau_1}$

an isotropic splitting $\sigma: TM \rightarrow E$, for $X, Y, Z \in \Gamma(TM)$ one has

$$\begin{aligned} H(X, Y, Z) &= \langle \llbracket \sigma(X), \sigma(Y) \rrbracket_E, \sigma(Z) \rangle_E = \iota_Z \sigma^*(\llbracket \sigma(X), \sigma(Y) \rrbracket_E) \\ &= \langle \rho_E^*(\sigma^*(\llbracket \sigma(X), \sigma(Y) \rrbracket_E)), \sigma(Z) \rangle_E . \end{aligned} \quad (4.38)$$

Hence the condition (4.36) has the interpretation of an invariance condition which itself imposes no restrictions on the representative H of the Ševera class; Remark 4.42 below makes this point clearer. Recall that a subbundle W is invariant with respect to an involutive subbundle $K \subset W$ if satisfies $\llbracket \Gamma(K), \Gamma(W) \rrbracket_E \subset \Gamma(W)$, which implies $\iota_X H = 0$, for all $X \in \Gamma(\rho(K))$, see [74]. Definition 4.35 is formulated in a way that avoids imposing any conditions on the representative H of the Ševera class of E .

Next we introduce a notion of when the adapted splittings are compatible.

Definition 4.39. Let V_1^+ be a D_1 -invariant generalised metric on $\underline{E}_1$ with isometry algebra $\mathfrak{k}_{\tau_1}$. Adapted splittings $\sigma_i: TM \rightarrow E$ with respect to K_i for $i = 1, 2$ are *compatible* if they satisfy

$$\llbracket \sigma_2(X), e \rrbracket = \mathbf{ad}_X^{\sigma_1} e , \quad (4.40)$$

for every $X \in \mathfrak{k}_2$ and $e \in \Gamma(E)$, where

$$\mathfrak{k}_2 := \varpi_{1*}^{-1}(\mathfrak{k}_{\tau_1}) \cap \Gamma(\rho(K_2)) . \quad (4.41)$$

Remark 4.42. Let $\sigma_1: TM \rightarrow E$ be an adapted splitting to K_1 . Then, under the isomorphism $(\sigma_1 + \frac{1}{2}\rho_E^*)^{-1}: E \rightarrow \mathbb{T}M$, the image of K_1 is $T\mathcal{F}_1$. Moreover, σ_1 composed with the isomorphism is the inclusion of TM into $\mathbb{T}M$, and yields a representative H_{σ_1} of the Ševera class of E . Meanwhile, an adapted splitting σ_2 for K_2 induces H_{σ_2} . The difference is given by

$$(\sigma_1 - \sigma_2)(X) = \rho_E^*(\iota_X B) \quad (4.43)$$

with $H_{\sigma_1} - H_{\sigma_2} = dB$. The adapted splittings σ_1 and σ_2 are compatible if

$$\begin{aligned} \llbracket \iota_X B, Y + \xi \rrbracket_{H_{\sigma_1}} - \text{pr}_2(\llbracket X, Y \rrbracket_{H_{\sigma_1}}) &= -\iota_Y d \iota_X B - \iota_Y \iota_X H_{\sigma_1} \\ &= -\iota_Y d \iota_X B - \iota_Y \iota_X dB = -\iota_Y \mathcal{L}_X B, \end{aligned} \quad (4.44)$$

for all $Y + \xi \in \Gamma(\mathbb{T}M_1)$, where we use that $\iota_X H_{\sigma_2} = \iota_X (H_{\sigma_1} - dB) = 0$ for the second equality. Thus the splittings are compatible if and only if $\mathcal{L}_X B = 0$ for every $X \in \mathfrak{k}_2$.

We are now ready to state our main result about the existence and uniqueness of geometrically T-dual backgrounds in the sense of Definition 4.25 through

Theorem 4.45. Let E be an exact Courant algebroids over M . Suppose E is endowed with involutive isotropic subbundles K_1 and K_2 of the same rank giving T-duality related Courant algebroids $\underline{E}_1$ and $\underline{E}_2$ with T-duality relation R . Assume further that the bisubmersion involutivity condition (4.19) holds, and that there are adapted splittings σ_1 and σ_2 of E which are compatible. Finally, assume that $\underline{E}_1$ is endowed with a D_1 -invariant generalised metric V_1^+ . Then the following are equivalent:

- (i) $K_1 \cap K_2^\perp \subseteq K_2$.
- (ii) $K_2 \cap K_1^\perp \subseteq K_1$.
- (iii) There exists a unique generalised metric V_2^+ on $\underline{E}_2$ such that R is a generalised isometry between V_1^+ and V_2^+ , i.e. $(\underline{E}_1, V_1^+)$ and $(\underline{E}_2, V_2^+)$ are geometrically T-dual.

Remark 4.46. We split the proof in three parts, beginning with Lemma 4.48 below which establishes the implications

$$(i) \iff (ii) \iff (iii). \quad (4.47)$$

The final implication requires the construction of the generalised metric V_2^+ on $\underline{E}_2$. The first step in this construction is to show that one can lift V_1^+ to $\widetilde{W}_1^+ \subset K_1^\perp \cap K_2^\perp$. Proposition 4.53 below establishes this result. The remainder of the proof is devoted to showing that the induced pre- K_2 -transverse generalised metric $W_2 := K_2 \oplus \widetilde{W}_1^+$ is a K_2 -transverse generalised metric, and hence reduces to a generalised metric V_2^+ .

Lemma 4.48. Suppose that E is an exact Courant algebroid over M such that $\underline{E}_1$ and $\underline{E}_2$ are T-duality related with T-duality relation R . Then the implications (4.47) hold.

Proof. To show that (i) implies (ii), we note that

$$K_2^\perp \cap K_1 \subseteq K_2 \implies K_2^\perp \cap K_1 \subseteq K_1 \cap K_2 . \quad (4.49)$$

Since $K_2^\perp \supset K_2$, it follows that

$$K_2^\perp \cap K_1 = K_2 \cap K_1 . \quad (4.50)$$

By Theorem 4.9, $K_2 \cap K_1$ has constant rank.

Using $\text{rk}(K_1) = \text{rk}(K_2)$ (condition (4.1)), we then obtain

$$\begin{aligned} \text{rk}(K_1 \cap K_2^\perp) &= \text{rk}(K_1) + \text{rk}(K_2^\perp) - \text{rk}(K_1 + K_2^\perp) \\ &= \text{rk}(K_2) + \text{rk}(K_2^\perp) - (\text{rk}(E_2) - \text{rk}(K_1^\perp \cap K_2)) \\ &= \text{rk}(E_2) - \text{rk}(E_2) + \text{rk}(K_1^\perp \cap K_2) = \text{rk}(K_1^\perp \cap K_2) . \end{aligned} \quad (4.51)$$

Hence

$$\text{rk}(K_1 \cap K_2) = \text{rk}(K_1 \cap K_2^\perp) = \text{rk}(K_1^\perp \cap K_2) . \quad (4.52)$$

One always has $K_1^\perp \cap K_2 \supseteq K_1 \cap K_2$. Since their ranks are the same, this becomes an equality. In particular, we obtain condition (ii).

The converse implication (ii) $\implies$ (i) follows from the same argument.

Finally, suppose that (i) does not hold. Then there is an element $k_1 \in K_1 \cap K_2^\perp$ such that $k_1 \notin K_2$. Then by Proposition 3.64, R cannot be a generalised isometry. $\square$

Proposition 4.53. Suppose that E is an exact Courant algebroid over M endowed with involutive isotropic subbundles K_1 and K_2 of the same rank, such that $\underline{E}_1$ and $\underline{E}_2$ are T-duality related. If either of the equivalent conditions (i) or (ii) of Theorem 4.45 holds, then there exists a splitting $s_0: K_1^\perp/K_1 \rightarrow K_1^\perp$ of the short

exact sequence

$$0 \longrightarrow K_1 \longrightarrow K_1^\perp \longrightarrow K_1^\perp/K_1 \longrightarrow 0 \quad (4.54)$$

such that $\text{im}(s_0) \subseteq K_2^\perp$, which is unique up to elements of $K_1 \cap K_2$.

Proof. Take an arbitrary splitting $s: K_1^\perp/K_1 \rightarrow K_1^\perp$ of the short exact sequence (4.54). If $\text{im}(s) \not\subseteq K_2^\perp$, then there exists $[e] \in K_1^\perp/K_1$ and $k_2 \in K_2$ such that $\langle s([e]), k_2 \rangle_E \neq 0$. We define the vector bundle map β_s covering the identity by

$$\beta_s = \flat_E \circ s: K_1^\perp/K_1 \longrightarrow K_2^*, \quad (4.55)$$

where $\flat_E: E \rightarrow E^*$ is the isomorphism induced by the pairing $\langle \cdot, \cdot \rangle_E$ on E :

$$\iota_{k_2} \beta_s([e_1]) := \langle s([e_1]), k_2 \rangle_E, \quad (4.56)$$

for $[e_1] \in K_1^\perp/K_1$ and $k_2 \in K_2$.

If $k_2 \in K_2 \cap K_1$, then

$$\langle s([e_1]), k_2 \rangle_E = 0 \quad (4.57)$$

since $s([e_1]) \in K_1^\perp$. Thus we can further define a map $\bar{\beta}_0$ by quotienting out this subspace:

$$\bar{\beta}_0: K_1^\perp/K_1 \longrightarrow (K_2/(K_2 \cap K_1))^*, \quad \iota_{[k_2]} \bar{\beta}_0([e_1]) := \langle s([e_1]), k_2 \rangle_E, \quad (4.58)$$

for $[e_1] \in K_1^\perp/K_1$ and $[k_2] \in K_2/(K_2 \cap K_1)$.

Similarly we define

$$\bar{\beta}: K_1 \longrightarrow (K_2/(K_2 \cap K_1))^*, \quad \iota_{[k_2]} \bar{\beta}(k_1) := \langle k_1, k_2 \rangle_E, \quad (4.59)$$

for $k_1 \in K_1$ and $[k_2] \in K_2/(K_2 \cap K_1)$, which is again well-defined.

We show that $\bar{\beta}$ is surjective: the kernel of $\bar{\beta}$ is given by

$$\ker(\bar{\beta}) = \{ k_1 \in K_1 \mid \langle k_1, k_2 \rangle_E = 0 \text{ for all } k_2 \in K_2 \} = K_2^\perp \cap K_1. \quad (4.60)$$

By assumption, $K_2^\perp \cap K_1 \subseteq K_2$, hence⁶

$$\ker(\bar{\beta}) = K_2^\perp \cap K_1 = K_2 \cap K_1 . \quad (4.61)$$

Recalling that $K_2 \cap K_1$ has constant rank and that $\text{rk}(K_1) = \text{rk}(K_2)$, it thus follows that

$$\begin{aligned} \text{rk}(\bar{\beta}) &= \text{rk}(K_1) - \text{rk}(K_2 \cap K_1) \\ &= \text{rk}(K_2) - \text{rk}(K_1 \cap K_2) = \text{rk}((K_2/(K_2 \cap K_1))^*) . \end{aligned} \quad (4.62)$$

Hence $\bar{\beta}$ is onto, and we can find a map $\alpha: K_1^\perp/K_1 \rightarrow K_1$ which fits into the diagram

$$\begin{array}{ccc} K_1^\perp/K_1 & \xrightarrow{\quad \alpha \quad} & K_1 \\ & \searrow \bar{\beta}_0 \quad \swarrow \bar{\beta} & \\ & (K_2/(K_2 \cap K_1))^* & \end{array} \quad (4.63)$$

We can now define a new splitting $s_0: K_1^\perp/K_1 \rightarrow K_1^\perp$ of the short exact sequence (4.54) by

$$s_0 := s - \alpha . \quad (4.64)$$

It follows that $\text{im}(s_0) \subseteq K_2^\perp$, since for every $k_2 \in K_2$ and $[e_1] \in K_1^\perp/K_1$ we can compute

$$\begin{aligned} \langle s_0([e_1]), k_2 \rangle_E &= \langle s([e_1]), k_2 \rangle_E - \langle \alpha([e_1]), k_2 \rangle_E \\ &= \iota_{[k_2]} \bar{\beta}_0([e_1]) - \iota_{[k_2]} \bar{\beta}(\alpha([e_1])) \\ &= \iota_{[k_2]} \bar{\beta}_0([e_1]) - \iota_{[k_2]} \bar{\beta}_0([e_1]) = 0 . \end{aligned} \quad (4.65)$$

Hence the new splitting is a map $s_0: K_1^\perp/K_1 \rightarrow K_2^\perp \cap K_1^\perp$. To show uniqueness up to elements of $K_2 \cap K_1$, note that there are two sources of non-uniqueness for s_0 : one from the choice of the initial splitting $s: K_1^\perp/K_1 \rightarrow K_1^\perp$, and one from the choice of α fitting the diagram (4.63).

⁶See the start of the proof of Lemma 4.48.

Since the map α is given by the right inverse of $\bar{\beta}$, it is unique up to elements of the kernel of $\bar{\beta}$, which by Equation (4.60) is $K_2 \cap K_1$. That is, if α and α' close the diagram (4.63), then there are induced splittings s_0 and s'_0 respectively defined by Equation (4.64), and

$$s_0 - s'_0 = \alpha - \alpha': K_1^\perp/K_1 \longrightarrow K_2 \cap K_1 . \quad (4.66)$$

Finally, suppose we started with different splittings $s, s': K_1^\perp/K_1 \rightarrow K_1^\perp$. Using the above construction, we obtain splittings $s_0, s'_0: K_1^\perp/K_1 \rightarrow K_2^\perp \cap K_1^\perp$. These splittings differ by elements of K_1 , hence

$$s_0 - s'_0: K_1^\perp/K_1 \longrightarrow K_2^\perp \cap K_1 = K_1 \cap K_2 , \quad (4.67)$$

giving the required uniqueness. $\square$

Proof of Theorem 4.45. Assume that condition (i) of Theorem 4.45 holds. By Theorem 3.84, there is a unique K_1 -transverse generalised metric W_1 on E such that $Q(K_1)$ is a regular transverse generalised isometry between W_1 and V_1^+ , using an arbitrary lift $\widetilde{W}_1^+$ of $W_1^+ = W_1/K_1$ to E . From Proposition 4.53 it follows that we can choose $\widetilde{W}_1^+$ such that $\widetilde{W}_1^+ \subseteq K_2^\perp \cap K_1^\perp$. Set $\widetilde{W}_2^+ := \widetilde{W}_1^+ \subseteq K_2^\perp$. By the uniqueness of the splitting $s: K_1^\perp/K_1 \rightarrow K_1^\perp$ up to elements in $K_1 \cap K_2$, this choice of $\widetilde{W}_2^+$ is unique up to elements of $K_1 \cap K_2$.

Thus we may uniquely define the bundle

$$W_2 = \widetilde{W}_2^+ \oplus K_2 . \quad (4.68)$$

The sum is direct: if $k_2 \in \widetilde{W}_2^+ \cap K_2$, then $k_2 \in \widetilde{W}_1^+$. Since $\widetilde{W}_1^+ \cap K_1 = \{0\}$, it follows that $k_2 \notin K_1$. But this contradicts condition (ii) unless $k_2 = 0$.

It follows that W_2 is a pre- K_2 -transverse generalised metric on E . This can be seen by noting firstly that $K_2 \subset W_2 \subset K_2^\perp$, since $\widetilde{W}_1^+ \subset K_2^\perp$. Moreover, for every $w_2 \in W_2$ such that $w_2 \notin K_2$, it follows that $\langle w_2, w_2 \rangle_E > 0$.

From Theorem 3.84 it follows that $Q(K_1)^\top$ is a transverse generalised isometry between V_1^+ and W_2 .

We shall now show that W_2 is a K_2 -transverse generalised metric which descends

to the quotient. Since V_1^+ is D_1 -invariant, for every $\underline{w} \in \Gamma(V_1^+)$ and $\underline{X} \in \mathfrak{k}_{V_1^+}$ it follows by definition that

$$\llbracket \sigma_1(\underline{X}), \underline{w}_1 \rrbracket_{E_1} - \rho_{E_1}^*(\sigma_1^*(\llbracket \sigma_1(\underline{X}), \sigma_1(\rho_{E_1}(\underline{w}_1)) \rrbracket_{E_1})) \in \Gamma(V_1^+). \quad (4.69)$$

Take a subset of sections $\{\underline{w}_j\}$ spanning V_1^+ pointwise and $\{\underline{X}_i\} \subset \mathfrak{k}_{V_1^+}$ spanning D_1 pointwise. The former lifts to a set of basic sections $\{w_j\}$ spanning $\widetilde{W}_1^+$ pointwise, the latter lifts to a set of projectable vector fields $\{X_i\} \subset \mathfrak{k}_2$ spanning $T\mathcal{F}_2$ pointwise. Since σ_1 and σ_2 are compatible, it follows that

$$\llbracket \sigma_2(X_i), w_j \rrbracket_E = \llbracket \sigma_1(X_i), w_j \rrbracket_E - \rho_E^*(\sigma_1^*(\llbracket \sigma_1(X_i), \sigma_1(\rho_E(w_j)) \rrbracket_E)) . \quad (4.70)$$

If $X \in \Gamma(TM_1)$ is projectable, then $\sigma_1(X)$ is a basic section, since σ_1 is an adapted splitting. Thus since the sections $\sigma_1(X_i)$ and w_j are basic, the right-hand side of Equation (4.70) is basic⁷ and descends to the quotient; by Equation (4.69), it moreover descends to a section of V_1^+ . It follows that the right-hand side of Equation (4.70) is a section of W_1 , hence $\llbracket \sigma_2(X_i), w_j \rrbracket_E \in \Gamma(\widetilde{W}_1^+) \oplus \Gamma(K_1)$.

We next show that in fact $\llbracket \sigma_2(X_i), w_j \rrbracket_E \in \Gamma(\widetilde{W}_1^+) \oplus \Gamma(K_1 \cap K_2)$. Suppose that $\llbracket \sigma_2(X_i), w_j \rrbracket_E \in \Gamma(K_1)$. Recall that $X_i \in \Gamma(\rho_E(K_2))$ and σ_2 is the adapted splitting to K_2 . Therefore $\sigma_2(X_i) \in \Gamma(K_2)$, and since $w_j \in \Gamma(K_1^\perp \cap K_2^\perp)$ it follows from Equation (2.38) that

$$\llbracket \sigma_2(X_i), w_j \rrbracket_E \in \Gamma(K_2^\perp \cap K_1) = \Gamma(K_1 \cap K_2), \quad (4.71)$$

with the last equality following from condition (i). We must therefore have

$$\llbracket \sigma_2(X_i), w_j \rrbracket_E \in \Gamma(\widetilde{W}_1^+ \oplus (K_1 \cap K_2)). \quad (4.72)$$

The final piece of the puzzle is found by noting that, since ρ_E is injective on K_2 as shown in Lemma 2.64, it follows that $\{\sigma_2(X_i)\}$ is a set of sections spanning K_2 pointwise. Let us denote these sections by $k_i := \sigma_2(X_i)$. Thus for any $k' \in \Gamma(K_2)$ and $w \in \Gamma(\widetilde{W}_1^+)$ there are expansions $k' = \sum_i f_i k_i$ and $w = \sum_j h_j w_j$ for some functions $f_i, h_j \in C^\infty(M)$ such that the sums are locally finite.

⁷Recall that $\llbracket \Gamma_{\text{bas}}(K_1^\perp), \Gamma_{\text{bas}}(K_1^\perp) \rrbracket_{E_1} \subset \Gamma_{\text{bas}}(K_1^\perp)$.

Using the anchored Leibniz rule (2.4) and Equation (2.10) we may then write

$$\begin{aligned} \llbracket k', w \rrbracket_E = \sum_{i,j} \left(f_i h_j \llbracket k_i, w_j \rrbracket_E - h_j (\rho_E(w_j) \cdot f_i) k_i \right. \\ \left. + (\rho_E(f_i k_i) \cdot h_j) w_j + h_j \langle k_i, w_j \rangle_E \mathcal{D}_E f_i \right). \end{aligned} \quad (4.73)$$

The first term lives in $\Gamma(\widetilde{W}_1^+ \oplus (K_1 \cap K_2))$ by Equation (4.72), the second term in $\Gamma(K_2)$, the third term in $\Gamma(\widetilde{W}_1^+)$, and the final term is zero since $\widetilde{W}_1^+ \subset K_2^\perp$. Hence

$$\llbracket \Gamma(K_2), \Gamma(\widetilde{W}_1^+) \rrbracket_E \subset \Gamma(\widetilde{W}_1^+) + \Gamma(K_2). \quad (4.74)$$

Thus

$$\begin{aligned} \llbracket \Gamma(K_2), \Gamma(W_2) \rrbracket_E &= \llbracket \Gamma(K_2), \Gamma(\widetilde{W}_2^+) \rrbracket_E + \llbracket \Gamma(K_2), \Gamma(K_2) \rrbracket_E \\ &\subset \Gamma(\widetilde{W}_2^+) + \Gamma(K_2) = \Gamma(W_2), \end{aligned} \quad (4.75)$$

where we have also used the property that K_2 is involutive.

Equation (4.75) shows that W_2 is K_2 -invariant, so by Proposition 2.56 it descends to a subbundle $V_2^+ := \natural_2(W_2)$ of $\underline{E}_2$. We know from Theorem 3.84 that V_2^+ defines a generalised metric on $\underline{E}_2$ such that $Q(K_2)$ is a regular transverse generalised isometry between W_2 and V_2^+ . Finally, since

$$Q(K_2) \diamond Q(K_1)^\top = \{ (\natural_1(e), e, e, \natural_2(e)) \mid e \in K_1^\perp \cap K_2^\perp \}, \quad (4.76)$$

and since $\widetilde{W}_1^\pm \subseteq K_1^\perp \cap K_2^\perp$, the rank condition (3.93) is satisfied. It follows that $Q(K_2) \circ Q(K_1)^\top = R$ is a generalised isometry between V_1^+ and V_2^+ . Thus $(\underline{E}_1, V_1^+)$ and $(\underline{E}_2, V_2^+)$ are geometrically T-dual. Uniqueness of V_2^+ follows by Corollary 3.67. $\square$

Remark 4.77 (Classification of T-duality). In Theorem 4.45 we are able to construct geometrically T-dual backgrounds because of the bisubmersion condition (4.19). This leads us to speculate that geometrically T-dual Courant algebroids can only be considered over Hausdorff Morita equivalent foliated manifolds, i.e. they might represent a subset of this equivalence class. This may lead to a classification

of T -dual backgrounds.

Geometric T -duality for Standard Courant Algebroids

We shall now discuss Theorem 4.45 when choosing an isomorphism of E with $\mathbb{T}M$. Recall that, in the assumptions of 4.45, we have two compatible adapted splittings σ_1 and σ_2 . Let $\Psi_1 = \sigma_1 + \frac{1}{2}\rho_E^*$ be the isomorphisms between $\mathbb{T}M$ and E induced by σ_1 . Hence, $\Psi_1^{-1}(K_1) = T\mathcal{F}_1$. Moreover, by applying Ψ_1^{-1} , Equation (4.43) can be written as $(\sigma_1^* \circ \sigma_2)(X) = -\iota_X B$, for all $X \in TM$, thus, $\Psi_1^{-1}(K_2) = e^{-B}(T\mathcal{F}_2)$ and $\Psi_1^{-1}(K_2^\perp) = e^B(TM) \oplus \text{Ann}(T\mathcal{F}_2)$. Therefore, the condition (i) yields that $T\mathcal{F}_1 \cap T\mathcal{F}_2 \subseteq \ker(B^\flat)$. Note that the same condition also follows from (ii).

Proposition 4.78. Let $(\mathbb{T}M, H)$ be the twisted standard Courant algebroid isomorphic to E via Ψ_1 , and suppose that the assumptions of Proposition 4.53 hold. Then $B \in \Omega^2(M)$ defined in Remark 4.42 takes the form

$$B = B_{\text{ver}} + B_{\text{hor}} + B_{\text{mix}} , \quad (4.79)$$

where

$$B_{\text{mix}} \in \Gamma((T^*\mathcal{F}_1 \cap \text{Ann}(T\mathcal{F}_2)) \wedge (\text{Ann}(T\mathcal{F}_1) \cap T^*\mathcal{F}_2)) \quad (4.80)$$

gives an isomorphism between $T\mathcal{F}_1 \cap \text{Ann}^*(T\mathcal{F}_2)$ and $\text{Ann}(T\mathcal{F}_1) \cap T^*\mathcal{F}_2$, whereas

$$B_{\text{ver}} \in \Gamma(\wedge^2 T^*\mathcal{F}_1) \quad \text{and} \quad B_{\text{hor}} \in \Gamma(\wedge^2 \text{Ann}(T\mathcal{F}_1)) \quad (4.81)$$

vanish on

$$\Gamma((T^*\mathcal{F}_1 \cap T^*\mathcal{F}_2) \wedge (T^*\mathcal{F}_1 \cap \text{Ann}(T\mathcal{F}_2))) \quad (4.82)$$

and

$$\Gamma((\text{Ann}(T\mathcal{F}_1) \cap \text{Ann}(T\mathcal{F}_2)) \wedge (\text{Ann}(T\mathcal{F}_1) \cap T^*\mathcal{F}_2)) , \quad (4.83)$$

respectively.

Proof. We find the conditions B must meet to satisfy the conditions (i) or (ii) of Theorem 4.45.

Write $N^*\mathcal{F} = \text{Ann}(T\mathcal{F})$. We can decompose

$$B = B_{\text{ver}} + B_{\text{hor}} + B_{\text{mix}} \quad (4.84)$$

where

$$B_{\text{ver}} \in \Gamma(\wedge^2 T^*\mathcal{F}_1) , \quad B_{\text{hor}} \in \Gamma(\wedge^2 N^*\mathcal{F}_1) \quad \text{and} \quad B_{\text{mix}} \in \Gamma(N^*\mathcal{F}_1 \wedge T^*\mathcal{F}_1) . \quad (4.85)$$

Recall that $\Psi_1^{-1}(K_2) = e^{-B}(T\mathcal{F}_2)$. Thus

$$\Psi_1^{-1}(K_2) \cap \Psi_1^{-1}(K_1^\perp) = \{ v_2 - \iota_{v_2} B \mid v_2 \in T\mathcal{F}_2 , \iota_{v_2} B \in \text{Ann}(T\mathcal{F}_1) \} . \quad (4.86)$$

Similarly

$$\Psi_1^{-1}(K_2^\perp) \cap \Psi_1^{-1}(K_1) = \{ v_1 \in T\mathcal{F}_1 \mid \iota_{v_1} B \in \text{Ann}(T\mathcal{F}_2) \} . \quad (4.87)$$

Conditions (i) and (ii) of Theorem 4.45 state that these must be respectively contained in K_1 and K_2 . The conditions given in the statement of the proposition ensure that this happens. For example, if $v_1 \in T\mathcal{F}_1$ but $v_1 \notin T\mathcal{F}_2$, then $v_1 \notin e^B(T\mathcal{F}_2)$. It follows that v_1 cannot be in $e^B(K_2^\perp)$. The only way this can happen is if $\iota_{v_1} B_{\text{mix}} \in T^*\mathcal{F}_2$ and is non-zero. Since this happens for all $v_1 \in T\mathcal{F}_1 \cap N\mathcal{F}_2$, B_{mix} gives an isomorphism between $T\mathcal{F}_1 \cap N\mathcal{F}_2$ and $N^*\mathcal{F}_1 \cap T^*\mathcal{F}_2$. The other conditions follow from similar considerations. $\square$

We shall now consider the setting of Example 4.22, since it is our main source of explicit examples.

Theorem 4.88. Let $(\mathbb{T}M_1, H_1)$ and $(\mathbb{T}M_2, H_2)$ be twisted standard Courant algebroids as in Example 4.22 and Proposition 4.23, isomorphic via $\Phi = \bar{\phi} \circ e^B$, and suppose that B decomposes as in Equation (4.79). Take a generalised metric V_1^+ on $\mathbb{T}\mathcal{Q}_1$ defined by $\underline{g}_1 \in \Gamma(\odot^2 T^*\mathcal{Q}_1)$ and $\underline{b}_1 \in \Omega^2(\mathcal{Q}_1)$ and define the subalgebras $\mathfrak{k}_{V_1^+}$

and $\mathfrak{k}_2$ by

$$\mathfrak{k}_{V_1^+} = \{ \underline{X} \in \Gamma(D_1) : \mathcal{L}_{\underline{X}}(\underline{g}_1 + \underline{b}_1) = 0 \} , \quad (4.89)$$

$$\mathfrak{k}_2 = (\varpi_1)_*^{-1}(\mathfrak{k}_{V_1^+}) \cap \Gamma(T\mathcal{F}_2) . \quad (4.90)$$

Suppose that $\mathfrak{k}_{V_1^+}$ spans pointwise D_1 , and that for each $X \in \mathfrak{k}_2$

$$\mathcal{L}_X B = 0 . \quad (4.91)$$

Then there exists a unique generalised metric V_2^+ on $\mathbb{T}\mathcal{Q}_2$ such that $(\mathbb{T}\mathcal{Q}_1, H_1, V_1^+)$ and $(\mathbb{T}\mathcal{Q}_2, H_2, V_2^+)$ are geometrically T-dual, i.e. R is a generalised isometry between V_1^+ and V_2^+ .

Proof. By Remarks 4.28 and 4.42, the conditions (4.91) ensure that the generalised metric $(\underline{g}_1, \underline{b}_1)$ is D_1 -invariant and the adapted splittings are compatible. The decomposition (4.79) ensures that conditions (i) and (ii) of Theorem 4.45 are satisfied. Hence the conclusion follows by Theorem 4.45. $\square$

Remark 4.92. The condition $\mathcal{L}_X B = 0$ for $X \in \mathfrak{k}_2$ together with the assumption of reducibility of E with respect to K_2 tells us that

$$\iota_X H_1 = \iota_X dB = -d(\iota_X B) , \quad (4.93)$$

similarly to Equation (1.5), i.e. it gives additional symmetries of the sigma-model Wess-Zumino functional for M , as discussed in Section 1.1.

Remark 4.94. To see why we named D_1 the distribution of T-duality directions, consider a point $(q_1, q_2) \in C \subset \mathcal{Q}_1 \times \mathcal{Q}_2$, and take $\underline{v} \in (D_1)_{q_1}$. Let $v \in (T\mathcal{F}_2)_m$ be such that $\varpi_{1*}(v) = \underline{v}$, where $\varpi_1(m) = q_1$. Denote $\varpi_2(m) = q_2$. It follows that $v \in (T_m\mathcal{F}_2 \cap \text{Ann}^*(T_m\mathcal{F}_1))$. By Proposition 4.78 we have

$$\iota_v B = \iota_v B_{\text{mix}} \in T_m^*\mathcal{F}_1 \cap \text{Ann}(T_m\mathcal{F}_2) . \quad (4.95)$$

Thus the corresponding element in $R_{(q_1, q_2)}$ is

$$(\natural_1(v), \natural_2(v)) = (\underline{v}, \natural_2(\iota_v B)) . \quad (4.96)$$

This is the usual exchange of tangent and cotangent (momentum and winding) directions seen in T-duality.

4.1.3 Local Geometric T-Duality

It is possible to generalise the considerations of Subsection 4.1.2 to the case in which a generalised metric has only locally defined Killing vectors. This will be useful for instance when looking at T-duality for non-principal bundles where the “action” is not globally defined, for instance in Sections 7.2.4 and 7.2.5.

Definition 4.97. A generalised metric V_1^+ on $\underline{E}_1$ is *locally D_1 -invariant* if there are local charts $\{U^{(i)}\}_{i \in I}$ covering $\mathcal{Q}_1$ and a locally finitely generating Lie subalgebra $\mathfrak{k}_{\tau_1}^{(i)} \subset \Gamma(D_1|_{U^{(i)}})$ such that

$$\mathrm{ad}_{\underline{X}^{(i)}}^{\sigma_1}(\underline{w}_1) \in \Gamma(V_1^+|_{U^{(i)}}) , \quad (4.98)$$

for every $\underline{X}^{(i)} \in \mathfrak{k}_{\tau_1}^{(i)}$, $\underline{w}_1 \in \Gamma(V_1^+|_{U^{(i)}})$ and $i \in I$.

Note that $\mathfrak{k}_{\tau_1}^{(i)} = \mathfrak{k}_{\tau_1}^{(j)}$ on overlaps $U^{(ij)} = U^{(i)} \cap U^{(j)}$. Hence the isometry algebra $\mathfrak{k}_{\tau_1}$ is well-defined, as is $\mathfrak{k}_2 := (\varpi_1)_*^{-1}(\mathfrak{k}_{\tau_1}) \cap \Gamma(\rho(K_2))$. If $U^{(i)}$ is a cover of $\mathcal{Q}_1$, then $\mathcal{U}^{(i)} := \varpi_1^{-1}(U^{(i)})$ is a cover of M , hence we can also define $\mathfrak{k}_2^{(i)} := \mathfrak{k}_2|_{\mathcal{U}^{(i)}}$.

Similarly, we can speak about when splittings are locally compatible.

Definition 4.99. Adapted splittings $\sigma_\ell : TM \rightarrow E$ with respect to K_ℓ for $\ell = 1, 2$ are *locally compatible* if they satisfy

$$[\![\sigma_2(X^{(i)}), e]\!] = \mathrm{ad}_{X^{(i)}}^{\sigma_1} e , \quad (4.100)$$

for every $X^{(i)} \in \mathfrak{k}_2^{(i)}$, $e \in \Gamma(E)$ and $i \in I$.

Theorem 4.101. Theorem 4.45 holds in the local case, i.e. we can replace compatibility of splittings and D_1 -invariance with their local counterparts.

Proof. The proof is exactly the same as in 4.45: the algebra $\mathfrak{k}_{\tau_1}$ only enters when one shows that the pre-transverse generalised metric W_2 is in fact a K_2 -transverse generalised metric. However, the condition $\llbracket K_2, W_2 \rrbracket \subset W_2$ can be checked locally, and $\sigma_2(\mathfrak{k}_2)$ spans K_2 locally. $\square$

4.1.4 The T-Duality Relation at the Level of Sections

Here we study the T-duality relation R at the level of sections. To proceed, let us characterise the sections of a Courant algebroid $E \rightarrow M$ which are invariant with respect to a distribution $D \subset TM$. In the setting of T-duality, $D_1 \subset TQ_1$ is the isometry distribution for a generalised metric on $\underline{E}_1$, which is induced by the reduction of Courant algebroids $Q(K_2) : E \rightarrow \underline{E}_2$. We will see that the D_1 -invariant sections are the sections of $\underline{E}_1$ which lift to basic sections of $K_2^\perp$ in the setting of Theorem 4.45. Thus we provide a notion inspired by Definition 4.35.

Definition 4.102. Let E be an exact Courant algebroid over M and let $D \subset TM$ be a distribution. Suppose that there exists a Lie subalgebra $\mathfrak{k}$ of $\Gamma(D)$ that spans D pointwise. Given a splitting $\sigma : TM \rightarrow E$, the space of *D-invariant sections* of E is given by

$$\Gamma_D(E) := \{ e \in \Gamma(E) \mid \text{ad}_X^\sigma e = 0 \text{ for all } X \in \mathfrak{k} \} . \quad (4.103)$$

Given an open cover $\{U^{(i)}\}_{i \in I}$ of M and a local presentation $\{\mathfrak{k}^{(i)}\}_{i \in I}$ of $\mathfrak{k}$, the space of *locally D-invariant sections* of E is

$$\Gamma_D^{(i)}(E) := \{ e \in \Gamma(E) \mid \text{ad}_{X^{(i)}}^\sigma e = 0 \text{ for all } X^{(i)} \in \mathfrak{k}^{(i)} \} . \quad (4.104)$$

The space of (locally) D -invariant sections is involutive, see Lemma B.14, and is a $C_D^\infty(M)$ -module, with $C_D^\infty(M) \cong C_{\text{bas}}^\infty(M, \mathcal{F})$, where $\mathcal{F}$ is the foliation of M that integrates the distribution D .

Remark 4.105. Definition 4.102 recovers the usual notion of $\mathfrak{g}$ -invariant sections for the action of a Lie algebra $\mathfrak{g}$. In that instance $\text{ad}_{\#_M(x)}^{\sigma_\chi} = \llbracket \chi(x), \cdot \rrbracket$ for $x \in \mathfrak{g}$, where $\chi : \mathfrak{g} \rightarrow \Gamma(E)$ is a trivially extended action in the sense of [51, Definition 2.12], with Lie algebra action on TM denoted by $\#_M : \mathfrak{g} \rightarrow \Gamma(TM)$, and σ_χ is a

splitting adapted to $\chi(\mathfrak{g})$ such that $\text{im}(\sigma_\chi \circ \#_M) = \text{im}(\chi)$. Further motivation for Definition 4.102 comes from the desire to have a notion of invariance even when there is no Lie or Courant algebra action on the Courant algebroid E .

Consider the T-duality setting when $K_1 \cap K_2 = \{0\}$, D_1 -invariant sections of $\underline{E}_1 \rightarrow \mathcal{Q}_1$ are lifted to K_2 -basic sections of $K_2^\perp$.

Lemma 4.106. Under the assumptions of Theorem 4.45, let $s: K_1^\perp/K_1 \rightarrow K_1^\perp$ be the unique⁸ splitting of the short exact sequence

$$0 \longrightarrow K_1 \longrightarrow K_1^\perp \longrightarrow K_1^\perp/K_1 \longrightarrow 0, \quad (4.107)$$

such that $\text{im}(s) \subset K_1^\perp \cap K_2^\perp$. If $\underline{e}^\circ \in \Gamma_{D_1}(\underline{E}_1)$, then $s(\underline{e}^\circ) \in \Gamma_{\text{bas}}(K_2^\perp)$, where $s(\underline{e}^\circ)$ is the lift of $\underline{e}^\circ$ to $\Gamma_{\text{bas}}(K_1^\perp)$ using the $C^\infty(\mathcal{Q}_1)$ -module isomorphism $\Gamma(\underline{E}_1) \cong \Gamma_{\text{bas}}(K_1^\perp)/\Gamma(K_1)$, as in Equation (2.45).

Proof. Denote for simplicity by $s: \Gamma(\underline{E}_1) \rightarrow \Gamma_{\text{bas}}(\text{im}(s))$ the map given by the composition of the lift from $\Gamma(\underline{E}_1)$ to $\Gamma_{\text{bas}}(K_1^\perp)/\Gamma(K_1)$ and the splitting of (4.107) given by Proposition 4.53, and note that $\mathfrak{b}_1 \circ s = \mathbb{1}_{\underline{E}_1}$. Since $\mathfrak{b}_1$ is a bracket homomorphism between $\llbracket \cdot, \cdot \rrbracket$ and $\llbracket \cdot, \cdot \rrbracket_1$, it then follows that

$$\mathfrak{b}_1(\llbracket s(\underline{\sigma}_1(\underline{X})), s(\underline{e}^\circ) \rrbracket - s\llbracket \underline{\sigma}_1(\underline{X}), \underline{e}^\circ \rrbracket_1) = 0, \quad (4.108)$$

for all $\underline{e}^\circ \in \Gamma_{D_1}(\underline{E}_1)$ and $\underline{X} \in \mathfrak{k}_{\tau_1}$, hence

$$\llbracket s(\underline{\sigma}_1(\underline{X})), s(\underline{e}^\circ) \rrbracket - s\llbracket \underline{\sigma}_1(\underline{X}), \underline{e}^\circ \rrbracket_1 \in \Gamma(K_1). \quad (4.109)$$

Let us show that

$$s(\underline{\rho}_1^* \underline{\sigma}_1^* \llbracket \underline{\sigma}_1(\underline{X}), \underline{\sigma}_1(\underline{\rho}_1(\underline{e}^\circ)) \rrbracket_1) = \rho^* \sigma_1^* \llbracket \sigma_1(X), \sigma_1(\rho(s(\underline{e}^\circ))) \rrbracket, \quad (4.110)$$

where X is the canonical lift of $\underline{X}$ to $\mathfrak{k}_2 \subset \Gamma(TM)$ (see Definition 4.39). The difference is again in $\Gamma(\ker(\mathfrak{b}_1)) = \Gamma(K_1)$. Under the anchor ρ the difference lands in $T\mathcal{F}_1 \cap T\mathcal{F}_2 = \{0\}$. But ρ is injective on K_1 by Lemma 2.64, hence the difference vanishes.

⁸By Proposition 4.53, s is unique because of the assumption that $K_1 \cap K_2 = \{0\}$.

We also have $s(\underline{\sigma}_1(\underline{X})) = \sigma_1(X)$, for all $\underline{X} \in \mathfrak{k}_{\tau_1}$. Putting everything together we get

$$\begin{aligned}
 \llbracket s(\underline{\sigma}_1(\underline{X})), s(\underline{e}^\circ) \rrbracket - s\llbracket \underline{\sigma}_1(\underline{X}), \underline{e}^\circ \rrbracket_1 &= \llbracket s(\underline{\sigma}_1(\underline{X})), s(\underline{e}^\circ) \rrbracket - s(\rho_1^* \underline{\sigma}_1^* \llbracket \underline{\sigma}_1(\underline{X}), \underline{\sigma}_1(\rho_1(\underline{e}^\circ)) \rrbracket_1) \\
 &= \llbracket \sigma_1(X), s(\underline{e}^\circ) \rrbracket - \rho^* \sigma_1^* \llbracket \sigma_1(X), \sigma_1(\rho(s(\underline{e}^\circ))) \rrbracket \\
 &= \llbracket \sigma_2(X), s(\underline{e}^\circ) \rrbracket ,
 \end{aligned} \tag{4.111}$$

where for the first equality we use the D_1 -invariance of $\underline{e}^\circ$, while for the last equality we use the compatibility of the splittings (see Definition 4.39). We have $\sigma_2(X) \in \Gamma(K_2)$, and $s(\underline{e}^\circ) \in \Gamma(K_1^\perp \cap K_2^\perp)$. Since $\llbracket \Gamma(K_2), \Gamma(K_2^\perp) \rrbracket \subset \Gamma(K_2^\perp)$, it follows from Equation (4.109) that

$$\llbracket \sigma_2(X), s(\underline{e}^\circ) \rrbracket \in \Gamma(K_2^\perp \cap K_1) = \Gamma(K_2 \cap K_1) = \{0\} . \tag{4.112}$$

Using the Leibniz rule for the Dorfman bracket, this extends to all elements of $\Gamma(K_2)$, and hence $s(\underline{e}^\circ)$ is basic with respect to K_2 . $\square$

Remark 4.113 (Local T-duality of Sections). These results also hold in the local case, that is, if τ_1 is locally D_1 -invariant in the sense of Definition 4.35. This is because, in the proof of Lemma 4.106, checking whether $s(\underline{e}^\circ)$ is basic with respect to K_2 for $\underline{e}^\circ \in \Gamma_{D_1}(\underline{E}_1)$ can be performed on locally defined sections of K_2 . Similarly, the invariance of $\delta(e)$ with respect to $T\mathcal{F}_2 = \rho(K_2)$ can be checked locally.

Correspondence Space

We shall discuss how the results of this subsection apply to the most prominent example of T-duality, that of two fiber bundles $\mathcal{Q}_1$ and $\mathcal{Q}_2$ over the same base manifold $\mathcal{B}$. This setup can be depicted in the commutative correspondence diagram

$$\begin{array}{ccc}
 & M & \\
 \varpi_1 \swarrow & & \searrow \varpi_2 \\
 \mathcal{Q}_1 & & \mathcal{Q}_2 \\
 \pi_1 \searrow & & \swarrow \pi_2 \\
 & \mathcal{B} &
 \end{array} \tag{4.114}$$

Because of the universal property of fibred products, we can take the correspondence space to be

$$M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2 .$$

In particular, $T\mathcal{F}_1 \cap T\mathcal{F}_2 = \{0\}$, with ϖ_1 and ϖ_2 the canonical projection maps. Since ρ is injective on K_1 and K_2 (see Lemma 2.64), it follows that $K_1 \cap K_2 = \{0\}$. The particular case of principal torus bundles is covered extensively in [28, 46, 53] and will be treated as a special case of our general formalism in Subsection 5.3.1.

Proposition 4.115. Suppose that the assumptions of Theorem 4.45 are satisfied, with T-duality relation $R: (\underline{E}_1, \tau_1) \dashrightarrow (\underline{E}_2, \tau_2)$. Assume further that there is a locally finite generating set of vector fields $\mathfrak{k}_2$ which are projectable by $\mathcal{F}_2$ such that

$$[\![\sigma_1(X), s(\underline{e}^\circ)]\!] = \mathbf{ad}_X^{\sigma_2} s(\underline{e}^\circ) \quad (4.116)$$

for every $X \in \mathfrak{k}_2$ and $\underline{e}^\circ \in \Gamma_{D_1}(\underline{E}_1)$, where $s: K_1^\perp/K_1 \rightarrow K_1^\perp$ is the unique splitting from Lemma 4.106. Then the T-dual generalised metric τ_2 is invariant with respect to the distribution

$$D_2 := T\mathcal{F}_2 = \varpi_{2*}(T\mathcal{F}_1) , \quad (4.117)$$

with Lie algebra $\mathfrak{k}_{\tau_2} := (\varpi_2)_*\mathfrak{k}_2$. Furthermore, there is an isomorphism $\Gamma_{D_1}(\underline{E}_1) \cong \Gamma_{D_2}(\underline{E}_2)$ of $C^\infty(\mathcal{B})$ -modules.

Proof. Let $(\mathbb{T}M, H_1)$ and $(\mathbb{T}M, H_2)$ be the twisted standard Courant algebroids which are isomorphic to E via the isomorphisms induced by the compatible adapted splittings σ_1 and σ_2 , respectively. Define $B \in \Omega^2(M)$ by

$$\sigma_1(X) - \sigma_2(X) = \rho^*(\iota_X B) \quad (4.118)$$

for $X \in \Gamma(TM)$. For every $\underline{Y}^\circ + \underline{\eta}^\circ \in \Gamma_{D_1}(\mathbb{T}\mathcal{Q}_1)$, let $\eta = \varpi_1^* \underline{\eta}^\circ$ and $Y + \eta = s(\underline{Y}^\circ + \underline{\eta}^\circ)$. In other words, Y is the unique lift of $\underline{Y}^\circ$ such that

$$\eta(Z) = B(Y, Z) , \quad (4.119)$$

for all $Z \in \Gamma(T\mathcal{F}_2)$. For $X \in \mathfrak{k}_2$, Equation (4.116) then reads

$$\begin{aligned} \iota_Y \iota_X dB + \iota_Y \mathcal{L}_X B &= \llbracket -\iota_X B, Y + \eta \rrbracket = \text{pr}_2(\llbracket \sigma_2(X), \sigma_2(Y) \rrbracket_{H_1}) \\ &= \text{pr}_2(\llbracket \sigma_1(X), \sigma_1(Y) \rrbracket_{H_2}) = \iota_Y \iota_X dB , \end{aligned} \quad (4.120)$$

hence $\iota_Y \mathcal{L}_X B = 0$ for every $X \in \mathfrak{k}_2$ and $\underline{Y}^\circ + \underline{\eta}^\circ \in \Gamma_{D_1}(\mathbb{T}\mathcal{Q}_1)$. Thus

$$\begin{aligned} 0 &= \mathcal{L}_X(\eta(Z) - B(Y, Z)) \\ &= \mathcal{L}_X(\eta(Z)) - (\mathcal{L}_X B)(Y, Z) - B([X, Y], Z) - B(Y, [X, Z]) \\ &= (\mathcal{L}_X \eta)(Z) + \eta([X, Z]) - B([X, Y], Z) - \eta([X, Z]) \\ &= -B([X, Y], Z) , \end{aligned} \quad (4.121)$$

for all $Z \in \Gamma(T\mathcal{F}_2)$. But $[X, Y] \in \Gamma(T\mathcal{F}_1)$ since Y is $\mathcal{F}_1$ -projectable, and B is non-degenerate on $T\mathcal{F}_1 \otimes T\mathcal{F}_2$ since $K_1 \cap K_2 = \{0\}$, hence $[X, Y] = 0$.

It therefore follows that

$$\text{ad}_{\frac{\sigma_2}{\omega_{2*}X}} \mathfrak{d}_2(s(\underline{Y}^\circ + \underline{\eta}^\circ)) = 0 . \quad (4.122)$$

By virtue of the generalised isometry R and the uniqueness of the lift, it thus follows that the map $\mathfrak{d}_2 \circ s: \Gamma_{D_1}(\mathbb{T}\mathcal{Q}_1) \rightarrow \Gamma_{D_2}(\mathbb{T}\mathcal{Q}_2)$ is an isomorphism. Hence we obtain the isomorphism of $C^\infty(\mathcal{B})$ -modules $\Gamma_{D_1}(\underline{E}_1) \cong \Gamma_{D_2}(\underline{E}_2)$. By taking $\underline{Y}^\circ + \underline{\eta}^\circ \in \Gamma(V_1^+)$, it then follows that V_2^+ is D_2 -invariant. $\square$

Proposition 4.123. Under the conditions of Proposition 4.115, there exists transitive Courant algebroids $\underline{\underline{E}}_1$ and $\underline{\underline{E}}_2$ over $\mathcal{B}$ together with a $C^\infty(\mathcal{B})$ -module isomorphism $\mathcal{F}: \Gamma(\underline{\underline{E}}_1) \rightarrow \Gamma(\underline{\underline{E}}_2)$ induced by the T-duality relation R .

Proof. Define a vector bundle $\underline{\underline{E}}_1$ over $\mathcal{B}$ by

$$\underline{\underline{E}}_1 = \underline{E}_1 / \underline{\mathcal{F}}_1 . \quad (4.124)$$

The space of sections of $\underline{\underline{E}}_1$ is identified with the space $\Gamma_{D_1}(\underline{E}_1)$ as a module over

$C_{D_1}^\infty(\mathcal{Q}_1) \cong C^\infty(\mathcal{B})$. This gives $\underline{\underline{E}}_1$ the structure of a Courant algebroid as follows. The involutivity of $\Gamma_{D_1}(\underline{E}_1)$ established in Lemma B.14 gives the bracket structure on $\underline{\underline{E}}_1$, while the pairing descends to $\mathcal{B}$ by Lemma B.6. The anchor $\underline{\underline{\rho}}_1 : \underline{\underline{E}}_1 \rightarrow T\mathcal{B}$ is defined by $\underline{\underline{\rho}}_1(\underline{e}) = \pi_{1*} \underline{\rho}_1(\underline{e}^\circ)$, where $\underline{\rho}_1(\underline{e}^\circ)$ is projectable to $\mathcal{B}$ since $\underline{e}^\circ \in \Gamma_{D_1}(\underline{E}_1)$. Since $\Gamma_{D_1}(\underline{E}_1)$ spans $\underline{E}_1$ pointwise, it follows that $\underline{\underline{\rho}}_1$ is surjective, and hence that $\underline{\underline{E}}_1$ is a transitive Courant algebroid.

By a completely analogous argument using the Hausdorff-Morita equivalent foliation $\underline{\mathcal{F}}_2$ of $\mathcal{Q}_2$, there is also an isomorphism $\Gamma(\underline{\underline{E}}_2) \cong \Gamma_{D_2}(\underline{E}_2)$, where

$$\underline{\underline{E}}_2 = \underline{E}_2 / \underline{\mathcal{F}}_2 \quad (4.125)$$

is a transitive Courant algebroid over $\mathcal{B}$. By Proposition 4.115, there is an isomorphism $\Gamma_{D_1}(\underline{E}_1) \cong \Gamma_{D_2}(\underline{E}_2)$. The chain of isomorphisms

$$\Gamma(\underline{\underline{E}}_1) \cong \Gamma_{D_1}(\underline{E}_1) \cong \Gamma_{D_2}(\underline{E}_2) \cong \Gamma(\underline{\underline{E}}_2) \quad (4.126)$$

defines the isomorphism $\mathcal{F} : \Gamma(\underline{\underline{E}}_1) \rightarrow \Gamma(\underline{\underline{E}}_2)$. □

4.2 T-duality for Torus Bundles

In this section, we will apply the constructions above to the well-known case of torus bundles over the same base, [46, 52, 53]. In Section 7.2.1, we will see an explicit example of the constructions herein and of later chapters.

4.2.1 The T-duality Relation

We briefly recall the definition of T-duality from [46] for principal torus bundles, and show that this nicely fits into our definition.

Definition 4.127. Let $\mathcal{Q}_1$ and $\mathcal{Q}_2$ be principal $\mathbb{T}^k$ -bundles over a common base manifold $\mathcal{B}$, and let $H_1 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_1)$ and $H_2 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_2)$ be $\mathbb{T}^k$ -invariant closed three-forms. Consider the fibred product $M := \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ giving the correspondence space

diagram

$$\begin{array}{ccc}
 & M & \\
 \varpi_1 \swarrow & & \searrow \varpi_2 \\
 \mathcal{Q}_1 & & \mathcal{Q}_2 \\
 \pi_1 \searrow & & \swarrow \pi_2 \\
 & \mathcal{B} &
 \end{array} \tag{4.128}$$

with M endowed with the closed three-form $\varpi_1^* H_1 - \varpi_2^* H_2$.

Then $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are *T-dual* if there is a $\mathbb{T}^{2k}$ -invariant two-form $B \in \Omega_{\mathbb{T}^{2k}}^2(M)$ such that

$$\varpi_1^* H_1 - \varpi_2^* H_2 = dB, \tag{4.129}$$

and the smooth skew-symmetric map

$$B: \mathfrak{k}_1 \otimes \mathfrak{k}_2 \longrightarrow \mathbb{R} \tag{4.130}$$

is non-degenerate, where $\mathfrak{k}_i = \ker(\varpi_{i*})$ for $i = 1, 2$.

Remark 4.131 (Correspondence Spaces as Bisubmersions). The commutative diagram (4.128) represents a particular case of a bisubmersion, as discussed in Remark 4.18. The Hausdorff-Morita condition (4.19) is satisfied, since the generators of the $\mathfrak{k}_1^k$ -action commute with the generators of the $\mathfrak{k}_2^k$ -action, where $\text{Lie}(\mathbb{T}^k) = \mathfrak{k}^k$ and the subscript $i = 1, 2$ indicates the manifold $\mathcal{Q}_i$ on which $\mathbb{T}^k$ acts. Here the $\mathbb{T}^k$ -action whose orbits are the integral manifolds of $\ker(\varpi_{2*})$ induces a $\mathbb{T}^k$ -action on $\mathcal{Q}_1$ which, in order to provide Definition 4.127, is supposed to be free and proper. The distribution induced by this $\mathbb{T}^k$ -action on $\mathcal{Q}_1$ defines the distribution of T-duality directions $D_1 = (\varpi_1)_* \mathfrak{k}_2$, whose integral manifolds are the fibres of $\pi_1: \mathcal{Q}_1 \rightarrow \mathcal{B}$. Similarly, $\mathcal{Q}_2$ inherits a $\mathbb{T}^k$ -action, and hence a distribution D_2 , whose foliation, given by the fibres of $\pi_2: \mathcal{Q}_2 \rightarrow \mathcal{B}$, is Hausdorff Morita equivalent to the foliation induced by D_1 .

Let us reinterpret this definition as a T-duality relation.

Lemma 4.132. Let $\mathcal{Q}_1$ and $\mathcal{Q}_2$ be T-dual in the sense of Definition 4.127. Then

there is a T-duality relation

$$R: (\mathbb{T}\mathcal{Q}_1, H_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, H_2) \quad (4.133)$$

supported on M , as in Definition 4.3.

Proof. It is enough to find a Courant algebroid E over M such that Definition 4.3 is satisfied. Take with $E = (\mathbb{T}M, H = \varpi_1^* H_1)$ over $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ along with $K_1 = \mathfrak{k}_1$ and $K_2 = e^{-B}(\mathfrak{k}_2)$. Since $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are T-dual in the sense of Definition 4.127, the $B \in \Omega_{\mathbb{T}^{2k}}^2(M)$ satisfying Equation (4.129) is non-degenerate on $\mathfrak{k}_1 \otimes \mathfrak{k}_2$. Thus $e^{-B}(\mathfrak{k}_2) \cap \mathfrak{k}_1 = \{0\}$ has constant rank. Hence we may form the Courant algebroid relation R fitting the diagram

$$\begin{array}{ccc} & \mathbb{T}M & \\ Q(K_1)^\top \nearrow & & \nwarrow Q(K_2) \\ \mathbb{T}\mathcal{Q}_1 & \xrightarrow{\quad R \quad} & \mathbb{T}\mathcal{Q}_2, \end{array} \quad (4.134)$$

supported on $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$, by Lemma 4.8. $\square$

Example 4.135. Let us write the relation explicitly in the case that $k = 1$, i.e. the circle bundle case. Take charts $\mathcal{U}^{(i)}$ for $\mathcal{B}$ which trivialise both $\mathcal{Q}_1$ and $\mathcal{Q}_2$, and let $U_\ell^{(i)} := \mathcal{U}^{(i)} \times \mathbb{S}^1$ be trivialising charts for $\mathcal{Q}_\ell$, for $\ell = 1, 2$. Let θ_1 and θ_2 be connections on $\mathcal{Q}_1$ and $\mathcal{Q}_2$, respectively. Set $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$ with the two-form⁹

$$B = \dot{\theta}_1 \wedge \ddot{\theta}_2, \quad (4.136)$$

which satisfies the conditions of Theorem 4.88.

Locally, if $H_1 = db_1$ for a Kalb-Ramond field $b_1 \in \Omega^2(U^{(i)})$, then R is given by

$$\Gamma(E_1 \times \overline{E}_2; R|_{U_C^{(i)}}) = \quad (4.137)$$

$$\text{Span}_{C^\infty(\mathcal{V}^{(i)})} \{ (\partial_\alpha, \partial_\alpha - (b_1)_{\alpha\theta} \partial_{\theta_2}), (dx^\alpha, dx^\alpha), (\partial_{\theta_1}, \theta_2), (\theta_1, \partial_{\theta_2}) \},$$

⁹Notation: from here and for the rest of the thesis, when in the correspondence space picture, we denote the pullback by ϖ_1 of an element $\omega \in \bigwedge^\bullet T^* \mathcal{Q}_1$ by $(\varpi_1)^* \omega_1 =: \dot{\omega}_1$, and similarly $(\varpi_2)^* \omega_2 =: \ddot{\omega}_2$ for $\omega_2 \in \bigwedge^\bullet T^* \mathcal{Q}_2$. We will use the same notation to denote the pullback by $\pi_i: \mathcal{Q}_i \rightarrow \mathcal{B}$ where no confusion should arise. The double pullback will be denoted by stacking this notation, thus if $\omega \in \bigwedge^\bullet T^* \mathcal{B}$, we have $(\pi_1 \circ \varpi_1)^* \omega =: \dot{\omega}$ and $(\pi_2 \circ \varpi_2)^* \omega =: \ddot{\omega}$ are elements of $\bigwedge^\bullet T^* M$.

where $U_C^{(i)} = U_1^{(i)} \times_{\mathcal{B}} U_2^{(i)}$ are trivialising charts, ∂_α with $\alpha = 1, \dots, \dim \mathcal{B}$ are horizontal coordinate vector fields on $U^{(i)}$, ∂_{θ_ℓ} are vector fields generating the circle action $\mathfrak{s}_\ell^1$ for $\ell = 1, 2$, and subscripts θ label the coordinates of the circle direction for both $\mathcal{Q}_1$ and $\mathcal{Q}_2$.

4.2.2 Geometric T-duality

Note that the converse of Lemma 4.132 is not true, since our notion of T-duality related Courant algebroids imposes no condition on B other than the constant rank condition from Proposition 4.23, which is trivially satisfied for any two-form $B \in \Omega^2(M)$ since $\mathfrak{k}_1 \cap \mathfrak{k}_2 = \{0\}$. We will see that, upon introducing geometric data, the possible choices for B coincide with the B -field choice in Definition 4.127. This is exemplified through

Proposition 4.138. If $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are T-dual in the sense of Definition 4.127, and there is a $\mathbb{T}_1^k$ -invariant generalised metric V_1^+ on $(\mathbb{T}\mathcal{Q}_1, \underline{H}_1)$, then there is a unique generalised metric V_2^+ such that $(\mathbb{T}\mathcal{Q}_1, \underline{H}_1, V_1^+)$ and $(\mathbb{T}\mathcal{Q}_2, \underline{H}_2, V_2^+)$ are geometrically T-dual.

Proof. If $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are T-dual in the sense of Definition 4.127, then B decomposes as in Equation (4.79). Furthermore, since V_1^+ is $\mathbb{T}^k$ -invariant, we define¹⁰ $\mathfrak{k}_{V_1^+} := \#_{\mathcal{Q}_1}(\mathfrak{k}_2^k) \subset \Gamma(D_1)$, which spans D_1 pointwise. Thus $\mathfrak{k}_2 = \#_M(\mathfrak{k}_2^k) \subset \Gamma(\mathfrak{k}_1)$, and since B is $\mathbb{T}^{2k}$ -invariant it follows that $\mathcal{L}_X B = 0$, for any $X \in \mathfrak{k}_2$. The conclusion then follows by Theorem 4.88. $\square$

In this picture, we can also provide a further characterisation of the components of the three-form $\underline{H}_1 \in \Omega_{\text{cl}}^3(\mathcal{Q}_1)$ through

Lemma 4.139. Let $(\mathbb{T}\mathcal{Q}_1, \underline{H}_1)$ and $(\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$ be twisted standard Courant algebroids with $\underline{H}_1 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_1)$ and $\underline{H}_2 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_2)$, and let them be endowed with generalised metrics V_1^+ and V_2^+ such that they are geometrically T-dual. Then $\varpi_1^* \underline{H}_1$ can be written as the sum of a doubly pulled back three-form $H_{\mathcal{B}} \in \Omega^3(\mathcal{B})$

¹⁰If a Lie group G acts on a manifold M , we denote by $\#_M: \mathfrak{g} \rightarrow \Gamma(TM)$ the corresponding action of its Lie algebra $\mathfrak{g}$ on M .

and a component determined by

$$\iota_{X_{v_2}} \dot{\underline{H}}_1 = \iota_{X_{v_2}} dB , \quad (4.140)$$

for any $X_{v_2} \in \Gamma(\mathfrak{t}_2^k)$.

Proof. Definition 4.127 implies that $\iota_{Y_{v_1}} \iota_{X_{v_1}} \dot{\underline{H}}_1 = 0$, for all $X_{v_1}, Y_{v_1} \in \Gamma(\mathfrak{t}_1)$, since B is $\mathbb{T}^{2k}$ -invariant and $\iota_{X_{v_1}} \ddot{\underline{H}}_2 = 0$, for all $X_{v_1} \in \Gamma(\mathfrak{t}_1)$. Hence only two components of $\dot{\underline{H}}_1$ are non-vanishing and, because of $\mathbb{T}^k$ -invariance of $\underline{H}_1$, one of them must be the double pullback of a three-form on $\mathcal{B}$. The other component is characterised as follows. Since any $\underline{X} \in \Gamma(D_1)$ lifts to a section $X_{v_2} \in \Gamma(\mathfrak{t}_2)$, and $\ddot{\underline{H}}_2$ is basic with respect to the orbits of $\mathbb{T}^k$ whose tangent distribution is $\mathfrak{t}_2$, that is it comes from the base $\mathcal{Q}_2$, Equation (4.140) follows. $\square$

Example 4.141 (B -fields from Connections, [46]). Let $\mathcal{Q}_1$ and $\mathcal{Q}_2$ be principal $\mathbb{T}^k$ -bundles over a common base manifold $\mathcal{B}$, and let them be endowed with closed three-forms $\underline{H}_1 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_1)$ and $\underline{H}_2 \in \Omega_{\mathbb{T}^k}^3(\mathcal{Q}_2)$. Choose connections $\theta_1 \in \Omega^1(\mathcal{Q}_1, \mathfrak{t}_1^k)$ and $\theta_2 \in \Omega^1(\mathcal{Q}_2, \mathfrak{t}_2^k)$, respectively, and suppose that

$$B = -\dot{\theta}_1 \wedge \ddot{\theta}_2 . \quad (4.142)$$

When Lemma 4.139 holds, we can write

$$\dot{\underline{H}}_1 = -\dot{\theta}_1 \wedge \ddot{c}_1(\mathcal{Q}_2) + \dot{\underline{H}}_{\mathcal{B}} , \quad (4.143)$$

where $c_1(\mathcal{Q}_2) = d\theta_2$ represents the Chern class of $\mathcal{Q}_2$. This can be shown by letting $X_{v_2} \in \Gamma(\mathfrak{t}_2)$ be a generator of the $\mathfrak{t}_2^k$ -action. Then

$$\iota_{X_{v_2}} \dot{\underline{H}}_1 = d(\ddot{\theta}_2) = \ddot{c}_1(\mathcal{Q}_2) , \quad (4.144)$$

and Equation (4.143) follows. A symmetric argument holds for $\ddot{\underline{H}}_2$. This is the key case considered in [46, 52] which provides criteria for the existence of T-dual pairs based on the components of the H -flux. It shows how T-duality in this case is characterised by an interchange of the Chern classes of the torus bundle with topological data associated to the H -flux.

The analogy with the correspondence space picture extends further when we consider the construction in [46, Section 3] of the isomorphism¹¹ of $\mathbb{T}^k$ -invariant sections $\mathcal{R}: \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_1) \rightarrow \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_2)$. In that construction a $\mathbb{T}^k$ -invariant section $X_1 + \xi_1 \in \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_1)$ is lifted to a $\mathbb{T}^{2k}$ -invariant section $\hat{X}_1 + \dot{\xi}_1 \in \Gamma_{\mathbb{T}^{2k}}(\mathbb{T}M)$ whose image under B -field transformation is basic:

$$(\dot{\xi}_1)(Y_2) + B(\hat{X}_1, Y_2) = 0 , \quad (4.145)$$

for all $Y_2 \in \Gamma(\mathfrak{t}_2^k)$. The non-degeneracy of B on $\mathfrak{k}_1 \otimes \mathfrak{k}_2$ ensures that the lift $\hat{X}_1$ satisfying Equation (4.145) is unique. Note that the lift $\hat{X}_1 + \dot{\xi}_1$ of $X_1 + \xi_1$ is the unique lift $s(X_1 + \xi_1)$ given by Proposition 4.53. There is thus a form $\xi_2 \in \Omega^1(\mathcal{Q}_2)$ such that $\ddot{\xi}_2 = \dot{\xi}_1 + \iota_{\hat{X}} B$. The isomorphism $\mathcal{R}$ is then constructed as the pushforward by ϖ_2 of $\hat{X}_1 + \dot{\xi}_1 + \iota_{\hat{X}_1} B$:

$$\mathcal{R}(X_1 + \xi_1) := \varpi_{2*} \hat{X}_1 + \xi_2 . \quad (4.146)$$

Proposition 4.123 is the generalisation of this result to arbitrary foliations in the correspondence space picture. We thus recover $\mathcal{R}$ through

Proposition 4.147. The isomorphism $\mathcal{F}$ given by Proposition 4.123 coincides with $\mathcal{R}$ defined by Equation (4.146) upon identifying $\Gamma(\underline{\underline{E}}_i)$ with $\Gamma_{D_i}(\mathbb{T}\mathcal{Q}_i)$ for $i = 1, 2$.

In the local picture, using (4.137), given a generalised metric $(g_1, 0)$, the components of the T -dual generalised metric (g_2, b_2) on $\mathcal{Q}_2$ can be read directly from the relation. For instance, given an element in $\partial_{\theta_1} + g_{\theta\theta}\theta_1 \in V^+ = \text{gr}(g_1)$, the corresponding element in R is given by

$$(\partial_{\theta_1} + g_{\theta\theta}\theta_1, \theta_2 + g_{\theta\theta}\partial_{\theta_2}) \in R^+ . \quad (4.148)$$

¹¹In [46] the isomorphism is denoted by ϕ , but here we use the symbol $\mathcal{R}$ to distinguish it from the already used ϕ in this thesis.

Continuing like this, one can read off the components of the T-dual metric as

$$\begin{aligned}
 (g_2)_{\theta\theta} &= \frac{1}{(g_1)_{\theta\theta}} \quad , \quad (g_2)_{\alpha\theta} = \frac{(b_1)_{\alpha\theta}}{(g_1)_{\theta\theta}} \quad , \\
 (g_2)_{\alpha\beta} &= (g_1)_{\alpha\beta} - \frac{(g_1)_{\alpha\theta} (g_1)_{\beta\theta} - (b_1)_{\alpha\theta} (b_1)_{\beta\theta}}{(g_1)_{\theta\theta}} \quad , \\
 (b_2)_{\alpha\theta} &= \frac{(g_1)_{\alpha\theta}}{(g_1)_{\theta\theta}} \quad , \quad (b_2)_{\alpha\beta} = (b_1)_{\alpha\beta} - \frac{(b_1)_{\alpha\theta} (g_1)_{\beta\theta} - (b_1)_{\beta\theta} (g_1)_{\alpha\theta}}{(g_1)_{\theta\theta}} \quad .
 \end{aligned} \tag{4.149}$$

These are the well-known Buscher rules of T-duality for circle bundles [39, 55].

Chapter 5

Relations for Generalised Riemannian Geometry

When considering quantization of the bosonic string, the full action given by the sum

$$S[\mathbb{X}] = S_g[\mathbb{X}] + S_H[\mathbb{X}] + S_{R_h}[\mathbb{X}] \quad (5.1)$$

of (1.1), (1.2) and (1.25) can be equivalently studied from the low energy effective action (1.26), which is equivalent to a generalised Einstein-Hilbert action (1.27) involving the *generalised scalar curvature*, and whose equations of motion involve the generalised Ricci tensor. The study of such generalised Riemannian geometric objects involves the study of generalised metrics and divergence operators on Courant algebroids.

5.1 Divergence Operators and Courant Algebroid Relations

In this section we shall discuss how Courant algebroid relations are instrumental to defining a robust notion of morphisms between divergence operators on Courant algebroids. We will define the notion of related divergence operators on Courant algebroids admitting a Courant algebroid relation.

5.1.1 Divergence Operators on Courant Algebroids

Let us recall the definition of divergence operator on a Courant algebroid [30, 62]. For further details on divergence operators on Courant algebroids, see [28, 49, 62] and references therein.

Definition 5.2. Let E be a Courant algebroid over a manifold M . A *divergence operator*, or simply a *divergence*, on E is an $\mathbb{R}$ -linear map

$$\operatorname{div} : \Gamma(E) \longrightarrow C^\infty(M) \quad (5.3)$$

satisfying an anchored Leibniz rule

$$\operatorname{div}(f e) = f \operatorname{div} e + \mathcal{L}_{\rho(e)} f , \quad (5.4)$$

for all $e \in \Gamma(E)$ and $f \in C^\infty(M)$.

Divergence operators on a Courant algebroid E form an affine space over $\Gamma(E)$: given any two divergences div and div' on E , there exists a uniquely determined section $e_{\operatorname{div}} \in \Gamma(E)$ such that

$$\operatorname{div} e - \operatorname{div}' e = \langle e_{\operatorname{div}}, e \rangle , \quad (5.5)$$

for all $e \in \Gamma(E)$.

Example 5.6 (Densities). Let E be a Courant algebroid over M and let μ be a nowhere-vanishing density on M . The divergence operator div_μ on E associated with μ is defined by

$$\operatorname{div}_\mu e := \mu^{-1} \mathcal{L}_{\rho(e)} \mu , \quad (5.7)$$

for all $e \in \Gamma(E)$.

Remark 5.8. Consider a Courant algebroid E over M endowed with a divergence operator div . Suppose that M admits a nowhere-vanishing density μ . Then there

is a section $e_{\text{div}} \in \Gamma(E)$ induced by the difference

$$\text{div } e - \text{div}_{\mu} e = \langle e_{\text{div}}, e \rangle , \quad (5.9)$$

for all $e \in \Gamma(E)$. The adjoint action

$$\text{ad}_{\text{div}} := \llbracket e_{\text{div}}, \cdot \rrbracket : \Gamma(E) \longrightarrow \Gamma(E) \quad (5.10)$$

associated with the divergence operator div is independent of the choice of density μ , see [49, Proposition 2.3].

The Dilaton

One application of divergence operators is to provide a definition of the missing piece of a generalised string background (g, H) on a Courant algebroid, namely the dilaton. We follow [28, 49] and [61, Definition 3.7] for the basic definitions and the main properties of dilatons.

Definition 5.11. A *dilaton* on an exact Courant algebroid E over M with generalised metric τ is a divergence div_{ϕ} such that

$$\text{div}_{\phi} e - \text{div}_{\mu_g} e = \langle \rho^*(d\phi), e \rangle , \quad (5.12)$$

for some $\phi \in C^{\infty}(M)$ and for all $e \in \Gamma(E)$, where (g, σ) is the pair of a Riemannian metric g with induced volume form μ_g on M and the isotropic splitting $\sigma : TM \rightarrow E$ associated with the generalised metric τ .¹ The function ϕ is a *dilaton field*.

Example 5.13. Let $H \in \Omega_{\text{cl}}^3(M)$ and let $\mathbb{T}M$ be an H -twisted standard Courant algebroid over an oriented manifold M , endowed with a generalised metric given by $V^+ = \text{gr}(g)$ where g is a Riemannian metric on M . Any divergence div on $\mathbb{T}M$ is associated with some section $X_{\text{div}} + \alpha_{\text{div}} \in \Gamma(\mathbb{T}M)$ by

$$\text{div} - \text{div}_{\mu_g} = \langle X_{\text{div}} + \alpha_{\text{div}}, \cdot \rangle , \quad (5.14)$$

where μ_g is the volume form associated to g . Then the divergence operator div is

¹This is the splitting such that the generalised metric is given by $(g, 0)$ in Proposition 3.41

compatible with V^+ , i.e. $\mathbf{ad}_{\text{div}}(\Gamma(V^+)) \subseteq \Gamma(V^+)$, if and only if

$$\mathcal{L}_{X_{\text{div}}}g = 0 \quad , \quad d\alpha_{\text{div}} - \iota_{X_{\text{div}}}H = 0 . \quad (5.15)$$

If we take

$$\text{div} = \text{div}_{e^{-2\phi} \mu_g} , \quad (5.16)$$

where $\phi \in C^\infty(M)$, then

$$X_{\text{div}} = 0 \quad , \quad \alpha_{\text{div}} = -2d\phi , \quad (5.17)$$

which are the usual equations for the appearance of the dilaton field ϕ in supergravity. Thus $\text{div}_{e^{-2\phi} \mu_g} = \text{div}_{-2\phi}$ in this example.

5.1.2 Related Divergence Operators

We will now give a notion of when two divergence operators are related by a Courant algebroid relation.

Definition 5.18. Let div_1 and div_2 be divergence operators on Courant algebroids E_1 and E_2 over manifolds M_1 and M_2 , respectively. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on a submanifold $C \subseteq M_1 \times M_2$. Then div_1 and div_2 are *R-related*, denoted $\text{div}_1 \sim_R \text{div}_2$, if $\text{div}_1 e_1 \sim_C \text{div}_2 e_2$ for every $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$. We will also use the notation $R: (E_1, \text{div}_1) \dashrightarrow (E_2, \text{div}_2)$ to describe the above setup.

Let us check that this notion of related divergence operators on Courant algebroids is well-defined.

Lemma 5.19. Relations between divergences in the sense of Definition 5.18 are well-defined: if $(e_1, e_2), (e'_1, e'_2) \in \Gamma(E_1 \times \overline{E}_2; R)$ such that $(e_1, e_2)_c = (e'_1, e'_2)_c$ for all $c \in C$, then $\text{div}_1 e_1 \sim_C \text{div}_2 e_2$ if and only if $\text{div}_1 e'_1 \sim_C \text{div}_2 e'_2$.

Proof. Define $\text{div}: \Gamma(E_1) \times \Gamma(E_2) \rightarrow C^\infty(M_1) \times C^\infty(M_2)$ by $\text{div} = (\text{div}_1, \text{div}_2)$, and extend this to $\widetilde{\text{div}}: \Gamma(E_1 \times \overline{E}_2) \rightarrow C^\infty(M_1 \times M_2)$. Suppose that $r, r' \in \Gamma(E_1 \times \overline{E}_2; R)$

with $r(c) = r'(c)$ for all $c \in C$. Then for any open subset $U \subset M_1 \times M_2$, there is a frame $\{r^i\} \subset \Gamma((E_1 \times \overline{E}_2)|_U; R)$ such that $r = \sum_i f_i r^i$ and $r' = \sum_i f'_i r^i$ with $f_i, f'_i \in C^\infty(U)$ satisfying $f_i(c) = f'_i(c)$ for all $c \in C \cap U$. Denoting the anchor of $E_1 \times \overline{E}_2$ by ρ , we then find on $C \cap U$ that

$$\widetilde{\text{div}} r = \sum_i (f_i \widetilde{\text{div}} r^i + \mathcal{L}_{\rho(r^i)} f_i) = \sum_i (f'_i \widetilde{\text{div}} r^i + \mathcal{L}_{\rho(r^i)} f'_i) = \widetilde{\text{div}} r', \quad (5.20)$$

where we use Lemma 3.12. Thus, on C , $\widetilde{\text{div}}$ is independent of the chosen section of $\Gamma(E_1 \times \overline{E}_2; R)$, and hence $\text{div}_1 \sim_R \text{div}_2$ is well-defined. $\square$

Example 5.21 (Classical Courant Algebroid Isomorphisms). Let $E_1 \rightarrow M_1$ and $E_2 \rightarrow M_2$ be Courant algebroids. Let $\Phi \in \text{Hom}(E_1, E_2)$ be a Courant algebroid isomorphism covering a diffeomorphism $\varphi \in \text{Diff}(M_1, M_2)$. If div_1 is a divergence operator on E_1 , then E_2 is endowed with a divergence operator div_2 given by

$$\text{div}_2 := (\varphi^{-1})^* \circ \text{div}_1 \circ \Phi^{-1}. \quad (5.22)$$

The divergence operators div_1 and div_2 are $\text{gr}(\Phi)$ -related.

Example 5.23 (Courant Algebroid Morphisms). Let $R: E_1 \rightarrowtail E_2$ be a Courant algebroid morphism over a smooth map $\varphi: M_1 \rightarrow M_2$. Then the condition of Definition 5.18 for div_1 and div_2 to be R -related divergences becomes

$$\text{div}_1 e_1 = \varphi^*(\text{div}_2 e_2), \quad (5.24)$$

for all $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$.

Consider the divergences given by

$$\text{div}_{\mu_i} e_i = \mu_i^{-1} \mathcal{L}_{\rho_i(e_i)} \mu_i, \quad (5.25)$$

where μ_i are nowhere-vanishing densities on M_i for $i = 1, 2$. Then Equation (5.24) implies that $\text{div}_{\mu_1} \sim_R \text{div}_{\mu_2}$ if and only if $\mu_1 = \varphi^* \mu_2$.

Example 5.26 (Reduction by Isotropic G-Actions). Let E be a Courant algebroid over M endowed with an isotropic action of a Lie algebra $\mathfrak{g}$, i.e. with a

linear map $\chi: \mathfrak{g} \rightarrow \Gamma(E)$ preserving the brackets such that $K = \chi(\mathfrak{g})$ is an isotropic subbundle of E . Let G be the connected Lie group integrating $\mathfrak{g}$, and suppose it acts freely and properly on M . Thus $\mathcal{Q} = M/G$ is a smooth manifold, with smooth quotient map $\varpi: M \rightarrow \mathcal{Q}$. Theorem 2.40 then constructs the reduced Courant algebroid $\underline{E}$ over $\mathcal{Q}$ such that $\Gamma(\underline{E}) \cong \Gamma_G(K^\perp) / \Gamma_G(K)$, where the subscript G denotes the G -invariant sections. By the construction in Section 3.1.2, there is a Courant algebroid morphism $Q(K): E \rightarrow \underline{E}$ over $\varpi: M \rightarrow \mathcal{Q}$.

A divergence operator div on E is G -equivariant if it obeys

$$\mathcal{L}_{\rho(\chi(x))} \text{div } e = \text{div} [\chi(x), e] \quad , \quad \text{div } \chi(x) = -\text{Tr}_{\mathfrak{g}}(\text{ad}_x) \quad , \quad (5.27)$$

for all $e \in \Gamma(E)$ and $x \in \mathfrak{g}$. Then by [49, Proposition 5.11], there exists a divergence operator $\underline{\text{div}}$ on the reduced Courant algebroid $\underline{E}$, since div restricts to a map $\Gamma_G(E) \rightarrow C_G^\infty(M)$ by the first condition of Equation 5.27 and $\text{div } \Gamma_G(K) = \{0\}$ by the second condition. In other words, the reduced divergence operator $\underline{\text{div}}$ satisfies

$$\varpi^*(\underline{\text{div}} \underline{e}) = \text{div } e \quad , \quad (5.28)$$

for any pair $(e, \underline{e}) \in \Gamma(E \times \underline{E}; Q(K))$. Thus, by Equation (5.24), we have $\text{div} \sim_{Q(K)} \underline{\text{div}}$.

Example 5.29 (Heterotic Reduction). Let us consider the same setting as in Example 5.26, with a few variations appropriate to applications in heterotic string theory [47]: suppose that $K = \chi(\mathfrak{g})$ satisfies $K \cap K^\perp = \{0\}$ and that div only satisfies the first condition of Equation (5.27). Then div still restricts to a map $\Gamma_G(E) \rightarrow C_G^\infty(M)$ and, because of our assumption on K , it descends to a divergence $\underline{\text{div}}$ on the reduced Courant algebroid $\underline{E} = K^\perp / G$ constructed in [47, 51]. Again the divergences div and $\underline{\text{div}}$ are $Q(K)$ -related.

Recall that divergences on a Courant algebroid E form an affine space over $\Gamma(E)$. This leads to

Lemma 5.30. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on $C \subseteq M_1 \times M_2$, and let $\text{div}_1 \sim_R \text{div}_2$ be R -related divergence operators. Let $\text{div}'_i - \text{div}_i = \langle e_{\text{div}_i}, \cdot \rangle_i$ for some $e_{\text{div}_i} \in \Gamma(E_i)$, for $i = 1, 2$. Then $\text{div}'_1 \sim_R \text{div}'_2$ if and only

if $(e_{\text{div}_1}, e_{\text{div}_2}) \in \Gamma(E_1 \times \overline{E}_2; R)$.

Proof. Let $r' = (e'_1, e'_2) \in \Gamma(E_1 \times \overline{E}_2; R)$. Since $\text{div}_1 \sim_R \text{div}_2$, it follows that $\text{div}'_1 \sim_R \text{div}'_2$ if and only if $\langle e_{\text{div}_1}, e'_1 \rangle_1 \sim_C \langle e_{\text{div}_2}, e'_2 \rangle_2$. Thus, for any $c = (m_1, m_2) \in C$, this requires

$$0 = \langle e_{\text{div}_1}, e'_1 \rangle_1(m_1) - \langle e_{\text{div}_2}, e'_2 \rangle_2(m_2) = \langle (e_{\text{div}_1}, e_{\text{div}_2}), r' \rangle(c). \quad (5.31)$$

Since r' was arbitrary, it follows that $(e_{\text{div}_1}, e_{\text{div}_2}) \in \Gamma(E_1 \times \overline{E}_2; R^\perp)$. Since R is a Dirac structure, then $R^\perp = R$. $\square$

When the relation R has trivial kernel and cokernel, there is a uniqueness result for divergences given by

Lemma 5.32. Let $R: E_1 \dashrightarrow E_2$ is a Dirac structure with trivial kernel $\text{Ker}(R)$ and cokernel $\text{CoKer}(R)$ (see Definition 3.21), supported on $C \subseteq M_1 \times M_2$ such that $\text{pr}_2(C) = M_2$. Let div_1 be a divergence on E_1 . Then there is at most one divergence div_2 on E_2 such that $\text{div}_1 \sim_R \text{div}_2$.

Proof. Suppose that divergence operators div_2 and div'_2 on E_2 are R -related to div_1 . Then there is a section $e \in \Gamma(E_2)$ such that $\text{div}_2 - \text{div}'_2 = \langle e, \cdot \rangle_2$. Thus at a point $m_2 \in M_2$ with $(m_1, m_2) \in C$ for some $m_1 \in M_1$, we get

$$0 = (\text{div}_1 e_1 - \text{div}_1 e_1)(m_1) = (\text{div}_2 e_2 - \text{div}'_2 e_2)(m_2) = \langle e, e_2 \rangle_2(m_2) \quad (5.33)$$

for all $e_2 \in (E_2)_{m_2}$, with $e_1 \in (E_1)_{m_1}$ such that $(e_1, e_2) \in R_{(m_1, m_2)}$. The Dirac condition, combined with the trivial kernel and cokernel condition, ensure that we can find such an element $e_1 \in (E_1)_{m_1}$ for all $m_2 \in M_2$ and $e_2 \in (E_2)_{m_2}$. It follows that $e = 0$. $\square$

Let us consider two R -related Courant algebroids E_1 and E_2 endowed, respectively, with divergence operators div_1 and div_2 . Suppose that M_1 and M_2 admit nowhere-vanishing densities μ_1 and μ_2 . Then we get sections e_{div_1} and e_{div_2} of E_1 and E_2 , respectively, defined as in Remark 5.8.

Proposition 5.34. Suppose that the divergence operators $(\text{div}_1, e_{\text{div}_1})$ and $(\text{div}_2, e_{\text{div}_2})$ on the Courant algebroids E_1 and E_2 are R -related, where R is a Dirac structure

in $E_1 \times \overline{E}_2$. Then the derivations $\mathbf{ad}_{\text{div}_1} := \llbracket e_{\text{div}_1}, \cdot \rrbracket_1$ and $\mathbf{ad}_{\text{div}_2} := \llbracket e_{\text{div}_2}, \cdot \rrbracket_2$ are R -related.

Proof. Since $e_{\text{div}_1} \sim_R e_{\text{div}_2}$, by definition $(e_{\text{div}_1}, e_{\text{div}_2}) \in \Gamma(E_1 \times \overline{E}_2; R)$. By definition of the Dorfman bracket on the product Courant algebroid $E_1 \times \overline{E}_2$ we get

$$\llbracket (e_{\text{div}_1}, e_{\text{div}_2}), (\cdot, \cdot) \rrbracket = (\llbracket e_{\text{div}_1}, \cdot \rrbracket_1, \llbracket e_{\text{div}_2}, \cdot \rrbracket_2) = (\mathbf{ad}_{\text{div}_1}, \mathbf{ad}_{\text{div}_2}) . \quad (5.35)$$

Since R is involutive, it follows that

$$\llbracket (e_{\text{div}_1}, e_{\text{div}_2}), (e'_1, e'_2) \rrbracket = (\mathbf{ad}_{\text{div}_1} e'_1, \mathbf{ad}_{\text{div}_2} e'_2) \in \Gamma(E_1 \times \overline{E}_2; R) , \quad (5.36)$$

for all $(e'_1, e'_2) \in \Gamma(E_1 \times \overline{E}_2; R)$. Hence $\mathbf{ad}_{\text{div}_1} \sim_R \mathbf{ad}_{\text{div}_2}$ in the sense of [1, Definition 6.1] extended to covariant differential operators on E_1 and E_2 . $\square$

Note that this result holds only if R is a Dirac structure, since it relies on the fact that the sections e_{div_1} and e_{div_2} are R -related.

Divergences from Connections

A broad class of divergence operators is obtained via Courant algebroid connections. A *Courant algebroid connection* ∇^E on a Courant algebroid $E \rightarrow M$ is an $\mathbb{R}$ -bilinear map

$$\nabla^E: \Gamma(E) \times \Gamma(E) \longrightarrow \Gamma(E) , \quad (5.37)$$

denoted $(e_1, e_2) \mapsto \nabla_{e_1}^E e_2$, which satisfies

$$\nabla_{f e_1}^E e_2 = f \nabla_{e_1}^E e_2 \quad , \quad \nabla_{e_1}^E (f e_2) = f \nabla_{e_1}^E e_2 + (\mathcal{L}_{\rho(e_1)} f) e_2 , \quad (5.38)$$

together with the metric compatibility condition

$$\langle \nabla_e^E e_1, e_2 \rangle + \langle e_1, \nabla_e^E e_2 \rangle = \mathcal{L}_{\rho(e)} \langle e_1, e_2 \rangle , \quad (5.39)$$

for all $e, e_1, e_2 \in \Gamma(E)$ and $f \in C^\infty(M)$.

Following [28, 62], the divergence operator of a Courant algebroid connection

$\nabla^E: \Gamma(E) \times \Gamma(E) \rightarrow \Gamma(E)$ is defined by

$$\operatorname{div}_{\nabla^E}(e) := \operatorname{Tr}(\nabla^E e), \quad (5.40)$$

for $e \in \Gamma(E)$, where the trace operator $\operatorname{Tr} \in \Gamma(\operatorname{End}(E)^*)$ is given on decomposable endomorphisms by $\operatorname{Tr}(\alpha \otimes e) := \alpha(e)$ for $\alpha \in \Gamma(E^*)$ and $e \in \Gamma(E)$. We shall discuss the circumstances under which divergences defined by Equation (5.40) are related as in Definition 5.18 if the underlying Courant algebroid connections are related.

Courant algebroid connections ∇^1 on E_1 and ∇^2 on E_2 are *R-related*, denoted $\nabla^1 \sim_R \nabla^2$, if $\nabla_{e_1}^1 e'_1 \sim_R \nabla_{e_2}^2 e'_2$ for any $(e_1, e_2), (e'_1, e'_2) \in \Gamma(E_1 \times \overline{E}_2; R)$; see [1, Section 6] for more details.

Proposition 5.41. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on $C \subseteq M_1 \times M_2$ with trivial kernel $\operatorname{Ker}(R)$ and cokernel $\operatorname{CoKer}(R)$, which is a Dirac structure in $E_1 \times \overline{E}_2$. Let $\nabla^1 \sim_R \nabla^2$ be *R*-related Courant algebroid connections on E_1 and E_2 . Then $\operatorname{div}_{\nabla^1} \sim_R \operatorname{div}_{\nabla^2}$.

Proof. Because of $C^\infty(M_i)$ -linearity in the first entry of ∇^i for $i = 1, 2$, we can define linear maps $A_{e_1}^1 = \nabla_{e_1}^1: \Gamma(E_1) \rightarrow \Gamma(E_1)$, for any $e_1 \in \Gamma(E_1)$, and $A_{e_2}^2 = \nabla_{e_2}^2: \Gamma(E_2) \rightarrow \Gamma(E_2)$, for any $e_2 \in \Gamma(E_2)$. Thus if $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$, then $A_{e_1}^1(e'_1) \sim_R A_{e_2}^2(e'_2)$ for any pair $(e'_1, e'_2) \in \Gamma(E_1 \times \overline{E}_2; R)$. Since R is a Dirac structure with trivial kernel and cokernel, there are *R*-related local frames $\{\psi_i^1\}$ for E_1 and $\{\psi_i^2\}$ for E_2 coming from a choice of any local frame for the bundle $R \rightarrow C$.² At any point $c \in C$, it follows that $(A_{e_1}^1)_k^j = (A_{e_2}^2)_k^j$ with respect to the *R*-related bases $\{\psi_i^1\}$ and $\{\psi_i^2\}$. Therefore $\operatorname{Tr}_1(\nabla^1 e_1)_{\operatorname{pr}_1(c)} = \operatorname{Tr}_2(\nabla^2 e_2)_{\operatorname{pr}_2(c)}$, for all $c \in C$. $\square$

Note that the trace operator in Equation (5.40) constrains the type of Courant algebroid relations that are suitable to bridge the notion of related connections with related divergences. On the other hand, for the relevant applications of this paper, the assumptions of Proposition 5.41 always hold since the T-duality relation will always be a generalised isometry, and hence has trivial kernel and cokernel.

²As in [1], we always assume that C is an embedded submanifold of $M_1 \times M_2$.

The Dilaton Shift

The natural step to take now is to see how dilatons, discussed in Subsection 5.1.1, behave when their corresponding divergences are related. In particular, we address how a generalised isometry yields a natural definition for the dilaton shift, see for instance [48].

For $i = 1, 2$, let $E_i \rightarrow M_i$ be exact Courant algebroids endowed with generalised metrics τ_i and divergence operators div_i . Define sections $e_{\text{div}_i} \in \Gamma(E_i)$ by $\langle e_{\text{div}_i}, \cdot \rangle_i = \text{div}_i - \text{div}_{\mu_{g_i}}$, where μ_{g_i} is the volume form on M_i induced by the generalised metric τ_i . Suppose that $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$ is a generalised isometry which is a Dirac structure in $E_1 \times \overline{E}_2$ such that $\text{div}_1 \sim_R \text{div}_2$.

For any $(e, {}^R e) \in \Gamma(E_1 \times \overline{E}_2; R)$, we then find

$$\langle e_{\text{div}_1}, e \rangle_1 + \text{div}_{\mu_{g_1}} e = \text{div}_1 e = \text{div}_2 {}^R e = \langle e_{\text{div}_2}, {}^R e \rangle_2 + \text{div}_{\mu_{g_2}} {}^R e . \quad (5.42)$$

Write $e_{\text{div}_2} - {}^R e_{\text{div}_1} =: e_{\text{dil}} \in \Gamma(E_2)$, where ${}^R e_{\text{div}_1}$ is the unique section of E_2 with $(e_{\text{div}_1}, {}^R e_{\text{div}_1}) \in \Gamma(E_1 \times \overline{E}_2; R)$. Since R is a Dirac structure, we then get

$$\begin{aligned} \langle e_{\text{div}_2}, {}^R e \rangle_2 + \text{div}_{\mu_{g_2}} {}^R e &= \langle {}^R e_{\text{div}_1}, {}^R e \rangle_2 + \langle e_{\text{dil}}, {}^R e \rangle_2 + \text{div}_{\mu_{g_2}} {}^R e \\ &= \langle e_{\text{div}_1}, e \rangle_1 + \langle e_{\text{dil}}, {}^R e \rangle_2 + \text{div}_{\mu_{g_2}} {}^R e . \end{aligned} \quad (5.43)$$

This leaves

$$\langle e_{\text{dil}}, {}^R e \rangle_2 = \text{div}_{\mu_{g_1}} e - \text{div}_{\mu_{g_2}} {}^R e , \quad (5.44)$$

which measures the failure of the divergences $\text{div}_{\mu_{g_1}}$ and $\text{div}_{\mu_{g_2}}$ from being R -related.

Definition 5.45. The *dilaton shift* is the section

$$e_{\text{dil}} = e_{\text{div}_2} - {}^R e_{\text{div}_1} \quad (5.46)$$

of E_2 , given by Equation (5.44) for $(e, {}^R e) \in \Gamma(E_1 \times \overline{E}_2; R)$.

5.1.3 Invariant Divergence Operators

Let us now describe an appropriate notion of invariant divergences on a Courant algebroid $E \rightarrow M$, with M endowed with a foliation $\mathcal{F}$ integrating a distribution D , using the notions developed in Section 4.1.4.

Definition 5.47. Let $D \subset TM$ be a distribution which is spanned pointwise by a (local) Lie subalgebra $\mathfrak{k} \subset \Gamma(D)$. A divergence operator div on E is (locally) compatible with D if

$$\mathcal{L}_X \text{div } e = \text{div } \text{ad}_X^\sigma e, \quad (5.48)$$

for every $e \in \Gamma(E)$ and $X \in \mathfrak{k}$ (respectively $X \in \mathfrak{k}^{(i)}$ for all $i \in I$).

Definition 5.47 gives, in this setting, the notion of equivariant divergence operator with respect to a group action on a Courant algebroid, see e.g. [49, Equation (13)].

Lemma 5.49. If div is a (locally) D -compatible divergence, then $\mathcal{L}_X \text{div } e^\circ = 0$ for all $e^\circ \in \Gamma_D(E)$. Conversely, if the space of (locally) D -invariant sections spans E pointwise and $\mathcal{L}_X \text{div } e^\circ = 0$ for all $X \in \mathfrak{k}$ and $e^\circ \in \Gamma_D(E)$, then div is (locally) compatible with D .

Proof. The first statement follows from Equation (5.48).

For the second statement, assume that E is spanned pointwise by $\Gamma_D(E)$, i.e. any section $e \in \Gamma(E)$ can be expressed as $e = \sum_i f_i e_i^\circ$ where $\{f_i\} \subset C^\infty(M)$ and $\{e_i^\circ\} \subset \Gamma_D(E)$, and that $\mathcal{L}_X \text{div } e^\circ = 0$ for all $X \in \mathfrak{k}$ and $e^\circ \in \Gamma_D(E)$. Then

$$\begin{aligned} \mathcal{L}_X \text{div } e &= \mathcal{L}_X \text{div} \left(\sum_i f_i e_i^\circ \right) \\ &= \sum_i \left((\mathcal{L}_X f_i) \text{div } e_i^\circ + f_i \mathcal{L}_X \text{div } e_i^\circ + \mathcal{L}_X \mathcal{L}_{\rho(e_i^\circ)} f_i \right) \\ &= \sum_i \left((\mathcal{L}_X f_i) \text{div } e_i^\circ + \mathcal{L}_{\rho(e_i^\circ)} \mathcal{L}_X f_i \right) \\ &= \text{div} \left(\sum_i (\mathcal{L}_X f_i) e_i^\circ \right). \end{aligned} \quad (5.50)$$

We now use the Leibniz rule for the σ -adjoint action (see Lemma B.3) to show that

$$\mathrm{ad}_X^\sigma e = \mathrm{ad}_X^\sigma \left(\sum_i f_i e_i^\circ \right) = \sum_i (\mathcal{L}_X f_i) e_i^\circ, \quad (5.51)$$

and the result follows. This calculation can also be carried out locally. $\square$

Example 5.52. Let E be an exact Courant algebroid over M endowed with the divergence operator div_μ associated to a nowhere-vanishing density μ on M which is D -invariant, i.e. $\mathcal{L}_X \mu = 0$ for all $X \in \mathfrak{k}$. Then div_μ is compatible with D in the sense of Definition 5.47.

For instance, let g be a D -invariant Riemannian metric on M with corresponding Hodge operator $\star_g$, and take $\mu = \mu_g = \star_g 1$. Then, since $\mathcal{L}_X \circ \star_g = \star_g \circ \mathcal{L}_X$, we see that μ_g is D -invariant as well, and

$$\mathcal{L}_X \mathrm{div}_{\mu_g} e^\circ = 0 \quad (5.53)$$

for any $e^\circ \in \Gamma_D(E)$ and $X \in \mathfrak{k}$.

Lemma 5.54. Let $E \rightarrow M$ be an exact Courant algebroid endowed with a D -invariant generalised metric, and any divergence operator div . Set $\mathrm{div} - \mathrm{div}_{\mu_g} = \langle e_{\mathrm{div}}, \cdot \rangle$, where μ_g is the volume form on M induced by the generalised metric. Then $\mathcal{L}_X \mathrm{div} e^\circ = 0$, for all $e^\circ \in \Gamma_D(E)$ and $X \in \mathfrak{k}$, if and only if $e_{\mathrm{div}} \in \Gamma_D(E)$. If $\Gamma_D(E)$ spans E pointwise, then div is moreover compatible with D .

Proof. For any $e^\circ \in \Gamma_D(E)$ and $X \in \mathfrak{k}$, by Lemma B.6,

$$\begin{aligned} 0 &= \mathcal{L}_X \mathrm{div} e^\circ = \mathcal{L}_X \mathrm{div}_{\mu_g} e^\circ + \mathcal{L}_X \langle e_{\mathrm{div}}, e^\circ \rangle \\ &= 0 + \langle \mathrm{ad}_X^\sigma e_{\mathrm{div}}, e^\circ \rangle + \langle e_{\mathrm{div}}, \mathrm{ad}_X^\sigma e^\circ \rangle \\ &= \langle \mathrm{ad}_X^\sigma e_{\mathrm{div}}, e^\circ \rangle \end{aligned} \quad (5.55)$$

This vanishes if and only if $e_{\mathrm{div}} \in \Gamma_D(E)$, establishing the first statement.

The second statement follows from Lemma 5.49. $\square$

5.1.4 T-Duality for Divergence Operators

We shall now prove a result similar to Theorem 4.45 for divergence operators, under the condition

$$K_1 \cap K_2 = \{0\} , \quad (5.56)$$

which we assume throughout this subsection.

Lemma 4.106 shows how D_1 -invariant sections of $\underline{E}_1$ can be lifted to K_2 -basic sections. This property is crucial for the behaviour of divergence operators under T-duality.

Theorem 5.57. Under the assumptions of Theorem 4.45, suppose that $D_1 \subset T\mathcal{Q}_1$ is spanned pointwise by a (local) Lie subalgebra $\mathfrak{k}_{\tau_1} \subset \Gamma(D_1)$. Assume further that $K_1 \cap K_2 = \{0\}$ and that $\Gamma_{D_1}(\underline{E}_1)$ spans $\underline{E}_1$ pointwise. If $\underline{E}_1$ admits a divergence div_1 which is (locally) compatible with D_1 , then there exists a unique divergence operator div_2 on $\underline{E}_2$ such that $\text{div}_1 \sim_R \text{div}_2$.

Proof. Let div_1 be a D_1 -compatible divergence on $\underline{E}_1$. By [5, Proposition 5.31], there is a unique splitting $s_1: K_1^\perp/K_1 \rightarrow K_1^\perp$ of the short exact sequence

$$0 \longrightarrow K_1 \longrightarrow K_1^\perp \longrightarrow K_1^\perp/K_1 \longrightarrow 0 \quad (5.58)$$

such that $\text{im}(s_1) \subset K_1^\perp \cap K_2^\perp$. Since $K_1 \cap K_2 = \{0\}$ and $\text{rk}(K_1) = \text{rk}(K_2)$, it follows that $\text{im}(s_1) = K_1^\perp \cap K_2^\perp$.

Define the map $\delta: \Gamma(\text{im}(s_1)) \rightarrow \varpi_1^*(C^\infty(\mathcal{Q}_1))$ by

$$\delta(s_1(e)) = \text{div}_1 \underline{e} , \quad (5.59)$$

where e is the element corresponding to $\underline{e} \in \Gamma(\underline{E}_1)$ using the identifications $\Gamma(\underline{E}_1) \cong \Gamma_{\text{bas}}(K_1^\perp)/\Gamma(K_1)$ and $C_{\text{bas}}^\infty(M, \mathcal{F}_1) \cong C^\infty(\mathcal{Q}_1)$. Note that δ behaves like a divergence operator, i.e. it satisfies a Leibniz-like rule, because $\varpi_{1*}(\rho(s_1(e))) = \underline{\rho}_1(\underline{e})$. Using Definition 5.47, the “divergence” δ satisfies $\mathcal{L}_X \delta(s_1(e)) = 0$, for any $X \in \mathfrak{k}_2$. One extends δ to the whole of $\Gamma(\text{im}(s_1))$ by the Leibniz rule.

Any section $e \in \Gamma(\text{im}(s_1)) \cap \Gamma_{\text{bas}}(K_2^\perp)$ can be written as $e = \sum_i f_i s_1(\underline{e}_i)$, where

$\underline{e}_i \in \Gamma_{D_1}(\underline{E}_1)$ and $f_i \in \varpi_2^*(C^\infty(\mathcal{Q}_2))$. Likewise any vector field $X \in \Gamma(\rho(K_2))$ can be written as $X = \sum_j g_j X_j$, where $X_j \in \mathfrak{k}_2$ and $g_j \in C^\infty(M)$. Thus

$$\begin{aligned} \mathcal{L}_X \delta(e) &= \sum_{i,j} \left(g_j \mathcal{L}_{X_j} (f_i \delta(s_1(\underline{e}_i))) + g_j \mathcal{L}_{X_j} \mathcal{L}_{\rho(s_1(\underline{e}_i))} f_i \right) \\ &= \sum_{i,j} \left(g_j (\mathcal{L}_{X_j} f_i) \delta(s_1(\underline{e}_i)) + g_j f_i \mathcal{L}_{X_j} \delta(s_1(\underline{e}_i)) \right. \\ &\quad \left. + g_j \mathcal{L}_{\rho(s_1(\underline{e}_i))} \mathcal{L}_{X_j} f_i + g_j \mathcal{L}_{[X_j, \rho(s_1(\underline{e}_i))]} f_i \right) = 0 . \end{aligned} \quad (5.60)$$

The final term vanishes since $[\varpi_{1*}(X_j), \rho_1(\underline{e}_i)] = 0$, hence $[X_j, \rho(s_1(\underline{e}_i))] \in \Gamma(\rho(K_1))$, whereas

$$[X_j, \rho(s_1(\underline{e}_i))] = \rho(\llbracket \sigma_2(X_j), s_1(\underline{e}_i) \rrbracket) \in \Gamma(\rho(K_2^\perp \cap K_1)) \subset \Gamma(\rho(K_2)) . \quad (5.61)$$

Hence the restriction of δ to $\Gamma(\text{im}(s_1)) \cap \Gamma_{\text{bas}}(K_2^\perp)$ maps to $\varpi_2^*(C^\infty(\mathcal{Q}_2))$.

By Proposition 4.53 there also exists a unique splitting $s_2 : K_2^\perp / K_2 \rightarrow K_2^\perp$ of the short exact sequence

$$0 \longrightarrow K_2 \longrightarrow K_2^\perp \longrightarrow K_2^\perp / K_2 \longrightarrow 0 \quad (5.62)$$

induced by $K_2^\perp$ and K_2 . As for s_1 , its image is $\text{im}(s_2) = K_1^\perp \cap K_2^\perp$. Thus we may define a divergence operator $\text{div}_2 : \Gamma(\underline{E}_2) \rightarrow C^\infty(\mathcal{Q}_2)$ by

$$\text{div}_2 \underline{e}_2 = \delta(e_2) , \quad (5.63)$$

where e_2 is the lift of $\underline{e}_2 \in \Gamma(\underline{E}_2)$ to $\Gamma_{\text{bas}}(K_2^\perp)$ using the unique splitting s_2 and the identifications $\Gamma(\underline{E}_2) \cong \Gamma_{\text{bas}}(K_2^\perp) / \Gamma(K_2)$ and $C_{\text{bas}}^\infty(M, \mathcal{F}_2) \cong C^\infty(\mathcal{Q}_2)$.

From Lemma 4.106 it follows that any K_2 -basic section e of $\text{im}(s_1)$ is written as $e = \sum_i f_i s_1(\underline{e}_i)$, where $\underline{e}_i \in \Gamma_{D_1}(\underline{E}_1)$ and $f_i \in \varpi_2^*(C^\infty(\mathcal{Q}_2))$. Hence e is also basic with respect to K_1 when $f_i \in \varpi_1^*(C^\infty(\mathcal{Q}_1)) \cap \varpi_2^*(C^\infty(\mathcal{Q}_2))$. Thus R -related sections $\psi_1 \sim_R \psi_2$ are given by $\psi_1 = \sum_i f_i \underline{e}_i$ and $\psi_2 = \sum_i \varpi_{2*}(\varpi_1^*(f_i)) \mathfrak{z}_2(s_1(\underline{e}_i))$, for $f_i \in C_{D_1}^\infty(\mathcal{Q}_1)$ and $\underline{e}_i \in \Gamma_{D_1}(\underline{E}_1)$. By the above construction, $\text{div}_1 \psi_1 = \text{div}_2 \psi_2$.

Uniqueness follows from Lemma 5.32.

The local version follows the same construction, by noting that the invariance of

$\delta(e)$ can be checked locally. $\square$

Correspondence Spaces

Using the results of Section 4.1.4, we can show that T-dual divergences induce isomorphic divergences, under $\mathcal{F}$, on the transitive Courant algebroids $\underline{E}_1$ and $\underline{E}_2$.

Lemma 5.64. Under the assumptions of Theorem 5.57, let $\underline{\text{div}}_1$ be a divergence on $\underline{E}_1$. Then there exists a divergence operator $\underline{\text{div}}_2$ on $\underline{E}_2$ such that

$$\underline{\text{div}}_1 \underline{e} = \underline{\text{div}}_2 \mathcal{F}(\underline{e}) , \quad (5.65)$$

for all $\underline{e} \in \Gamma(\underline{E}_1)$.

Proof. The pullback $\pi_1^* \underline{\text{div}}_1$ defines a D_1 -compatible divergence on $\underline{E}_1$. By Theorem 5.57, there is a T-dual divergence $\underline{\text{div}}_2$ on $\underline{E}_2$. By Lemma 5.49, it follows that $\underline{\text{div}}_2$ is D_2 -compatible, and hence defines a divergence $\underline{\text{div}}_2 := \pi_{2*} \underline{\text{div}}_2$ on $\underline{E}_2$ which by construction satisfies Equation (5.65). $\square$

5.2 Generalised Ricci Flow and Courant Algebroid Relations

In this section we discuss the problem of whether the generalised Ricci flow is compatible with geometric T-duality of Definition 4.25, as well as Theorems 4.45 and 5.57. In other words, provided with the short-time existence of solutions of generalised Ricci flow for $(\underline{E}_1, V_1^+, \text{div}_1)$, as developed in [61], we investigate whether the T-dual generalised metric V_2^+ of Theorem 4.45 and divergence div_2 of Theorem 5.57 on $\underline{E}_2$ at a given time is a solution of generalised Ricci flow as well. Our starting point will be the introduction of a suitable notion of relation for generalised Ricci tensors, which enables a proof of the compatibility of T-duality with generalised string background equations.

5.2.1 Generalised Ricci Tensor and Scalar Curvature

Let us introduce the main object of our study in this section, the *generalised Ricci tensor*, following the definition from [49, Definition 3.1].

Definition 5.66. Let E be a Courant algebroid over M endowed with a generalised metric V^+ and a divergence operator div . The *generalised Ricci tensor* of the pair (V^+, div) is the map

$$\text{GRic}_{V^+, \text{div}} : \Gamma(V^+) \times \Gamma(V^-) \longrightarrow C^\infty(M) \quad (5.67)$$

given by

$$\text{GRic}_{V^+, \text{div}}(v^+, w^-) := \text{div} \llbracket v^+, w^- \rrbracket^+ - \mathcal{L}_{\rho(w^-)} \text{div} v^+ - \text{Tr}_{V^+}(\llbracket \cdot, w^- \rrbracket^-, v^+ \rrbracket^+) , \quad (5.68)$$

for $v^+ \in \Gamma(V^+)$ and $w^- \in \Gamma(V^-)$, where $\llbracket \cdot, \cdot \rrbracket^\pm$ denotes the projection of the Dorfman bracket to the subbundle $V^\pm$. We will equivalently use the notation $\text{GRic}_{\tau, \text{div}}$, where τ is the automorphism of E defined by the generalised metric V^+ .

By interchanging the roles of V^+ and V^- , we similarly define $\text{GRic}_{V^-, \text{div}}$.

It is shown in [49, Proposition 3.2] that $\text{GRic}_{V^+, \text{div}} \in \Gamma(V^{+*} \otimes V^{-*})$, i.e. $\text{GRic}_{V^+, \text{div}}$ is a covariant tensor on the subbundles V^+ and V^- . There are many definitions of generalised Ricci tensor available in the literature, however it has been shown in [63] that all of them are equivalent. We choose the present formulation in order to discuss how generalised Ricci tensors relate via generalised isometries which are also relations between divergence operators.

Remark 5.69. When $E = (\mathbb{T}M, H)$ is a H -twisted standard Courant algebroid over M , for $H \in \Omega_{\text{cl}}^3(M)$, the generalised Ricci tensor is related to the classical Ricci tensor Ric_g of the Riemannian metric g induced by V^+ , as well as the Kalb-Ramond flux H and the divergence div_{μ_g} : If $\text{div} - \text{div}_{\mu_g} = \langle e, \cdot \rangle$ with $e = e^+ + e^- \in \Gamma(V^+ \oplus V^-)$, we denote $\varphi^\pm = g(\rho(e^\pm), \cdot)$. Then for $X, Y \in \Gamma(TM)$, it follows from [28, Proposition 3.30] that

$$\text{GRic}_{V^\pm, \text{div}}(X^\pm, Y^\mp) = \left(\text{Ric}_g - \frac{1}{4} H^2 \mp \frac{1}{2} d^* H \pm \nabla^\pm \varphi^\pm \right)(X, Y) , \quad (5.70)$$

where $X^\pm = X \pm g(X, \cdot)$, $H^2(X, Y) = g(\iota_X H, \iota_Y H)$, d^* is the codifferential corresponding to the exterior differential d , and

$$\nabla^\pm = \nabla^g \pm \frac{1}{2} g^{-1} H \quad (5.71)$$

are the Bismut connections³ with ∇^g the Levi-Civita connection of g .

Following [61] we introduce the *full generalised Ricci tensor* $\overline{\text{GRic}}_{V^+, \text{div}}$ defined by

$$\overline{\text{GRic}}_{V^+, \text{div}} := \text{GRic}_{V^+, \text{div}} - \text{GRic}_{V^-, \text{div}} . \quad (5.72)$$

In [28, Definition 3.52] a pair (V^+, div) on a Courant algebroid E is called a *generalised Einstein pair* if $\overline{\text{GRic}}_{V^+, \text{div}} = 0$.

The full generalised Ricci tensor also allows for a definition of the generalised scalar curvature in this setting [61, Definition 4.9].

Definition 5.73. The *generalised scalar curvature* of the pair (V^+, div) is the function $\text{GR}_{V^+, \text{div}} \in C^\infty(M)$ given by

$$\text{GR}_{V^+, \text{div}} := \text{Tr} \left(\tau \overline{\text{GRic}}_{V^+, \text{div}} \right) , \quad (5.74)$$

where τ is the automorphism of E associated with the generalised metric V^+ .

A relevant property of the generalised scalar curvature, proven in [49, 61], is that it can be written as

$$\text{GR}_{V^+, \text{div}} = \mu^{-1/2} \Delta_{V^+} \mu^{1/2} + \langle e_{\text{div}}, e_{\text{div}} \rangle + 2 \text{div}_\mu e_{\text{div}} , \quad (5.75)$$

for some everywhere-positive density μ on M , where $e_{\text{div}} \in \Gamma(E)$ is given by $\langle e_{\text{div}}, \cdot \rangle = \text{div} - \text{div}_\mu$, while Δ_{V^+} is the Laplace operator defined by the generalised metric V^+ and a Levi-Civita Courant algebroid connection ∇^E on E for V^+ . Note that

$$\Delta_{V^+} f = \text{div}_{\nabla^E} (\tau(\mathcal{D}f)) , \quad (5.76)$$

³That is, $\nabla^\pm$ is the unique metric compatible connection with skew-symmetric torsion $\pm H$.

for all $f \in C^\infty(M)$, and that the generalised scalar curvature does not depend on the choice of density μ .

Remark 5.77. In the setting of Remark 5.69, on $(\mathbb{T}M, H)$ with generalised metric V^+ and associated Riemannian metric g , as well as with the dilaton field $\phi \in C^\infty(M)$, the generalised scalar curvature is given by

$$\text{GR}_{V^+, \text{div}_e^{-2\phi} \mu_g} = R_g - \frac{1}{12} \|H\|_g^2 - 4 e^\phi \Delta_g e^{-\phi} . \quad (5.78)$$

Here R_g is the classical scalar curvature and Δ_g the Laplace operator associated with g , while the Hodge norm of the Kalb-Ramond flux $H \in \Omega_{\text{cl}}^3(M)$ is given by

$$\|H\|_g^2 \mu_g = H \wedge \star_g H . \quad (5.79)$$

This realises the connection between the Einstein-Hilbert functional (1.27) and the low energy string effective action (1.26). When M is compact, the corresponding Einstein-Hilbert functional (1.27) recovers the familiar action functional for $\mathcal{N} = 0$ supergravity in the string frame.

Remark 5.77 together with the discussion below Equation (1.27) motivates

Definition 5.80. The *generalised string background equations* for a pair (V^+, div) on E are given by

$$\text{GRic}_{V^+, \text{div}} = 0 \quad , \quad \text{GR}_{V^+, \text{div}} = 0 . \quad (5.81)$$

A pair (V^+, div) satisfying this equation is called a *generalised Einstein pair*.

5.2.2 Related Generalised Ricci Tensors

In order to discuss Courant algebroid relations between generalised Ricci tensors, we weaken the notion of [1, Definition 3.7] by considering tensors defined on subbundles of related Courant algebroids.

Definition 5.82. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on $C \subseteq M_1 \times M_2$. Let $\mathcal{T}_i \in \Gamma(V_i^{(1)*} \otimes \cdots \otimes V_i^{(k)*})$, for $i = 1, 2$, be covariant k -tensors defined on subbundles $V_i^{(j)} \subset E_i$ for $j = 1, \dots, k$. Then $\mathcal{T}_1$ and $\mathcal{T}_2$ are

R -related, denoted $\mathcal{T}_1 \sim_R \mathcal{T}_2$, if for every $(c_1, c_2) \in C$ and every k -tuple $(e_1^{(j)}, e_2^{(j)}) \in R_{(c_1, c_2)} \cap (V_1^{(j)} \times V_2^{(j)})_{(c_1, c_2)}$ they satisfy

$$\mathcal{T}_1(e_1^{(1)}, \dots, e_1^{(k)})(c_1) = \mathcal{T}_2(e_2^{(1)}, \dots, e_2^{(k)})(c_2) . \quad (5.83)$$

With this notion we can now establish when generalised Ricci tensors are related by a generalised isometry.

Lemma 5.84. Let $R: (E_1, V_1^+) \dashrightarrow (E_2, V_2^+)$ be a generalised isometry supported on $C \subseteq M_1 \times M_2$ which is a Dirac structure in $E_1 \times \overline{E}_2$. Let div_1 and div_2 be divergence operators on E_1 and E_2 , respectively, such that $\text{div}_1 \sim_R \text{div}_2$. Then

$$(i) \quad \text{div}_1 \llbracket v_1^+, v_1^- \rrbracket_1^+ \sim_C \text{div}_2 \llbracket v_2^+, v_2^- \rrbracket_2^+ ,$$

$$(ii) \quad \mathcal{L}_{\rho_1(v_1^-)} \text{div}_1 v_1^+ \sim_C \mathcal{L}_{\rho_2(v_2^-)} \text{div}_2 v_2^+ ,$$

$$(iii) \quad \text{Tr}_{V_1^+}(\llbracket \llbracket \cdot, v_1^- \rrbracket_1^-, v_1^+ \rrbracket_1^+) \sim_C \text{Tr}_{V_2^+}(\llbracket \llbracket \cdot, v_2^- \rrbracket_2^-, v_2^+ \rrbracket_2^+) ,$$

for any $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$ which under Definition 3.56 decomposes in $\Gamma(R)$ as

$$(e_1, e_2)|_C = (v_1^+, v_2^+) + (v_1^-, v_2^-) , \quad (5.85)$$

with $(v_1^\pm, v_2^\pm) \in \Gamma(R^\pm)$.

Proof. (i) follows from the fact that if $e_1 \sim_R e_2$, then the corresponding projections satisfy $e_1^\pm \sim_R e_2^\pm$ by Equation (3.57).

(ii) follows from Lemma 3.12, since R is compatible with the anchors.

(iii) follows from the same argument as for (i), and the fact that the trace is compatible with R in the following sense. At any point $c = (c_1, c_2) \in C$, let $A_i: (V_i^+)_{c_i} \rightarrow (V_i^+)_{c_i}$ for $i = 1, 2$ be linear maps such that $A_1(v_1) \sim_R A_2(v_2)$ for every pair (v_1, v_2) such that $v_1 \sim_R v_2$. Then with respect to any given bases $\{v_1^{(i)}, \dots, v_n^{(i)}\}$ of V_i^+ such that $v_j^{(i)} \sim_R v_j^{(i)}$, the corresponding matrix elements are equal, $(A_1)_k^j = (A_2)_k^j$, hence $\text{Tr}_{V_1^+}(A_1) = \text{Tr}_{V_2^+}(A_2)$. $\square$

Corollary 5.86. If $R: (E_1, V_1^+) \dashrightarrow (E_2, V_2^+)$ is a generalised isometry supported on $C \subseteq M_1 \times M_2$ which is a Dirac structure in $E_1 \times \overline{E}_2$, and $\text{div}_1 \sim_R \text{div}_2$, then $\text{GRic}_{V_1^+, \text{div}_1} \sim_R \text{GRic}_{V_2^+, \text{div}_2}$ and $\text{GR}_{V_1^+, \text{div}_1} \sim_C \text{GR}_{V_2^+, \text{div}_2}$.

Proof. This follows immediately from Lemma 5.84 and from

$$\text{GRic}_{V_1^-, \text{div}_1} \sim_R \text{GRic}_{V_2^-, \text{div}_2} , \quad (5.87)$$

which can be shown using Lemma 5.84 by exchanging the roles of V_i^+ and V_i^- for $i = 1, 2$. $\square$

Note that we do not need to require that the divergence operators are compatible with the generalised metrics. Moreover, the same proof carries over when considering divergence operators coming from related Courant algebroid connections, as in Proposition 5.41.

5.2.3 T-Duality for Generalised Ricci Tensors and String Backgrounds

The relevant application of Proposition 5.86 for us is when R is a T-duality relation.

Corollary 5.88. Under the assumptions of Theorem 5.57, if $\underline{E}_1$ admits a divergence div_1 which is (locally) compatible with D_1 , then the generalised Ricci tensors $\text{GRic}_{V_1^+, \text{div}_1}$ and $\text{GRic}_{V_2^+, \text{div}_2}$ are T-duality related, where div_2 is the divergence given by Theorem 5.57. Similarly, the generalised scalar curvatures $\text{GR}_{V_1^+, \text{div}_1}$ and $\text{GR}_{V_2^+, \text{div}_2}$ are T-duality related.

Recalling Definition 5.80, we can then immediately infer that generalised T-dualities preserve solutions of generalised string background equations.

Theorem 5.89. Under the assumptions of Theorem 5.57, if a D_1 -invariant pair (V_1^+, div_1) is a solution of the generalised string background equations on $\underline{E}_1$, then the T-dual pair (V_2^+, div_2) is a solution of the generalised string background equations on $\underline{E}_2$.

Proof. The first statement follows from Corollary 5.88 and Definition 5.82 since if $\text{GRic}_{V_1^+, \text{div}_1} = 0$ and $\text{GR}_{V_1^+, \text{div}_1} = 0$ on $\mathcal{Q}_1$, then the property $\text{pr}_i(\underline{C}) = \varpi_i(M) = \mathcal{Q}_i$ for $i = 1, 2$ ensures that $\text{GRic}_{V_2^+, \text{div}_2}$ and $\text{GR}_{V_2^+, \text{div}_2}$ vanish on all of $\mathcal{Q}_2$. $\square$

5.2.4 Generalised Ricci Flow and T-Duality

We are now ready to discuss the interplay between geometric T-duality, introduced in Subsection 4.1.2, and the generalised Ricci flow, inspired by the treatment of [49].

Let E be a Courant algebroid over M endowed with a generalised metric V^+ and a divergence operator div . The *generalised Ricci flow* on E , given by [61, Definition 7.1], is the system of second order non-linear partial differential equations

$$\frac{\partial}{\partial t}\tau = -2\text{GRic}_{\tau,\text{div}} \quad , \quad \frac{\partial}{\partial t}\text{div} = -\mathcal{D}\text{GR}_{\tau,\text{div}} \quad , \quad (5.90)$$

where τ is the automorphism of E whose $+1$ -eigenbundle is V^+ , and $t \in [0, \infty)$ is the flow parameter which we call ‘time’.

In the setting of geometric T-duality described in Subsections 4.1.2 and 5.1.4, whereby a generalised isometry R plays the role of the T-duality relation, we consider a D_1 -invariant pair (τ_1, div_1) of a generalised metric and divergence on $\underline{E}_1$, yielding a pair (τ_2, div_2) on the T-dual Courant algebroid $\underline{E}_2$. These pairs give *a priori* unrelated generalised Ricci flows. We will now show that they are indeed related.

Theorem 5.91. Under the assumptions of Theorems 4.45 and 5.57, suppose that a unique solution $(\tau_1(t), \text{div}_1(t))$ of the generalised Ricci flow on $\underline{E}_1$ with initial condition $(\tau_1(0), \text{div}_1(0)) = (\tau_1, \text{div}_1)$ exists for $t \in [0, T)$, for some $T \in \mathbb{R}_{>0}$, and that the Lie algebra of isometries $\mathfrak{k}_{\tau_1}$ consists of complete vector fields. Then there is a unique T-dual family of pairs $(\tau_2(t), \text{div}_2(t))$ that is a solution of the generalised Ricci flow on $\underline{E}_2$ for all $t \in [0, T)$.

Proof. By assumption, any Killing vector X for τ_1 integrates to a one-parameter family of Courant algebroid automorphisms Φ_s such that $\Phi_s^* \tau_1 = \tau_1$ and $\Phi_s^* \text{div}_1 = \text{div}_1$. If $(\tau_1(t), \text{div}_1(t))$ is a solution of the generalised Ricci flow, then by the naturality of the generalised Ricci tensor and generalised scalar curvature, it follows that $(\Phi_s^* \tau_1(t), \Phi_s^* \text{div}_1(t))$ is also a solution of the generalised Ricci flow. By the uniqueness of the solution, it follows that Φ_s is a one-parameter group of isometries of $(\tau_1(t), \text{div}_1(t))$ for all $t \in [0, T)$. Hence the vector field X is Killing for $\tau_1(t)$, and we may take $\mathfrak{k}_{\tau_1(t)} := \mathfrak{k}_{\tau_1}$, for which $\text{div}_1(t)$ is compatible. By Theorems 4.45 and 5.57, we may construct the unique T-dual pair $(\tau_2(t), \text{div}_2(t))$ for all $t \in [0, T)$, which by Corollary 5.88 satisfies the generalised Ricci flow. $\square$

In this sense, we may therefore say that the T-duality relation R is *compatible* with the generalised Ricci flow. In particular, it follows from Theorem 5.89 that fixed points of the generalised Ricci flow, i.e. solutions to the generalised string background equations of Definition 5.80, are preserved under T-duality.

The proof of Theorem 5.91 assumes the existence and uniqueness of the generalised Ricci flow, together with completeness of the vector fields generating isometries of τ_1 , which as shown in [61] are guaranteed on a compact manifold. This leads to

Corollary 5.92. Let $\mathcal{Q}_1$ be a compact manifold with a D_1 -invariant pair (τ_1, div_1) . Then there is a unique pair $(\tau_2(t), \text{div}_2(t))$ solving the generalised Ricci flow on $\underline{E}_2$ which is T-dual to the generalised Ricci flow solution $(\tau_1(t), \text{div}_1(t))$ on $\underline{E}_1$, for all times $t \in [0, T)$.

Note that the compactness of $\mathcal{Q}_2$ is not required.

Local T-Duality of the Generalised Ricci flow

The proof of Theorem 5.91 requires that the Killing vector fields for τ_1 integrate to a flow on $\underline{E}_1$, thus this cannot be done in the local case. We shall now give the appropriate statement of the T-duality of generalised Ricci flows in the local case. The construction of the T-dual generalised Ricci tensor can be done immediately as a consequence of Theorem 4.101 and Remark 4.113. In the course of the proof of Theorem 5.91, we assumed that the generators of $\mathfrak{k}_{\tau_1}$ were given by complete vector fields integrating to isometries, and that the generalised Ricci flow solution is unique, in order to assert that $\mathfrak{k}_{\tau_1(0)} \subset \mathfrak{k}_{\tau_1(t)}$. Dropping the former assumptions and simply assuming the latter condition yields

Theorem 5.93. Let $(\tau_1(t), \text{div}_1(t))$ be a solution of the generalised Ricci flow on $\underline{E}_1$ for $t \in [0, T)$ with initial condition $\tau_1(0) = \tau_1$ and $\text{div}_1(0) = \text{div}_1$. Let $\{U^{(i)}\}_{i \in I}$ be an open cover of $\mathcal{Q}_1$ and suppose that $(\tau_1(t), \text{div}_1(t))$ is locally D_1 -invariant, where D_1 is generated by vector fields $\{X_1^{(i)}, \dots, X_{r_1}^{(i)}\}_{i \in I}$, for all $t \in [0, T)$. Then there is a unique family $(\tau_2(t), \text{div}_2(t))$ which is T-dual to $(\tau_1(t), \text{div}_1(t))$ and which is a solution to the generalised Ricci flow on $\underline{E}_2$.

5.3 T-duality for Torus Bundles: Divergences and Generalised Riemannian Geometry

Here we see how notions of T-duality for divergences and generalised Riemannian geometry in the torus bundle case developed in [28] can be described using the language of Courant algebroid relations developed above.

5.3.1 Divergences

We recall the notion of T-duality from [28, Definition 10.13].

Definition 5.94. Suppose that $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are T-dual in the sense of Definition 4.127, with isomorphism $\mathcal{R}: \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_1) \rightarrow \Gamma_{\mathbb{T}^k}(\mathbb{T}\mathcal{Q}_2)$. A $\mathbb{T}^k$ -invariant pair (τ_2, div_2) on $\mathbb{T}\mathcal{Q}_2$ is *T-dual* via $\mathcal{R}$ to a $\mathbb{T}^k$ -invariant pair (τ_1, div_1) on $\underline{E}_1$ if

$$\mathcal{R} \circ \pi_{1*} \tau_1 = \pi_{2*} \tau_2 \circ \mathcal{R} \quad , \quad \pi_{1*} \text{div}_1 = \mathcal{R}^* \pi_{2*} \text{div}_2 \quad , \quad (5.95)$$

where $\pi_i: \mathcal{Q}_i \rightarrow \mathcal{B}$ with $i = 1, 2$ are the bundle projections.

Here the notion of a $\mathbb{T}^k$ -invariant pair coincides with the notion of a D_1 -invariant metric and D_1 -compatible divergence operator. The T-duality relation R encapsulates this definition, through

Proposition 5.96. If $(\mathbb{T}\mathcal{Q}_2, \tau_2, \text{div}_2)$ is T-dual to $(\mathbb{T}\mathcal{Q}_1, \tau_1, \text{div}_1)$ in the sense of Definition 5.94, then the generalised isometry $R: (\underline{E}_1, \tau_1) \dashrightarrow (\underline{E}_2, \tau_2)$ is such that $\text{div}_1 \sim_R \text{div}_2$.

Proof. The result follows immediately from Proposition 4.147 and Lemma 5.64. $\square$

Example 5.97 (Circle Bundles). Let $\mathcal{Q}_i$ be circle bundles over a common base $\mathcal{B}$, for $i = 1, 2$, and let $M = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$. Let θ_i be connections on the S^1 -bundles $\pi_i: \mathcal{Q}_i \rightarrow \mathcal{B}$, and ∂_{θ_i} the global vector fields defining the circle fibres. Let g_1 be a metric on $\mathcal{Q}_1$ of the form

$$g_1 = g \theta_1 \otimes \theta_1 + \dot{g}_{\mathcal{B}} \quad , \quad (5.98)$$

where $g \in C^\infty(\mathcal{B})$ is an everywhere-positive function, and $g_{\mathcal{B}}$ is a Riemannian metric on $\mathcal{B}$. Then g_1 is ∂_{θ_1} -invariant (i.e. S^1 -invariant), and hence we may construct the metric

$$g_2 = g^{-1} \theta_2 \otimes \theta_2 + \ddot{g}_{\mathcal{B}} \quad (5.99)$$

on $\mathcal{Q}_2$ such that the T-duality relation $R: (\mathbb{T}\mathcal{Q}_1, 0, g_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, 0, g_2)$ is a generalised isometry.

The corresponding densities are

$$\mu_{g_1} = \sqrt{g} |\theta_1 \wedge \dot{\mu}_{g_{\mathcal{B}}}| \quad , \quad \mu_{g_2} = \sqrt{g^{-1}} |\theta_2 \wedge \ddot{\mu}_{g_{\mathcal{B}}}| \quad (5.100)$$

and one calculates

$$\operatorname{div}_{\mu_{g_1}} e_1 = \sqrt{g^{-1}} \mathcal{L}_{\rho_1(e_1)} \sqrt{g} \quad , \quad \operatorname{div}_{\mu_{g_2}} e_2 = \sqrt{g} \mathcal{L}_{\rho_2(e_2)} \sqrt{g^{-1}} \quad , \quad (5.101)$$

for all $e_i \in \Gamma(\mathbb{T}\mathcal{Q}_i)$. If $X \in \Gamma(T\mathcal{B})$, then⁴ $X \sim_R X$, but $\operatorname{div}_{\mu_{g_1}} X \neq \operatorname{div}_{\mu_{g_2}} X$ in general, and hence the divergence operators are not R -related. By Equation (5.44), this necessitates a dilaton shift.

Indeed, by imposing Equation (5.44) for $X + \xi \in \Gamma(\mathbb{T}\mathcal{B})$, we find

$$\langle Y_{\text{dil}} + \eta_{\text{dil}}, X + \xi + \partial_2 + \theta_2 \rangle_2 = \sqrt{g^{-1}} X(\sqrt{g}) - \sqrt{g} X(\sqrt{g^{-1}}) = X(\log g) \quad . \quad (5.102)$$

This gives

$$Y_{\text{dil}} = 0 \quad , \quad \eta_{\text{dil}} = d \log g \quad , \quad (5.103)$$

which is the usual dilaton shift. A similar argument can be used for the dilaton shift of any T-dual pair of torus bundles.

⁴With a slight abuse of notation, here we consider the horizontal lift of X to $\mathcal{Q}_i$ with respect to θ_i .

5.3.2 Generalised Riemannian Geometry and the Ricci-Buscher Rules

Let us discuss the generalised Ricci tensors for T-duality between torus bundles, as formulated in [62, Proposition 6.3]. In this case, given two T-dual $\mathbb{T}^k$ -equivariant exact Courant algebroids $\underline{E}_1$ and $\underline{E}_2$ endowed with T-dual pairs (V_1^+, div_1) and (V_2^+, div_2) , respectively, as in Definition 5.94, their generalised Ricci tensors satisfy

$$\pi_{1*} \text{GRic}_{V_1^+, \text{div}_1} = \mathcal{R}^* \pi_{2*} \text{GRic}_{V_2^+, \text{div}_2} . \quad (5.104)$$

This is equivalent to $\text{GRic}_{V_1^+, \text{div}_1} \sim_R \text{GRic}_{V_2^+, \text{div}_2}$, where R is the generalised isometry which appears in Proposition 5.96.

Let us explore how this simple relationship for the generalised Ricci tensors allows us to relate the classical Ricci tensors of T-dual Riemannian manifolds, akin to the Buscher rules. Let $\mathcal{Q}_1$ and $\mathcal{Q}_2$ be geometrically T-dual circle bundles over a base $\mathcal{B}$, with T-duality relation $R: (\mathbb{T}\mathcal{Q}_1, H_1, \tau_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, H_2, \tau_2)$. For any open cover $U^{(i)}$ trivialising $\mathcal{Q}_1$, the Kalb-Ramond flux on $\mathcal{Q}_1$ is given locally by $H_1 = db_1$ for some two-form $b_1 \in \Omega^2(U^{(i)})$. If $\tau_1 = (g_1, 0)$ for an S^1 -invariant Riemannian metric g_1 on $\mathcal{Q}_1$, then the T-dual generalised metric $\tau_2 = (g_2, b_2)$ is given by the Buscher rules (4.149).

On $\mathcal{Q}_1$ we take the dilaton⁵

$$e = 4 \, d \log (g_1)_{\theta\theta} = \frac{4}{(g_1)_{\theta\theta}} \partial_\alpha (g_1)_{\theta\theta} \, dx^\alpha , \quad (5.105)$$

which is invariant. This ensures that $\text{div}_1 := \text{div}_{4 \log (g_1)_{\theta\theta}} \sim_R \text{div}_{\mu_{g_2}} =: \text{div}_2$. For this choice $\varphi^+ = \frac{1}{2} e$.

If ∇^+ is the Bismut connection for (g_1, H_1) , then its Christoffel symbols are given by

$$\Gamma^{+i}_{jk} = \Gamma^i_{jk} + \frac{1}{2} (H_1)^i_{jk} , \quad (5.106)$$

where Γ^i_{jk} are the Christoffel symbols for the Levi-Civita connection ∇^{g_1} of g_1 , and

⁵We adhere to the usual Einstein summation convention over repeated upper and lower indices throughout.

indices are raised using the metric g_1 in the standard way. Hence

$$\frac{1}{2} (g_1)_{\theta\theta} \nabla^+ \varphi^+ = \left(\partial_\alpha \partial_\beta (g_1)_{\theta\theta} - \frac{1}{(g_1)_{\theta\theta}} \partial_\alpha (g_1)_{\theta\theta} \partial_\beta (g_1)_{\theta\theta} \right) dx^\beta \otimes dx^\alpha \quad (5.107)$$

$$- \partial_\alpha (g_1)_{\theta\theta} \left(\Gamma^\alpha_{ij} dx^i \odot dx^j + \frac{1}{2} (H_1)^\alpha_{ij} dx^i \wedge dx^j \right). \quad (5.108)$$

Let Ric_{g_1} be the Ricci tensor of g_1 , with coordinate frame components $(\text{Ric}_{g_1})_{ij} := \text{Ric}_{g_1}(\partial_i, \partial_j)$. Then the generalised Ricci tensor of (τ_1, div_1) is given from Equation (5.70) by

$$\begin{aligned} (\text{GRic}_1)_{ij} &:= \text{GRic}_{g_1, \text{div}_1}((\partial_i)^+, (\partial_j)^-) \\ &= (\text{Ric}_{g_1})_{ij} - \frac{1}{4} (H_1^2)_{ij} - \frac{1}{2} (d^* H_1)_{ij} \\ &\quad + \frac{2}{(g_1)_{\theta\theta}} \partial_i \partial_j (g_1)_{\theta\theta} - \frac{2}{(g_1)_{\theta\theta}^2} \partial_i (g_1)_{\theta\theta} \partial_j (g_1)_{\theta\theta} \\ &\quad - \frac{2}{1 (g_1)_{\theta\theta}} \partial_\alpha (g_1)_{\theta\theta} \Gamma^\alpha_{ij} - \frac{1}{(g_1)_{\theta\theta}} \partial_\alpha (g_1)_{\theta\theta} (H_1)^\alpha_{ij}, \end{aligned} \quad (5.109)$$

where $(H_1^2)_{ij} = (H_1)_{ikl} (H_1)_j{}^{kl}$ and $(d^* H_1)_{ij} = -g_1^{kl} \nabla_k^{g_1} (H_1)_{lij}$.

From the local form of the relation R given by Equation (4.137), together with Corollary 5.88, we obtain

$$\text{GRic}_{(g_2, b_2), \text{div}_2}((\partial_\alpha - (b_1)_{\alpha\theta} \partial_{\theta_2})^+, (\partial_\beta - (b_1)_{\beta\theta} \partial_{\theta_2})^-) = \text{GRic}_{g_1, \text{div}_1}((\partial_\alpha)^+, (\partial_\beta)^-),$$

$$\text{GRic}_{(g_2, b_2), \text{div}_2}((\partial_\alpha - (b_1)_{\alpha\theta} \partial_{\theta_2})^+, (\partial_{\theta_2})^-) = \text{GRic}_{g_1, \text{div}_1}((\partial_\alpha)^+, -(\frac{1}{(g_1)_{\theta\theta}} \partial_{\theta_1})^-),$$

$$\text{GRic}_{(g_2, b_2), \text{div}_2}((\partial_{\theta_2})^+, (\partial_\alpha - (b_1)_{\alpha\theta} \partial_{\theta_2})^-) = \text{GRic}_{g_1, \text{div}_1}((\frac{1}{(g_1)_{\theta\theta}} \partial_{\theta_1})^+, (\partial_\alpha)^-),$$

$$\text{GRic}_{(g_2, b_2), \text{div}_2}((\partial_{\theta_2})^+, (\partial_{\theta_2})^-) = \text{GRic}_{g_1, \text{div}_1}((\frac{1}{(g_1)_{\theta\theta}} \partial_{\theta_1})^+, -(\frac{1}{(g_1)_{\theta\theta}} \partial_{\theta_1})^-). \quad (5.110)$$

In the coordinate frame $\{\partial_\alpha, \partial_\theta\}_{\alpha=1, \dots, \dim \mathcal{B}}$, using Equation 5.70 the Ricci tensor

Ric_{g_2} of g_2 is thus given by

$$(\text{Ric}_{g_2})_{\theta\theta} = -\frac{1}{(g_1)_{\theta\theta}^2} (\text{GRic}_1)_{\theta\theta} + \frac{1}{4} (H_2^2)_{\theta\theta} + \frac{1}{2} (d^* H_2)_{\theta\theta} , \quad (5.111)$$

$$(\text{Ric}_{g_2})_{\alpha\theta} = -\frac{1}{(g_1)_{\theta\theta}} (\text{GRic}_1)_{\alpha\theta} + \frac{1}{4} (H_2^2)_{\alpha\theta} + \frac{1}{2} (d^* H_2)_{\alpha\theta} + \frac{(b_1)_{\alpha\theta}}{(g_1)_{\theta\theta}^2} (\text{GRic}_1)_{\theta\theta} , \quad (5.112)$$

$$\begin{aligned} (\text{Ric}_{g_2})_{\alpha\beta} &= (\text{GRic}_1)_{\alpha\beta} + \frac{1}{4} (H_2^2)_{\alpha\beta} + \frac{1}{2} (d^* H_2)_{\alpha\beta} \\ &\quad + \frac{(b_1)_{\beta\theta}}{(g_1)_{\theta\theta}} (\text{GRic}_1)_{\alpha\theta} - \frac{(b_1)_{\beta\theta} (b_1)_{\alpha\theta}}{(g_1)_{\theta\theta}^2} (\text{GRic}_1)_{\theta\theta} - \frac{(b_1)_{\alpha\theta}}{(g_1)_{\theta\theta}} (\text{GRic}_1)_{\theta\beta} . \end{aligned} \quad (5.113)$$

5.3.3 Generalised Ricci Flow

The (local) T-duality of the generalised Ricci flow can be formulated in this setting. Suppose that $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are geometrically T-dual circle bundles over the same base, with T-duality relation $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$. If $(\tau_1(t), \text{div}_1(t))$ is a family of $\mathbb{T}^k$ -invariant pairs on $\mathbb{T}\mathcal{Q}_1$ then, for each t there is a unique T-dual pair $(\tau_2(t), \text{div}_2(t))$ such that $R: (\underline{E}_1, \tau_1(t)) \dashrightarrow (\underline{E}_2, \tau_2(t))$ is a generalised isometry and $\text{GRic}_{\tau_1(t), \text{div}_1(t)} \sim_R \text{GRic}_{\tau_2(t), \text{div}_2(t)}$. From Theorem 5.93 with $U^{(1)} = \mathcal{Q}_1$, it follows that if $(\tau_1(t), \text{div}_1(t))$ solves the generalised Ricci flow on $\underline{E}_1$, then so does $(\tau_2(t), \text{div}_2(t))$ on $\underline{E}_2$. This is the content of [28, Theorem 10.19].

In the case that $\mathcal{B}$ is compact, starting with a $\mathbb{T}^k$ -invariant pair (τ_1, div_1) , since the vector fields generating $\mathfrak{k}_{\tau_1} = \mathfrak{t}^k$ are complete we may apply Corollary 5.92.

We will explore explicit examples of the T-duality of the generalised Ricci flow in Chapter 7.

Chapter 6

Relations for Generalised Complex Geometry

In this chapter we explore how one can obtain a sensible notion of relating spinors between manifolds via Courant algebroid relations. This is done by considering spinor modules for the Clifford algebra generated by the relation R itself. As Lagrangian subspaces are in one-to-one correspondence with pure spinors, we describe also how to relate Lagrangian subspaces between relations, and show how this relates to the notion of related spinors. For specific relations, for instance when R is a generalised isometry, this becomes a one-to-one correspondence.

6.1 Linear Algebra

We begin with a discussion on the linear algebra of spinors, Lagrangians, and their relations

6.1.1 Linear Relations

Let $(W_1, \langle \cdot, \cdot \rangle_1)$ and $(W_2, \langle \cdot, \cdot \rangle_2)$ be finite-dimensional vector spaces endowed with a symmetric, non-degenerate, bilinear pairing, and let $W_1 \times \overline{W}_2$ be the product vector space $W_1 \times W_2$ endowed with the pairing $\langle \cdot, \cdot \rangle_1 - \langle \cdot, \cdot \rangle_2$.

Definition 6.1. A *linear relation* between $(W_1, \langle \cdot, \cdot \rangle_1)$ and $(W_2, \langle \cdot, \cdot \rangle_2)$, denoted $W_1 \sim_U W_2$, is a Lagrangian subspace $U \subset W_1 \times \overline{W}_2$.

Example 6.2. Let $f: W_1 \rightarrow W_2$ be a linear map such that $\langle f(w), f(w') \rangle_2 = \langle w, w' \rangle$ for every $w, w' \in W_1$. Then $\text{gr}(f) \subset W_1 \times \overline{W_2}$ defines a Lagrangian subspace, and hence a linear relation.

Example 6.3. Let V_1 and V_2 be finite-dimensional vector spaces and let $J: V_1 \rightarrow V_2$ be a linear map. Then the vector spaces $\mathbb{V}_i := V_i \oplus V_i^*$, for $i = 1, 2$, endowed with the pairing

$$\langle v + \alpha, w + \beta \rangle_i = \iota_v \beta + \iota_w \alpha, \quad (6.4)$$

for all $v, w \in V_i$ and $\alpha, \beta \in V_i^*$, are related by

$$U_J := \{ (v_1 + \alpha_1, v_2 + \alpha_2) \in \mathbb{V}_1 \times \mathbb{V}_2 : v_2 = J(v_1), \alpha_1 = J^*(\alpha_2) \}, \quad (6.5)$$

which is a Lagrangian subspace of $\mathbb{V}_1 \times \overline{\mathbb{V}_2}$.

We shall now introduce linear relations between Lagrangian subspaces. We define the kernel K_U and cokernel CK_U of a linear relation U analogously to 3.21.

Definition 6.6. Let $W_1 \sim_U W_2$ and let $V_i \subset W_i$ be a Lagrangian subspace, for $i = 1, 2$. Define

$$U|_{V_i} := \{ (w_1, w_2) \in U \mid w_i \in V_i \}, \quad (6.7)$$

for $i = 1, 2$. Then U is a linear relation between the Lagrangian subspaces V_1 and V_2 , denoted $V_1 \sim_U V_2$, if

$$U|_{V_1} = U \cap (V_1 \times V_2) + CK_U, \quad \text{and} \quad U|_{V_2} = U \cap (V_1 \times V_2) + K_U, \quad (6.8)$$

as a vector subspaces.

Note that the inclusion $\supseteq$ always holds for in 6.8.

Remark 6.9. If $W_1 \sim_U W_2$ and $V_1 \sim_U V_2$ for $V_1 \subset W_1$ and $V_2 \subset W_2$ Lagrangian, then if $(v_1, w_2) \in U|_{V_1}$, there is some $v_2 \in V_2$ such that $(v_1, v_2) \in U$.

Note that if $U = \text{gr}(f)$ for some map $f: W_1 \rightarrow W_2$, then $CK_U = \{0\}$, $K_U = \text{gr}(f|_{\ker(f)})$, and $\ker(\text{gr}(f)) = \ker(f)$. For Lagrangian subspaces $V_i \subset W_i$, the condition that $U|_{V_1} = U \cap (L_1 \times L_2)$ is equivalent to $f(L_1) \subseteq L_2$. The condition that

$U|_{V_2} = U \cap (L_1 \times L_2)$ is equivalent to $f^{-1}(L_2) \subseteq L_1 + \ker(f)$. In the case that f is an isomorphism these conditions reduce to $f(L_1) = L_2$, and (6.8) becomes $\text{gr}(f)|_{V_1} = \text{gr}(f)|_{V_2} = \text{gr}(f) \cap (V_1 \times V_2)$.

6.1.2 Clifford Algebras

Recall that $\mathbb{V} = V \oplus V^*$ has a Clifford algebra structure generated by the relation $\mathbf{v} \cdot \mathbf{v} = \langle \mathbf{v}, \mathbf{v} \rangle$, for $\mathbf{v} \in \mathbb{V}$. An isomorphism of Clifford algebras is a $\phi: A \rightarrow B$ such that $\phi(a \cdot a') = \phi(a) \cdot \phi(a')$. Also recall that $\bigwedge^\bullet V^*$ is a $\text{Cl}(\mathbb{V})$ -module via the action

$$(v + \nu) \cdot \alpha = \iota_v \alpha + \nu \wedge \alpha. \quad (6.10)$$

A *Clifford module* over $\text{Cl}(\mathbb{V})$ is a vector space $\mathbf{S}$ together with an algebra homomorphism $\Psi: \text{Cl}(\mathbb{V}) \rightarrow \text{End}(\mathbf{S})$. The elements of $\mathbf{S}$ are called *spinors*. A Clifford module $\mathbf{S}$ is called a *spinor module* if it is irreducible, i.e. if there are no non-trivial proper sub-modules $\mathbf{S}' \subset \mathbf{S}$ such that $\mathbf{S}'$ is also a Clifford module.

Remark 6.11 (Spinor Lines). Let $\text{Lag}(\mathbb{V})$ be the set of Lagrangian subspaces of $\mathbb{V}$. For any $L \in \text{Lag}(\mathbb{V})$, the subspace $\Omega^L \subset \mathbf{S}$ defined by

$$\Omega^L := \{ \phi \in \mathbf{S} \mid \mathbf{v} \cdot \phi = 0, \forall \mathbf{v} \in L \} \quad (6.12)$$

is 1-dimensional. Moreover, for any $\phi \in \mathbf{S}$, we define its *null space* $N_\phi \subset \mathbb{V}$ by

$$N_\phi := \{ \mathbf{v} \in \mathbb{V} \mid \mathbf{v} \cdot \phi = 0 \}. \quad (6.13)$$

Note that N_ϕ is always an isotropic subspace. We thus call $\phi \in \mathbf{S}$ a *pure spinor* if N_ϕ is a Lagrangian subspace. Notice that if ϕ is a pure spinor, then so is $c\phi$ for any $c \in \mathbb{R}$. Thus each pure spinor has an associated *spinor line* $\Omega^{N_\phi} \subset \mathbf{S}$. Hence there is a one-to-one correspondence between Lagrangian subbundles $L \in \text{Lag}(\mathbb{V})$ and pure spinor lines $\Omega^L \subset \mathbf{S}$.

Example 6.14. $\mathbf{S} = \bigwedge^\bullet V^*$ defines a spinor module for $\text{Cl}(\mathbb{V})$, with $\Psi(\mathbf{v})(\mathbf{s}) = \mathbf{v} \cdot \mathbf{s}$, for every $\mathbf{v} \in \mathbb{V}$ and $\mathbf{s} \in \mathbf{S}$. Indeed, if W is any vector space with split-signature bilinear form, and $L \subset W$ is any Lagrangian subspace of W , then $\mathbf{S} = \bigwedge^\bullet L^*$ defines a spinor module for $\text{Cl}(W)$, with Ψ similarly defined.

Consider $1 \in \bigwedge^\bullet V^*$, the unit spinor. Then

$$N_1 = \{v + \nu : \iota_v 1 + \nu \wedge 1 = 0\} = V, \quad (6.15)$$

and hence 1 is a pure spinor, with associated Lagrangian V . The associated line is $\Omega^V = \mathbb{R}$ as a subspace of $\bigwedge^\bullet V^*$. At the opposite extreme, let $\lambda \in \bigwedge^{\text{top}} V^*$. Then $N_\lambda = V^*$ and $\Omega^{V^*} = \bigwedge^{\text{top}} V^*$. See [29, Section 2.5] for a full description of pure spinors associated to Lagrangian subbundles of $\mathbb{V}$.

6.1.3 Linear Clifford Relations

For V, W vector spaces, we can define two distinct $\text{Cl}(\mathbb{V}) \times \text{Cl}(\mathbb{W})$ -module structures on the product $\bigwedge^\bullet V^* \times \bigwedge^\bullet W^*$ through

$$(\mathbf{v}, \mathbf{w}) \cdot (\nu, \mu) = (\mathbf{v} \cdot \nu, \pm \mathbf{w} \cdot \mu), \quad (6.16)$$

for $\mathbf{v} \in \mathbb{V}$ and $\mathbf{w} \in \mathbb{W}$. One then has

$$(\mathbf{v}, \mathbf{w})^2 \cdot (\nu, \mu) = (\mathbf{v}^2 \cdot \nu, \mathbf{w}^2 \cdot \mu), \quad (6.17)$$

and so defines a Clifford action if and only if $\mathbf{v}^2 = \langle \mathbf{v}, \mathbf{v} \rangle_{\mathbb{V}} = \langle \mathbf{w}, \mathbf{w} \rangle_{\mathbb{W}} = \mathbf{w}^2$.

Let $U \subset \mathbb{V} \times \overline{\mathbb{W}}$ be isotropic. Then, for each $u = (\mathbf{v}, \mathbf{w}) \in U$, one has that $\langle \mathbf{v}, \mathbf{v} \rangle_{\mathbb{V}} = \langle \mathbf{w}, \mathbf{w} \rangle_{\mathbb{W}}$. It follows that U has a Clifford algebra structure defined by $u^2 = \langle \mathbf{v}, \mathbf{v} \rangle_{\mathbb{V}} = \langle \mathbf{w}, \mathbf{w} \rangle_{\mathbb{W}}$. Denote by $\text{Cl}^+(U)$ the algebra $\text{Cl}(U)$ acting on $\bigwedge^\bullet V^* \times \bigwedge^\bullet W^*$ with $+$ parity in Equation (6.16), and by $\text{Cl}^-(U)$ the algebra with opposite parity. In general, $\bigwedge^\bullet V^* \times \bigwedge^\bullet W^*$ is a reducible representation of $\text{Cl}^+(U)$ or $\text{Cl}^-(U)$.

Definition 6.18. A subspace $S_U \subset \bigwedge^\bullet V^* \oplus \bigwedge^\bullet W^*$ is a U -Clifford relation of parity $i \in \{-, +\} = \mathbb{Z}_2$ if S_U is a spinor module for $\text{Cl}^i(U)$. Elements $\nu \in \bigwedge^\bullet V^*$ and $\mu \in \bigwedge^\bullet W^*$ are said to be U -related, denoted $\nu \sim_U \mu$, if $(\nu, \mu) \in S_U$.

When the kernels of U are non-trivial, there is more freedom in choosing S_U to account for the presence of kernels. To address this freedom, we can consider only the injective part of the relation. Recall the kernel K_U of R and define $\underline{\text{CK}}_U =$

$(\{0\} \times \mathbb{W}) \cap (U/K_U)$. We define

$$U^\times = (U/K_U)/\underline{\text{CK}}_U \subset \mathbb{V}/\ker(U) \times \mathbb{W}/\text{coker}(U). \quad (6.19)$$

We denote by $[\mathbf{v}, \mathbf{w}]$ the equivalence class of $(\mathbf{v}, \mathbf{w}) \in U$, and by $[(\mathbf{v}, \mathbf{w})]$ the projection under the quotient $U \rightarrow U^\times$. Note that this projection descends to the Clifford algebra, since for any $u \in U$ and k, k' in K_U or CK_U , we have $\langle u+k, u+k' \rangle = \langle u, u \rangle$. The quotient is then taken component wise on the tensor product, i.e. $[u \otimes u'] = [u] \otimes [u']$. We thus have the Clifford algebra $\text{Cl}(U^\times)$.

Suppose that there exists an action of $\text{Cl}(U^\times)$ on $\bigwedge^\bullet V^* \times \bigwedge^\bullet W^*$. We say that this action is *compatible with $\text{Cl}^i(U)$* if for each $[(\mathbf{v}, \mathbf{w})] \in \text{Cl}(U^\times)$ there is a $(\mathbf{v}, \mathbf{w}) \in \text{Cl}^i(U)$ with $[(\mathbf{v}, \mathbf{w})] = [\mathbf{v}, \mathbf{w}]$ such that

$$(\mathbf{v}, \mathbf{w}) \cdot (\alpha, \beta) = [\mathbf{v}, \mathbf{w}] \cdot (\alpha, \beta). \quad (6.20)$$

Definition 6.21. A subspace $S_U \subset \bigwedge^\bullet V^* \oplus \bigwedge^\bullet W^*$ is a *weak U -Clifford relation of parity i* if there exists an action of $\text{Cl}(U^\times)$ on $\bigwedge^\bullet V^* \oplus \bigwedge^\bullet W^*$ compatible with $\text{Cl}^i(U)$ for which S_U is a spinor module.

Let us show that the above definitions lead to a notion of “isomorphism” of spinor modules.

Proposition 6.22. Let p_i be the projections from $\mathbb{V} \times \mathbb{W}$ to the first and second factor for $i = 1, 2$ respectively, and denote by the same symbols the projections from $\bigwedge^\bullet V^* \times \bigwedge^\bullet W^*$. Let S_U be a weak U -Clifford relation of parity j , with $p_i(U) \neq 0$ and $p_i(S_U) \neq \{0\}$ for $i = 1, 2$. Then

$$S_U \cap (\{0\} \times \bigwedge^\bullet W^*) = S_U \cap (\bigwedge^\bullet V^* \times \{0\}) = \{(0, 0)\}, \quad (6.23)$$

Proof. Suppose $(0, \mu) \in S_U$ for some $\mu \in \bigwedge^\bullet W^*$. By the invariance of S_U , we have

$$S' := \text{Cl}^j(U) \cdot (0, \mu) \subset S. \quad (6.24)$$

The set S' is invariant under $\text{Cl}^j(U)$, thus by the irreducibility of S , either $S' = S$, or $S' = \{(0, 0)\}$. By the assumptions $p_2(U) \neq \{0\}$ and $p_1(S) \neq \{0\}$, the latter holds,

hence $\mu = 0$. The other equality holds similarly. $\square$

Corollary 6.25. If $\mathbf{S}_U$ is a weak U -Clifford relation with $p_i(U) \neq 0$ and $p_i(\mathbf{S}_U) \neq \{0\}$ for $i = 1, 2$, and $(\nu, \mu), (\nu, \mu') \in \mathbf{S}_U$, then $\mu = \mu'$.

Proposition 6.26. A U -Clifford relation of parity i defines a weak U -Clifford relation.

Proof. Suppose that $\mathbf{S}_U$ is a U -Clifford relation in the sense of Definition 6.18. For $(k, 0) \in \mathbf{K}_U$ and $(\alpha, \beta) \in \mathbf{S}_U$, we have

$$(k, 0) \cdot (\alpha, \beta) = (k \cdot \alpha, 0). \quad (6.27)$$

By Proposition 6.22, we have that $k \cdot \alpha = 0$, and hence $\mathbf{K}_U$ acts trivially on $\mathbf{S}_U$, and similarly for $\underline{\mathbf{C}}\mathbf{K}_U$. Hence $\mathbf{S}_U$ descends to a spinor module for $\mathbf{Cl}(U^\times)$. $\square$

Example 6.28. Suppose that $\bar{f}: \mathbb{V} \rightarrow \mathbb{W}$ is an isomorphism, given by $\bar{f} = f^{-1} + f^*$, where $f: W \rightarrow V$ is an isomorphism. Let $\mathbf{S}_{\text{gr}(\bar{f})}$ be a $\text{gr}(\bar{f})$ -Clifford relation of parity $+$. Then, noting that $\mathbf{Cl}(\text{gr}(\bar{f}))$ is the graph of the induced isomorphism from $\mathbf{Cl}(\mathbb{V})$ to $\mathbf{Cl}(\mathbb{W})$,

$$(v + \nu, \bar{f}(v + \nu)) \cdot (\alpha, \beta) = (\iota_v \alpha + \nu \wedge \alpha, \iota_{f^{-1}(v)} \beta + f^*(\nu) \wedge \beta), \quad (6.29)$$

for every $(\alpha, \beta) \in \mathbf{S}_R$, and $v + \nu \in \mathbb{V}$. Note that if α is a 0-form, then β is also a 0-form, otherwise there is $w \in W$ such that $f(w) \cdot \alpha = 0$, contradicting Corollary 6.25. It follows that, on 0-forms, $\mathbf{S}_R$ is generated by $(1, \kappa)$ for some $\kappa \in \mathbb{R}^\times$. Similarly, if $\alpha \sim_R \beta$ and $\alpha \in V^*$, then $\beta \in W^*$, and by the above we have $\beta(w) = \kappa \alpha(f(w))$. Let $w \in W$ be such that $\alpha(f(w)) = 1$. Then

$$(\alpha, \beta) = (\alpha, f^* \alpha) \cdot (f(w), w) \cdot (\alpha, \beta) = (\alpha, f^* \alpha) \cdot (1, \beta(w)) = (\alpha, \kappa f^* \alpha). \quad (6.30)$$

Thus $\alpha \sim_R \kappa f^* \alpha$, and hence, by Corollary 6.25, $\beta = \kappa f^* \alpha$. By induction, it follows that $\mathbf{S}_{\text{gr}(\bar{f})} = \text{gr}(\kappa f^*)$.

Remark 6.31. In light of the above example, note that if $\mathbf{S}_U \subset \bigwedge^\bullet V^* \oplus \bigwedge^\bullet W^*$ is

a U -Clifford relation (regardless of parity), then for each $\kappa \in \mathbb{R} \setminus \{0\}$

$$\mathbf{S}_U^\kappa = \{ (\alpha, \kappa \alpha_U) : (\alpha, \alpha_U) \in \mathbf{S}_U \} \quad (6.32)$$

is also a U -Clifford relation of the same parity. This is similar to the ambiguity addressed in [31, Part II] regarding the choice of connection on the spinor bundle. In order to remove this ambiguity, let $n = \dim V$ and consider

$$\det(\mathbf{S}_U^\kappa)^* = \{ (\mu, \frac{1}{\kappa^{2n}} \mu_U) : (\mu, \mu_U) \in \det \mathbf{S}_U^* \}. \quad (6.33)$$

It follows that for every $\kappa \in \mathbb{R} \setminus \{0\}$, $\mathbf{S}_U \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}} = \mathbf{S}_U^\kappa \otimes |\det(\mathbf{S}_U^\kappa)^*|^{\frac{1}{2n}}$. Hence

Definition 6.34. If $\mathbf{S}_U$ is a weak U -Clifford relation of parity i for $U \subset \mathbb{V} \times \mathbb{W}$, the *canonical U -Clifford relation of parity i* is

$$\mathcal{S}_U = \mathbf{S}_U \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}} \quad (6.35)$$

where $n = \dim \mathbf{S}_U$. As a spin module, $\text{Cl}(U)$ acts trivially on the second component of $\mathcal{S}_U$.

If $\Omega_{\mathbb{V}} \subset \bigwedge^\bullet V^*$ and $\Omega_{\mathbb{W}} \subset \bigwedge^\bullet W^*$ are pure spinor lines, we say that $\Omega_{\mathbb{V}}$ and $\Omega_{\mathbb{W}}$ are *U -related* if

$$(\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}} \subset \mathcal{S}_U, \quad (6.36)$$

and denote this as $\Omega_{\mathbb{V}} \sim_U \Omega_{\mathbb{W}}$.

Remark 6.37. Note that Definition 6.34 does not depend on the choice of $\mathbf{S}_U$. Indeed, suppose there exists some U -Clifford relation $\mathbf{S}_U$. Then for every $(u_1, u_2) \otimes (\mu_1, \mu_2) \in (\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}}$, and $\kappa \in \mathbb{R} \setminus \{0\}$,

$$(u_1, u_2) \otimes (\mu_1, \frac{1}{\kappa} \mu_2) = (u_1, \frac{1}{\kappa} u_2) \otimes (\mu_1, \mu_2) \in (\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}} \quad (6.38)$$

Hence $(\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det(\mathbf{S}_U^\kappa)^*|^{\frac{1}{2n}} \subset (\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2n}}$. The opposite inclusion

holds similarly, hence

$$(\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det \mathbf{S}_U^*|^{\frac{1}{2^n}} = (\Omega_{\mathbb{V}} \times \Omega_{\mathbb{W}}) \otimes |\det(\mathbf{S}_U^\kappa)^*|^{\frac{1}{2^n}}. \quad (6.39)$$

Proposition 6.40. Let $U: \mathbb{V} \dashrightarrow \mathbb{W}$ be a linear relation, and let $L_1 \subset \mathbb{V}$ and $L_2 \subset \mathbb{W}$ be Lagrangian subspaces, with $\Omega_1 \subset \wedge^\bullet V^*$ and $\Omega_2 \subset \wedge^\bullet W^*$ the corresponding pure spinor lines. Suppose there is a U -Clifford relation $\mathbf{S}_U$. If $\Omega_1 \sim_U \Omega_2$, then $L_1 \sim_R L_2$.

Proof. If $\Omega_1 \sim_U \Omega_2$, take $(\phi_1, \phi_2) \in \Omega_1 \times \Omega_2$ such that $(\phi_1, \phi_2) \in \mathbf{S}_U$ and consider $(\mathbf{1}, \mathbf{w}) \in U|_{L_1}$. Then

$$(\mathbf{1}, \mathbf{w}) \cdot (\phi_1, \phi_2) = (0, \mathbf{w} \cdot \phi_2). \quad (6.41)$$

Since $(\phi_1, \phi_2) \in \mathbf{S}_U$, by Proposition 6.22 it follows that $\mathbf{w} \cdot \phi_2 = 0$ and hence $\mathbf{w} \in L_2$, i.e. $U|_{L_1} \subset L_1 \times L_2$. Similarly, $U|_{L_2} \subset L_1 \times L_2$, and hence $L_1 \sim_U L_2$. $\square$

Remark 6.42. Suppose that $L_1 \sim_R L_2$ and that either $p_1(R) \supseteq L_1$ or $p_2(R) \supseteq L_2$. Assume without loss of generality the latter holds. By Proposition 6.22, there is some ϕ'_2 such that $(\phi_1, \phi'_2) \in \mathbf{S}_U$, where ϕ_1 is a generating spinor for L_1 . Since $p_2(R) \supseteq L_2$, for each $\mathbf{1}_2 \in L_2$, there is $\mathbf{1}_1 \in L_1$ such that $(\mathbf{1}_1, \mathbf{1}_2) \in U \cap (L_1 \times L_2)$, and hence

$$(\mathbf{1}_1, \mathbf{1}_2) \cdot (\phi_1, \phi'_2) = (0, \mathbf{1}_2 \cdot \phi'_2), \quad (6.43)$$

must vanish. Thus $\phi'_2 \in \Omega_2$ so that $\Omega_1 \sim_U \Omega_2$. When U is a Lagrangian subspace with trivial kernel and cokernel, this holds automatically.

Remark 6.44. We remark that in the weak case, given $(\mathbf{1}, \mathbf{w}) \in U|_{L_1}$, and $[\mathbf{1}, \mathbf{w}] \in U|_{L_1}^\times$, in general we do not have

$$[\mathbf{1}, \mathbf{w}] \cdot (\phi_1, \phi_2) = (\mathbf{1}, \mathbf{w}) \cdot (\phi_1, \phi_2). \quad (6.45)$$

In this light, when $U: \mathbb{V} \dashrightarrow \mathbb{W}$ and $\mathbb{V}$ (respectively $\mathbb{W}$) is equipped with a Lagrangian subbundle L_1 (resp. L_2), we say that $\mathbf{S}_U$ is an L_1 -weak (resp. L_2 -weak)

U-Clifford relation if S_U is a weak U -Clifford relation satisfying the above (resp. the above with L_2 in place of L_1). Then the conclusion of 6.40 hold in the weak case.

Remark 6.46. Thus far, we have been working over the field $\mathbb{R}$. Everything discussed up to this point in this chapter holds over $\mathbb{C}$. Let us introduce some notation in the complex case. If E is a real vector bundle, its complexification $E \otimes \mathbb{C}$ is denoted by $E^{\mathbb{C}}$. For a subbundle $L \subset E^{\mathbb{C}}$, we denote the complex conjugate of L by $\overline{L}$.

6.2 Dirac Structures and Generalised Complex Geometry

6.2.1 Spinors and Courant Algebroids

We follow e.g. [31, Section 7.1] for this brief review of spinors on general Courant algebroids, which follows closely the discussion in the Introduction for the generalised tangent bundle TM .

Let $(E, \langle, \rangle)$ be rank $2n$ pseudo-Euclidean vector bundle¹ over M with split signature pairing, where $n = \dim M$. The pairing defines the relation $e \otimes e - \langle e, e \rangle$ on the tensor space of E , and hence defines the Clifford algebra $\text{Cl}(E)$. Let $S \rightarrow M$ be a vector bundle whose fiber over $p \in M$ defines a Clifford module over $\text{Cl}(E)_p = \text{Cl}(E_p)$, i.e. there is an algebra homomorphism $\Psi_p: \text{Cl}(E_p) \rightarrow \text{End}(S_p)$. If S_p defines an irreducible Clifford module for every p , and Ψ is smooth, we call S a *spinor bundle*.

The Clifford bundle $\text{Cl}(E)$ comes with a $\mathbb{Z}_2$ grading into elements of even and odd degree, which induces a grading on $\text{End}(S)$. We may thus define the graded commutator of operators in $\text{End}(S)$, i.e. if $A, B \in \text{End}(S)$ are of degree $|A|$ and $|B|$ respectively, and $\omega \in S$,

$$[A, B] \cdot \omega = A \cdot B \cdot \omega - (-1)^{|A||B|} B \cdot A \cdot \omega. \quad (6.47)$$

The role of the twisted differential d_H from Equation (1.18) is taken by

Definition 6.48. A *Dirac generating operator* D for the pseudo-Euclidean vector

¹i.e. $\langle, \rangle$ is symmetric and non-degenerate.

bundle $E \rightarrow M$ is an odd operator on $\Gamma(\mathbf{S})$ satisfying

1. $[\mathbf{D}, f] \in \Gamma(E)$, for all $f \in C^\infty(M)$, i.e., $[\mathbf{D}, f]$ is the operator given by the Clifford action of some section of E ;
2. $[[\mathbf{D}, e_1], e_2] \in \Gamma(E)$, for all $e_1, e_2 \in \Gamma(E)$;
3. $\mathbf{D}^2 \in C^\infty(M)$, i.e., $\mathbf{D}^2$ acts as a multiplication by a function.

Example 6.49. The twisted differential dH 1.18 on a manifold M is an example of a Dirac generating operator for $\mathbb{T}M$ on $\Omega^\bullet(M)$.

The following general result, shown in [30], shows how a Dirac generating operator defines a Courant algebroid.

Theorem 6.50 (Corollary 2.3, [30]). Let $\mathbf{D}$ be a Dirac generating operator for the pseudo-Euclidean vector bundle $E \rightarrow M$. Then there is a canonical Courant algebroid structure on E whose anchor and Dorfman bracket are respectively given by

$$\mathcal{L}_{\rho(e)}f := [[\mathbf{D}, f], e], \quad (6.51)$$

for all $e \in \Gamma(E)$, $f \in C^\infty(M)$, and

$$[[e_1, e_2]]_E := [[\mathbf{D}, e_1], e_2], \quad (6.52)$$

for all $e_1, e_2 \in \Gamma(E)$.

A converse to the above Theorem, first stated in [30], is given by

Theorem 6.53 (Theorem 41, [31]). Let E be a regular Courant algebroid with split signature pairing. Every spinor bundle $\mathbf{S}$ over $\text{Cl}(E)$ admits locally a Dirac generating operator.

Since any vector bundle with split signature pairing admits a spinor bundle, we can thus construct a Dirac generating operator over any regular split signature Courant algebroid. We will make use of this result in Section 6.5.

6.2.2 Dirac Structures

Here we describe some of the main properties of Dirac structures on a Courant algebroid, as introduced in Definition 3.1, and their reduction. Dirac structures play an important role in the theory of generalised complex geometry, as will be elucidated in 6.2.3.

Let $(E \rightarrow M, \llbracket \cdot, \cdot \rrbracket, \langle \cdot, \cdot \rangle, \rho)$ be a Courant algebroid. Recall that a Dirac structure is a maximally isotropic, involutive subbundle $L \subset E$, possibly supported on a submanifold $N \subset M$, such that $\rho(L) \subseteq TN$.

Remark 6.54. Notice that $(L, \rho|_L, \llbracket \cdot, \cdot \rrbracket_L)$ defines a Lie algebroid, since, for any $e_1, e_2 \in \Gamma(L)$, we have

$$\llbracket e_1, e_2 \rrbracket + \llbracket e_2, e_1 \rrbracket = \mathcal{D}\langle e_1, e_2 \rangle = 0, \quad (6.55)$$

i.e. the restriction of the Dorfman bracket to L is skew-symmetric and satisfies the Jacobi identity. For a Dirac structure L supported on a submanifold N the restriction of the anchor ρ is defined in order to provide the data for Lie algebroid L over N .

Example 6.56. Let M be a smooth manifold endowed with a regular foliation $\mathcal{F}$. Then the standard Courant algebroid TM admits a Dirac structure $L = T\mathcal{F} \oplus \text{Ann}(T\mathcal{F})$ characterised by the involutivity condition

$$\llbracket X + \alpha, Y + \beta \rrbracket = [X, Y] + \iota_X d\beta - \iota_Y d\alpha \in \Gamma(L), \quad (6.57)$$

for all $X + \alpha, Y + \beta \in \Gamma(L)$. Notice that the bracket above is skew-symmetric and satisfies the Jacobi identity, i.e. is a Lie bracket. Thus L together with pr_1 and the restriction of the Dorfman bracket to L defines a Lie algebroid.

Example 6.58 (Pre-Symplectic structures). Let M be an even-dimensional manifold endowed with a 2-form ω . Then the subbundle

$$L^\omega = \text{gr}(\omega) = \{v + \iota_v \omega : v \in TM\} \quad (6.59)$$

is an almost Dirac structure on the standard Courant algebroid $TM \oplus T^*M$. Fur-

thermore, L^ω is involutive, i.e. a Dirac structure, if and only if $d\omega = 0$. In other words, L^ω is a Dirac structure if and only if (M, ω) is a pre-symplectic manifold.

Dirac Reduction

The reduction of subbundles of a Courant algebroid given by Proposition 2.56 can be applied to Dirac structures $L \subset E$.

Corollary 6.60. If L in Proposition 2.56 is a Dirac structure, then $\underline{L}$ is a Dirac structure for the reduced Courant algebroid.

Proof. Consider any $\underline{e}, \underline{e}' \in \Gamma(\underline{L})$. Then they can be lifted to $e, e' \in \Gamma_{\text{bas}}(L \cap K^\perp)$, because $L \cap K^\perp$ has constant rank. Then

$$[[e, e']]_E \in \Gamma_{\text{bas}}(L \cap K^\perp) , \quad (6.61)$$

since L is involutive and

$$[[\Gamma_{\text{bas}}(K^\perp), \Gamma_{\text{bas}}(K^\perp)]]_E \subseteq \Gamma_{\text{bas}}(K^\perp) . \quad (6.62)$$

Therefore

$$\natural([e, e'])_E := [\underline{e}, \underline{e}']_{\underline{E}} \in \Gamma(\underline{L}) . \quad (6.63)$$

Moreover $\underline{L}$ is maximally isotropic by construction. $\square$

6.2.3 Generalised Complex Structures

Generalised complex structures originated as the generalisation of ordinary complex structures from the tangent bundle TM to the generalised tangent bundle $\mathbb{T}M$. For this brief review, we follow [29, Section 4].

Definition 6.64. Let $(E, [\cdot, \cdot], \langle \cdot, \cdot \rangle, \rho)$ be an exact Courant algebroid over M with $\text{rk}(E) = 4n$, where $\dim_{\mathbb{R}}(M) = 2n$. A *generalised almost complex structure* on E is a morphism $\mathcal{J} \in \text{Aut}(E)$, covering the identity, such that $\mathcal{J}^2 = -\mathbb{1}_E$ and

$$\langle \mathcal{J}(e), \mathcal{J}(e') \rangle = \langle e, e' \rangle , \quad (6.65)$$

for all $e, e' \in E$. Furthermore, $\mathcal{J}$ is a *generalised complex structure* if either of its $\pm i$ -eigenbundles $L, \bar{L} \subset E^{\mathbb{C}}$ is involutive with respect to the Dorfman bracket $\llbracket \cdot, \cdot \rrbracket$.

Remark 6.66. Notice that this definition is tantamount to having a reduction of the structure group $O(2n, 2n)$ of E to $U(n, n)$. Moreover, the integrability condition for one of the eigenbundles implies the integrability of the other one, since they are complex conjugate. Hence both eigenbundles L and $\bar{L}$ are Dirac structures, since they are maximally isotropic by construction. As discussed in [29], generalised complex structures are in one-to-one correspondence with complex Dirac structures satisfying the condition $L \cap \bar{L} = \{0\}$.

Example 6.67 (Complex manifolds). Let M be endowed with an almost complex structure J . Then the standard Courant algebroid $\mathbb{T}M = TM \oplus T^*M$ inherits a generalised almost complex structure

$$\mathcal{J}_J = \begin{pmatrix} -J & 0 \\ 0 & J^{\mathfrak{t}} \end{pmatrix} \quad (6.68)$$

where $J^{\mathfrak{t}}: T^*M \rightarrow T^*M$ is the transpose of J . Notice that the orthogonality of $\mathcal{J}$ is equivalent to the condition $\mathcal{J}^{\mathfrak{t}} = -\mathcal{J}$. Notice that $\mathcal{J}$ is a generalised complex structure if and only if J is a complex structure on M . This is shown as follows. The $+i$ -eigenbundle L of $\mathcal{J}$ defined in Equation (6.68) is given by

$$L = T_{0,1} \oplus \text{Ann}(T_{0,1}) = T_{0,1} \oplus T_{1,0}^* \subset \mathbb{T}M^{\mathbb{C}}, \quad (6.69)$$

which is maximally isotropic, where $T_{0,1}$ is the $-i$ -eigenbundle of J , given by

$$T_{0,1} = \{v + iJ(v) : v \in TM\} \subset TM^{\mathbb{C}}, \quad (6.70)$$

and $T_{1,0}$ the $+i$ -eigenbundle. Then, if L is involutive, for any $X + \xi, Y + \eta \in \Gamma(L)$, we have

$$\llbracket X + \xi, Y + \eta \rrbracket = [X, Y] + \mathcal{L}_X \eta - \iota_Y d\xi \in \Gamma(T_{0,1} \oplus T_{1,0}^*) \quad (6.71)$$

which implies that $[X, Y] \in \Gamma(T_{0,1})$ for any $X, Y \in \Gamma(T_{0,1})$. Conversely, if J is

integrable, we have that the Dorfman bracket (6.71) can be written as

$$\llbracket X + \xi, Y + \eta \rrbracket = [X, Y] + \iota_X \bar{\partial} \eta - \iota_Y \bar{\partial} \xi \quad (6.72)$$

where ∂ is the Dolbeault differential induced by the integrability of J . Thus, by definition of ∂ , we have that $\iota_X \bar{\partial} \eta - \iota_Y \bar{\partial} \xi \in \Gamma(T_{1,0}^*)$ and, hence, L is involutive with respect to the Dorfman bracket.

Example 6.73 (Symplectic manifolds). Let (M, ω) be a symplectic manifold. Then ω determines a generalised complex structure on $\mathbb{T}M$ given by

$$\mathcal{J}_\omega = \begin{pmatrix} 0 & -\omega^{-1} \\ \omega & 0 \end{pmatrix} \quad (6.74)$$

whose $+i$ -eigenbundle is

$$L = \{ v - i \iota_v \omega : v \in TM \} \subset \mathbb{T}M^{\mathbb{C}}. \quad (6.75)$$

We can show that L is involutive as follows. For all $X, Y \in \Gamma(TM)$ we have

$$\begin{aligned} \llbracket X - i(\iota_X \omega), Y - i(\iota_Y \omega) \rrbracket &= [X, Y] - i \mathcal{L}_X(\iota_Y \omega) + i(\iota_Y d\iota_X \omega) \\ &= [X, Y] - i(\iota_{[X, Y]} \omega) - i(\iota_Y \iota_X d\omega), \end{aligned} \quad (6.76)$$

which yields that L is involutive because ω is closed.

Let $L \subset E$ be a Lagrangian subbundle. Recall from Remark 6.11 that at each $p \in M$ there is a pure spinor line bundle Ω_p associated to L_p given by

$$\Omega_p = \{ \phi \in \mathbf{S} : \Psi(e)\phi = 0, \forall e \in L_p \}. \quad (6.77)$$

This defines a line bundle over M , which equivalently defines $\mathcal{J}$ when $L \cap \bar{L} = \{0\}$. We thus may interchangeably use $\mathcal{J}, L$ or Ω when discussing generalised complex structures. The spinor line Ω defines a filtration, where $\Omega^n = \Omega$ and $\Omega^{n-k} = (\wedge^k \bar{L}) \cdot \Omega^n$. If L defines an almost generalised complex structure $\mathcal{J}$, then this

becomes a grading

$$\bigwedge^\bullet T^*M \otimes \mathbb{C} = \Omega^n \oplus \Omega^{n-1} \oplus \dots \oplus \Omega^0 \quad (6.78)$$

since $L \cap \bar{L} = \{0\}$.

The H -twisted integrability of an almost generalised complex structure on $\mathbb{T}M$ (or a maximal isotropic in general) can be stated in terms of the Dirac generating operator d_H , where $\mathcal{J}$ is integrable if and only if $d_H(\Gamma(\Omega^n)) \subset \Gamma(\Omega^{n-1})$, see [29, Theorem 3.38]. If $\mathcal{J}$ is integrable, then $d_H(\Gamma(\Omega^k)) \subset \Gamma(\Omega^{k-1}) \oplus \Gamma(\Omega^{k+1})$, and we have a decomposition $d_H = \partial + \bar{\partial}$ where [29, Theorem 4.23]

$$\partial: \Gamma(\Omega^k) \rightarrow \Gamma(\Omega^{k-1}), \quad \bar{\partial}: \Gamma(\Omega^k) \rightarrow \Gamma(\Omega^{k+1}). \quad (6.79)$$

There is also the notion of the type of a generalised complex structure $\mathcal{J}$ on a $2n$ manifold M , an integer $k \in \{0, \dots, n\}$ at each point, defining a lower semicontinuous function.

Definition 6.80. The *type* of $\mathcal{J}$, denoted $\text{type}(\mathcal{J})$, is defined by the codimension of $\rho(L)$ in M , i.e. $k = n - \text{rank}(\rho(L))$.

In the standard Courant algebroid case, for any $m \in M$ an element ϕ of Ω_m can be written as [29, Theorem 4.8]

$$\phi = c \exp(B + i\omega) \wedge \Theta \quad (6.81)$$

for B, ω real two-forms, and $\Theta = \theta_1 \wedge \dots \wedge \theta_k$ for some linearly independent complex forms $\theta_1, \dots, \theta_k$. The condition that $L \cap \bar{L} = \{0\}$ can be stated as the requirement that $\omega^{n-k} \wedge \Theta \wedge \bar{\Theta} \neq 0$. The integer k is again the type of the generalised complex structure $\mathcal{J}$.

Reduction of Generalised Complex Structures

In this subsection we assume that the exact Courant algebroid $E \rightarrow M$ satisfies the assumptions of the reduction Theorem 2.40.

Proposition 6.82 (Proposition 6.1, [50]). Let E be a reducible exact Courant algebroid over M endowed with a generalised complex structure $\mathcal{J}$, with $+i$ -eigenbundle $L \subset E^{\mathbb{C}}$, such that $\mathcal{J}(K) \cap K^{\perp} \subset K$ and $\text{rk}(\mathcal{J}(K) \cap K^{\perp})$ is constant. Suppose that

$$\llbracket \Gamma(K), \Gamma(L \cap K^{\perp}) \rrbracket \subset \Gamma(L + K) . \quad (6.83)$$

Then the reduced Courant algebroid $\underline{E}$ over $\mathcal{Q}$ inherits a generalised complex structure $\underline{\mathcal{J}}$ with $+i$ -eigenbundle $\underline{L} = \mathfrak{q}(L \cap K^{\perp} + K)$.

Proof. By Proposition 2.56, the condition (6.83) ensures the existence of a Dirac structure $\underline{L} = \mathfrak{q}(L \cap K^{\perp} + K) \subset \underline{E}^{\mathbb{C}}$. In order to show that $\underline{L}$ is the $+i$ -eigenbundle of a generalised complex structure, we need that $\underline{L} \cap \overline{\underline{L}} = \{0\}$. This follows from the assumptions that $\mathcal{J}(K) \cap K^{\perp} \subset K$ and has constant rank by using [51, Lemma 5.1]. $\square$

6.2.4 Generalised Kähler Structures

Here we introduce the notion of a generalised Kähler structure on a Courant algebroid E .

Definition 6.84 (Generalised Kähler Structures). A *generalised almost Kähler structure* on a Courant algebroid E is a pair of generalised almost complex structures $(\mathcal{J}^+, \mathcal{J}^-)$

1. $\mathcal{J}^+ \mathcal{J}^- = \mathcal{J}^- \mathcal{J}^+$,
2. $\tau = -\mathcal{J}^+ \mathcal{J}^-$ defines a generalised metric.

We sometimes denote the pair $(\mathcal{J}^+, \mathcal{J}^-)$ by $\mathcal{K}$. The generalised almost Kähler structure is a *generalised Kähler structure* when both $\mathcal{J}^{\pm}$ are generalised complex structures.

A *generalised Calabi-Yau structure* is a generalised Kähler structure whose associated line bundles are holomorphically trivial, i.e. there are non-vanishing sections $\varphi^{\pm}$ of $\Omega^{\pm}$ for the line bundles associated to $\mathcal{J}^{\pm}$ such that $d_H \omega = 0$.

Example 6.85. A classical Kähler manifold is a triple (g, J, ω) of a Riemannian metric, complex structure and symplectic form such that $\omega = gJ$. Taking $\mathcal{J}^+ = \mathcal{J}_J$ and $\mathcal{J}^- = \mathcal{J}_{\omega}$, it follows that $(\mathcal{J}^+, \mathcal{J}^-)$ is a generalised Kähler structure on $(\mathbb{T}M, 0)$.

If the classical Kähler structure is Calabi-Yau, then the associated holomorphic volume form Θ and Kähler form satisfy $\omega^n = i^n n! / 2^n \Theta \wedge \bar{\Theta}$, and hence define a generalised Calabi-Yau structure.

Note there is the following result on an equivalent definition of generalised Kähler structures [29, Theorem 6.37].

Theorem 6.86 ([29]). Let M be $2n$ -manifold and take a closed three-form $H \in \Omega^3(M)$. The bi-Hermitian structure (g, b, J_+, J_-) of a Riemannian metric g on M , $b \in \Omega^2(M)$, and complex structures $J_\pm$ on M preserving g is equivalent to a generalised almost Kähler structure by

$$\mathcal{J}_\pm = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \begin{pmatrix} J_+ \pm J_- & -(\omega_+^{-1} \mp \omega_-^{-1}) \\ \omega_+ \mp \omega_- & -(J_+^* \pm J_-^*) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -b & 1 \end{pmatrix}. \quad (6.87)$$

Then $\mathcal{K} = (\mathcal{J}_+, \mathcal{J}_-)$ is a generalised Kähler structure on $(\mathbb{T}M, H)$ if and only if

$$-d_+^c \omega_+ = d_-^c \omega_- = H + db, \quad (6.88)$$

where $\omega_\pm = J_\pm g$.

6.3 Related Dirac Structures

We introduce the notion of relation between Dirac structures descending from Courant algebroid relations, following Section 6.1.1.

Definition 6.89 (Dirac Relations). Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on a submanifold $C \subset M_1 \times M_2$ and let L_1 and L_2 be almost Dirac structures for E_1 and E_2 , respectively. Define the restriction of R to L_i by

$$R|_{L_i} := \{ (e_1, e_2) \in R : e_i \in L_i \}. \quad (6.90)$$

Then R is a *relation between the Dirac structures L_1 and L_2* , or a *Dirac relation*, if

$$R|_{L_1} = R \cap (L_1 \times L_2)|_C + \text{CK}_R, \quad R|_{L_2} = R \cap (L_1 \times L_2)|_C + \text{K}_R, \quad (6.91)$$

and are regular vector sub-bundles supported on C . Again, we note that the inclusion $\supseteq$ always holds.

Remark 6.92 (Dirac and Lie Algebroid Relations). Let $(A_1, [\cdot, \cdot]_1, \rho_1)$ and $(A_2, [\cdot, \cdot]_2, \rho_2)$ be Lie algebroids over M_1 and M_2 , respectively. Then the product vector bundle $A_1 \times A_2$ over $M_1 \times M_2$ is endowed with a Lie algebroid structure given by the Lie bracket $([\cdot, \cdot]_1, [\cdot, \cdot]_2)$ and anchor map (ρ_1, ρ_2) . Following Meinrenken [77, Sections 2.3], a *Lie algebroid relation* R between the Lie algebroids A_1 and A_2 is a Lie subalgebroid $R \subset A_1 \times A_2$ supported on a submanifold $C \subset M_1 \times M_2$.

Suppose $\text{CK}_R = \text{K}_R = \{0_R\}$. Since L_1 and L_2 are Dirac structures, then the Dirac relation $R|_{L_1} = R|_{L_2}$ such that $L_1 \sim_R L_2$ is a Lie algebroid relation between L_1 and L_2 . Recall that the Lie algebroid structure for L_1 and L_2 is induced by the Courant structure on E_1 and E_2 respectively.

Example 6.93 (Foliated manifolds). Let $(M_1, \mathcal{F}_1)$ and $(M_2, \mathcal{F}_2)$ be foliated manifolds and consider a foliation-preserving diffeomorphism $\phi \in \text{Diff}(M_1, M_2)$, i.e. $\phi_*(T\mathcal{F}_1) = T\mathcal{F}_2$. Then ϕ induces the Courant algebroid isomorphism

$$\bar{\phi} := \phi_* + (\phi^{-1})^*: \mathbb{T}M_1 \longrightarrow \mathbb{T}M_2 \quad (6.94)$$

covering ϕ . We now show that the induced Courant algebroid relation $\text{gr}(\bar{\phi})$ supported on $\text{gr}(\phi)$ is a Dirac relation between the Dirac structures L_1 and L_2 induced by the foliations, i.e. for

$$L_i := T\mathcal{F}_i \oplus \text{Ann}(T\mathcal{F}_i) , \quad (6.95)$$

for $i = 1, 2$. Notice that the kernel and cokernel of $\text{gr}(\bar{\phi})$ are trivial. Then

$$\text{gr}(\bar{\phi})|_{L_1} = \{ (X + \xi, \phi_*(X) + (\phi^{-1})^*(\xi)) : X + \xi \in T\mathcal{F}_1 \oplus \text{Ann}(T\mathcal{F}_1) \} . \quad (6.96)$$

Since $\phi_*(T\mathcal{F}_1) = T\mathcal{F}_2$ we have that $\text{gr}(\bar{\phi})|_{L_1} \subseteq (L_1 \times L_2) \cap \text{gr}(\bar{\phi})$. Similarly, $\text{gr}(\bar{\phi})|_{L_2} \subseteq (L_1 \times L_2) \cap \text{gr}(\bar{\phi})$, hence $\text{gr}(\bar{\phi})$ is a Dirac relation between L_1 and L_2 .

Example 6.97 (Dirac reduction). Consider the reduction relation $Q(K): E \rightarrow \underline{E}$ by an isotropic subbundle $K \subset E$, as in Section 3.1.2, with $Q(K)$ supported on

$\text{gr}(\varpi) \subseteq M \times \mathcal{Q}$. Let us find ourselves in the setting of Corollary 6.60, hence E is endowed with a Dirac structure L that reduces to $\underline{L} = \mathfrak{h}(L \cap K^\perp + K)$. We shall show that $Q(K)$ is an almost Dirac relation between L and $\underline{L}$ in the sense of Definition 6.89. Recall that $K_{Q(K)} = K$ and $\text{CK}_{Q(K)} = \{0\}$. For $m \in M$, and corresponding point $c = (m, \varpi(m)) \in \text{gr}(\varpi)$, we have

$$Q(K)_c \cap (L_m \times \underline{L}_{\varpi(m)}) = \{ (e, \mathfrak{h}(e)) : e \in K_m^\perp \cap L_m \} . \quad (6.98)$$

while

$$Q(K)|_{L(m, \varpi(m))} = \{ (e, \mathfrak{h}(e)) : e \in K_m^\perp \cap L_m \} , \quad (6.99)$$

giving equality. Furthermore, for L to be reducible, $L \cap K^\perp$ has to be constant rank, which means that $Q(K)|_L$ is a regular vector subbundle supported on $\text{gr}(\varpi)$.

Similarly, we have that

$$Q(K)|_{\underline{L}(m, \varpi(m))} = \{ (e, \mathfrak{h}(e)) \in Q(K)_{(m, \varpi(m))} : \mathfrak{h}(e) \in \underline{L}_{\varpi(m)} \} , \quad (6.100)$$

where, by definition of $\mathfrak{h}$ and $\underline{L}$, $e \in K_m^\perp \cap L_m + K_m$. Again by the regularity of $K^\perp \cap L$, we have that $Q(K)|_{\underline{L}} \subset (L \times \underline{L})|_{\text{gr}(\varpi)} + K_{Q(K)}$ is a regular vector subbundle, thus showing that $L \sim_{Q(K)} \underline{L}$.

6.3.1 Dirac Structures and T-duality

Since we have at our disposal Dirac relations, we can now discuss T-duality for Dirac structures. To this, as in Definition 4.35, we need the following

Definition 6.101. Let $\underline{\sigma}_1 : T\mathcal{Q}_1 \rightarrow \underline{E}_1$ be the splitting of the exact Courant algebroid $\underline{E}_1$ over $\mathcal{Q}_1$ induced by an isotropic splitting $\sigma_1 : TM \rightarrow E$ of E which is adapted to $K_1 \subset E$. A Dirac structure $\underline{L}_1$ on $\underline{E}_1$ is *invariant with respect to D_1* , or simply *D_1 -invariant*, if there are generators $\{\underline{X}_1, \dots, \underline{X}_{r_1}\} \subset \Gamma(T\mathcal{Q}_1)$ of the $C^\infty(\mathcal{Q}_1)$ -submodule $\Gamma(D_1)$ such that

$$\text{ad}_{\underline{X}_k}^{\underline{\sigma}_1}(\underline{e}_1) \in \Gamma(\underline{L}_1) , \quad (6.102)$$

for $k = 1, \dots, r_1$ and $\underline{e}_1 \in \Gamma(\underline{L}_1)$.

Then, we have an analogous result for Dirac structures to Theorem 4.45.

Proposition 6.103. Let the assumptions of Theorem 4.45 and let $\underline{L}_1$ be a D_1 -invariant Dirac structure on $\underline{E}_1$. Then there exists a (unique) Dirac structure $\underline{L}_2$ on $\underline{E}_2$ such that the T-duality relation R is a Dirac relation.

Proof. The construction of the T-dual Dirac structure $\underline{L}_2$ on $\underline{E}_2$ follows the same steps as in the construction of the T-dual generalised metric V_2^+ of Theorem 4.45. $\square$

6.4 Spinors and Courant Algebroid relations

We now explain how the space of spinors between related Courant algebroids can be related using the language of spinor representation of the relation, and how this approach can be used to related differential graded operators.

6.4.1 Clifford Algebra Relations and Differential Graded Operators

Let us extend the notion of Courant algebroid relations to the spinor modules between related Courant algebroids, using the ideas of Sections 6.1.2.

Let $E_1 \rightarrow M_1$, $E_2 \rightarrow M_2$ be pseudo-Euclidean vector bundles and R an isotropic subbundle of $E_1 \times \overline{E}_2$ supported on $C \subset M_1 \times M_2$ such that $\rho(R^\perp) \subseteq TC$. As in Definition 6.21, we can define the space $R^\times$ by quotienting the kernel and cokernel of R , analogously to the quotient (6.19).

Definition 6.104. A vector subbundle $\mathcal{S}_R \subset (\wedge^\bullet T^*M_1 \times \wedge^\bullet T^*M_2)|_C$ is a R -Clifford relation of parity $i \in \mathbb{Z}_2$ if it is a spinor module of the typical fiber of $\text{Cl}^i(R)$, where the Clifford action is defined as in (6.16) with the fibers of the bundle $(\text{Cl}(E_1) \times \text{Cl}(E_2))|_C$ acting on the fibers of $(\wedge^\bullet T^*M_1 \times \wedge^\bullet T^*M_2)|_C$.

If there exists an action of $\text{Cl}(R^\times)$ on $(\wedge^\bullet T^*M_1 \times \wedge^\bullet T^*M_2)|_C$ compatible with $\text{Cl}^i(R)$ and a subbundle $\mathcal{S}_R \subset (\wedge^\bullet T^*M_1 \times \wedge^\bullet T^*M_2)|_C$ such that $\mathcal{S}_R$ is a spinor module for $\text{Cl}^i(R^\times)$, then $\mathcal{S}_R$ is said to be a *weak R -Clifford relation of parity $i \in \mathbb{Z}_2$* .

Note that Proposition 6.22 and Corollary 6.25 have their smoothly varying counterparts in this case, i.e.

Proposition 6.105. Let p_i be the projections from $E_1 \times E_2$ to the first and second factor for $i = 1, 2$ respectively, and denote by the same symbols the projections from $\wedge^\bullet T^*M_1 \times \wedge^\bullet T^*M_2$. Let S_R be an R -Clifford relation of parity j , with $p_i(R) \neq 0$ and $p_i(S_R) \neq \{0\}$ for $i = 1, 2$. Then, for each $c \in C$

$$(S_R)_c \cap (\{0\} \times \wedge^\bullet T^*M_2)_c = (S_R)_c \cap (\wedge^\bullet T^*M_2 \times \{0\})_c = \{(0, 0)_c\}, \quad (6.106)$$

Corollary 6.107. Let S_R is an R -Clifford relation, with $p_i(R) \neq 0$ and $p_i(S_R) \neq \{0\}$ for $i = 1, 2$. For every $c \in C$, if $(\nu, \mu)_c, (\nu, \mu')_c \in (S_R)_c$, then $\mu = \mu'$.

Example 6.108. Let M be foliated by $\mathcal{F}$, with smooth leaf space $\mathcal{Q}$. Let $Q(\mathcal{F}): \mathbb{T}M \rightarrow \mathbb{T}\mathcal{Q}$ be the reduction relation given by Section 3.1.2, covering the surjective submersion $\varpi: M \rightarrow \mathcal{Q}$. Consider

$$S_{Q(\mathcal{F})} = \{(\varpi^*\alpha, \alpha)|_{\text{gr}(\varpi)} : \alpha \in \wedge^\bullet T^*\mathcal{Q}\}, \quad (6.109)$$

supported on $\text{gr}(\varpi)$. Recall that for each $X + \eta \in \Gamma(\mathbb{T}\mathcal{Q})$, the section $\dot{X} + \dot{\eta} \in \Gamma(\mathbb{T}M)$ is such that $(\dot{X} + \dot{\eta}, X + \eta) \in \Gamma(E \times \overline{E}; Q(\mathcal{F}))$, where $\dot{\eta} = \varpi^*\eta$ and $\dot{X}$ is any lift of X , i.e. $\varpi_*\dot{X} = X$. Then, on C ,

$$\begin{aligned} (\dot{X} + \dot{\eta}, X + \eta) \cdot (\varpi^*\alpha, \alpha) &= (\iota_{\dot{X}}(\varpi^*\alpha) + \dot{\eta} \wedge \varpi^*\alpha, \iota_X\alpha + \eta \wedge \alpha) \\ &= (\varpi^*(\iota_X\alpha + \eta \wedge \alpha), \iota_X\alpha + \eta \wedge \alpha) \in S_{Q(\mathcal{F})}. \end{aligned} \quad (6.110)$$

We therefore see that $S_{Q(\mathcal{F})}$ is invariant under the action of $\text{Cl}(Q(\mathcal{F}))$, providing it a spin representation. To show that it is an irreducible representation, suppose that $S' \subset S_{Q(\mathcal{F})}$ is invariant. Then S' has the form

$$S' = \{(\varpi^*\alpha, \alpha)|_{\text{gr}(\varpi)} : \alpha \in S\} \quad (6.111)$$

where $S \subset \wedge^\bullet T^*\mathcal{Q}$ is a vector subbundle. By invariance of S' and projecting to the second factor, it follows that S is invariant under the action of $\text{Cl}(\mathbb{T}\mathcal{Q})$. But $\wedge^\bullet T^*\mathcal{Q}$ is a spinor module for $\text{Cl}(\mathbb{T}\mathcal{Q})$, and hence $S = \{0\}$ or $S = \wedge^\bullet T^*\mathcal{Q}$. Thus $S_{Q(\mathcal{F})}$ is a $Q(\mathcal{F})$ -Clifford relation of parity $+$.

Example 6.112. Let $(M, \mathcal{F})$ be a foliation with compact, oriented fibers, with smooth leaf space $\mathcal{Q}$. Since the leaves of $\mathcal{F}$ are compact and oriented, there is a map

$$\begin{aligned} \varpi_*: \Omega^\bullet(M) &\rightarrow \Omega^\bullet(\mathcal{Q}) \\ \alpha &\mapsto \int_{\mathcal{F}} \alpha \end{aligned} \tag{6.113}$$

given by the fiber integration of forms over the fibers of $\mathcal{F}$, inducing a vector bundle morphism $\wedge^\bullet T^*M \rightarrow \wedge^\bullet T^*\mathcal{Q}$ denoted by the same symbol. Consider $\text{gr}(\varpi_*)$ as a subspace of $(\wedge^\bullet T^*M \times \wedge^\bullet T^*\mathcal{Q})|_{\text{gr}(\varpi)}$. This is not a spinor module for $Q(\mathcal{F})$. Indeed take $\alpha \in \wedge^\bullet T^*\mathcal{Q}$ and note that $\varpi_*(\varpi^*(\alpha)) = 0$, thus $(\varpi^*\alpha, 0) \in \text{gr}(\varpi_*)$. By Proposition 6.105, it follows that $\text{gr}(\varpi_*)$ is not a spinor module for $Q(\mathcal{F})$.

Consider instead a fiber volume form $\mu \in \det(T^*\mathcal{F})$, and consider $S_\mu := \text{gr}(\varpi_*|_A)$, where A is the subspace of forms $\varpi^*\alpha \wedge \mu$ for $\alpha \in \wedge^\bullet T^*\mathcal{Q}$, supported on $\text{gr}(\varpi)$. This is not an invariant subspace for the action of $\text{Cl}(Q(\mathcal{F}))$, since if $X \in T\mathcal{F} = \text{Ker}(Q(\mathcal{F}))$, then $(X, 0) \cdot (\varpi^*\alpha \wedge \mu, \alpha) = (\varpi^*\alpha \wedge \iota_X \mu, 0) \notin S_\mu$. However, we may define an action of $\text{Cl}(Q(\mathcal{F})^\times)$ on $(\wedge^\bullet T^*M \times \wedge^\bullet T^*\mathcal{Q})|_{\text{gr}(\varpi)}$. Note that $Q(\mathcal{F})^\times$ is the set of pairs $(X + \xi, \underline{X} + \underline{\xi})$, where $X \in TM/T\mathcal{F}$, $\xi \in \text{Ann}(T\mathcal{F})$, and $\underline{X}$ and $\underline{\xi}$ are canonically associated to X and ξ respectively. Then, by considering $D = \text{Ann}(\mu) \cong TM/T\mathcal{F}$, we have $X_D \in TM$ associated to X , and hence we define

$$\begin{aligned} (X + \xi, \underline{X} + \underline{\xi}) \cdot (\varpi^*\alpha \wedge \mu, \varpi_*(\varpi^*\alpha \wedge \mu)) &:= \\ (X_D + \xi, \underline{X} + \underline{\xi}) \cdot (\varpi^*\alpha \wedge \mu, \varpi_*(\varpi^*\alpha \wedge \mu)) \end{aligned} \tag{6.114}$$

for the usual action of $\text{Cl}^j(Q(\mathcal{F}))$ on $(\wedge^\bullet T^*M \times \wedge^\bullet T^*\mathcal{Q})|_{\text{gr}(\varpi)}$, where $j = \dim(\mathcal{F}) \bmod 2$. Then, since $\varpi_*X_D = \underline{X}$,

$$\begin{aligned} (X + \xi, \underline{X} + \underline{\xi}) \cdot (\varpi^*\alpha \wedge \mu, \varpi_*(\varpi^*\alpha \wedge \mu)) &= \\ = (\varpi^*(\iota_{\underline{X}}\alpha + \xi \wedge \alpha) \wedge \mu, (-1)^j(\underline{X} + \underline{\xi}) \cdot \varpi_*(\varpi^*\alpha \wedge \mu)) &\tag{6.115} \\ = (\varpi^*(\iota_{\underline{X}}\alpha + \xi \wedge \alpha) \wedge \mu, \varpi_*(\varpi^*(\iota_{\underline{X}}\alpha + \xi \wedge \alpha) \wedge \mu)) &\in S_\mu. \end{aligned}$$

where we use Appendix C to pass the action through the fiber integration in the

second factor. Thus S_μ is a weak $Q(\mathcal{F})$ -Clifford relation of parity $\dim(\mathcal{F}) \bmod 2$. Note that $S_{Q(\mathcal{F})}$ is also a spin module for $\text{Cl}(Q(\mathcal{F})^\times)$, and so is isomorphic to S_μ , with the isomorphism given explicitly by

$$(\varpi^* \alpha, \alpha)|_{\text{gr}(\varpi)} \mapsto \left(\varpi^* \alpha \wedge \mu, \left(\int_{\mathcal{F}} \mu \right) \alpha \right) \Big|_{\text{gr}(\varpi)}. \quad (6.116)$$

We can define the smooth version of the canonical Clifford relation, allowing us to relate spinor lines, as in Definition 6.34.

Definition 6.117. If S_R is a weak R -Clifford relation of parity i for $R \subset E_1 \times E_2$, the *canonical R -Clifford relation of parity i* is the vector bundle

$$\mathcal{S}_R = S_R \otimes |\det S_R^*|^{\frac{1}{2^n}} \quad (6.118)$$

over $C \subset M_1 \times M_2$, where $n = \dim V = \dim W$. As a spin module, the typical fiber of $\text{Cl}^i(R)$ acts trivially on the second component of S_R .

If $\Omega_1 \subset \wedge^{\bullet} T^* M_1$ and $\Omega_2 \subset \wedge^{\bullet} T^* M_2$ are pure spinor line bundles, we say that Ω_1 and Ω_2 are *R -related* if

$$(\Omega_1 \times \Omega_2)|_C \otimes |\det S_R^*|^{\frac{1}{2^n}} \subset \mathcal{S}_R, \quad (6.119)$$

is a vector subbundle and denote this as $\Omega_1 \sim_R \Omega_2$.

Again, the smoothly varying version of Proposition 6.40 follows.

Proposition 6.120. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on C , with spin modules S_i for E_i for $i = 1, 2$, and an R -Clifford relation $S_R \subset (S_1 \times S_2)|_C$. Let $L_1 \subset E_1$ and $L_2 \subset E_2$ be Lagrangian subbundles, and let $\Omega_1 \subset S_1$ and $\Omega_2 \subset S_2$ be the corresponding pure spinor lines. If $\Omega_1 \sim_R \Omega_2$, then $L_1 \sim_R L_2$.

Remark 6.121. The smoothly varying cases of Remarks 6.42 and 6.44 also follow. The condition that $p_1(R) \supseteq L_1$ or $p_2(R) \supseteq L_2$ is satisfied for instance when R is a generalised isometry, whose kernel and cokernel must be trivial. It is also satisfied when R is the reduction relation $Q(K)$, since $p_2(Q(K)) = \underline{E}$.

Remark 6.122. Let us discuss Example 6.112 in relation to [78], where the theory of reduction of pure spinors is discussed. Therein, Drummond considers a Courant

algebroid reduction by an isotropic subbundle K . A pure spinor α gives a representative of a Lagrangian subbundle L . The push-forward of α is non-zero if and only if $L \cap K = 0$. However, from [50, Proposition 4.1], we know that any Lagrangian subbundle with $L \cap K^\perp$ having constant rank has an associated reduced Lagrangian $\underline{L}$. In the case that $L \cap K \neq 0$, Drummond resolves this issue by considering a perturbation L_D of L by a subbundle $D \subset \mathbb{T}M$. The spinor associated to L_D is then given by $\mu_D \cdot \alpha$, for some nowhere vanishing $\mu_D \in \det(D)$. The spinor $\varpi_*(\mu_D \cdot \alpha)$ then generates the Lagrangian $\underline{L}$. For instance, in the case that $\alpha = \varpi^* \underline{\alpha}$ for some $\underline{\alpha} \in \Omega^{\text{top}}(\mathcal{Q})$, we have $L = T\mathcal{F} \oplus \text{Ann}(T\mathcal{F})$. Then $D = T^*\mathcal{F}$ and so $\alpha_D = \mu \wedge \alpha$ for some $\mu \in \det(T^*\mathcal{F})$. Then the pushforward of α_D by ϖ_* is non-zero. Applying this construction to all $\alpha \in \Omega^\bullet(\mathcal{Q})$ (note that the Lagrangian L will change, but we may use the same D), one obtains the space $S' \subset \Omega^\bullet(M) \times \Omega^\bullet(\mathcal{Q})$ of pairs

$$S' = \{ (\alpha_D, (\int_{\mathcal{F}} \mu) \alpha) : \alpha \in \Omega^\bullet(\mathcal{Q}) \} \quad (6.123)$$

and hence $S' = S_\mu$, where S_μ is as in Example 6.112 and action given by (6.114). For $L_1 \subset E_1$ Lagrangian, S' is an L_1 -weak $Q(\mathcal{F})$ -Clifford relation (see Remark 6.44) if and only if $\Omega_1 \subset p_1(S')$ for the spinor line Ω_1 associated to L_1 .

We highlight that, while Drummond perturbs the Lagrangian L to L_D to obtain a reduced spinor such that $\Omega \sim_R {}^R\Omega$, this can be achieved instead by considering an isomorphic spinor module for $\text{Cl}(Q(\mathcal{F})^\times)$. For instance, the case that $L \cap K = 0$ requires the spinor module S' , while the case that $K \subset L$ can be considered using $S_{Q(\mathcal{F})}$ as in Example 6.108.

Definition 6.124. Let $\mathcal{F}$ be a foliation of M . A form $\omega \in \Omega^\bullet(M)$ is said to be *invariant along $\mathcal{F}$* if there is a Lie subalgebra $\mathfrak{f} \subset \Gamma(T\mathcal{F})$ spanning $T\mathcal{F}$ pointwise such that for each $X \in \mathfrak{f}$ we have $\mathcal{L}_X \omega = 0$. A pure spinor line bundle $\Omega \subset \bigwedge^\bullet T^*M$ is said to be *invariant along $\mathcal{F}$* if there is a Lie subalgebra $\mathfrak{f} \subset \Gamma(T\mathcal{F})$ spanning $T\mathcal{F}$ pointwise such that for each $X \in \mathfrak{f}$ we have $\mathcal{L}_X \Gamma(\Omega) \subset \Gamma(\Omega)$. Note that if ω is invariant along $\mathcal{F}$ then the line Ω_ω generated by ω is invariant.

Remark 6.125 (Invariant Differential Forms). In the T-duality setting, we denote by $\Omega_{D_1}^\bullet(\mathcal{Q}_1)$ the space of D_1 -invariant differential forms on $\mathcal{Q}_1$, i.e. forms $\underline{\omega} \in \Omega^\bullet(\mathcal{Q}_1)$ such that $\mathcal{L}_{\underline{X}} \underline{\omega} = 0$ for all $\underline{X} \in \mathfrak{k}_{\tau_1}$. Note that for any $X \in \Gamma(T\mathcal{Q}_1)$,

$e = Y + \eta \in \Gamma(\mathbb{T}\mathcal{Q}_1, H)$ and $\omega \in \Omega^\bullet(\mathcal{Q}_1)$, we have

$$\begin{aligned} \mathcal{L}_X((Y + \eta) \cdot \omega) &= \mathcal{L}_X(\iota_Y \omega) + \mathcal{L}_X(\eta \wedge \omega) \\ &= \iota_Y \mathcal{L}_X \omega + \iota_{[X, Y]} \omega + (\mathcal{L}_X \eta) \wedge \omega + \eta \wedge \mathcal{L}_X \omega \\ &= ([X, Y] + \mathcal{L}_X \eta) \cdot \omega + (Y + \eta) \cdot \mathcal{L}_X \omega, \end{aligned} \quad (6.126)$$

i.e.

$$\mathcal{L}_X((Y + \eta) \cdot \omega) = \text{ad}_X^\sigma(Y + \eta) \cdot \omega + (Y + \eta) \cdot \mathcal{L}_X \omega. \quad (6.127)$$

Example 6.128. Let $\underline{\alpha} \in \Omega^\bullet(\mathcal{Q})$ be a nowhere vanishing form on $\mathcal{Q}$, and consider the pure line $\Omega \subset \wedge^\bullet T^*M$ generated by $\alpha = \varpi^* \underline{\alpha}$. Then Ω is invariant, since for $X \in T\mathcal{F}$ and $f \in C^\infty(M)$,

$$\begin{aligned} \mathcal{L}_X f(\varpi^* \underline{\alpha}) &= \iota_X d(f \varpi^* \underline{\alpha}) + d(f \iota_X \varpi^* \underline{\alpha}) \\ &= \iota_X d f \wedge \varpi^* \underline{\alpha} + f \iota_X \varpi^*(d \underline{\alpha}) + 0 \\ &= X(f) \varpi^* \underline{\alpha} \in \Omega. \end{aligned} \quad (6.129)$$

For $S_{Q(\mathcal{F})}$ as in Example 6.108, denote by $\underline{\Omega}$ the spinor line bundle associated to $\underline{\alpha}$. For each $u \in (\Omega \times \underline{\Omega})|_{\text{gr}(\varpi)}$, we have

$$u_c = u_{(m, \varpi(m))} = (f \alpha, \underline{g} \underline{\alpha})_c = (f(m) \alpha_m, \underline{g}(q) \underline{\alpha}_q) = (f(m) \alpha_m, \frac{g(q)}{f(m)} f(m) \underline{\alpha}_q) \in S_{Q(\mathcal{F})}^\kappa \quad (6.130)$$

where $f \in C^\infty(M)$, $\underline{g} \in C^\infty(\mathcal{Q})$ and $\kappa(c) = \frac{\varpi^* g(m)}{f(m)}$. Thus

$$\begin{aligned} (\Omega \times \underline{\Omega})|_{\text{gr}(\varpi)} \otimes (\det((S_{Q(\mathcal{F})})^*))^{1/2^n} &= (\Omega \times \underline{\Omega})|_{\text{gr}(\varpi)} \otimes (\det((S_{Q(\mathcal{F})}^\kappa)^*))^{1/2^n} \\ &\subset S_{Q(\mathcal{F})}^\kappa \otimes (\det((S_{Q(\mathcal{F})})^*))^{1/2^n} = \mathcal{S}_{Q(\mathcal{F})}, \end{aligned} \quad (6.131)$$

hence $\Omega \sim_{Q(\mathcal{F})} \underline{\Omega}$.

Relations for Dirac Generating Operators and the Fourier-Mukai Transform

Let $R \subset E_1 \times \overline{E}_2$ be an Lagrangian subbundle, and D_i be Dirac generating operators (DGOs) on S_i , the spin module over $\text{Cl}(E_i)$ for $i = 1, 2$. Suppose that $S_R \subset (S_1 \times S_2)|_C$ is an R -Clifford relation of parity j .

Definition 6.132. S_R is said to be a (D_1, D_2) -DGO relation if $(D_1 \times (-1)^j D_2)(S_R) \subset S_R$.

Example 6.133. Let $S_{Q(\mathcal{F})}$ be as in Example 6.108, in the case of reduction of a possibly twisted Courant algebroid $(\mathbb{T}M, H)$ for some $H \in \Omega_{\text{cl}}^3(M)$ and consider the twisted differential d_H . By Example 2.71, it follows that H is basic with respect to $\mathcal{F}$, hence there is a $\underline{H} \in \Omega_{\text{cl}}^3(\mathcal{Q})$ such that $\varpi^* \underline{H} = H$. We can thus define the twisted differential $d_{\underline{H}}$ on $\mathcal{Q}_1$, and we have

$$(d_H, d_{\underline{H}})(\varpi^* \alpha, \alpha) = (\varpi^*(d_{\underline{H}} \alpha), d_{\underline{H}} \alpha) \in S_{Q(\mathcal{F})}, \quad (6.134)$$

hence $S_{Q(\mathcal{F})}$ is a $(d_H, d_{\underline{H}})$ -DGO relation for $Q(\mathcal{F})$.

Similarly, for any $\mathcal{F}$ -invariant $\mu \in \det(T^* \mathcal{F})$, we get that S_μ of Example 6.112 is a $(d_H, d_{\underline{H}})$ -DGO relation.

Example 6.135 (The Fourier-Mukai Transform). In the setting of T-duality of torus bundles described in [46, 52] there is an accompanying isomorphism of cohomology $\varrho: H^\bullet(\mathcal{Q}_1, \underline{H}_1) \rightarrow H^\bullet(\mathcal{Q}_2, \underline{H}_2)$, given by

$$\varrho(\omega) := \int_{\mathbb{T}^k} e^B \varpi_1^* \omega \quad (6.136)$$

where the integration is along the fibers of $\mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2 \rightarrow \mathcal{Q}_2$. The map ϱ satisfies the crucial properties

$$d_{\underline{H}_2} \varrho(\omega) = (-1)^k \varrho(d_{\underline{H}_1} \omega), \quad \mathcal{R}(e) \cdot \varrho(\omega) = (-1)^k \varrho(e \cdot \omega), \quad (6.137)$$

so that ϱ is an isomorphism between the differential complexes $(\Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_1), d_{\underline{H}_1})$ and $(\Omega_{\mathbb{T}^k}^\bullet(\mathcal{Q}_2), d_{\underline{H}_2})$, as well being an isomorphism of Clifford modules. This is the Fourier-Mukai transform.

By Lemma 4.132, there is a T-duality relation $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$ supported on $C = \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$. Because of the second equation in (6.137), it follows that $\text{gr}(\varrho) \subset \Omega^\bullet(\mathcal{Q}_1) \times \Omega^\bullet(\mathcal{Q}_2)$ is invariant under the action of $\text{Cl}(R)$: for $c = (q_1, q_2) \in C$, if $(e, {}^R e) \in R_c$, and $\omega \in \bigwedge_{q_1}^\bullet(T^*\mathcal{Q}_1)$, then

$$(e, {}^R e) \cdot (\omega, \varrho(\omega)) = (e \cdot \omega, (-1)^k {}^R e \cdot \varrho(\omega)) = (e \cdot \omega, \varrho(e \cdot \omega)) \in \text{gr}(\varrho), \quad (6.138)$$

where k is the dimension of the fibers $\mathbb{T}^k$, and hence defines the parity of a representation of $\text{Cl}(R)$. Thus, $\text{gr}(\varrho)$ is a spinor representation of $\text{Cl}(R)$, and hence an R -Clifford module of parity k . Now suppose that $\text{gr}(\varrho)$ is reducible, i.e. there is some $\text{Cl}(R)$ -invariant subbundle $S' \subset \text{gr}(\varrho)$. It follows that $S' = \text{gr}(\varrho|_{\Omega_1})$ for some subspace $\Omega_1 \subset \bigwedge^\bullet T^*\mathcal{Q}_1$. By the invariance of S' under $\text{Cl}(R)$, it follows that Ω_1 is invariant under $\text{Cl}(\mathbb{T}\mathcal{Q}_1)$. By the irreducibility of $\bigwedge^\bullet T^*\mathcal{Q}_1$ as a $\text{Cl}(\mathbb{T}\mathcal{Q}_1)$ module, we have that $\Omega_1 = \{0\}$ or $\Omega_1 = \Omega^\bullet(\mathcal{Q}_1)$, and hence $\text{gr}(\varrho)$ is irreducible. The property (6.137) shows that $\text{gr}(\varrho)$ is a $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation.

We shall now show how this definition yields a Courant algebroid relation for structures defined as in Theorem 6.50.

Proposition 6.139. Let $R \subset E_1 \times \overline{E}_2$ be a Lagrangian subbundle supported on $C \subset M_1 \times M_2$, with trivial kernels. Suppose that $\text{Cl}(R)$ has a faithful, irreducible spinor representation S_R , and that S_R is a (D_1, D_2) -DGO relation. Then R is a Courant algebroid relation $R: E_1 \dashrightarrow E_2$, where the Courant algebroid structure on E_i is given by the Dirac generating operator D_i for $i = 1, 2$.

Proof. Let $c = (m_1, m_2) \in C$. According to Theorem 6.50, for every $s \in S$,

$$[[r, r']]_c \cdot s \in S. \quad (6.140)$$

for $r = (e_1, e_2), r' = (e'_1, e'_2) \in \Gamma(R)$ for sections $e_i, e'_i \in \Gamma(E_i)$. Since $\ker(R) = \{0\}$, there is some $([e_1, e'_1]_1^R)_{m_2}$ such that $(([e_1, e'_1]_1)_{m_1}, ([e_1, e'_1]_1^R)_{m_2}) \in R_c$. One must have that $(([e_1, e'_1]_1)_{m_1}, ([e_1, e'_1]_1^R)_{m_2}) \cdot s \in S_c$. It follows that for every $s = (s_1, s_2) \in S_c$,

$$(0 \cdot s_1, (([e_1, e'_1]_1^R)_{m_2} - ([e_2, e'_2]_2)_{m_2}) \cdot s_2) \in S_c. \quad (6.141)$$

By Proposition 6.105, the second factor must be zero. Since S is a faithful representation, $(\llbracket e_1, e'_1 \rrbracket_1^R)_{m_2} = (\llbracket e_2, e'_2 \rrbracket_2)_{m_2}$. Since this happens for all $c \in C$, R is involutive and thus defines a Courant algebroid relation. $\square$

6.4.2 Integrability of Almost Dirac Structures

Let $L \subset (\mathbb{T}M, H)$ be a Lagrangian subbundle, and let Ω be the corresponding spinor line. Recall that by [29, Proposition 3.44], L is a Dirac structure (i.e. it is integrable) if and only if Ω satisfies $d_H(\Omega) \subset L^* \cdot \Omega$, i.e. there is some section $X + \xi$ of $\mathbb{T}M$ such that for any non-vanishing section $\omega \in \Gamma(\Omega)$,

$$d_H \omega = (X + \xi) \cdot \omega. \quad (6.142)$$

Suppose $R: (\mathbb{T}M_1, H_1) \dashrightarrow (\mathbb{T}M_2, H_2)$ is a Courant algebroid relation with R -Clifford relation S_R , and suppose that S_R is a (d_{H_1}, d_{H_2}) -DGO relation. Suppose that L_i is an almost Dirac structure on $\mathbb{T}M_i$ with spinor line Ω_i , for $i = 1, 2$, and that $\Omega_1 \sim_R \Omega_2$. By Proposition 6.120, $L_1 \sim_R L_2$. Then, if L_1 is integrable, for every $\omega_1 \in \Omega_1$ there is a section $X_1 + \xi_1 \in \Gamma(\mathbb{T}M_1)$ such that

$$d_{H_1} \omega_1 = (X_1 + \xi_1) \cdot \omega_1. \quad (6.143)$$

Let $\omega_2 \in \Omega_2$, with $(\omega_1, \omega_2) \in S_R$. If $p_1(R) \supseteq L_1$, there is $X_2 + \xi_2 \in \mathbb{T}M_2$ be such that $X_1 + \xi_1 \sim_R X_2 + \xi_2$. Then, since $d_{H_1} \omega_1 \sim_R d_{H_2} \omega_2$ and $(X_1 + \xi_1) \cdot \omega_1 \sim_R (X_2 + \xi_2) \cdot \omega_2$, by Corollary 6.107, it follows that

$$d_{H_2} \omega_2 = (X_2 + \xi_2) \cdot \omega_2 \quad (6.144)$$

and hence L_2 is integrable. Thus we find

Proposition 6.145. If $R: (\mathbb{T}M_1, H_1) \dashrightarrow (\mathbb{T}M_2, H_2)$ is a Courant algebroid relation with R -Clifford relation S_R , and take an almost Dirac structure L_i on $\mathbb{T}M_i$ with associated spinor line Ω_i , for $i = 1, 2$. Suppose that $\Omega_1 \sim_R \Omega_2$ and that $p_i(R) \supseteq L_i$. Then L_1 is integrable if and only if L_2 is integrable

6.5 Spinor Relations for Geometric T-Duality

We shall now show a converse to Proposition 6.139. By this, we mean to show how, in the T-duality setting, the T-duality relation gives rise to a DGO-relation. To do this, we will restrict to the case that $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are both fibre bundles over a manifold $\mathcal{B}$. We will construct a spinor module S_R for $\text{Cl}(R)$, and then construct the canonical Dirac generating operator for this spin module. These constructions are outlined in [31]. We will then show that this DGO is in fact given by $(d_{\underline{H}_1}, d_{\underline{H}_2})$.

6.5.1 The T-duality Spinor Relation S_R

The construction of S_R is by construction of a Courant algebroid $R_{\mathcal{B}}$ over $\mathcal{B}$, whose typical fiber is isomorphic to R .

Proposition 6.146. There is a regular Courant algebroid $R_{\mathcal{B}}$ over $\mathcal{B}$, whose fiber is isomorphic (as a vector bundle) to the fiber of R .

Proof. $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are both fiber bundles over the base manifold $\mathcal{B}$, fitting the diagram

$$\begin{array}{ccc}
 & M & \\
 \varpi_1 \swarrow & & \searrow \varpi_2 \\
 \mathcal{Q}_1 & & \mathcal{Q}_2 \\
 \pi_1 \searrow & & \swarrow \pi_2 \\
 & \mathcal{B} &
 \end{array} \tag{6.147}$$

If $(e_1, e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$, then for $f_i \in C^\infty(\mathcal{Q}_i)$, we have that $(f_1 e_1, f_2 e_2) \in \Gamma(E_1 \times \overline{E}_2; R)$ if and only if $f_1(m_1) = f_2(m_2)$ for all $(m_1, m_2) \in C$. In other words, there is a function² $f \in C^\infty(\mathcal{B})$ such that $\pi_i^* f = f_i$. Thus $\Gamma(E_1 \times E_2; R)$ is a locally free finitely generated $C^\infty(\mathcal{B})$ -module, hence, by the Serre-Swan Theorem, there exists a vector bundle over $\mathcal{B}$ whose space of sections is given by $\Gamma(E_1 \times E_2; R)$. This bundle inherits the structure of a Courant algebroid in the following way. Let $c \in (q_1, q_2) \in C = M$, which has a corresponding point $b \in \mathcal{B}$ (since M and $\mathcal{B}$ fit into the diagram (6.147), b is defined as $b = \pi_i(\varpi_i(q_i))$). We denote a point $(R_{\mathcal{B}})_b$ by the same symbol as the corresponding point $(e, {}^R e) \in R_c$, where context is clear. By the construction of R , this is independent of the point $c \in C$ corresponding to

²Note that $p_i(C) = \mathcal{Q}_i$.

b. The anchor is given by

$$\rho_{R_{\mathcal{B}}}(e, {}^R e)_b := (\pi_1)_* \rho_1(e)_{q_1} = (\pi_2)_* \rho_2({}^R e)_{q_2} . \quad (6.148)$$

Here, $\rho_{R_{\mathcal{B}}}$ is well defined since $\rho(e, {}^R e) = (\rho_1(e), \rho_2(e_R)) \in TC$, and so project to the same vector in $T\mathcal{B}$ under pushforward by $\pi_1 \circ p_1$ and $\pi_2 \circ p_2$. The regularity of $\rho_{R_{\mathcal{B}}}$ follows by the regularity of ρ_i and the fact that $p_i(R) = \underline{E}_i$ for $i = 1, 2$. The other structures are given by the usual structures on R . $\square$

For regular Courant algebroids, there is a result of Chen *et al.* [79, Lemma 1.2] stating that a regular Courant algebroid E admits an isomorphism

$$E \cong F \oplus G \oplus (\ker \rho)^\perp \quad (6.149)$$

where $F = \text{im}(\rho)$, and G is such that $\ker \rho = G \oplus (\ker \rho)^\perp$ and is orthogonal to F . This provides the splitting

$$R_{\mathcal{B}} \cong F \oplus G \oplus F^*, \quad (6.150)$$

where $F \cong T\mathcal{B}$ and $F^* \cong T^*\mathcal{B}$. Note that for a choice of G , for $X \in T\mathcal{B}$ there are unique lifts $\dot{X} \in T\mathcal{Q}_1$ and $\ddot{X} \in T\mathcal{Q}_2$ such that $X_F := (\dot{X}, \ddot{X}) \in F$. For $\omega \in T^*\mathcal{B}$, we simply set $\omega_{F^*} := (\pi_1^* \omega, \pi_2^* \omega) \in F^*$. For $g, g' \in G$, the Courant algebroid structures in this splitting are then given by

$$\langle X_F + g + \omega_{F^*}, Y_F + g' + \chi_{F^*} \rangle = \frac{1}{2}(\iota_X \omega + \iota_Y \chi) + \langle g, g' \rangle_R \quad (6.151)$$

$$\rho_{R_{\mathcal{B}}}(X_F, g, \omega_{F^*}) = X \quad (6.152)$$

$$\llbracket X_F + g + \omega_{F^*}, Y_F + g' + \chi_{F^*} \rrbracket = \quad (6.153)$$

$$(\llbracket \dot{X} + \ell + \pi_1^* \omega, \dot{Y} + \ell' + \pi_1^* \chi \rrbracket_1, \llbracket \ddot{X} + {}^R \ell + \pi_2^* \omega, \ddot{Y} + {}^R \ell' + \pi_2^* \chi \rrbracket_2)$$

where $g = (\ell, {}^R \ell), g' = (\ell', {}^R \ell') \in R$

Moreover, recall the following results of Alekseev-Xu [30, Theorem 4.1]: If E is a regular Courant Algebroid, then every spinor bundle $\mathbf{S}$ over $\text{Cl}(E)$ admits locally a

Dirac generating operator which can be expressed by means of a Courant algebroid connection, as discussed below. On the weighted spinor bundle, there is a canonical DGO [31, Theorem 52]. Let us first discuss how this works in the case of principal circle bundles.

Example 6.154. Let us consider the T-duality setting for circle bundles. For any open $U \subset \mathcal{B}$, sections of $R_{\mathcal{B}}|_U$ are

$$\{ X, \xi, (\partial_{\theta_1}, \theta_2), (\theta_1, \partial_{\theta_2}) \}_{C^\infty(\mathcal{B})} , \quad (6.155)$$

for all $X \in \Gamma(TU)$, $\xi \in \Omega^1(U)$ and for some connections θ_1, θ_2 on $\mathcal{Q}_1 \rightarrow \mathcal{B}, \mathcal{Q}_2 \rightarrow \mathcal{B}$ respectively, pulled back to $\mathcal{B}$ and restricted to U . Here $\{ \cdot \}_{C^\infty(\mathcal{B})}$ denotes the $C^\infty(\mathcal{B})$ -span of the set. Thus

$$\ker \rho|_U = \{ \xi, (\partial_{\theta_1}, \theta_2), (\theta_1, \partial_{\theta_2}) \}_{C^\infty(\mathcal{B})} , \quad (\ker \rho)^\perp|_U = \{ \xi \} , \quad (6.156)$$

and hence

$$R_{\mathcal{B}}|_U = \{ \xi \} \oplus \{ X \} \oplus \{ (\partial_{\theta_1}, \theta_2), (\theta_1, \partial_{\theta_2}) \}_{C^\infty(\mathcal{B})} =: F^*|_U \oplus F|_U \oplus G|_U . \quad (6.157)$$

In order to define the bracket of two sections of F , the projection to G is calculated from the relation R . The choice of splitting $E \cong F \oplus G \oplus F^*$ is equivalent to a choice of lift of $X \in T\mathcal{B}$ to $(\dot{X}, \ddot{X}) \in R$. For the circle (or, more generally, principal) bundle case, this choice of lift is equivalent to the choice of connections θ_1, θ_2 , and the lifts are thus (X^{h_1}, X^{h_2}) , the horizontal lifts with respect to these connections. The requirement that $(X^{h_1}, X^{h_2}) \in R$ is the condition that $\underline{\dot{H}}_1 - \underline{\ddot{H}}_2 = d(\dot{\theta}_1 \wedge \ddot{\theta}_2)$, as discussed in [46].

Let us recall from (4.143) that $\underline{H}_i = \dot{h} + c_{3-i} \wedge \theta_i$, for $i = 1, 2$, where c_j is the Chern class associated with the connection θ_j for $j = 1, 2$, i.e. $c_j = d\theta_j$, and

$h \in \Omega^3(\mathcal{B})$. Then the bracket of two elements of F is

$$\begin{aligned} \llbracket X, Y \rrbracket &= [X, Y]_{\mathcal{B}} + \text{Pr}_{F^*}(H_1(X^{h_1}, Y^{h_1}, \cdot), H_2(X^{h_2}, Y^{h_2}, \cdot)) \\ &\quad + \text{Pr}_G([X^{h_1}, Y^{h_1}] + H_1(X^{h_1}, Y^{h_1}, \cdot), [X^{h_2}, Y^{h_2}] + H_2(X^{h_2}, Y^{h_2}, \cdot)) \quad (6.158) \\ &= [X, Y]_{\mathcal{B}} + h(X, Y, \cdot) + c_1(X, Y)(\partial_{\theta_1}, \theta_2) + c_2(X, Y)(\theta_1, \partial_{\theta_2}), \end{aligned}$$

where $[\cdot, \cdot]_{\mathcal{B}}$ denotes the Lie bracket on $\mathcal{B}$.

A spin bundle for $\text{Cl}(G)$ is generated by³ $(\wedge^{\bullet}(\partial_{\theta_1}, \theta_2)) \cdot (\theta_1, 1) = \{(\theta_1, 1), (1, \theta_2)\}$. The spinor bundle for $\text{Cl}(R_{\mathcal{B}})$ is then

$$S_R = \wedge^{\bullet} F^* \otimes S_G. \quad (6.159)$$

This also defines the spin bundle S_R for R .

We now extend the above example to the general case where $\mathcal{Q}_1$ and $\mathcal{Q}_2$ are fibred over the same manifold $\mathcal{B}$ with surjective submersions $\pi_i(\mathcal{Q}_i) = \mathcal{B}$. By Proposition 6.146, there is a regular Courant algebroid $R_{\mathcal{B}}$ over $\mathcal{B}$ that is fiber-wise isomorphic to R . Given a splitting $R_{\mathcal{B}} \cong F \oplus G \oplus F^*$, define $G_1^* = p_{T^*} p_1(G) \subset T^* \mathcal{Q}_1$, where p_{T^*} is the projection $\mathbb{T}M \rightarrow T^*M$. Consider the spinor $u = (\lambda, 1)$, where $\lambda \in \wedge^{\text{top}} G_1^*$ (this can always be done locally, provided the typical fiber of D_1 is orientable), which is acted upon by $\text{Cl}(G)$. This is a pure spinor, with Lagrangian subbundle $L \subset G$. Choose the dual L^* such that $\rho_2(p_2(L^*)) = 0$. Then we have a spin bundle $S_G = (\wedge^{\bullet} L^*) \cdot u$. Note that $S_G \subset (\wedge^{\bullet} T^* \mathcal{Q}_1 \times \wedge^{\bullet} T^* \mathcal{Q}_2)|_C$. The bundle

$$S_R = \wedge^{\bullet} F^* \otimes S_G \quad (6.160)$$

over $\mathcal{B}$ is a spin representation for the Clifford bundle $\text{Cl}(R_{\mathcal{B}})$, with action

$$(X_F + g + \chi_{F^*}) \cdot (\omega_{F^*} \otimes \mathbf{s}) = (X_F + \chi_{F^*}) \cdot \omega_{F^*} \otimes \mathbf{s} + (-1)^{|\omega_{F^*}|} \omega_{F^*} \otimes g \cdot \mathbf{s} \quad (6.161)$$

for $X_F + g + \omega_{F^*} \in R_{\mathcal{B}}$, $\chi_{F^*} \in \wedge^{\bullet} F^*$, and $\mathbf{s} \in S_G$. These actions are performed using

³Following Section 1.2 of [80], $(\theta_1, 1)$ is a pure spinor, with Lagrangian subbundle $L = \text{span}\{(\theta_1, \partial_{\theta_2})\}$, and thus defines a spin module $S_G = \text{Cl}(G)/(\text{Cl}(G)L)$. A choice of subbundle $L^* = \text{span}\{(\partial_{\theta_1}, \theta_2)\}$ transverse to L provides an isomorphism $S_G \cong \wedge^{\bullet} L^* \cdot (\theta_1, 1)$

the Clifford action (6.16) of $\text{Cl}(R)$ on $(\wedge^\bullet T^* \mathcal{Q}_1 \times \wedge^\bullet T^* \mathcal{Q}_2)|_C$ by pulling S_R back by (π_1^*, π_2^*) , where we recall that $C \cong \mathcal{Q}_1 \times_{\mathcal{B}} \mathcal{Q}_2$. By construction, S_R is also a spin representation for $\text{Cl}(R)$, with the same action. We leave the pullback by (π_1^*, π_2^*) implicit in the following.

Proposition 6.162. S_R is an R -Clifford relation of parity $\frac{1}{2} \dim G \pmod 2$.

Proof. By [31, Remark 57], it follows that S_R is an irreducible spin representation for $\text{Cl}(R_{\mathcal{B}})$, and hence for $\text{Cl}(R)$. $\square$

With the construction of the spinor bundle S_R , we have the following result about transporting spinor line bundles to the T-dual manifold.

Theorem 6.163. Let $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$ be a T-duality relation supported on $C \subset \mathcal{Q}_1 \times \mathcal{Q}_2$. Given a D_1 -invariant pure spinor line $\Omega \subset \wedge^\bullet T^* \mathcal{Q}_1$, then there is a pure spinor line ${}^R\Omega \subset \wedge^\bullet T^* \mathcal{Q}_2$ such that $\Omega \sim_R {}^R\Omega$.

Proof. Let $\phi \in \Gamma(\Omega)$ be a section of Ω . Then, for $e \in \Gamma(L^\phi)$ and $X \in \Gamma(T\mathcal{Q}_1)$, by Equation (6.127),

$$0 = \mathcal{L}_X(e \cdot \phi) = \text{ad}_X^\sigma e \cdot \phi + e \cdot \mathcal{L}_X \phi. \quad (6.164)$$

It follows that Ω is invariant if and only if L^ϕ is invariant, i.e. $\text{ad}_X \Gamma(L^\phi) \subset \Gamma(L^\phi)$ for each $X \in \Gamma(D_1)$. Thus, according to Lemma 4.106, we may take a spanning set of invariant sections for L^ϕ , which can be carried to $\underline{E}_2$, and define $L^{R\phi}$ as the span of these sections. It follows that $L^{R\phi}$ is Lagrangian and that $L^\phi \sim_R L^{R\phi}$. There is a line bundle ${}^R\Omega$ associated with $L^{R\phi}$. By Remark 6.42, for each point $c = (q_1, q_2) \in C$ it follows that $\Omega_{q_1} \sim_{R_c} {}^R\Omega_{q_2}$. Since this is true for all $c \in C$, it follows that $\Omega \sim_R {}^R\Omega$. $\square$

6.5.2 The Differential Graded Operator on S_R

Before discussing the general theory, let us continue the discussion of Example 6.154.

Example 6.165. For $w \in \wedge^\bullet F^*$ and $s \in \{(\theta_1, 1), (1, \theta_2)\}$, we define the map

$$\overline{W}(w \otimes s) = w \wedge (c_2, c_2) \otimes (\partial_{\theta_1}, \theta_2) \cdot s - w \wedge (c_1, c_1) \otimes (\theta_1, \partial_{\theta_2}) \cdot s \quad (6.166)$$

One can then define the DGO on S_R , as in [31, Theorem 61], as

$$D(\mathbf{w} \otimes \mathbf{s}) = (d^F \omega) \otimes \mathbf{s} - ((h, h) \wedge \omega) \otimes \mathbf{s} + (-1)^{|\omega|+1} \overline{W}(\omega \otimes \mathbf{s}), \quad (6.167)$$

and extended through the Leibniz rule to the whole of S_R . After skew-symmetrisation and pulling back from $\mathcal{B}$ to $C \subset \mathcal{Q}_1 \times \mathcal{Q}_2$ by (π_1^*, π_2^*) , this reduces to the DGO $(d_{\underline{H}_1}, d_{\underline{H}_2})$ on S_R . It follows that S_R is a $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation for R .

In order to write a DGO on S_R , following [79, Section 1.3], we introduce the maps

$$W: \Gamma(F) \times \Gamma(F) \rightarrow \Gamma(G), \quad W(f_1, f_2) = \text{Pr}_G(\llbracket f_1, f_2 \rrbracket) \quad (6.168)$$

$$\mathcal{H}: \Gamma(F) \times \Gamma(F) \times \Gamma(F) \rightarrow C^\infty(\mathcal{B}), \quad \mathcal{H}(f_1, f_2, f_3) = \langle \text{Pr}_{F^*} \llbracket f_1, f_2 \rrbracket, f_3 \rangle_F, \quad (6.169)$$

$$\nabla: \Gamma(F) \times \Gamma(G) \rightarrow \Gamma(G), \quad \nabla_f g = \text{Pr}_G \llbracket f, g \rrbracket, \quad (6.170)$$

for all $f_1, f_2, f_3, f \in \Gamma(F)$, $g \in \Gamma(G)$. Note that, since $F = T\mathcal{B}$ and $F^* = T^*\mathcal{B}$, we have $\mathcal{H} = (\dot{h}, \ddot{h})$ for some $h \in \Omega^3(\mathcal{B})$. The canonical DGO is then given by [31, Theorem 61]

$$D(\mathbf{w} \otimes \mathbf{s}) = d^F \mathbf{w} \otimes \mathbf{s} + \nabla^{0, S_G}(\mathbf{s}) \wedge \mathbf{w} - (\mathcal{H} \wedge \mathbf{w}) \otimes \mathbf{s} + (-1)^{|\mathbf{w}|+1} \overline{W}(\mathbf{w} \otimes \mathbf{s}) + \frac{1}{4}(-1)^{|\mathbf{w}|+1} \mathbf{w} \otimes C\mathbf{s}. \quad (6.171)$$

Here ∇^{0, S_G} is an F -connection on S_G compatible with ∇ , i.e.

$$\nabla_X^{0, S_G}(g \cdot s) = \nabla_X g \cdot s + g \cdot \nabla_X^{0, S_G} s, \quad (6.172)$$

and C is the 3-form on G given by $(g_1, g_2, g_3) \mapsto \langle \llbracket g_1, g_2 \rrbracket, g_3 \rangle$ for $g_1, g_2, g_3 \in G$. $\overline{W}$ is defined as

$$\overline{W}(\mathbf{w} \otimes \mathbf{s}) = \frac{1}{2} \sum_{i, j, \mu} \langle W(f_i, f_j), g_\mu \rangle \tilde{f}_i \wedge \tilde{f}_j \wedge \mathbf{w} \otimes \tilde{g}_\mu \cdot \mathbf{s}, \quad (6.173)$$

where f_i is a (local) basis for F , $\tilde{f}_i$ its dual, g_μ is a basis for G , and $\tilde{g}_\mu$ its dual, i.e.

$$\langle f_i, \tilde{f}_j \rangle = \delta_{ij} \text{ and } \langle g_\mu, \tilde{g}_\nu \rangle = \delta_{\mu\nu}.$$

Proposition 6.174. Let $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1) \rightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$ be a T-duality relation, and let $R_{\mathcal{B}}$ be the regular Courant algebroid constructed in Proposition 6.146. Let $\mathbf{D}$ be the Dirac generating operator given by (6.171). Let $\lambda \in \det(G_1^*)$ be such that the spinor line Ω_λ generated by λ is D_1 -invariant, i.e. for all $X \in \mathfrak{k}_{\tau_1}$, we have $\mathcal{L}_X \lambda \in \Omega_\lambda$. Then $\mathbf{D} = (d_{\underline{H}_1}, d_{\underline{H}_2})$.

Proof. Let $\mathbf{w} = (\dot{\omega}, \ddot{\omega}) \in \wedge^\bullet F^*$ and $\mathbf{s} = (\ell, {}^R\ell) \cdot u \in \wedge^\bullet L^* \cdot u$, where $u = (\lambda, 1)$ for $\lambda \in \wedge^{\text{top}} G_1^*$ and is D_1 -invariant. In the following calculations, we omit the parity of the relation for convenience. We consider each term in (6.171): The first term simply becomes

$$d^F(\dot{\omega}, \ddot{\omega}) = (d\dot{\omega}, d\ddot{\omega}). \quad (6.175)$$

For the second term, take an F -connection on S_G given on $u = (\lambda, 1)$ by

$$\nabla^{0, S_G}(\lambda, 1) = (0, 0), \quad (6.176)$$

and extend to the whole of S_G by compatibility with ∇ .

Let $f_i = (\dot{X}_i, \ddot{X}_i)$, $\phi_i = (\dot{\alpha}_i, \ddot{\alpha}_i)$ be a basis for F, F^* respectively, which are dual, with $i = 1, \dots, \dim \mathcal{B}$. Let $n = \dim \mathcal{F}_1 = \text{rank}(L)$ and let $g_\mu = (a_\mu, b_\mu)$ be a basis for $G = L \oplus L^*$, with $(g_\mu)_{\mu \leq n} \in L$ and $(g_\mu)_{\mu > n} \in L^*$. Let $\tilde{g}_\mu = (\tilde{a}_\mu, \tilde{b}_\mu)$ be the dual basis, seen as a subset of $G = L^* \oplus L$. Consider $(\ell, {}^R\ell) \in L^*$ such that $\ell \in \mathfrak{k}_{\tau_1}$, and let $\mathbf{s} = (\ell, {}^R\ell) \cdot u$. Then

$$\begin{aligned} \nabla(\ell, {}^R\ell) &= \sum_i \phi_i \wedge \nabla_{f_i}(\ell, {}^R\ell) \\ &= \sum_i \phi_i \wedge \text{Pr}_G([\dot{X}_i, \ell]_1, [\ddot{X}_i, {}^R\ell]_2) \\ &= \sum_{\mu \leq n, i} \phi_i \wedge (\tilde{a}_\mu (\iota_{[\dot{X}_i, \ell]} a_\mu), \tilde{b}_\mu (\iota_{b_\mu} \mathcal{L}_{\ddot{X}_i} {}^R\ell)) + \sum_{\mu > n, i} \phi_i \wedge ((\iota_{a_\mu} \iota_{\dot{X}_i} \iota_\ell \underline{H}_1) \tilde{a}_\mu, 0) \end{aligned} \quad (6.177)$$

and hence

$$\begin{aligned}
 \nabla^{0,S_G} \mathbf{s} &= \sum_i \phi_i \wedge \nabla_{f_i}^{0,S_G} \mathbf{s} \\
 &= \sum_i \phi_i \wedge \nabla_{f_i}(\ell, {}^R\ell) \cdot u \\
 &= \sum_{\mu \leq n, i} \phi_i \wedge (\iota_{[\dot{X}_1, \ell]} a_\mu \iota_{\tilde{a}_\mu} \lambda, \tilde{b}_\mu \iota_{b_\mu} \mathcal{L}_{\dot{X}_i} {}^R\ell) + \sum_{\mu > n, i} \phi_i \wedge ((\iota_{a_\mu} \iota_{\dot{X}_i} \iota_\ell \underline{H}_1) \tilde{a}_\mu \wedge \lambda, 0) \\
 &= \sum_{\mu \leq n, i} \phi_i \wedge (\iota_{[\dot{X}_1, \ell]} \lambda, \tilde{b}_\mu \iota_{b_\mu} \mathcal{L}_{\dot{X}_i} {}^R\ell) \\
 &= \sum_{\mu \leq n, i} \phi_i \wedge (\iota_{\dot{X}_i} (\iota_\ell d\lambda + d\iota_\ell \lambda), \tilde{b}_\mu \iota_{b_\mu} \mathcal{L}_{\dot{X}_i} {}^R\ell) \\
 &= \sum_{\mu \leq n, i} \phi_i \wedge (\iota_{\dot{X}_i} (\mathcal{L}_\ell \lambda), \tilde{b}_\mu \iota_{b_\mu} \mathcal{L}_{\dot{X}_i} {}^R\ell)
 \end{aligned} \tag{6.178}$$

The term involving $\underline{H}_1$ vanishes since $\lambda \in \bigwedge^{\text{top}} G_1^\star$ and hence $\tilde{a}_\mu \wedge \lambda = 0$ for all μ . Since Ω_λ is D_1 -invariant, it follows that $\iota_{\dot{X}_i}(\mathcal{L}_\ell \lambda) = 0$ and so $\nabla^{0,S_G} \mathbf{s} = 0$ for $\mathbf{s} = (\ell, {}^R\ell) \cdot u$.

The third term is

$$\begin{aligned}
 -(\mathcal{H} \wedge \mathbf{w}) \otimes \mathbf{s} &= -\frac{1}{3} \sum_{i,j,k} ((\underline{H}_1)_{ijk} \alpha_i \wedge \alpha_j \wedge \alpha_k \wedge \dot{\omega}, (\underline{H}_2)_{ijk} \alpha_i \wedge \alpha_j \wedge \alpha_k \wedge \ddot{\omega}) \otimes \mathbf{s} \\
 &= -(\dot{h} \wedge \dot{\omega}, \ddot{h} \wedge \ddot{\omega}) \otimes \mathbf{s}
 \end{aligned} \tag{6.179}$$

The fourth term we calculate as follows. Projecting with p_2 ,

$$\begin{aligned}
 & p_2 \overline{W}(\mathbf{w} \otimes \mathbf{s}) \\
 &= \frac{1}{2} p_2 \sum_{i,j,\mu} \langle R((X_i, X_i), (X_j, X_j)), (a_\mu, b_\mu) \rangle (\alpha_i, \alpha_i) \wedge (\alpha_j, \alpha_j) \wedge (\omega, \omega) \\
 & \quad \otimes (\tilde{a}_\mu, \tilde{b}_\mu) \cdot (\ell, {}^R \ell) \cdot (\lambda, 1) \\
 &= \frac{1}{2} \sum_{i,j,\mu} \langle \llbracket X_i, X_j \rrbracket_1, b_\mu \rangle \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{b}_\mu \cdot {}^R \ell \\
 &= \frac{1}{2} \sum_{\mu \leq n, i, j} (\underline{H}_2)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{b}_\mu \wedge {}^R \ell + \frac{1}{2} \sum_{\mu > n, i, j} b_\mu ([X_i, X_j]_1) \alpha_i \wedge \alpha_j \wedge \omega \otimes \iota_{\tilde{b}_\mu} {}^R \ell \\
 &= \frac{1}{2} \sum_{\mu \leq n, i, j} (\underline{H}_2)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{b}_\mu \wedge {}^R \ell + \frac{1}{2} \sum_{i,j} \alpha_i \wedge \alpha_j \wedge \omega \otimes {}^R \ell ([X_i, X_j]_1) \\
 &= \frac{1}{2} \sum_{\mu \leq n, i, j} (\underline{H}_2)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{a}_\mu \wedge {}^R \ell - \frac{1}{2} \sum_{i,j} \alpha_i \wedge \alpha_j \wedge \omega \otimes d^R \ell(X_i, X_j)
 \end{aligned} \tag{6.180}$$

Similarly, projecting with p_1

$$p_1 \overline{W}(\mathbf{w} \otimes \mathbf{s}) \tag{6.181}$$

$$= \frac{1}{2} \sum_{i,j,\mu} \langle \llbracket X_i, X_j \rrbracket_2, a_\mu \rangle \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{a}_\mu \cdot \iota_\ell \lambda \tag{6.182}$$

$$= \frac{1}{2} \sum_{\mu > n, i, j} (\underline{H}_1)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{a}_\mu \wedge \iota_\ell \lambda + \frac{1}{2} \sum_{\mu \leq n, i, j} a_\mu ([X_i, X_j]) \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{a}_\mu \cdot \iota_\ell \lambda \tag{6.183}$$

$$= \frac{1}{2} \sum_{\mu > n, i, j} (\underline{H}_1)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \omega \otimes \tilde{a}_\mu \wedge \iota_\ell \lambda - \frac{1}{2} \sum_{i,j} \alpha_i \wedge \alpha_j \wedge \omega \otimes d(\iota_\ell \lambda)(X_i, X_j). \tag{6.184}$$

On the skew-symmetrised product $\mathbf{w} \wedge \mathbf{s}$, we obtain, for instance

$$\begin{aligned} p_2 \overline{W}(\mathbf{w} \wedge \mathbf{s}) &= \frac{1}{2}(-1)^{|\omega|} \sum_{\mu > n, i, j} (\underline{H}_2)_{ij\mu} \alpha_i \wedge \alpha_j \wedge \tilde{a}_\mu \wedge \omega \wedge \iota_\ell \lambda \\ &\quad - \frac{1}{2} \sum_{i, j} \omega \wedge \alpha_i \wedge \alpha_j \wedge d(\iota_\ell \lambda)(X_i, X_j). \end{aligned} \quad (6.185)$$

Thus, by a similar calculation it follows that the third, fourth and fifth term combine to give

$$\begin{aligned} & -(\mathcal{H} \wedge \mathbf{w}) \wedge \mathbf{s} + (-1)^{|\omega|+1} \overline{W}(\mathbf{w} \wedge \mathbf{s}) + \frac{1}{4}(-1)^{|\omega|+1} \mathbf{w} \wedge C\mathbf{s} \\ &= (-\underline{H}_1 \wedge \omega \wedge \iota_\ell \lambda + (-1)^{|\omega|} \omega \wedge d\iota_\ell \lambda, -\underline{H}_2 \wedge \omega \wedge {}^R\ell + (-1)^{|\omega|} \omega \wedge d^R\ell) \end{aligned} \quad (6.186)$$

Hence,

$$\begin{aligned} D(\mathbf{w} \wedge \mathbf{s}) &= (d\dot{\omega} \wedge \iota_\ell \lambda + (-1)^{|\omega|} \omega \wedge d\iota_\ell \lambda - \underline{H}_1 \wedge \omega \wedge \iota_\ell \lambda, \\ &\quad d\ddot{\omega} \wedge {}^R\ell + (-1)^{|\omega|} \omega \wedge d^R\ell - \underline{H}_2 \wedge \omega \wedge {}^R\ell) \\ &= (d_{\underline{H}_1}, d_{\underline{H}_2})(\mathbf{w} \wedge \mathbf{s}). \end{aligned} \quad (6.187)$$

Note that $\mathfrak{k}_{\tau_1}$ spans $p_1(L^*)$ pointwise, hence, by Leibniz rule, this can be extended to the whole of L^* , which can then be extended to the whole of $\mathbf{S}_G$, and hence $\mathbf{S}_R$. $\square$

Corollary 6.188. Under the assumptions of Proposition 6.174, $\mathbf{S}_R$ is a $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation.

Example 6.189 (The Fourier-Mukai Revisited). Continuing from Example 6.135, since both $\text{gr}(\varrho)$ and $\mathbf{S}_R$ are spinor bundles for $\text{Cl}(R)$, it follows that they are isomorphic. The isomorphism becomes equality upon a rescaling of λ (or equivalently a rescaling of the measure in the Fourier-Mukai transform). The fact that $\mathbf{S}_R$ is a $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation is equivalent to the fact that $d_{\underline{H}_2} \varrho(\omega) = (-1)^k \varrho(d_{\underline{H}_2} \omega)$. Note that the existence of λ satisfying the assumptions of Proposition 6.174 is guaranteed: if θ_1 is a connection on the principal $\mathbb{T}^k$ -bundle $\mathcal{Q}_1 \rightarrow \mathcal{B}$, and t_i is a basis of $\mathfrak{t}_1^k$, then we can write $\theta_1 = \sum_i \theta_1^i \otimes t_i$. Then $\lambda = \theta_1^1 \wedge \dots \wedge \theta_1^k$ is invariant.

6.5.3 T-duality Relations and Supergravity

As discussed, the bosonic field content for the Type II supergravity on a manifold M is given by (g, H, ϕ, u) for a Riemannian metric g on M , closed three-form H (or Kalb-Ramond field), $\phi \in C^\infty(M)$ (the dilaton), and $F \in \Omega^\bullet(M)$ an inhomogenous degree form of even or odd degree closed under the twisted differential d_H (serving as a collection of Ramond-Ramond fields). F is also required to satisfy an additional self-duality condition, which we discuss below. The equations of motion for Type II supergravity can then be stated on the twisted standard Courant algebroid $(\mathbb{T}M, H)$, equipped with the generalised metric V_g^+ , divergence $\text{div} = \text{div}_\phi$, and spinor bundle $\Omega^\bullet(M)$, and are given by

$$\text{GRic}_{g, \text{div}_\phi}(u_+, v_-) = \frac{i}{8\nu}((u_+ \cdot F, v_- \cdot F)), \quad \forall u_+ \in V_g^+, v_- \in V_g^- \quad (6.190)$$

$$\text{GR}_{g, \text{div}_\phi} = 0. \quad (6.191)$$

Here $((\cdot, \cdot))$ is the pairing on spinors, discussed below, and $\text{GRic}_{g, \text{div}}$ is the generalised Ricci tensor associated to the pair (V_g^+, div_ϕ) , as in Definition 5.66 and Equation (5.70). Finally, $\nu = 1$ for the type II A theory and $\nu = i$ for the type II B theory.

Pairing on Spinors

We define the pairing on the space $\tilde{\mathcal{S}} = \mathcal{S} \otimes (\det \mathcal{S})^{-1/2}$ by

$$((\cdot, \cdot)): \tilde{\mathcal{S}} \otimes \tilde{\mathcal{S}} \rightarrow \mathbb{C}, \quad ((s \otimes \mu, s' \otimes \mu)) = \langle (\vartheta s \wedge s')^{\text{top}}, \mu^2 \rangle, \quad (6.192)$$

where $\langle \cdot, \cdot \rangle$ denotes the contraction between $\det \mathcal{S}$ and $(\det \mathcal{S})^{-1} \cong \det \mathcal{S}^*$, and ϑ is the operator $\vartheta s = i^{|s|} s$. Recall that $S_R = \wedge^\bullet F^* \otimes S_G$, where $S_G = (\wedge^\bullet L^*) \cdot (\lambda, 1)$. We consider $\tilde{\mathcal{S}}_R \subset ((\wedge^\bullet T^* \mathcal{Q}_1) \otimes (\det T \mathcal{Q}_1)^{1/2}) \times ((\wedge^\bullet T^* \mathcal{Q}_2) \otimes (\det T \mathcal{Q}_2)^{1/2})$ as

$$\tilde{\mathcal{S}}_R = \wedge^\bullet F^* \otimes (\wedge^{\text{top}} F^*)^{-1/2} \otimes (\wedge^\bullet L^*) \cdot (\lambda, 1) \otimes \left(((\wedge^{\text{top}} L^*) \cdot (\lambda, 1))^* \cdot (\lambda', 1) \right)^{1/2}, \quad (6.193)$$

where $\lambda' \in \det(T\underline{\mathcal{F}}_1)$ is such that $\lambda' \cdot \lambda = 1$. Since L^* is such that $\rho_2(p_2(L^*)) = 0$, it follows that $((\bigwedge^{\text{top}} L^*) \cdot (\lambda, 1))^*$ is generated by $(1, {}^R\lambda)$ for ${}^R\lambda \in \bigwedge^{\text{top}} T\underline{\mathcal{F}}_2$ satisfying $(\lambda, {}^R\lambda) \in \text{Cl}(R)$. We then say that $\tilde{s}_1 \approx_R \tilde{s}_2$ if $(\tilde{s}_1, \tilde{s}_2) \in \tilde{\mathcal{S}}$.

Proposition 6.194. If $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$ is a T-duality relation with R -Clifford relation $\mathcal{S}_R$ given by (6.160) and $\tilde{\mathcal{S}}_R$ as in (6.193). Then for any $\tilde{s}_1 \approx_R \tilde{s}_2$, $\tilde{s}'_1 \approx_R \tilde{s}'_2$, we have that

$$((s_1, s'_1))_1 = i^{-n}(-1)^{n^2-n|s'_1|}((s_2, s'_2))_2, \quad (6.195)$$

in $\mathbb{C}$, where $((\cdot, \cdot))_i$, for $i = 1, 2$, are the pairings on spinors on each related Courant algebroid.

Proof. Let $c \in C$. Using the splitting (6.150) for R , we can write $s_1 = \dot{\omega} \wedge (\ell \cdot \lambda)$, $s_2 = \dot{\eta} \wedge (m \cdot \lambda)$, $s'_1 = \ddot{\omega} \wedge {}^R\ell$, and $s'_2 = \ddot{\eta} \wedge {}^Rm$. Then, we can write $\mu_2 \in (\det T^*\mathcal{Q}_2)^{-1/2}$ as $\mu_2 = \ddot{\mu}_B \otimes \sqrt{{}^R\lambda}$, for $\mu_B \in (\det T^*\mathcal{B})^{-1/2}$ and $\sqrt{{}^R\lambda} \in p_2(((\bigwedge^{\text{top}} L^*) \cdot (\lambda, 1))^{1/2})$ such that $\sqrt{{}^R\lambda}^2 = {}^R\lambda$. Assume that $|\omega| + |\eta| = \dim \mathcal{B} =: n_B$ and $|{}^R\ell| + |{}^Rm| = \dim \mathcal{F}_1 =: n_{\mathcal{F}_1}$, so that the pairing is non-zero. Then

$$\begin{aligned} ((s_2 \otimes \mu_2, s'_2 \otimes \mu_2)) &= \langle i^{|\omega|+|\ell|} \ddot{\omega} \wedge {}^R\ell \wedge \ddot{\eta} \wedge {}^Rm, \mu_2 \rangle \\ &= i^{|\omega|+|\ell|} (-1)^{|\ell||\eta|} \langle \omega \wedge \eta, \mu_B^2 \rangle \langle {}^R\ell \wedge {}^Rm, {}^R\lambda \rangle. \end{aligned} \quad (6.196)$$

Similarly, we have

$$\begin{aligned} ((s_1 \otimes \dot{\mu}_B \otimes \sqrt{\lambda'}, s'_1 \otimes \dot{\mu}_B \otimes \sqrt{\lambda'})) &= i^{|\omega|+n-|\ell|} \langle \dot{\omega} \wedge (\ell \cdot \lambda) \wedge \dot{\eta} \wedge (m \cdot \lambda), \dot{\mu}_B^2 \otimes \lambda' \rangle \\ &= i^{|\omega|+n-|\ell|} (-1)^{(n-|\ell|)|\eta|} \langle \dot{\omega} \wedge \dot{\eta} \wedge (\ell \cdot \lambda) \wedge (m \cdot \lambda), \dot{\mu}_B^2 \otimes \lambda' \rangle \\ &= i^{|\omega|+n-|\ell|} (-1)^{(n-|\ell|)|\eta|} (-1)^{|\ell|(n-|\ell|)} \langle \omega \wedge \eta, \mu_B^2 \rangle \langle (\ell \cdot m \cdot \lambda) \lambda, \lambda' \rangle. \end{aligned} \quad (6.197)$$

We have that $\ell \cdot m \cdot \lambda \sim_R (-1)^{n(n+1)/2} {}_{LR\lambda}({}^R\ell \wedge {}^Rm)$ in $\text{Cl}(G_c)$. Since both sides are real numbers, and for real numbers $a, b \in \mathbb{R}$ we have $(a, b) \in \text{Cl}(G)$ if and only if

$a = b$, it follows that

$$((\tilde{s}_1, \tilde{s}'_1)) = c((\tilde{s}_2, \tilde{s}'_2)) \quad (6.198)$$

where

$$\begin{aligned} c &= i^{|\omega|+|\ell|-|\omega|-n+|\ell|} (-1)^{|\ell||\eta|-n|\eta|+|\ell||\eta|-n|\ell|+|\ell|^2} \\ &= i^{-n} (-1)^{|\ell|-n(|s'_1|+|\ell|-n)-n|\ell|+|\ell|^2} \\ &= i^{-n} (-1)^{n^2-n|s'_1|}. \end{aligned} \quad (6.199)$$

□

Self-Dual Spinors and T-duality

Suppose we have a T-duality Clifford relation $\mathbf{S}$ of parity i . Consider a basis $\{r_1, \dots, r_n\}$ for R^+ such that $\{p_1(r_1), \dots, p_1(r_n)\} = \{e_1, \dots, e_n\}$ is an orthonormal basis for V_1^+ . Then $\{p_2(r_1), \dots, p_2(r_n)\} = \{f_1, \dots, f_n\}$ is an orthonormal basis for V_2^+ . Define $\mathcal{R}_{V_1^+} = 2^{n/2} e_1 \cdot \dots \cdot e_n$ and $\mathcal{R}_{V_2^+} = 2^{n/2} f_1 \cdot \dots \cdot f_n$. Then, for all $(u_1, u_2) \in \mathbf{S}$,

$$(\mathcal{R}_{V_1^+} u_1, (-1)^{in} \mathcal{R}_{V_2^+} u_2) \in \mathbf{S}. \quad (6.200)$$

Thus, if u_1 satisfies some self-duality condition $\mathcal{R}_{V_1^+} u_1 = u_1$, then

$$(\mathcal{R}_{V_1^+} u_1, (-1)^{in} \mathcal{R}_{V_2^+} u_2) = (u_1, (-1)^{in} \mathcal{R}_{V_2^+} u_2) \implies (-1)^{in} \mathcal{R}_{V_2^+} u_2 = u_2, \quad (6.201)$$

by Corollary 6.107. In type II supergravity, $i = 1$ and $n = 10$, hence u_2 also satisfies a self-duality condition.

In the context of generalised geometry, we replace the spinor space $\Omega^\bullet(\mathcal{Q}_i)$ with the space $\tilde{\Omega}^\bullet(\mathcal{Q}_i) = \Omega^\bullet(\mathcal{Q}_i) \otimes (\det T\mathcal{Q}_1)^{1/2}$ in equations (6.190) and (6.191) for $i = A, B$, giving the *modified type II supergravity equations*, see [49, Section 7.1].

Proposition 6.202. Let $(g_A, \underline{H}_A, \phi_A, F_A)$ and $(g_B, \underline{H}_B, \phi_B, F_B)$ be backgrounds on T-dual principal circle bundles $\mathcal{Q}_A, \mathcal{Q}_B \rightarrow \mathcal{B}$ over the same base $\mathcal{B}$, respectively, and

let

$$R: (\mathbb{T}\mathcal{Q}_A, \underline{H}_A, V_{g_A}^+, \text{div}_{\phi_A}) \dashrightarrow (\mathbb{T}\mathcal{Q}_B, \underline{H}_B, V_{g_B}^+, \text{div}_{\phi_B}) \quad (6.203)$$

be a T-duality relation with R -Clifford relation $\mathbf{S}_R$ such that $F_A \approx_R F_B$. Then $(g_A, \underline{H}_A, \phi_A, F_A)$ satisfies the equations of motion (6.190)-(6.191) for $\nu = 1$ if and only if $(g_B, \underline{H}_B, \phi_B, F_B)$ satisfy (6.190)-(6.191) for $\nu = i$.

Proof. In the type II A case, F is even and hence $|(u_A)_\pm \cdot F_A| = 1 \pmod{2}$. Thus, with $n = 1$ we have

$$(((u_A)_+ \cdot F_A, (v_A)_- \cdot F_A)) = \frac{1}{i}(((u_B)_+ \cdot F_B, (v_B)_- \cdot F_B)). \quad (6.204)$$

We know from Corollary 5.86 that the generalised Ricci tensors are preserved by R . Thus if the background $(g_A, \underline{H}_A, \phi_A, F_A)$ satisfies the type II A case, then

$$\begin{aligned} \text{GRic}_B((u_B)_+, (v_B)_-) &= \text{GRic}_A((u_A)_+, (v_A)_-) \\ &= \frac{i}{8}(((u_A)_+ \cdot F_A, (v_A)_- \cdot F_A)) \\ &= \frac{1}{8}(((u_B)_+ \cdot F_B, (v_B)_- \cdot F_B)), \end{aligned} \quad (6.205)$$

and hence $(g_B, \underline{H}_B, \phi_B, F_B)$ is a solution of the type II B theory, where we note that since $\mathbf{S}_R$ is degree shifting, F_B is odd. Conversely, if the type II B equations of motion are satisfied, then since $|(u_B)_\pm \cdot F_B| = 0 \pmod{2}$,

$$\begin{aligned} \text{GRic}_A((u_A)_+, (v_A)_-) &= \text{GRic}_B((u_B)_+, (v_B)_-) \\ &= \frac{1}{8}(((u_B)_+ \cdot F_B, (v_B)_- \cdot F_B)) \\ &= \frac{-1}{8i}(((u_B)_+ \cdot F_B, (v_B)_- \cdot F_B)). \end{aligned} \quad (6.206)$$

as required. We have already seen that the self-duality condition is satisfied. Finally, since $\mathbf{S}_R$ is a $(d_{\underline{H}_A}, d_{\underline{H}_B})$ -DGO relation, it follows that $d_{\underline{H}_A} F_A = 0$ if and only if $d_{\underline{H}_B} F_B = 0$. $\square$

6.6 Generalised Complex Relations

In this section, we consider how generalised complex structures behave under Courant algebroid relations. We will see that a generalised complex relation has an induced grading, and we will write an explicit formula for the type change of generalised complex structures in T-duality. Thus suppose that E_1 and E_2 are Courant algebroids respectively endowed with generalised almost complex structures $\mathcal{J}_1$ and $\mathcal{J}_2$. The product Courant algebroid $E_1 \times E_2$ is endowed with the generalised almost complex structure $\mathcal{J} = \mathcal{J}_1 \times \mathcal{J}_2$.

As before, when working with generalised complex structures, one introduces the complexified Courant algebroid in order to discuss the $\pm i$ -eigenbundles. All the results on Dirac relations and spinor relations of the previous sections hold in the complex case too.

Definition 6.207. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation supported on $C \subseteq M_1 \times M_2$, and let $\mathcal{J}_1, \mathcal{J}_2$ be generalised almost complex structures on E_1, E_2 respectively. R is a *generalised almost complex relation* if $\mathcal{J}(R) = R$. If $\mathcal{J}_1$ and $\mathcal{J}_2$ are generalised complex structures, then R is a *generalised complex relation*.

A first instance of this definition appeared in [81] with C being the graph of a smooth map.

Example 6.208. Let $R = \text{gr}(\Phi): E_1 \rightarrowtail E_2$ be a classical Courant algebroid morphism supported on $\text{gr}(\phi) \subset M_1 \times M_2$, for some vector bundle morphism $\Phi: E_1 \rightarrow E_2$ covering a smooth map $\phi: M_1 \rightarrow M_2$, where E_1 and E_2 are endowed with the generalised complex structures $\mathcal{J}_1$ and $\mathcal{J}_2$, respectively. Then for any $e_1 \in E_1$, an element $(e_1, e_2) \in E_1 \times E_2$ is also in R if and only if $(e_1, e_2) = (e_1, \Phi(e_1)) \in R$. Thus

$$\mathcal{J}(e_1, \Phi(e_1)) = (\mathcal{J}_1(e_1), \mathcal{J}_2(\Phi(e_1))) \quad (6.209)$$

which, by Definition (6.207), is again in R if and only if

$$\mathcal{J}_2 \circ \Phi = \Phi \circ \mathcal{J}_1, \quad (6.210)$$

as expected for a well-defined notion of morphism of generalised complex structures.

This condition is equivalent to

$$\Phi(L_1) = L_2 \quad (6.211)$$

where L_1 and L_2 are the $+i$ -eigenbundles of $\mathcal{J}_1$ and $\mathcal{J}_2$, respectively.

Example 6.212. In the above setting, let $\mathcal{J}_1$ be given by a symplectic structure ω on M_1 . Then $L_1 = \{v - i\iota_v\omega : v \in TM^\mathbb{C}\}$. Since $\Phi = e^B \circ \bar{\phi}$,

$$\begin{aligned} \Phi(L_1) &= \{ \phi_*v + B(\phi_*v) - (\phi^{-1})^*(i\iota_v\omega) : v \in TM_1 \} \\ &= \{ w + B(w) - i(\iota_w(\phi^{-1})^*\omega) : w \in TM_2 \} \end{aligned} \quad (6.213)$$

Thus $\mathcal{J}_2$ is a B -twist of a symplectic structure.

Remark 6.214. If $R: (E_1, \mathcal{J}_1) \dashrightarrow (E_2, \mathcal{J}_2)$ is a generalised complex relation, and $(k, 0) \in K_R$ then $(\mathcal{J}_1(k), 0) \in K_R$, and hence $\mathcal{J}_1$ preserves $\text{Ker}(R)$. Similarly $\mathcal{J}_2$ preserves $\text{CoKer}(R)$.

Definition 6.215. A Courant Algebroid relation $R: (E_1, \mathcal{J}_1) \dashrightarrow (E_2, \mathcal{J}_2)$ is a *generalised almost complex relation* if at each $c \in C$ there is the splitting

$$R_c^\mathbb{C} = R_c^i \oplus R_c^{-i}, \quad (6.216)$$

where $R_c^i = R_c^\mathbb{C} \cap (L_1 \times L_2)_c$ and $R_c^{-i} = R_c^\mathbb{C} \cap (\bar{L}_1 \times \bar{L}_2)_c$, for any $c \in C$. Again, we drop the adjective “almost” when $\mathcal{J}_1, \mathcal{J}_2$ are generalised almost complex structures.

Remark 6.217. Note that since R is a real vector bundle (meaning $\overline{R^\mathbb{C}} = R^\mathbb{C}$), then $\overline{R^i} \subset R^{-i}$. Similarly, $\overline{R^{-i}} \subset R^i$ and hence $R^{-i} \subset \overline{R^i}$. Thus $\overline{R^i} = R^{-i}$ and $\overline{R^{-i}} = R^i$, hence $\text{rank}(R^i) = \text{rank}(R^{-i}) = \frac{1}{2}\text{rank}(R)$.

Similarly to Proposition 3.58 giving equivalence of definitions of generalised isometries, we have

Proposition 6.218. The Definitions 6.207 and 6.215 are equivalent.

Proof. If $\mathcal{J}(R_c^\mathbb{C}) = R_c^\mathbb{C}$, then $\mathcal{J}(r) = \mathcal{J}(r^i + r^{-i}) = ir^i - ir^{-i} \in R_c^\mathbb{C}$, hence $R_c^\mathbb{C} \subset R^i \oplus R^{-i}$. The converse also follows. $\square$

From Proposition 6.145, we get

Proposition 6.219. Let $R: (E_1, \mathcal{J}_1) \dashrightarrow (E_2, \mathcal{J}_2)$ be a generalised almost complex relation, with $p_2(R) \supset L_2$ (respectively $p_1(R) \supset L_1$). If $\mathcal{J}_1$ (respectively $\mathcal{J}_2$) is a generalised complex structure (i.e. L_1 (respectively L_2) is integrable), then $\mathcal{J}_2$ (respectively $\mathcal{J}_1$) is a generalised complex structure, hence R is a generalised complex relation.

Remark 6.220. Note that if $R: (E_1, L_1) \dashrightarrow (E_2, L_2)$ is a generalised almost complex relation, then one automatically has $L_1 \sim_R L_2$ and $\bar{L}_1 \sim_R \bar{L}_2$. The converse is also true, since we have that $\overline{R|_{L_1}} = R|_{\bar{L}_1}$, and hence

$$R^{\mathbb{C}}|_{\bar{L}_1} = R^{\mathbb{C}} \cap (\bar{L}_1 \times \bar{L}_2) + \text{CK}_R \quad (6.221)$$

since R is real, and noting that CK_R is also real. Thus there is a third equivalent definition, that is, R is a generalised almost complex relation if and only if $L_1 \sim_R L_2$ for the Lagrangian subbundles generated by $\mathcal{J}_1$ and $\mathcal{J}_2$. Therefore, following Remark 6.121, we have

Proposition 6.222. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation, and let $\mathcal{J}_i$ be almost generalised complex structures on E_i , with associated pure spinor line $\Omega^{\mathcal{J}_i}$ for $i = 1, 2$. If $\Omega^{\mathcal{J}_1} \sim_R \Omega^{\mathcal{J}_2}$ then R is a generalised almost complex relation.

For Courant algebroids equipped with Lagrangian structures $(E_1, L_1), (E_2, L_2)$, and a Courant algebroid relation $R: E_1 \dashrightarrow E_2$, recall that there are the subbundles K_R, CK_R , hence we may define the subbundle

$$(R_c^i)^{\times} = (R_c^{\mathbb{C}} \cap (L_1 \times L_2)_c + (K_R^{\mathbb{C}})_c + (\text{CK}_R^{\mathbb{C}})_c) / (K_R^{\mathbb{C}})_c / (\text{CK}_R^{\mathbb{C}})_c \quad (6.223)$$

and similarly for $(R_c^{-i})^{\times}$.

Definition 6.224. A Courant Algebroid relation $R: (E_1, \mathcal{J}_1) \dashrightarrow (E_2, \mathcal{J}_2)$ is a *weak generalised almost complex relation* if at each $c \in C$ there is the splitting

$$(R_c^{\mathbb{C}})^{\times} = (R_c^i)^{\times} \oplus (R_c^{-i})^{\times}, \quad (6.225)$$

for any $c \in C$.

It follows that if R is a generalised almost complex relation, then it is necessarily a weak generalised almost complex relation.

The grading induced by generalised complex structure (6.78) similarly induces a grading of the Clifford relation resulting in

Proposition 6.226. Let $R: (E_1, \mathcal{J}_1) \dashrightarrow (E_2, \mathcal{J}_2)$ be a generalised almost complex relation, with R -Clifford relation $\mathbf{S}_R$. There is the grading

$$\mathcal{S}_R = \mathcal{S}_R^n \oplus \mathcal{S}_R^{n-1} \oplus \dots \oplus \mathcal{S}_R^0 \quad (6.227)$$

where $\mathcal{S}_R^i = ((\Omega_1^i \times \Omega_2^i) \otimes |\det \mathbf{S}_R^*|^{\frac{1}{2^n}}) \cap \mathcal{S}_R$, for Ω_i^k as in (6.78), for $k = 1, \dots, n$ and $i = 1, 2$.

Proof. Since $L_1 \sim_R L_2$, we have that $\bar{L}_1 \sim_R \bar{L}_2$. Defining $\mathcal{S}_R^n = (\Omega_1 \times \Omega_2) \otimes |\det \mathbf{S}_R^*|^{\frac{1}{2^n}}$

$$\mathcal{S}_R^{n-k} = \{ (l_1, l_2) \cdot \mathcal{S}_R^n : (l_1, l_2) \in \wedge^k((L_1 \times L_2) \cap R) \}. \quad (6.228)$$

Since $\mathcal{S}_R$ is invariant under the spinor action of R , it follows that $\mathcal{S}_R^{n-k} \subset \mathcal{S}_R$. The projections $p_i(\mathcal{S}_R^{n-k}) = \Omega_i^{n-k}$ ensure that this sum (6.227) is disjoint, and since R is a Dirac relation between L_1 and L_2 , this ensures equality. $\square$

Let $\partial_i, \bar{\partial}_i$ be the operators associated with the generalised complex structures $\mathcal{J}_i$. Then

Proposition 6.229. Let $R: E_1 \dashrightarrow E_2$ be a generalised complex relation, with R -Clifford relation $\mathbf{S}_R$ such that $\mathbf{S}_R$ is a (d_{H_1}, d_{H_2}) -DGO relation. Then R

$$(\partial_1, \partial_2)(\mathbf{S}_R) \subset \mathbf{S}_R, \quad (\bar{\partial}_1, \bar{\partial}_2)(\mathbf{S}_R) \subset \mathbf{S}_R. \quad (6.230)$$

Proof. Let $(\rho_1, \rho_2) \in \mathcal{S}_R^k$, where we suppress the determinant notation. Then $r = (d_{H_1}\rho_1, d_{H_2}\rho_2) \in \mathcal{S}_R$, and so there is a decomposition $r = r_0 + \dots + r_n$ from (6.227). Since $d_{H_i}(\Gamma(\Omega_i^k)) \subset \Gamma(\Omega_i^{k-1}) \oplus \Gamma(\Omega_i^{k+1})$, it follows that $r = r_{k-1} + r_{k+1}$, and hence

$$(\partial_1, \partial_2): \mathcal{S}_R^k \rightarrow \mathcal{S}_R^{k-1}, \quad (\bar{\partial}_1, \bar{\partial}_2): \mathcal{S}_R^k \rightarrow \mathcal{S}_R^{k+1}, \quad (6.231)$$

and the result follows. $\square$

Generalised Complex Reduction

Consider the reduction of a generalised almost complex structure $\mathcal{J}$ to a generalised almost complex structure $\underline{\mathcal{J}}$. Then, by Remark 6.214, $Q(K)$ is a generalised almost complex relation from $\mathcal{J}$ to $\underline{\mathcal{J}}$ if and only if $\mathcal{J}(K) = K$. This can be formulated in terms of the subbundle L corresponding to $\mathcal{J}$, where

$$Q(K) = Q(K)^i \oplus Q(K)^{-i} \iff (K^\perp)^\mathbb{C} = (L \cap (K^\perp)^\mathbb{C}) \oplus (\bar{L} \cap (K^\perp)^\mathbb{C}). \quad (6.232)$$

The final condition is equivalent to $\mathcal{J}(K^\perp) = K^\perp$, and hence $\mathcal{J}(K) = K$.

When $\mathcal{J}(K) \neq K$, set $L_K = (L \cap (K^\perp)^\mathbb{C} + K^\mathbb{C})/K^\mathbb{C}$ and $\bar{L}_K = (\bar{L} \cap (K^\perp)^\mathbb{C} + K^\mathbb{C})/K^\mathbb{C}$. Thus

$$(Q(K)_c^i)^\times = \{ (\ell, \mathcal{J}_{q,m}^{-1}(\ell) : \ell \in L_K \} \quad (6.233)$$

and similarly for the complex conjugate. Thus $Q(K)$ is a weak generalised almost complex relation if and only if $(K^\perp)^\mathbb{C}/K^\mathbb{C} = L_K \oplus \bar{L}_K$. Note that if L is maximally isotropic then L_K is maximally isotropic in $(K^\perp)^\mathbb{C}/K^\mathbb{C}$. Similarly $\bar{L}_K$ is maximally isotropic. The condition that their intersection is zero is given by

$$(L \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}) \cap (\bar{L} \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}) \subset K^\mathbb{C}. \quad (6.234)$$

It follows that, when the above is satisfied, $(K^\perp)^\mathbb{C}/K^\mathbb{C} = L_K \oplus \bar{L}_K$ and $Q(K)$ is a weak generalised almost complex relation. In [51, Lemma 5.1], (6.234) is shown to be equivalent to $\mathcal{J}(K) \cap K^\perp \subset K$. This condition appears in [50, Proposition 6.1], generalising [51, Theorem 5.2].

Consider a generalised almost complex structure $\mathcal{J}$ on E . The associated spinor is given by $\phi = \exp(\alpha) \wedge \Theta$. Let us examine how the reduction of spinors proceeds in the case of symplectic and complex generalised complex structures

Example 6.235. Suppose $\phi = \exp(\alpha)$ and α is such that $\iota_Y \iota_Z \alpha = 0$ for all $Y, Z \in T\mathcal{F}^\mathbb{C}$, but there is some $X \in T\mathcal{F}^\mathbb{C}$ such that $\iota_X \alpha \neq 0$ and such that $\iota_X \alpha$ is real. Then

$$(X - \iota_X \alpha) \cdot \exp(\alpha) = 0 \quad (6.236)$$

so that $X - \iota_X \alpha \in L$. Moreover, since $\iota_Y \iota_X \alpha = 0$ for each $Y \in T\mathcal{F}^\mathbb{C}$, it follows that $X - \iota_X \alpha \in L \cap (K^\perp)^\mathbb{C}$. Then $\iota_X \alpha \in L \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}$, and $\overline{\iota_X \alpha} \in \overline{L} \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}$. Thus, since $\iota_X \alpha$ is real, we get $\iota_X \alpha \in (L \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}) \cap (\overline{L} \cap (K^\perp)^\mathbb{C} + K^\mathbb{C})$, but does not live in $K^\mathbb{C}$. Hence such an α cannot reduce to a generalised almost complex structure.

Assume that $\mathcal{F}$ has compact orientable fibers. Suppose that $\phi = \exp(i\omega)$. By the above arguments, there is no $X \in T\mathcal{F}^\mathbb{C}$ such that $\iota_X \alpha \in \text{Ann}(T\mathcal{F})$ and is non-zero. It follows that ω decomposes as $\omega = \omega_Q + \omega_{\mathcal{F}}$, where $\omega_Q \in \wedge^2 \text{Ann}(T\mathcal{F})$ and for every pointwise linearly independent $X, Y \in T\mathcal{F}$, we have $\iota_X \iota_Y \omega_{\mathcal{F}} \neq 0$. Take $\mu = \omega_{\mathcal{F}}^k$, where $k = \frac{1}{2} \dim \mathcal{F}$, and take S_μ as in Example 6.112. Then $\Omega^\phi \subset A$, and if ϕ is $T\mathcal{F}$ -invariant in the sense of Definition 6.124, we have that ω_Q is basic, and $\int_{\mathcal{F}} \exp(\omega_{\mathcal{F}})$ is a basic function. Thus $(\exp(\omega), (\int_{\mathcal{F}} \exp(\omega_{\mathcal{F}})) \exp(\omega_Q)) \in S_\mu$ and hence $\Omega^\phi \sim_{Q(K)} \underline{\Omega}^\phi$, where $\underline{\phi} = \exp(\omega_Q)$ and $\varpi^* \underline{\omega}_Q = \omega_Q$.

Example 6.237. Similarly, let $\phi = \Theta = \theta_1 \wedge \dots \wedge \theta_n$, for some linearly independent θ_i , such that $\theta_1, \dots, \theta_n, \bar{\theta}_1, \dots, \bar{\theta}_n$ are also linearly independent. Suppose that there is some $X \in TM$ with $X \notin T\mathcal{F}$ and some $i \in 1, \dots, n$ such that $\iota_X \theta_i = 0$, and that there is some $a \in \mathbb{C}$ such that $X + a\bar{X} := Y \in K^\mathbb{C}$.⁴ It then follows that $X - Y \in L \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}$ and $X - Y \in \overline{L} \cap (K^\perp)^\mathbb{C} + K^\mathbb{C}$. Since $X \notin T\mathcal{F}^\mathbb{C}$, it follows that $X - Y \notin T\mathcal{F}^\mathbb{C} = K^\mathbb{C}$, and so in order for such a Θ to reduce to an generalised almost complex structure, such an X cannot exist.

By the above arguments it follows that we can write $\Theta = \theta_1 \wedge \dots \wedge \theta_k \wedge \theta_{k+1} \wedge \dots \wedge \theta_n$ where $\theta_i \in \text{Ann}(T\mathcal{F})$ for $i = k+1, \dots, n$, and $k = \frac{1}{2} \dim \mathcal{F}$. Suppose that ϕ is invariant along $T\mathcal{F}$, and take $\mu = \theta_{k+1} \wedge \dots \wedge \theta_n$, and consider

$$S'_\mu = \{ (\mu \wedge \varpi^* \underline{\beta}, \left(\int_{\mathcal{F}} \mu \wedge \bar{\mu} \right) \underline{\beta}) : \underline{\beta} \in T^* \mathcal{Q} \}. \quad (6.238)$$

Then $\Omega^\phi \sim_{Q(K)} \underline{\Omega}^\phi$ for $\underline{\phi} = (\int_{\mathcal{F}} \mu \wedge \bar{\mu}) \underline{\Theta}$, where $\varpi^* \underline{\Theta} = \theta_1 \wedge \dots \wedge \theta_k$. Thus ϕ reduces to a generalised almost complex structure of type $n - k$, i.e. an ordinary almost complex structure on $\mathcal{Q}$ (recall that $\dim \mathcal{Q} = 2n - 2k$).

⁴Locally, such a θ_i might look like $dx^i + i dy^i$, where x^i is a coordinate in the base and y^i a coordinate in the fiber. Then $X = \partial_{x^i} + i \partial_{y^i}$ and $a = -1$.

6.6.1 T-Duality and Type Change of Generalised Complex Structures

Again, let us consider the case when $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1) \rightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$ is a T-duality relation, and hence a generalised isometry. The following arguments can be made for a generalised almost complex structure, but for simplicity we assume integrability of the generalised almost complex structures. By a D_1 -invariant generalised complex structure $\mathcal{J}_1$ on $\mathcal{Q}_1$ we mean a generalised complex structure whose $+i$ -eigenbundle L_1 (and hence $-i$ -eigenbundle $\bar{L}_1$) (and hence associated spinor $\phi_{\mathcal{J}_1}$) is invariant. An immediate consequence of Theorem 6.163 and Remark 6.220 is

Proposition 6.239. If $\mathcal{J}_1$ is an D_1 -invariant generalised complex structure on $\mathcal{Q}_1$, then there is a generalised complex structure $\mathcal{J}_2$ on $\mathcal{Q}_2$ such that $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1, \mathcal{J}_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2, \mathcal{J}_2)$ is a generalised complex relation.

Proof. The existence of an almost generalised complex structure $\mathcal{J}_2$ is given by Theorem 6.163. From Section 6.5, there is an R -Clifford relation S_R which is a $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation. The integrability of $\mathcal{J}_2$ is then given by Proposition 6.145. $\square$

The type of a generalised complex structure $\mathcal{J}$ can be determined by the pure spinor line bundle it generates. For instance, if $\mathcal{J}_1$ is a generalised complex structure on $\mathbb{T}\mathcal{Q}_1$ and is of type k , then

$$\phi_{\mathcal{J}_1} = \exp(B_1 + i\omega_1) \wedge \Theta_1, \quad (6.240)$$

where B_1, ω_1 are real two-forms, and $\Theta_1 = \theta_1 \wedge \dots \wedge \theta_k$ for some linearly independent complex forms θ_i .

Proposition 6.241. Let $\alpha = B_1 + i\omega_1$, and let $\text{rank } D_1 = d$, and $\dim \mathcal{B} = b$. At a point $c = (q_1, q_2) \in C$, there are the decompositions

$$\alpha = \dot{\alpha}_{\mathcal{B}} + \alpha_1 + \alpha_{\text{mix}}, \quad (6.242)$$

$$\Theta_1 = \dot{\mu}_{\mathcal{B}} \wedge \mu_1 \wedge \mu_{\text{mix}}, \quad (6.243)$$

for $\alpha_{\mathcal{B}} \in \wedge^2 T_b^* \mathcal{B}$, $\alpha_1 \in (\wedge^2 T^* \mathcal{F}_1)_{q_1}$, $\alpha_{\text{mix}} \in (T^* \mathcal{F}_1 \wedge \pi_1^* T^* \mathcal{B})_{q_1}$, $\mu_{\mathcal{B}} \in \wedge^{k_1} T^* \mathcal{B}$, $\mu_1 \in (\wedge^{k_2} T^* \mathcal{F}_1)_{q_1}$, and $\mu_{\text{mix}} \in (\wedge^{k_3} (T^* \mathcal{F}_1 + T^* \mathcal{B}))_{q_1}$ complex forms. The type of the complex structure $\mathcal{J}_2$ T-dual to $\mathcal{J}_1$ is

$$\text{type}(\mathcal{J}_2) = \text{type}(\mathcal{J}_1) - k_3 + m_3 \quad (6.244)$$

where m_3 is the smallest integer such that $\alpha_{\text{mix}}^{m_3} \neq 0$. The spinor associated to $\mathcal{J}_2$ is given by

$$\phi_{\mathcal{J}_2} = \exp(B_2 + i\omega_2) \wedge \Theta_2. \quad (6.245)$$

where B_2, ω_2, Θ_2 are real, real and complex forms respectively given by Equation (6.257) below.

Proof. We can write

$$\alpha_{\text{mix}} = b_{\alpha}^1 \wedge f_{\alpha}^1 + \dots + b_{\alpha}^{m_3} \wedge f_{\alpha}^{m_3}, \quad (6.246)$$

$$\mu_{\text{mix}} = (b_{\mu}^1 + f_{\mu}^1) \wedge \dots \wedge (b_{\mu}^{k_3} + f_{\mu}^{k_3}), \quad (6.247)$$

for $b_{\alpha}^i, b_{\mu}^j \in T^* \mathcal{B}$ and $f_{\alpha}^i, f_{\mu}^j \in G_1^*$ for $i = 1, \dots, m_3$ and $j = 1, \dots, k_3$, where $G_1^* = p_{T^*} p_1(G)$ as in the construction of $\mathbf{S}_R$. Note that $k = k_1 + k_2 + k_3$ is the type of $\mathcal{J}_1$.

Let $\hat{\mu}_1, \hat{f}_{\alpha}^i, \hat{\alpha}_1, \hat{f}_{\mu}^i \in p_1(G) \cap \wedge^{\bullet} T_{q_2} \mathcal{Q}_2$ be such that

$$\hat{\mu}_1 \cdot \mu_1 = \hat{f}_{\alpha}^i \cdot f_{\alpha}^i = \hat{\alpha}_1 \cdot \alpha_1 = \hat{f}_{\mu}^i \cdot f_{\mu}^i = 1, \quad (6.248)$$

while other possible actions between these elements are zero, and moreover $\hat{\mu}_1 \cdot \bar{\mu}_1 = 0$. Let also $\hat{\alpha}_{\text{mix}}$ and $\hat{\mu}_{\text{mix}}$ be given by exchanging f_{α}^i and f_{μ}^j with their hatted versions in α_{mix} and μ_{mix} respectively, for $i = 1, \dots, m_3$ and $j = 1, \dots, k_3$. Then, with $m_2 \in \mathbb{N}$ the largest natural number such that $\alpha_2^{m_1} \neq 0$, set

$$\lambda = \frac{1}{m_2!} \alpha_1^{m_2} \wedge f_{\alpha}^1 \wedge \dots \wedge f_{\alpha}^{m_3} \wedge \mu_1 \wedge \bar{\mu}_1 \wedge f_{\mu}^1 \wedge \dots \wedge f_{\mu}^{k_3}. \quad (6.249)$$

Since $\phi_{\mathcal{J}_1}$ defines a D_1 -invariant generalised complex structure, it follows that λ defines a D_1 -invariant fiber volume form on D_1 . The above considerations, along

with the following Lemma, allow us to factorise $\phi_{\mathcal{J}_1} = \hat{\phi} \cdot \lambda$.

Lemma 6.250. For the above defined forms, we have

1. $\exp(\hat{\alpha}_1) \cdot \frac{1}{m_2!} \alpha_1 = \exp(\alpha_1)$,
2. $\exp(\hat{\alpha}_{\text{mix}}) \cdot (f_\alpha^1 \wedge \dots \wedge f_\alpha^{m_3}) = (\ddot{b}_\alpha^1 + f_\alpha^1) \wedge \dots \wedge (\ddot{b}_\alpha^{m_3} + f_\alpha^{m_3})$,
3. $\hat{\mu}_1 \cdot (\mu_1 \wedge \bar{\mu}_1) = \bar{\mu}_1$,
4. $\hat{\mu}_{\text{mix}} \cdot (f_\mu^1 \wedge \dots \wedge f_\mu^{k_3}) = \exp(\ddot{b}_\mu^1 \wedge f_\mu^1 + \dots + \ddot{b}_\mu^{k_3} \wedge f_\mu^{k_3})$

Proof. (1) Firstly, for each $j \leq m_2$

$$\hat{\alpha}_1^j \cdot \alpha_1^{m_2} = m_2 j \hat{\alpha}_1^{j-1} \cdot \alpha_1^{m_2-1} = \dots = \frac{m_2! j!}{(m_2 - j)!} \alpha_1^{m_2-j}. \quad (6.251)$$

Thus

$$\begin{aligned} e^{\hat{\alpha}_1} \cdot \frac{1}{m_2!} \alpha_1^{m_2} &= (1 + \hat{\alpha}_1 + \dots + \frac{1}{m_2!} \hat{\alpha}_1^{m_2}) \cdot \frac{1}{m_2!} \alpha_1^{m_2} \\ &= \frac{1}{m_2!} \alpha_1^{m_2} + \frac{m_2}{m_2!} \alpha_1^{m_2-1} + \dots + \frac{1}{m_2! j!} \frac{m_2! j!}{(m_2 - j)!} \alpha_1^{m_2-j} + \dots + \frac{m_2!^2}{m_2!^2} \\ &= \exp(\alpha_1), \end{aligned} \quad (6.252)$$

(2)

$$\begin{aligned} \exp(\hat{\alpha}_{\text{mix}}) \cdot (f_\alpha^1 \wedge \dots \wedge f_\alpha^{m_3}) &= e^{\ddot{b}_\alpha^1 \wedge \hat{f}_\alpha^1} \wedge \dots \wedge e^{\ddot{b}_\alpha^{m_3} \wedge \hat{f}_\alpha^{m_3}} \cdot (f_\alpha^1 \wedge \dots \wedge f_\alpha^{m_3}) \\ &= (e^{\ddot{b}_\alpha^1 \wedge \hat{f}_\alpha^1} \cdot f_\alpha^1) \wedge \dots \wedge (e^{\ddot{b}_\alpha^{m_3} \wedge \hat{f}_\alpha^{m_3}} \cdot f_\alpha^{m_3}) \\ &= ((1 + \ddot{b}_\alpha^1 \wedge \hat{f}_\alpha^1) \cdot f_\alpha^1) \wedge \dots \wedge ((1 + \ddot{b}_\alpha^{m_3} \wedge \hat{f}_\alpha^{m_3}) \cdot f_\alpha^{m_3}) \\ &= (\ddot{b}_\alpha^1 + f_\alpha^1) \wedge \dots \wedge (\ddot{b}_\alpha^{m_3} + f_\alpha^{m_3}) \end{aligned} \quad (6.253)$$

where we use the fact that $\hat{f}_\alpha^i \cdot \ddot{b}_\alpha^j = 0$ for all i, j , and $\hat{f}_\alpha^i \cdot f_\alpha^j = 0$ when $i \neq j$ to commute the exponentials.

(3) Follows immediately

(4)

$$\begin{aligned}
 \hat{\mu}_{\text{mix}} \cdot (f_\mu^1 \wedge \dots \wedge f_\mu^{k_3}) &= (\ddot{b}_\mu + \hat{f}_\mu^1) \wedge \dots \wedge (\ddot{b}_\mu^{k_3} + \hat{f}_\mu^{k_3}) \cdot (f_\mu^1 \wedge \dots \wedge f_\mu^{k_3}) \\
 &= ((\ddot{b}_\mu + \hat{f}_\mu^1) \cdot f_\mu^1) \wedge \dots \wedge ((\ddot{b}_\mu^{k_3} + \hat{f}_\mu^{k_3}) \cdot f_\mu^{k_3}) \\
 &= (1 + \ddot{b}_\mu \wedge f_\mu^1) \wedge \dots \wedge (1 + \ddot{b}_\mu^{k_3} \wedge f_\mu^{k_3}) \\
 &= \exp(\ddot{b}_\mu^1 \wedge f_\mu^1 + \dots + \ddot{b}_\mu^{k_3} \wedge f_\mu^{k_3})
 \end{aligned} \tag{6.254}$$

□

By Proposition 6.105, there is a $\phi_{\mathcal{J}_2}$ such that $(\phi_{\mathcal{J}_1}, \phi_{\mathcal{J}_2}) \in (\mathbf{S}_R)_c$. From the results of the above Lemma, since $(\phi_{\mathcal{J}_1}, \phi_{\mathcal{J}_2}) = (\hat{\phi}, \phi_{\mathcal{J}_2}) \cdot (\lambda, 1)$, where

$$\hat{\phi} = e^{\dot{\alpha}_{\mathcal{B}} + \hat{\alpha}_1 + \hat{\alpha}_{\text{mix}}} \wedge \dot{\mu}_{\mathcal{B}} \wedge \hat{\mu}_1 \wedge \hat{\mu}_{\text{mix}} \tag{6.255}$$

Note that if $\omega = \dot{\beta}$ for $\beta \in T_b^* \mathcal{B}$, then $(\dot{\beta}, \ddot{\beta}) \in \text{Cl}(F^*)_c$ by the splitting (6.150). It follows that

$$\begin{aligned}
 \phi_{\mathcal{J}_2} &= \exp(\ddot{\alpha}_{\mathcal{B}}) \wedge \exp(\alpha_2) \wedge (\ddot{b}_\alpha^1 + \tilde{f}_\alpha^1) \wedge \dots \wedge (\ddot{b}_\alpha^{m_3} + \tilde{f}_\alpha^{m_3}) \wedge \\
 &\quad \mu_{\mathcal{B}} \wedge \bar{\mu}_2 \wedge \exp(\ddot{b}_\mu^1 \wedge \tilde{f}_\mu^1 + \dots + \ddot{b}_\mu^{k_3} \wedge \tilde{f}_\mu^{k_3}) \\
 &= \exp(B_2 + i\omega_2) \wedge \Theta_2
 \end{aligned} \tag{6.256}$$

for $\tilde{f}_\alpha^j$ and $\tilde{f}_\mu^k$ such that $\hat{f}_\alpha^j \sim_R \tilde{f}_\alpha^j$ and $\hat{f}_\mu^k \sim_R \tilde{f}_\mu^k$ in $\text{Cl}(R)_c$, for $j = 1, \dots, m_3$ and $k = 1, \dots, k_3$. In the above, we have made the definitions

$$\begin{aligned}
 B_2 &:= \text{Re}(\ddot{\alpha}_{\mathcal{B}} + \alpha_2 + \alpha_{\mu_{\text{mix}}}), \\
 \omega_2 &:= \text{Im}(\ddot{\alpha}_{\mathcal{B}} + \alpha_2 + \alpha_{\mu_{\text{mix}}}), \\
 \Theta_2 &:= \mu_{\alpha_{\text{mix}}} \wedge \mu_{\mathcal{B}} \wedge \bar{\mu}_2,
 \end{aligned} \tag{6.257}$$

where

$$\alpha_{\mu_{\text{mix}}} = \dot{b}_\mu^1 \wedge \tilde{f}_\mu^1 + \dots + \dot{b}_\mu^{k_3} \wedge \tilde{f}_\mu^{k_3}, \quad (6.258)$$

$$\mu_{\alpha_{\text{mix}}} = (\ddot{b}_\alpha^1 + \tilde{f}_\alpha^1) \wedge \dots \wedge (\ddot{b}_\alpha^{m_3} + \tilde{f}_\alpha^{m_3}). \quad (6.259)$$

Hence the type of $\mathcal{J}_2$ is $k_1 + k_2 + m_3$, and the result follows. $\square$

Remark 6.260. Let $A \in \varpi_1^* T_{q_1}^* \mathcal{Q}_1 \wedge \varpi_2^* T_{q_2}^* \mathcal{Q}_2$ be given by

$$A = \dot{f}_\alpha^1 \wedge \ddot{f}_\alpha^1 + \dots + \dot{f}_\alpha^{m_3} \wedge \ddot{f}_\alpha^{m_3} + A_{\alpha_{1,2}} + A_{\mu_{1,2}} + \dot{f}_\mu^1 \wedge \ddot{f}_\mu^1 + \dots + \dot{f}_\mu^{k_3} \wedge \ddot{f}_\mu^{k_3} + \text{c.c.} \quad (6.261)$$

where we do not give the expression for $A_{\alpha_{1,2}}$ and $A_{\mu_{1,2}}$, but are two forms such that such that $\hat{\alpha}_1 \sim_R \alpha_2$ and $\hat{\mu}_1 \sim_R \mu_2$ respectively. Here c.c. denotes the complex conjugate. Write $A = A_{\alpha_{\text{mix}}} + A_{\alpha_{1,2}} + A_{\mu_{1,2}} + A_{\mu_{\text{mix}}} + \text{c.c.}$. For $j \in \mathbb{N}$, consider

$$\begin{aligned} & (A + \dot{\alpha}_1 + \dot{\alpha}_{\text{mix}})^j \wedge \dot{\mu}_1 \wedge \dot{\mu}_{\text{mix}} \\ &= \sum_{a+b+c+d=j} (A_{\alpha_{\text{mix}}} + \alpha_{\text{mix}})^a \wedge (A_{\alpha_{1,2}} + \alpha_1)^b \wedge (\text{c.c.})^d \wedge \mu_1 \wedge \mu_{\text{mix}} \end{aligned} \quad (6.262)$$

Since λ defines a fiber volume form on the leaves of $\mathcal{F}_2$, we have $d = 2k_2 + k_3 + 2m_2 + m_3$. Then, as $\text{type}(\mathcal{J}_2) = \text{type}(\mathcal{J}_1) - k_3 + m_3$, we have

$$\text{type}(\mathcal{J}_2) = \text{type}(\mathcal{J}_1) + 2(m_2 + k_2 + m_3) - d. \quad (6.263)$$

If we take $j = m_2 + k_2 + m_3$, (6.262) includes (a multiple of) the term

$$(A_{\alpha_{\text{mix}}} + \alpha_{\text{mix}})^{m_3} \wedge (A_{\alpha_{1,2}} + \alpha_1)^{m_2} \wedge \overline{A}_{\mu_{1,2}}^{k_2} \wedge \mu_1 \wedge \mu_{\text{mix}}. \quad (6.264)$$

which is non-zero and lies in $\det(T^* \dot{\mathcal{F}}_2) \wedge \varpi^* \wedge^\bullet T^* \mathcal{Q}_2$, and in fact gives the smallest j such that this holds. This is in agreement with [46, Example 4.3].

Example 6.265. Continuing the analogy with [46], the expression (6.256) allows us to easily see what happens in specific instances of generalised complex structures on $\mathcal{Q}_1$. Suppose that $\dim(\mathcal{Q}_1) = 2n$, and $\mathcal{F}_1$ has codimension n , i.e. $\text{rank}(D_1) = n$.

If $\mathcal{J}_1$ is purely complex, and the fibers $\mathcal{F}_1$ are also complex, then $\phi_{\mathcal{J}_1}$ has the form

$$\phi_{\mathcal{J}_1} = \dot{\mu}_{\mathcal{B}} \wedge \mu_1, \quad (6.266)$$

that is, there is no μ_{mix} component. Thus, the T-dual spinor $\phi_{\mathcal{J}_2}$ corresponding to the complex structure $\mathcal{J}_2$ is given by

$$\phi_{\mathcal{J}_2} = \ddot{\mu}_{\mathcal{B}} \wedge \bar{\mu}_2 \quad (6.267)$$

Thus $\mathcal{J}_2$ is also a purely complex structure.

Conversely, if $\mathcal{J}_1$ is a complex structure given by J_1 and the the fibers of $\mathcal{F}_1$ are real (i.e. $J_1(T\mathcal{F}_1) \cap T\mathcal{F}_1 = \{0\}$), then

$$\phi_{\mathcal{J}_1} = \mu_{\text{mix}} \implies \phi_{\mathcal{J}_2} = e^{\text{Re}(\alpha_{\mu_{\text{mix}}}) + i \text{Im}(\alpha_{\mu_{\text{mix}}})}. \quad (6.268)$$

Thus $\mathcal{J}_2$ is symplectic with a potential B -field twist. Moreover, since $\alpha_{\mu_{\text{mix}}}$ has always a component in $T\mathcal{B}$, the fibers $T\mathcal{F}_2$ define Lagrangian submanifolds with respect to this symplectic structure.

If $\mathcal{J}_1$ is purely symplectic, and the leaves of $\mathcal{F}_1$ are symplectic, then $\phi_{\mathcal{J}_1} = \exp(\dot{\alpha}_{\mathcal{B}} + \alpha_1)$, and hence $\phi_{\mathcal{J}_2} = \exp(\ddot{\alpha}_{\mathcal{B}} + \alpha_2)$ is still symplectic.

If $\mathcal{J}_2$ is symplectic but the leaves of $\mathcal{F}_1$ are Lagrangian, then $\phi_{\mathcal{J}_1} = \exp(\alpha_{\text{mix}})$, and hence $\phi_{\mathcal{J}_2} = \mu_{\alpha_{\text{mix}}}$, and hence has complex type.

We thus recover the results of [46, Table 1 of Example 4.3], which we summarize here

Structure on $\mathcal{Q}_1$	Leaves of $\mathcal{F}_1$	Structure on $\mathcal{Q}_2$	Leaves of $\mathcal{F}_2$
Complex	Complex	Complex	Complex
Complex	Real	Symplectic	Lagrangian
Symplectic	Symplectic	Symplectic	Symplectic
Symplectic	Lagrangian	Complex	Real

6.7 Generalised Kähler Relations

Definition 6.269. Let $R: E_1 \dashrightarrow E_2$ be a Courant algebroid relation, and $\mathcal{K}_1 = (\mathcal{J}_1^+, \mathcal{J}_1^-)$, $\mathcal{K}_2 = (\mathcal{J}_2^+, \mathcal{J}_2^-)$ be generalised almost Kähler structures on E_1, E_2 respectively. Then R is a *generalised almost Kähler relation* if R is a generalised almost complex relation for $(E_1, \mathcal{J}_1^+) \dashrightarrow (E_2, \mathcal{J}_2^+)$ and $(E_1, \mathcal{J}_1^-) \dashrightarrow (E_2, \mathcal{J}_2^-)$. If $\mathcal{K}_1, \mathcal{K}_2$ are generalised Kähler structures, then R is a *generalised Kähler relation*. If $\mathcal{K}_1, \mathcal{K}_2$ are generalised Calabi-Yau structures, then R is a *generalised Calabi-Yau relation*.

Proposition 6.270. A generalised almost Kähler relation R has decomposes into 4 isotropic subbundles

$$R_c^{\mathbb{C}} = R^{i,i} \oplus R^{i,-i} \oplus R^{-i,i} \oplus R^{-i,-i} \quad (6.271)$$

where $R_c^{i,i} = R_c \cap ((L_1^+ \cap L_1^-) \times (L_2^+ \cap L_2^-))_c$, $R_c^{i,-i} = R_c^{\mathbb{C}} \cap ((L_1^+ \cap \bar{L}_1^-) \times (L_2^+ \cap \bar{L}_2^-))_c$ and so on, for $L_i^{\pm}$ the $\pm i$ -eigenbundle of $\mathcal{J}_i^{\pm}$ respectively, for $i = 1, 2$.

Recall that for a generalised almost Kähler structure, there is the decomposition of the generalised metric $C^{\pm}$ as $(V^+)^{\mathbb{C}} = (L^+ \cap L^-) \oplus (\bar{L}^+ \cap \bar{L}^-)$ and $(V^-)^{\mathbb{C}} = (L^+ \cap \bar{L}^-) \oplus (\bar{L}^+ \cap L^-)$. From this, we get

Lemma 6.272. Let $R: (E_1, \mathcal{K}_1) \dashrightarrow (E_2, \mathcal{K}_2)$ be a generalised almost Kähler relation, then R is a generalised isometry between the generalised metrics V_1^+ and V_2^+ defined by $\mathcal{K}_1$ and $\mathcal{K}_2$ respectively.

Proof. This follows since, from the above, we get

$$R_c^{\mathbb{C}} = (R_c^{i,i} \oplus R_c^{-i,-i}) \oplus (R_c^{i,-i} \oplus R_c^{-i,i}) = (R_c^+)^{\mathbb{C}} \oplus (R_c^-)^{\mathbb{C}} = (R_c^+ \oplus R_c^-)^{\mathbb{C}}, \quad (6.273)$$

and hence $R_c = R_c^+ \oplus R_c^-$. □

Note also, from the definition of a generalised almost Kähler relation, that for the generalised metrics τ_1, τ_2 defined by $\mathcal{K}_1, \mathcal{K}_2$ respectively,

$$(\tau_1 \times \tau_2)(R) = (-\mathcal{J}_1^+ \mathcal{J}_1^- \times -\mathcal{J}_2^+ \mathcal{J}_2^-)(R) = R \quad (6.274)$$

and hence R is a generalised isometry.

Since for a generalised almost Kähler relation, R is necessarily a generalised isometry, by Proposition 3.64, we have that K_R and CK_R are trivial.

Remark 6.275 (Relating Associated Bi-Hermitian Structures). Since a generalised almost Kähler relation $R: (E_1, \mathcal{K}_1) \dashrightarrow (E_2, \mathcal{K}_2)$ is a generalised isometry, there is the splitting

$$R = R^+ \oplus R^- \quad (6.276)$$

where $R^\pm = R \cap (V_1^\pm \times V_2^\pm)$ for the generalised metrics $V_1^\pm, V_2^\pm$ associated to $\mathcal{K}_1, \mathcal{K}_2$ respectively. Moreover, since $\mathcal{J}_i^\pm$ commute with the metrics, it follows that $(\mathcal{J}_1^+ \times \mathcal{J}_2^+)(R^\pm) = R^\pm$. Define $T^\pm = \rho(R^\pm) \subset TC$, so that $\rho: R^\pm \rightarrow T^\pm$ are isomorphisms covering the identity. The bi-Hermitian structures associated with $\mathcal{K}_i$, given by (g_i, b_i, I_i^+, I_i^-) , act on T^+ and T^- , and since $\mathcal{J}_1^\pm \times \mathcal{J}_2^\pm$ preserves $R^\pm$ respectively, we have that

$$\rho \circ (\mathcal{J}_1^\pm \times \mathcal{J}_2^\pm) \circ (\rho|_{R^\pm})^{-1}: T^\pm \rightarrow T^\pm \quad (6.277)$$

is an isomorphism. In other words, $(I_1^\pm \times I_2^\pm)(T^\pm) = T^\pm$. In this sense, we say that the bi-Hermitian structures are R -related.

Example 6.278 (bi-Hermitian for torus bundles). In the setting of T-duality for principal torus bundles, there is the following formula for the dual bi-Hermitian structure [46, Example 4.7]

$$I_2^\pm = \rho_2^\pm \mathcal{R}(\rho_1^\pm)^{-1} I_1^\pm (\rho_2^\pm \mathcal{R}(\rho_1^\pm)^{-1})^{-1}. \quad (6.279)$$

Then for $(X_1, X_2) \in T^+$

$$\begin{aligned}
 (I_1^+ \times I_2^+)(X_1, X_2) &= (\rho_1^+ \mathcal{J}_1^+ (\rho_1^+)^{-1} X_1, \rho_2^+ \mathcal{R} (\rho_1^+)^{-1} I_1^+ \rho_1^+ \mathcal{R}^{-1} (\rho_2^+)^{-1} X_2) \\
 &= (\rho_1^+ \mathcal{J}_1^+ X_1^+, \rho_2^+ \mathcal{R} (\rho_1^+)^{-1} \rho_1^+ \mathcal{J}_1^+ (\rho_1^+)^{-1} \rho_1^+ \mathcal{R}^{-1} (\rho_2^+)^{-1} X_2) \\
 &= (\rho_1^+ \mathcal{J}_1^+ X_1^+, \rho_2^+ \mathcal{R} \mathcal{J}_1^+ \mathcal{R}^{-1} (\rho_2^+)^{-1} X_2) \\
 &= (\rho_1^+ \times \rho_2^+)(\mathcal{J}_1^+ X_1^+, \mathcal{R} \mathcal{J}_1^+ X_1^+),
 \end{aligned} \tag{6.280}$$

where $X_i^+ = (\rho_i^+)^{-1} X_i$ for $i = 1, 2$. Since $\mathcal{R}$ is induced by the relation R , it follows that $(I_1^+ \times I_2^+)(T^+) = T^+$, and similarly for T^- .

Recall that for a generalised almost Kähler structure, there is the bigrading

$$\bigwedge^\bullet T^* M \otimes \mathbb{C} = \bigoplus_{p,q} \Omega^{p,q}. \tag{6.281}$$

Following from Proposition 6.229, we get

Proposition 6.282. There is a bigrading

$$\mathcal{S}_R = \bigoplus_{p,q} \mathcal{S}_R^{p,q} \tag{6.283}$$

where $\mathcal{S}_R^{p,q} = ((\Omega_1^{p,q} \times \Omega_2^{p,q}) \otimes |\det \mathbf{S}_R^*|^{\frac{1}{2^n}}) \cap \mathcal{S}_R$

6.7.1 T-duality

Suppose that $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2)$ is a T-duality relation. By a D_1 -invariant generalised Kähler structure on $\mathcal{Q}_1$ we mean a generalised Kähler structure both of whose generalised complex structures are D_1 -invariant. Then the following follows immediately from Proposition 6.239.

Proposition 6.284. If $\mathcal{Q}_1$ has a D_1 -invariant generalised Kähler structure $\mathcal{K}_1$, then there is a generalised Kähler structure $\mathcal{K}_2$ on $\mathcal{Q}_2$ such that $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1, \mathcal{K}_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2, \mathcal{K}_2)$ is a generalised Kähler relation.

Proposition 6.285. Let $R: \mathbb{T}\mathcal{Q}_1 \dashrightarrow \mathbb{T}\mathcal{Q}_2$ be a T-duality relation. If $\mathcal{CY}_1$ is a D_1 -invariant generalised Calabi-Yau structure on $\mathcal{Q}_1$, then there is a generalised

Calabi-Yau structure $\mathcal{CY}_2$ on $\mathcal{Q}_2$ such that $R: (\mathbb{T}\mathcal{Q}_1, \underline{H}_1, \mathcal{CY}_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, \underline{H}_2, \mathcal{CY}_2)$ is a generalised Calabi-Yau relation.

Proof. Since $\mathcal{CY}_1$ is, in particular, a generalised Kähler structure, the existence of a generalised Kähler structure $\mathcal{K}_2$ on $\mathcal{Q}_2$ is given by Proposition 6.284. The integrability of the generalised Kähler structures is determined by the $(d_{\underline{H}_1}, d_{\underline{H}_2})$ -DGO relation $\mathbf{S}_R$, and since there is a nowhere vanishing closed section $\phi_1 \in \Omega_1^+$ such that $d_{\underline{H}_1}\phi_1 = 0$, it follows that for ϕ_2 such that $(\phi_1, \phi_2) \in \mathbf{S}_R$, we have $d_{\underline{H}_2}\phi_2 = 0$, and hence is generalised Calabi-Yau. $\square$

Example 6.286 (Mirror Symmetry, [46]). Consider the setting of a Calabi-Yau manifold $\mathcal{Q}_1$, with T-duality taken along a special Lagrangian fibration $\mathcal{F}_1$, i.e. the symplectic form ω_1 vanishes along $\mathcal{F}_1$ and the complex structure J_1 is real, and $\dim \mathcal{F}_1 = \frac{1}{2} \dim \mathcal{Q}_1$. This induces a generalised Calabi-Yau structure with $\mathcal{J}_1^+ = \mathcal{J}_{\omega_1}$ and $\mathcal{J}_1^- = \mathcal{J}_{J_1}$. From Table 6.265, the T-dual generalised Calabi-Yau structure has swapped symplectic and complex structure, $\mathcal{J}_2^+ = \mathcal{J}_{J_2}$ and $\mathcal{J}_2^- = \mathcal{J}_{\omega_2}$. The bigrading of Proposition 6.285 allows us to associated $\Omega_1^{p,q}$ with $\Omega_2^{p,q}$.

For an ordinary Calabi-Yau structure, there is an isomorphism of the ordinary (p, q) decomposition with the decomposition into $\Omega^{p,q}$ given by

$$\Omega_I^{p-q} \cap \Omega_\omega^{n-p-q} = e^{i\omega} \cdot e^{-\frac{1}{i}\omega^{-1}} \cdot \wedge^{p,q} T^*M \otimes \mathbb{C}. \quad (6.287)$$

Thus, in T-duality we may associate $\wedge^{p,q} T^*\mathcal{Q}_1$ and $\wedge^{n-p,q} T^*\mathcal{Q}_2$, giving the usual Hodge diamond of Betti numbers in mirror symmetry.

In Section 7.2.6, we will see explicit examples of generalised Kähler relations, including between generalised Kähler structures not coming from ordinary Kähler structures.

Chapter 7

Applications and Examples

In this Section we will look at some specific examples of how the constructions of the previous Chapters apply to various examples of T-duality. We have seen throughout how the case of principal torus bundles falls under our regime. We will see how Poisson-Lie T-duality can be viewed as a Courant algebroid relation, and how the additional structures are transported by the Courant algebroid relation. We also include examples of T-duality coming from general fiber bundles, without a principal bundle structure on $\mathcal{Q}_i$. This includes T-duality for the Klein bottle, and its 3-dimensional counterpart, where we see how the H -flux continues to play an important role in the topology change. Explicit examples of the generalised Ricci flow and how it transforms under T-duality will be discussed. We will also see an example of T-duality between generalised Kähler geometries not coming from ordinary Kähler manifolds, which relates harmonic functions on a space to two symplectic structures on the dual manifold.

7.1 Poisson-Lie T-duality

Construction of the Relation

As discussed in the Introduction, a particularly relevant example of T-duality comes from the description of Poisson-Lie T-duality, detailed in [27]. In the setting of [1, Section 4.4], let E be a G -equivariant exact Courant algebroid over a manifold M with a free and proper action of a connected Lie group G , whose Lie algebra $\mathfrak{g}$ is endowed with a split signature pairing $\langle \cdot, \cdot \rangle_{\mathfrak{g}}$ and an extended action $\chi: \mathfrak{g} \rightarrow \Gamma(E)$.

Suppose that $K = \chi(\mathfrak{g})$ satisfies $K \cap K^\perp = \{0\}$, and let $\varpi_G: M \rightarrow \mathcal{B} = M/G$ be the quotient map.

Let $\mathfrak{h} \subset \mathfrak{g}$ be a Lie subalgebra which integrates to a closed connected subgroup $H \subset G$ such that $K_H = \chi(\mathfrak{h})$ is an isotropic subbundle of E . Let $\varpi_H: M \rightarrow \mathcal{Q} = M/H$ be the quotient map. Then there is a Courant algebroid morphism $R(H): \underline{E}_H \rightarrow \underline{E}$ over φ between the reduced Courant algebroids

$$\underline{E} = K^\perp / G \quad , \quad \underline{E}_H = \frac{K_H^\perp}{K_H} / H \quad , \quad (7.1)$$

where $\underline{E} \rightarrow \mathcal{B}$ is a transitive Courant algebroid such that $\underline{E} / \text{im}(\underline{\rho}^*) = TM/G$ and $\varphi: \mathcal{Q} \rightarrow \mathcal{B}$ is the canonical surjective submersion induced by the subgroup inclusion $H \subset G$, which fits into the commutative diagram

$$\begin{array}{ccc} & M & \\ \varpi_H \swarrow & \downarrow \varpi_G & \\ \mathcal{Q} & & \mathcal{B} \\ & \searrow \varphi & \end{array} \quad (7.2)$$

If $\mathfrak{h}$ is a Lagrangian subalgebra, i.e. $\mathfrak{h}^\perp = \mathfrak{h}$, then $K_H^\perp = K_H \oplus K^\perp$ so that $K_H^\perp / K_H \cong K^\perp$ and $\underline{E}_H \cong K^\perp / H$. Then the reduced Courant algebroid $\underline{E}_H \rightarrow \mathcal{Q}$ is exact and is naturally the Courant algebroid pullback $\underline{E}_H \cong \varphi^* \underline{E}$ of $\underline{E} \rightarrow M$. It follows that $R(H) = \text{gr}(\Phi_H)$ for a classical Courant algebroid morphism $\Phi_H: \underline{E}_H \rightarrow \underline{E}$ over φ [1, Example 4.31]. If $\mathfrak{h}'$ is any other Lagrangian subalgebra of $\mathfrak{g}$ with integrating Lie group H' , then also $R(H') = \text{gr}(\Phi_{H'})$ for a classical Courant algebroid morphism $\Phi_{H'}: \underline{E}_{H'} \rightarrow \underline{E}$ over the surjective submersion $\varphi': \mathcal{Q}' \rightarrow \mathcal{B}$, where $\mathcal{Q}' = M/H'$.

The composition

$$R_{H,H'} := \text{gr}(\Phi_{H'})^\top \circ \text{gr}(\Phi_H) : \underline{E}_H \dashrightarrow \underline{E}_{H'} \quad , \quad (7.3)$$

is defined and supported on the fibred product $\mathcal{Q} \times_{\mathcal{B}} \mathcal{Q}'$; it is equal to the fibred product $\underline{E}_H \times_{\underline{E}} \underline{E}_{H'}$. Note that if $\mathfrak{h} \cap \mathfrak{h}' = \{0\}$, then $(\mathfrak{g}, \mathfrak{h}, \mathfrak{h}')$ is a Manin triple and (H, H') is a dual pair of Poisson-Lie groups. Generally, the relation $R_{H,H'}$ is

called a *Poisson-Lie T-duality relation* (with spectators $\mathcal{B}$), which we may view diagrammatically as

$$\begin{array}{ccc}
 \underline{E}_H & \overset{R_{H,H'}}{\dashrightarrow} & \underline{E}_{H'} \\
 \searrow \Phi_H & & \swarrow \Phi_{H'} \\
 & \underline{E} &
 \end{array} \tag{7.4}$$

Importantly, $R_{H,H'}$ coincides with the T-duality relation $R = Q(K_{H'}) \circ Q(K_H)$ found in this thesis. When $\mathbf{G}$ is abelian, it coincides with the T-duality relation for torus bundles of Lemma 4.132.

Relations for Divergences and Generalised Ricci Tensors

In the spirit of [1, Example 5.9], if div is a $\mathbf{G}$ -equivariant divergence on E as in Example 5.29, then $\underline{E}$ inherits a divergence operator $\underline{\text{div}}$, so that $\text{div}_H := \Phi_H^*(\underline{\text{div}})$ and $\text{div}_{H'} := \Phi_{H'}^*(\underline{\text{div}})$ define divergences on $\underline{E}_H$ and $\underline{E}_{H'}$ respectively. It then follows that $\text{div}_H \sim_{R_{H,H'}} \text{div}_{H'}$.

Let $V^+ \subset E$ be a $\mathbf{G}$ -invariant generalised metric. Its reduction by both the $\mathfrak{h}$ -action and the $\mathfrak{h}'$ -action yields the generalised metrics V_H^+ and $V_{H'}^+$ on $\underline{E}_H$ and $\underline{E}_{H'}$ respectively. As shown in [1], $R_{H,H'}$ is a generalised isometry between V_H^+ and $V_{H'}^+$. It then follows that $\text{GRic}_{V_H^+, \text{div}_H} \sim_{R_{H,H'}} \text{GRic}_{V_{H'}^+, \text{div}_{H'}}$.

Relations for Spinors

Consider the setting of Poisson-Lie T-duality “without spectators”, i.e. when $\mathcal{B} = \{*\}$. Let $\mathfrak{g}$ be a Lie bialgebra with a non-degenerate pairing, $\mathbf{G}$ a connected Poisson-Lie group integrating $\mathfrak{g}$, and let $H_1, H_2 \subset \mathbf{G}$ be such that $\mathfrak{h}_1, \mathfrak{h}_2 \subset \mathfrak{g}$ are Lagrangian. There is then the T-duality relation $R: E_1 \dashrightarrow E_2$, for exact Courant algebroids E_1 over $\mathbf{G}/H_2$ and E_2 over $\mathbf{G}/H_1$. Assuming that $R: E_1 \dashrightarrow E_2$ is a generalised isometry, the existence of an invariant volume form λ on $\mathbf{G}/H_1$ is equivalent to $\mathfrak{h}_1$ being a unimodular Lie algebra. We may thus construct the $R_{H,H'}$ -Clifford spinor relation $S_{R_{H,H'}}$. Note that the unimodularity condition coincides with the existence of an invariant divergence on $\mathbf{G}/H_1$ [49, Example 2.5].

7.2 Explicit Examples

We have already described the behavior of the T-duality relation for classic classes of examples such as principal torus bundles throughout the thesis and Poisson-Lie T-duality above. In this final section, after an explicit example using the machinery of the torus bundle setting applied to Lens spaces, we look at some explicit examples that are not necessarily covered by these cases discussed thus far.

7.2.1 Lens Spaces

Consider a three-dimensional S^1 -bundle $\pi_1: \mathcal{Q}_1 \rightarrow S^2$. Choose a chart with coordinates (x, y, z) such that z is adapted to the circle fibres and (x, y) are pullback coordinates from the two-sphere S^2 . Pick a connection

$$\theta_1 = dz + m x dy \quad (7.5)$$

on $\mathcal{Q}_1$ and an integral volume form

$$\underline{H}_1 = k dx \wedge dy \wedge dz , \quad (7.6)$$

with $m, k \in \mathbb{Z}$, such that

$$\int_{\mathcal{Q}_1} \underline{H}_1 = k \quad \text{and} \quad \int_{S^2} c_1(\mathcal{Q}_1) = m , \quad (7.7)$$

where $c_1(\mathcal{Q}_1) = d\theta_1 = m dx \wedge dy$ represents the Chern class of $\mathcal{Q}_1$.

On $\mathcal{Q}_1$ we take as metric

$$\underline{g}_1 = \dot{g}_{S^2} + g_0 \theta_1 \otimes \theta_1 , \quad (7.8)$$

where g_{S^2} is the standard round metric on S^2 , and $g_0 \in C^\infty(S^2)$. Then $\frac{\partial}{\partial z}$, a local expression of the generator of the $\mathfrak{u}(1)$ -action, is a Killing vector field for $\underline{g}_1$.

Consider another three-dimensional S^1 -bundle $\pi_2: \mathcal{Q}_2 \rightarrow S^2$ endowed with connection

$$\theta_2 = d\tilde{z} + n x dy \quad (7.9)$$

and Chern number $n \in \mathbb{Z}$, where $\tilde{z}$ is the coordinate adapted to the fibres. Let $M = \mathcal{Q}_1 \times_{S^2} \mathcal{Q}_2$ with projections $\varpi_i: M \rightarrow \mathcal{Q}_i$, for $i = 1, 2$, and coordinates¹ $(x, y, z, \tilde{z})$ pulled back from $\mathcal{Q}_1$ and $\mathcal{Q}_2$. In order to satisfy the conditions of Theorem 4.88, we consider the two-form

$$B = -\dot{\theta}_1 \wedge \ddot{\theta}_2 = x \, dy \wedge (n \, dz - m \, d\tilde{z}) - dz \wedge d\tilde{z} \quad (7.10)$$

for our B -field transformation.

For $(\mathbb{T}M, H)$ to be reducible with respect to $K_2 = e^{-B}\mathfrak{k}_2$, we have that

$$\dot{H}_1 - dB = k \, dx \wedge dy \wedge dz + m \, dx \wedge dy \wedge d\tilde{z} - n \, dz \wedge dx \wedge dy \quad (7.11)$$

$$= (k - n) \, dx \wedge dy \wedge dz + m \, dx \wedge dy \wedge d\tilde{z} \quad (7.12)$$

has to be basic with respect to the fibres of ϖ_2 , whose coordinate in this adapted frame is z . This happens if and only if $n = k$. Thus $\mathcal{Q}_2$ has three-form $\underline{H}_2$ and Chern form $c_1(\mathcal{Q}_2)$ satisfying

$$\int_{\mathcal{Q}_2} \underline{H}_2 = m \quad \text{and} \quad \int_{S^2} c_1(\mathcal{Q}_2) = k, \quad (7.13)$$

as well as metric obtained from Example 5.97 as

$$\underline{g}_2 = \ddot{g}_{S^2} + g_0^{-1} \theta_2 \otimes \theta_2, \quad (7.14)$$

which is again S^1 -invariant.

Hence the lens space $\mathbb{L}(m, k)$ is T-dual to the lens space $\mathbb{L}(k, m)$, realising a topology change between T-dual spaces, see e.g. [53]. In particular, the Hopf fibration $\mathbb{L}(1, 0) \simeq S^3$ without H -flux is T-dual to the trivial circle bundle $\mathbb{L}(0, 1) \simeq S^2 \times S^1$ with H -flux whose integral is 1. Moreover, the Hopf fibration with H -flux $\mathbb{L}(1, 1)$ is self-T-dual.

¹We omit the pullback notation for the coordinates.

The Classical Ricci Tensor

Consider the metric on the Lens space $\mathbf{L}(0, k) \simeq \mathbf{S}^2 \times \mathbf{S}^1$ with H -flux whose integral is k . Thus $\underline{H} = k\mu_{\mathbf{S}^2} \wedge dz$, where $\mu_{\mathbf{S}^2}$ is the standard volume element on $\mathbf{S}^2$, and z is the coordinate of the circle $\mathbf{S}^1$. Take again the metric

$$\underline{g}_1 = \dot{g}_{\mathbf{S}^2} + g_0 dz \otimes dz, \quad (7.15)$$

for some function $g_0 \in C^\infty(\mathbf{S}^2)$. As we have seen, there is then the T-duality relation

$$R: (\mathbf{TL}(0, k), \underline{H}, \underline{g}_1) \dashrightarrow (\mathbf{TL}(k, 0), 0, \underline{g}_2). \quad (7.16)$$

From the Ricci-Buscher rules (5.111)-(5.113), the Ricci tensor for $\underline{g}_2$ can be calculated. For instance, the component in the circle direction can be calculated to be

$$\begin{aligned} \text{Ric}_{\theta_k \theta_k}(\underline{g}_2) &= -\frac{1}{g_0^2} \text{GRic}_{zz} \\ &= -\frac{1}{g_0^2} \text{Ric}_{zz}(\underline{g}_1) + \frac{1}{2g_0^2} k^2 - \frac{1}{g_0^3} ((\partial_1 g_0)^2 + (\partial_2 g_0)^2) \\ &= \frac{1}{2g_0^2} (\partial_1^2 g_0 + \partial_2^2 g_0) - \frac{3}{4g_0^3} ((\partial_1 g_0)^2 + (\partial_2 g_0)^2) + \frac{1}{2g_0^2} k^2. \end{aligned} \quad (7.17)$$

7.2.2 Hamilton's Cigar Soliton/Witten's Black Hole

Let $\mathcal{Q}_1 = \mathbb{R}^2 \setminus \{0\}$ with the cigar metric

$$g_1 = \frac{dx \otimes dx + dy \otimes dy}{1 + x^2 + y^2}, \quad (7.18)$$

where $(x, y) \in \mathcal{Q}_1$. In geodesic coordinates this can be written as

$$g_1 = ds \otimes ds + \tanh^2(s) d\theta \otimes d\theta. \quad (7.19)$$

Then $\partial_\theta = \frac{\partial}{\partial \theta}$ is a Killing vector for this metric, and it generates a circular foliation $\underline{\mathcal{F}}_1$ with leaf space $\mathcal{Q}_1/\underline{\mathcal{F}}_1 = \mathbb{R}_{>0}$.

The family of metrics

$$g_1(t) = \frac{\cosh^2(s)}{e^{4t} + \sinh^2(s)} ds \otimes ds + \frac{\sinh^2(s)}{e^{4t} + \sinh^2(s)} d\theta \otimes d\theta \quad (7.20)$$

on $\mathcal{Q}_1$ is a Ricci soliton with initial condition $g_1(0) = g_1$. That is, it is a solution to the Ricci flow

$$\frac{\partial}{\partial t} g_1(t) = -2 \operatorname{Ric}_{g_1(t)} \quad (7.21)$$

which is self-similar, evolving the metric g_1 purely by pullback under the one-parameter family of diffeomorphisms $(\sinh(s), \theta) \mapsto (e^{2t} \sinh(s), \theta)$. This Ricci soliton is called Hamilton's cigar [82]. In string theory it appears as the Euclidean Witten black hole background, which is a fixed point of the renormalisation group flow of a gauged Wess-Zumino-Witten model (see e.g. [83]). In this subsection we demonstrate the behaviour under geometric T-duality of this classic solution to the Ricci flow.

Let $\mathcal{Q}_2 = \mathbb{R} \times \mathbb{R}_{>0}$, with coordinates (z, s) , and foliation $\underline{\mathcal{F}}_2$ given by the projection to the second factor, so that $\mathcal{Q}_2/\underline{\mathcal{F}}_2 = \mathbb{R}_{>0}$. Let $M = \mathcal{Q}_1 \times_{\mathbb{R}_{>0}} \mathcal{Q}_2$, with foliations $\mathcal{F}_1$ generated by $\partial_z = \frac{\partial}{\partial z}$ and $\mathcal{F}_2$ generated by ∂_θ , and take

$$B = d\theta \wedge dz .$$

Denoting $\partial_s = \frac{\partial}{\partial s}$, we obtain the subbundle $R \subset \mathbb{T}\mathcal{Q}_1 \times \mathbb{T}\mathcal{Q}_2$ whose sections are given by

$$\Gamma(R) = \operatorname{Span}_{C^\infty(\mathbb{R}_{>0})} \{ (\partial_s, \partial_s), (ds, ds), (\partial_\theta, dz), (d\theta, \partial_z) \} . \quad (7.22)$$

Since ∂_θ is a Killing vector of $g_1(t)$ for all t , by Theorem 5.93 we get the generalised isometry

$$R: (\mathbb{T}\mathcal{Q}_1, 0, g_1(t)) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, 0, g_2(t)) , \quad (7.23)$$

with the T-dual family of metrics $g_2(t)$ on $\mathcal{Q}_2$ given by

$$g_2(t) = \frac{\cosh^2(s)}{e^{4t} + \sinh^2(s)} ds \otimes ds + \frac{e^{4t} + \sinh^2(s)}{\sinh^2(s)} dz \otimes dz , \quad (7.24)$$

and R -related divergence operators $\operatorname{div}_{\mu_{g_1(t)}} \sim_R \operatorname{div}_\phi(t)$ where

$$\operatorname{div}_\phi(t) = \operatorname{div}_{\mu_{g_2(t)}} + \langle d\phi, \cdot \rangle_2 = \operatorname{div}_{\mu_{g_2(t)}} - 4 \langle \varphi^+, \cdot \rangle_2 , \quad (7.25)$$

with

$$\phi = -\log \left(\frac{e^{4t} + \sinh^2(s)}{\sinh^2(s)} \right) , \quad \varphi^+ = -\frac{1}{4} d\phi = -\frac{e^{4t} \cosh(s)}{2 \sinh^3(s) + 2 e^{4t} \sinh(s)} ds .$$

Then the pair $(g_2(t), \operatorname{div}_\phi(t))$ solves the system

$$\frac{\partial}{\partial t} g_2(t) = -2 \operatorname{Ric}_{g_2(t)} - 2 \nabla^{g_2(t)} \varphi^+ , \quad \frac{\partial}{\partial t} \operatorname{div}_\phi(t) = -d \left(R_{g_2(t)} - 4 e^\phi \Delta_{g_2(t)} e^{-\phi} \right) . \quad (7.26)$$

Analogously to the original cigar soliton, the T-dual solution is a *generalised* Ricci soliton [61, 84], i.e. a self-similar solution of the generalised Ricci flow (7.26), evolving the metric g_2 and dilaton field ϕ purely by pullback under the one-parameter family of Courant algebroid automorphisms given by the diffeomorphisms $(\sinh(s), z) \mapsto (e^{2t} \sinh(s), z)$.

7.2.3 Three-Dimensional Hyperbolic Space

In this subsection we shall present a new example of geometric T-duality that will illustrate how the notion of relations between generalised Ricci tensors aids as a computational tool as well, in addition to yielding an explicit T-dual solution of the generalised Ricci flow.

Geometric T-Duality

Consider the pair of trivial circle bundles $\mathcal{Q}_1, \mathcal{Q}_2$ over $\mathbb{R}_{>0}^2$, with coordinates

$$\mathcal{Q}_i = S^1 \times \mathbb{R}_{>0}^2 = \{ (\vartheta_i, r, z) \mid \vartheta \in [0, 2\pi) , r, z > 0 \} . \quad (7.27)$$

Denote by ∂_1 the coordinate vector field on $\mathcal{Q}_1$ in the ϑ_1 -direction, and similarly for ∂_2 on $\mathcal{Q}_2$. The foliations $\mathcal{F}_i$ generated by ∂_i induce a Hausdorff-Morita equivalence between $(\mathcal{Q}_1, \mathcal{F}_1)$ and $(\mathcal{Q}_2, \mathcal{F}_2)$, with correspondence diagram

$$\begin{array}{ccc}
 & M & \\
 \varpi_1 \swarrow & & \searrow \varpi_2 \\
 \mathcal{Q}_1 & & \mathcal{Q}_2 \\
 \pi_1 \searrow & & \swarrow \pi_2 \\
 & \mathbb{R}_{>0}^2 &
 \end{array} \tag{7.28}$$

where $M = \mathcal{Q}_1 \times_{\mathbb{R}_{>0}^2} \mathcal{Q}_2$.

On $\mathcal{Q}_1$, take the hyperbolic metric g_1 and closed three-form H_1 given by

$$g_1 := \lambda^2 g_{\mathbb{H}^3} = \frac{\lambda^2}{z^2} (r^2 d\vartheta_1 \otimes d\vartheta_1 + dr \otimes dr + dz \otimes dz) , \tag{7.29}$$

$$H_1 := h \mu_{\mathbb{H}^3} = h r d\vartheta_1 \wedge dr \wedge dz , \tag{7.30}$$

where $\lambda, h \in \mathbb{R}$.

Introduce the two-form

$$B = d\vartheta_1 \wedge (d\vartheta_2 + \frac{1}{2} h r^2 dz) \tag{7.31}$$

on M , such that $dB = \dot{H}_1$. Then B is non-degenerate as a map from $T\mathcal{F}_1 \otimes T\mathcal{F}_2$ to $\mathbb{R}$. With $K_1 = T\mathcal{F}_1$ and $K_2 = e^{-B}(T\mathcal{F}_2)$, it follows that $K_1 \cap K_2 = \{0\}$ and $K_2 \cap K_1^\perp \subset K_1$. By 4.9 we obtain the diagram of standard Courant algebroids

$$\begin{array}{ccc}
 & (\mathbb{T}M, \dot{H}_1) & \\
 Q(K_1) \swarrow & & \searrow Q(K_2) \\
 (\mathbb{T}\mathcal{Q}_1, H_1) & \xrightarrow{\quad R \quad} & (\mathbb{T}\mathcal{Q}_2, 0)
 \end{array} \tag{7.32}$$

where we may explicitly write the relation R from its generating sections as the $C^\infty(\mathbb{R}_{>0}^2)$ -module

$$\begin{aligned}
 \Gamma(R) = \text{Span}_{C^\infty(\mathbb{R}_{>0}^2)} \{ & (\partial_r, \partial_r) , (\partial_1, d\vartheta_2 + \frac{hr^2}{2} dz) , (\partial_z, \partial_z - \frac{hr^2}{2} \partial_2) , \\
 & (dr, dr) , (d\vartheta_1, \partial_2) , (dz, dz) \} .
 \end{aligned} \tag{7.33}$$

Note that ∂_1 is a Killing vector for g_1 , and $\mathcal{L}_{\dot{\partial}_1} B = \mathcal{L}_{\ddot{\partial}_2} B = 0$, where $\dot{\partial}_1$ denotes the vector field generating the quotient ϖ_2 such that $(\varpi_1)_*(\dot{\partial}_1) = \partial_1$, and similarly for $\dot{\partial}_2$. Thus taking the isometry algebra $\mathfrak{k}_{g_1} = \text{Span}_{\mathbb{R}} \{ \partial_{\theta_1} \}$, it follows that g_1 defines a $T\mathcal{F}_1$ -invariant generalised metric on $\mathbb{T}\mathcal{Q}_1$. From Theorem 4.45 it follows that

$$R: (\mathbb{T}\mathcal{Q}_1, H_1, g_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, 0, g_2) \quad (7.34)$$

is a generalised isometry, where the T-dual metric on $\mathcal{Q}_2$ is given by

$$g_2 = \frac{z^2}{\lambda^2 r^2} d\vartheta_2 \otimes d\vartheta_2 + \frac{\lambda^2}{z^2} dr \otimes dr + \frac{h z^2}{\lambda^2} dz \odot d\vartheta_2 + \left(\frac{\lambda^2}{z^2} + \frac{h^2 r^2 z^2}{4 \lambda^2} \right) dz \otimes dz . \quad (7.35)$$

By Proposition 4.115, g_2 is then $T\mathcal{F}_2$ -invariant, with isometry algebra $\mathfrak{k}_{g_2} = \text{Span}_{\mathbb{R}} \{ \partial_{\theta_2} \}$.

Generalised Ricci Tensor

The Ricci tensor of $g_1 = \lambda^2 g_{\mathbb{H}^3}$ is

$$\text{Ric}_{g_1} = -2 g_{\mathbb{H}^3} , \quad (7.36)$$

hence g_1 is an Einstein metric. We further find

$$H_1^2 = \frac{2 h^2}{\lambda^4} g_{\mathbb{H}^3} , \quad d^* H_1 = 0 . \quad (7.37)$$

Then the Ricci tensor for g_2 can be calculated from the Ricci-Buscher rules of Subsection 5.3.2 as

$$\begin{aligned} (\text{Ric}_{g_2})_{rr} &= -\frac{h^2}{2 \lambda^4 z^2} - \frac{2}{r^2} , & (\text{Ric}_{g_2})_{zz} &= -\frac{2}{r^2} , & (\text{Ric}_{g_2})_{\vartheta\vartheta} &= \frac{h^2 z^2}{2 \lambda^8 r^2} - \frac{2 z^4}{\lambda^4 r^4} , \\ (\text{Ric}_{g_2})_{rz} &= \frac{2}{r z} - \frac{h^2 r}{2 \lambda^4} , & (\text{Ric}_{g_2})_{r\vartheta} &= -\frac{h}{\lambda^4 r} , & (\text{Ric}_{g_2})_{z\vartheta} &= 0 . \end{aligned} \quad (7.38)$$

Generalised Ricci Flow

Consider now the geometry with the dilaton turned off on $\mathcal{Q}_1$, i.e. $\text{div}_1 = \text{div}_{\mu_{g_1}}$. Allowing λ and h to depend on time, the generalised Ricci flow on $\mathbb{T}\mathcal{Q}_1$ is the

one-parameter family of pairs $(g_1(t), \text{div}_1(t))$ with $g_1(t) = \lambda(t)^2 g_{\mathbb{H}^3}$ and $\text{div}_1(t) = \text{div}_{\mu_{g_1(t)}} = \text{div}_{\mu_{g_{\mathbb{H}^3}}}$ such that

$$\frac{\partial}{\partial t} \lambda(t) = \frac{2}{\lambda(t)} + \frac{h(t)^2}{2\lambda(t)^5} \quad , \quad \frac{\partial}{\partial t} h(t) = 0 \quad , \quad \frac{\partial}{\partial t} \text{div}_1(t) = d \left(\frac{6}{\lambda(t)^2} + \frac{h(t)^2}{2\lambda(t)^6} \right) = 0 . \quad (7.39)$$

The metric $g_2(t)$ given by Equation (7.35) is then the solution of the system

$$\frac{\partial}{\partial t} g_2(t) = -2 \text{Ric}_{g_2(t)} - 2 \nabla^{g_2(t)} d \log(z^2/r^2), \quad (7.40)$$

$$\frac{\partial}{\partial t} \text{div}_\phi(t) = -d \left(R_{g_2(t)} - \frac{4z^2}{r^2} \Delta_{g_2(t)}(r^2/z^2) \right), \quad (7.41)$$

with dilaton field

$$\phi = \log(z^2/r^2) .$$

Thus the generalised Ricci flow becomes the evolution of the scale factor $\lambda(t)$, with constant Kalb-Ramond flux h . For large $\lambda(t)$ the solution asymptotes to $\pm \sqrt{\lambda_0^2 + 2t}$, where $\lambda_0 = \lambda(0)$, and therefore the flow diverges for long times independently of h . In particular, the behaviour for large $\lambda(t)$ is essentially constant. For small $\lambda(t)$ the flow diverges as $\pm \sqrt[6]{\lambda_0^6 + 3h^2 t}$. In particular, the flow has no fixed points for any $h \in \mathbb{R}$, and so the solution does not converge to any generalised string background.

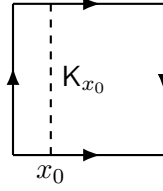
7.2.4 Klein Bottle

In this subsection we consider an example of geometric T-duality between non-principal S^1 -bundles.

Consider the Klein bottle $\mathbf{K}$ which is the compact non-orientable surface defined by

$$\mathbf{K} = \{ (x, y) \in [0, 1] \times [0, 1] \} / \sim , \quad (7.42)$$

where $(x, 0) \sim (x, 1)$ and $(0, y) \sim (1, 1 - y)$. We regard $\mathbf{K}$ as a circle bundle over S^1 , with S^1 fibres given by $\mathbf{K}_{x_0} = \{ (x_0, y) \}$, which we may depict as



Correspondence Space

Let $K_1 = K_2 = K$ be Klein bottles, viewed as S^1 -bundles over a common circle S^1 , with projection maps $\pi_i: K_i \rightarrow S^1$ given by $\pi_i(x, y) = x$. Their fibred product is given by

$$M = K_1 \times_{S^1} K_2 = \{((x, y), (x, z)) \in [0, 1]^{\times 4}\} / \sim, \quad (7.43)$$

with

$$\begin{aligned} ((x, 0), (x, z)) &\sim ((x, 1), (x, z)) \quad , \quad ((x, y), (x, 0)) \sim ((x, y), (x, 1)) \quad , \\ ((0, y), (0, z)) &\sim ((1, 1 - y), (0, z)) \sim ((1, 1 - y), (1, 1 - z)) \quad . \end{aligned} \quad (7.44)$$

This is a T^2 -bundle over S^1 , which may be more succinctly described as

$$M = \{(x, y, z) \in [0, 1]^{\times 3}\} / \sim \quad (7.45)$$

with

$$(x, 0, z) \sim (x, 1, z) \quad , \quad (x, y, 0) \sim (x, y, 1) \quad , \quad (0, y, z) \sim (1, 1 - y, 1 - z) \quad , \quad (7.46)$$

and projection map $\hat{\pi}: M \rightarrow S^1$ given by $\hat{\pi}(x, y, z) = x$. Denote the quotients $\varpi_i: M \rightarrow K_i$, given by projection.

We cover M with open charts given by

$$U^{(1)} = \{(x, y, z) \in M \mid x \in (0, 1)\} \quad , \quad U^{(2)} = \{(x, y, z) \in M \mid x \neq \tfrac{1}{2}\} \quad , \quad (7.47)$$

$$\varphi^{(1)}: U^{(1)} \longrightarrow (0, 1) \times [0, 1]^{\times 2} \quad , \quad \varphi^{(2)}: U^{(2)} \longrightarrow (0, 1) \times [0, 1]^{\times 2} \quad , \quad (7.48)$$

where

$$\varphi^{(1)}(x, y, z) = (x, y, z) \quad , \quad \varphi^{(2)}(x, y, z) = \begin{cases} (x + \frac{1}{2}, y, z) & , \quad x \in [0, \frac{1}{2}) \\ (x - \frac{1}{2}, 1 - y, 1 - z) & , \quad x \in (\frac{1}{2}, 1] \end{cases} . \quad (7.49)$$

The intersection can be written as a disjoint union

$$U^{(1)} \cap U^{(2)} = \{ (x, y, z) \in M \mid x \in (0, \frac{1}{2}) \} \sqcup \{ (x, y, z) \in M \mid x \in (\frac{1}{2}, 1) \} , \quad (7.50)$$

and we denote the first disjoint factor by $W^{(1)}$, and the second by $W^{(2)}$. On the intersection, the transition functions are given by

$$\varphi^{(2)} \circ (\varphi^{(1)})^{-1} = \begin{cases} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & \text{on } W^{(1)} \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} & \text{on } W^{(2)} \end{cases} . \quad (7.51)$$

It follows that M is an orientable $\mathbb{T}^2$ -bundle over $\mathbb{S}^1$.

The foliations $\mathcal{F}_1$ and $\mathcal{F}_2$ of M are inherited from the fibration structures of the bundles $\mathbf{K}_2$ and $\mathbf{K}_1$ respectively, while the quotient maps ϖ_1 and ϖ_2 are given by the projections from M to $\mathbf{K}_1$ and $\mathbf{K}_2$ respectively.

Reduction

Consider the untwisted standard Courant algebroid $(\mathbb{T}M, 0)$ over M . Define $K_1 = T\mathcal{F}_1 \subset \mathbb{T}M$. Let $f^{(i)}(x)$ be smooth bump functions with support in $U^{(i)}$ such that the vector fields $Z^{(1)} = f^{(1)} \partial_z$ and $Z^{(2)} = f^{(2)} \partial_z$, which can be extended globally as spanning sections of K_1 , are not simultaneously zero. Similarly, define $Y^{(i)} = f^{(i)} \partial_y$, and also $\lambda^{(i)} = f^{(i)} dy$. Note that ∂_x and dx are already globally defined.

As global sections of $K_1^\perp$ we then take

$$\Gamma_{\text{bas}}(K_1^\perp) = \text{Span}_{C^\infty(\mathbb{S}^1)} \{ \partial_x, Y^{(1)}, Y^{(2)}, Z^{(1)}, Z^{(2)}, dx, \lambda^{(1)}, \lambda^{(2)} \} . \quad (7.52)$$

One easily sees that these are basic, for instance

$$\begin{aligned} [Z^{(i)}, \partial_x] &= -(\partial_x f^{(i)}) \partial_z \in \Gamma(K_1), \quad [Z^{(i)}, Y^{(j)}] = 0, \quad \mathcal{L}_{Z^{(i)}} dx = d\iota_{Z^{(i)}} dx = 0 \\ \mathcal{L}_{Z^{(i)}} \lambda^{(j)} &= \iota_{Z^{(i)}} d(f^{(j)} dx) = Z^{(i)}(f^{(j)}) dx - df^{(j)}(\iota_{Z^{(i)}} dx) = 0. \end{aligned} \quad (7.53)$$

Thus the basic sections span $K_1^\perp$ pointwise.

Alternatively, since $H = 0$, consider the standard splitting $\sigma_1: TM \rightarrow TM$ given by the inclusion $X \mapsto X + 0$. Then $\sigma_1(X)$ is K_1 -basic whenever X is $\mathcal{F}_1$ -projectable, hence σ_1 gives a K_1 -adapted splitting. In either case, the assumptions of Theorem 2.40 hold, and we may form the reduced Courant algebroid

$$\frac{K_1^\perp}{K_1} / \mathcal{F}_1 = (\mathbb{T}K_1, 0). \quad (7.54)$$

In the same way, we can take $K_2 = T\mathcal{F}_2 \subset TM$, and form the reduced Courant algebroid $(\mathbb{T}K_2, 0)$ over K_2 .

T-Duality

Consider the two-form B defined on each chart $U^{(i)}$ as

$$B = dy \wedge dz. \quad (7.55)$$

It satisfies $(\varphi^{(2)} \circ (\varphi^{(1)})^{-1})^* B = B$ on $U^{(1)} \cap U^{(2)}$, and hence gives rise to a globally defined two-form $B \in \Omega^2(M)$.

The two-form B is non-degenerate on $T\mathcal{F}_1 \otimes T\mathcal{F}_2$, and so it follows that $K_1 \cap e^{-B}(K_2) = \{0\}$ and $e^B(K_1) \cap K_2^\perp \subset K_2$. By Theorem 4.9 we may form the T-duality relation R supported on $C = M$. Its sections given by

$$\Gamma((\mathbb{T}K_1 \times \mathbb{T}K_2)|_{U^{(i)}}; R) = \text{Span}_{C^\infty(\mathcal{V}^{(i)})} \{ (\partial_x, \partial_x), (dx, dx), (\partial_y, dz), (dy, \partial_z) \}. \quad (7.56)$$

Consider the flat metric on K_1 induced by the standard flat metric

$$g = dx \otimes dx + dy \otimes dy \quad (7.57)$$

on $\mathbb{R}^2$, which is given locally by the same formula. On $\varpi_1(U^{(1)})$ and $\varpi_1(U^{(2)})$, the vector field ∂_y is a local Killing vector of g generating the isometry distribution D_1 . By Theorem 4.101, the relation

$$R: (\mathbb{TK}_1, 0, g) \dashrightarrow (\mathbb{TK}_2, 0, g) \quad (7.58)$$

is a generalised isometry. This gives the geometric self-duality of the Klein bottle.

Generalised Ricci Flow

The generalised Ricci flow is the same on both manifolds K_1 and K_2 . It is given by the usual Ricci flow of the flat metric, i.e. the trivial stationary flow.

7.2.5 Klein Bottle Fibration

In this subsection we extend the example of geometric T-duality for non-principal circle bundles from Subsection 7.2.4 to include Kalb-Ramond flux, and hence topology change.

Let

$$\mathcal{Q}_1 = K_1 \times S^1, \quad (7.59)$$

where K is the Klein bottle as in Subection 7.2.4, and $S^1 = [0, 1]/(0 \sim 1)$. We regard $\mathcal{Q}_1$ as a circle bundle over a torus T^2 , with open charts

$$U^{(1)} = \{ (x, y, z) \in \mathcal{Q}_1 \mid x \in (0, 1) \} \quad , \quad U^{(2)} = \{ (x, y, z) \in \mathcal{Q}_1 \mid x \neq \tfrac{1}{2} \} \quad , \quad (7.60)$$

$$\varphi^{(1)}: U^{(1)} \longrightarrow (0, 1) \times [0, 1]^{\times 2} \quad , \quad \varphi^{(2)}: U^{(2)} \longrightarrow (0, 1) \times [0, 1]^{\times 2} \quad , \quad (7.61)$$

where

$$\varphi^{(1)}(x, y, z) = (x, y, z) \quad , \quad \varphi^{(2)}(x, y, z) = \begin{cases} (x + \frac{1}{2}, y, z) & \text{on } W^{(1)} \\ (x - \frac{1}{2}, 1 - y, z) & \text{on } W^{(2)} \end{cases} \quad , \quad (7.62)$$

with $W^{(1)} \sqcup W^{(2)} = U^{(1)} \cap U^{(2)}$, similarly to Subsection 7.2.4.

Topological T-Duality

Let

$$H_1 = \frac{\pi}{2} \sin(\pi x) dx \wedge dy \wedge dz \quad (7.63)$$

be a local representative of the non-trivial cohomology class in $H^3(\mathcal{Q}_1, \mathbb{Z}) = \mathbb{Z}_2$.

Let us work out the T-dual manifold $\mathcal{Q}_2$. We construct the correspondence space M as a circle bundle over $\mathcal{Q}_1$, with projection map $\varpi_1 : M \rightarrow \mathcal{Q}_1$, using $U^{(i)}$ as trivialising charts. The additional circle coordinate is denoted by $\tilde{y}$. We seek a B -field containing the component $dy \wedge d\tilde{y}$, and satisfying $dB = \varpi_1^* H_1$. Thus locally we take

$$B = dy \wedge (d\tilde{y} - \frac{1}{2} \cos(\pi x) dz) \quad .$$

This form is indicative of a topology change on M , which arises from a local splitting of the tangent bundle $TM = \mathbf{Ver}(M) \oplus \mathbf{Hor}(M)$ into vertical and horizontal parts: the vertical subbundle $\mathbf{Ver}(M)$ is given locally by the span of $\partial_{\tilde{y}}$, while the horizontal subbundle $\mathbf{Hor}(M)$ is given by the span of ∂_x , ∂_y and $\partial_z + \frac{1}{2} \cos(\pi x) \partial_{\tilde{y}}$. It suggests charts for M given by

$$\tilde{\varphi}^{(1)}(x, y, \tilde{y}, z) = (x, y, \tilde{y}, z) \quad (7.64)$$

$$\tilde{\varphi}^{(2)}(x, y, \tilde{y}, z) = \begin{cases} (x + \frac{1}{2}, y, \tilde{y}, z) & \text{on } W^{(1)} \\ (x - \frac{1}{2}, 1 - y, 1 - \tilde{y}, z + \tilde{y}) & \text{on } W^{(2)} \end{cases} \quad , \quad (7.65)$$

which is well-defined provided we identify

$$(0, y, \tilde{y}, z) \sim (1, 1 - y, 1 - \tilde{y}, z + \tilde{y})$$

on M . This also ensures that the two-form B is well-defined.

Upon quotienting M by the foliation $\mathcal{F}_2$ generated by ∂_y , we obtain the leaf space

$$\mathcal{Q}_2 = [0, 1]^{\times 3} / \sim ,$$

where

$$(x, 0, z) \sim (x, 1, z) \quad , \quad (x, \tilde{y}, 0) \sim (x, \tilde{y}, 1) \quad , \quad (0, \tilde{y}, z) \sim (1, 1 - \tilde{y}, z + \tilde{y}) . \quad (7.66)$$

This too is a circle bundle over $\mathbb{T}^2$.

We thus get the T-duality relation

$$R : (\mathbb{T}\mathcal{Q}_1, H_1) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, 0) . \quad (7.67)$$

On $\mathcal{V}^{(i)} = U^{(i)} \times \mathbb{S}^1$, which defines a trivialising chart for M , it is locally generated by

$$\begin{aligned} \Gamma((\mathbb{T}\mathcal{Q}_1 \times \mathbb{T}\mathcal{Q}_2)|_{\mathcal{V}^{(i)}}; R) = \text{Span}_{C^\infty(\mathcal{V}^{(i)})} \{ & (\partial_x, \partial_x) , (\partial_y, d\tilde{y} - \tfrac{1}{2} \cos(\pi x) dz) , \\ & (\partial_z, \partial_z + \tfrac{1}{2} \cos(\pi x) \partial_{\tilde{y}}) , (dx, dx) , (dy, \partial_{\tilde{y}}) , (dz, dz) \} . \end{aligned} \quad (7.68)$$

The general framework for topological T-duality between affine torus bundles over a common base $\mathcal{B}$ was considered in [85] using the language of local coefficient systems. In this setting T-duality is stated after a choice of local coefficient system $\mathbb{Z}_\xi$, i.e. an element ξ of $\mathbf{H}^1(\mathcal{B}, \mathbb{Z}_2)$. In our present example, a representative of ξ is equivalent to a choice of charts. Since the torus $\mathbb{T}^2$ has twisted cohomology group $\mathbf{H}^2(\mathbb{T}^2, \mathbb{Z}_\xi) = \mathbb{Z}_2$, there are two circle bundles over $\mathbb{T}^2$ for this choice of charts. Turning on the Kalb-Ramond flux H_1 on $\mathcal{Q}_1$ (of which there is only one cohomologically non-trivial choice since $\mathbf{H}^3(\mathcal{Q}_1, \mathbb{Z}) = \mathbb{Z}_2$), it follows that T-duality passes from the trivial to the non-trivial element of $\mathbf{H}^2(\mathbb{T}^2, \mathbb{Z}_\xi)$. This can also be formulated in terms of twisted connections, also considered in [85] where T-duality for twisted

cohomology was treated, whereby the one-form $d\tilde{y} - \frac{1}{2} \cos(\pi x) dz$ defines a twisted connection on the double cover $\tilde{\mathcal{Q}}_2$ of $\mathcal{Q}_2$ with non-trivial curvature.

Compared to [85], the advantage of the formulation considered here is that it does not consider the affine structure of the torus bundle, nor even the group structure on the fibers, only the fibration structure itself. It also permits consideration of the T-duality of the geometric structures, as we illustrate in Subsection 7.2.5 below, which was not considered in the approach of [85].

Geometric T-Duality

Let g_1 be the flat metric on $\mathcal{Q}_1$ which is locally given by

$$g_1 = dx \otimes dx + dy \otimes dy + dz \otimes dz .$$

Then the T-duality relation R is a generalized isometry between g_1 and the T-dual metric g_2 on $\mathcal{Q}_2$ given by

$$g_2 = dx \otimes dx + d\tilde{y} \otimes d\tilde{y} - \cos(\pi x) d\tilde{y} \odot dz + \left(1 + \frac{1}{4} \cos^2(\pi x)\right) dz \otimes dz . \quad (7.69)$$

On $\mathcal{Q}_1$ we further find

$$H_1^2 = \frac{\pi^2}{2} \sin^2(\pi x) g_1 \quad , \quad d^* H_1 = \frac{\pi^2}{2} \cos(\pi x) dy \wedge dz .$$

Using the Ricci-Buscher rules of Subsection 5.3.2, from this we can calculate the non-zero components of the Ricci tensor of g_2 to be

$$(\text{Ric}_{g_2})_{yy} = \frac{\pi^2}{8} \sin^2(\pi x) , \quad (7.70)$$

$$(\text{Ric}_{g_2})_{zy} = \frac{\pi^2}{4} \cos(\pi x) + \frac{\pi^2}{16} \sin^2(\pi x) \cos(\pi x) , \quad (7.71)$$

$$(\text{Ric}_{g_2})_{xx} = -\frac{\pi^2}{8} \sin^2(\pi x) , \quad (7.72)$$

$$(\text{Ric}_{g_2})_{zz} = -\frac{\pi^2}{8} \sin^2(\pi x) + \frac{\pi^2}{4} \cos^2(\pi x) + \frac{\pi^2}{32} \sin^2(\pi x) \cos^2(\pi x) . \quad (7.73)$$

The generalised Ricci flow for this example will be a highly complicated non-

linear system due to the trigonometric functions appearing. We do not attempt to solve it here, as it requires some involved numerical computation which is beyond the scope of this thesis.

7.2.6 Generalised Kähler Relations

Example 7.74 (Semi-flat Calabi-Yau Metrics, [28]). Let $\mathcal{Q}_1 = D \times \mathbb{T}^k$ where $D \subset \mathbb{R}^k$ is the unit disc. Take coordinates $\{x^1, \dots, x^k\}$ on D , and $\{y^1, \dots, y^k\}$ on $\mathbb{T}^k$. For any $A > 0$ there is a unique solution $u: D \rightarrow \mathbb{R}$ to the Monge-Ampère equation

$$\det(u_{ij}) = A. \quad (7.75)$$

with $u|_{\partial D} = 0$ and $u_{ij} > 0$, where $u_{ij} = \frac{\partial^2 u}{\partial x^i \partial x^j}$. We can define a $\mathbb{T}^k$ -invariant metric

$$g = \sum_{i,j} u_{ij} (dx^i \otimes dx^j + dy^i \otimes dy^j) \quad (7.76)$$

With complex structure $J(\partial_{x^i}) = \partial_{y^i}$, the metric g is Kähler, with Kähler form

$$\omega = \frac{i}{2} \sum_{i,j} u_{ij} dz^i \wedge d\bar{z}^j = \sum_{i,j} u_{ij} dx^i \wedge dy^j \quad (7.77)$$

where $z^i = x^i + iy^i$. Thus we can define a generalised Kähler structure $\mathcal{K}_1 = (\mathcal{J}_J, \mathcal{J}_\omega)$, with generalised metric given by g . This has Ricci form equal to

$$\text{Ric}(\omega) = -i\bar{\partial}\partial \log \det(u_{ij}) = 0, \quad (7.78)$$

and hence g is Ricci flat. The associated spinors are $\phi_J = dz^1 \wedge \dots \wedge dz^k$, and $\phi_\omega = \exp(i\omega)$, which are both closed, and hence $\mathcal{K}_1$ defines a generalised Calabi-Yau structure.

Consider $\mathcal{Q}_2 = D \times \tilde{\mathbb{T}}^k$, with coordinates on $\tilde{\mathbb{T}}^k$ given by $\{\tilde{y}^1, \dots, \tilde{y}^k\}$, and define the correspondence space $M = \mathcal{Q}_1 \times_D \mathcal{Q}_2$. We take as B -field on M the two-form $B = dy^1 \wedge d\tilde{y}^1 + \dots + dy^k \wedge d\tilde{y}^k$, and hence we may write the T-duality relation

$$R = \text{Span}_{C^\infty(D)} \{ (\partial_{x^i}, \partial_{x^i}), (dx^i, dx^i), (\partial_{y^i}, d\tilde{y}^i), (dy^i, \partial_{\tilde{y}^i}) \} \quad (7.79)$$

Since $\mathcal{J}_J$ and $\mathcal{J}_\omega$ are both T^k -invariant, we can transport them via Proposition 6.284 to a T-dual generalised Kähler structure. To see what the T-dual structure looks like, we can apply the results of Proposition 6.241. There are two equivalent approaches for this, which illustrate two ways of viewing the T-dual structure. Firstly, we can write

$$\phi_J = dz^1 \wedge \dots \wedge dz^k = \mu_{\text{mix}} \quad (7.80)$$

$$\phi_\omega = \exp(i\omega) = \exp(\alpha_{\text{mix}}) \quad (7.81)$$

where

$$\alpha_{\text{mix}} = \sum_{i,j} u_{ij} dx^i \wedge dy^j = dx^1 \wedge f^1 + \dots + dx^k \wedge f^k \quad (7.82)$$

for $f^i = \sum_j u_{ij} dy^j$. Then $\hat{f}^i = \sum_j u^{ij} \partial_{\tilde{y}^j}$. Thus, we get $\tilde{f}^i = \sum_j u^{ij} d\tilde{y}^j$. Hence, the structures $\tilde{\phi}_J, \tilde{\phi}_\omega$ dual to ϕ_J, ϕ_ω are given by

$$\tilde{\phi}_J = \exp(idx^1 \wedge d\tilde{y}^1 + \dots + idx^k \wedge d\tilde{y}^k) \quad (7.83)$$

$$\tilde{\phi}_\omega = (dx^1 + i \sum_j u^{1j} d\tilde{y}^j) \wedge \dots \wedge (dx^k + i \sum_j u^{kj} d\tilde{y}^j) \quad (7.84)$$

Thus $\tilde{\phi}_J, \tilde{\phi}_\omega$ correspond to the symplectic structure $\tilde{\omega}$ and complex structure $\tilde{J}$ respectively, given by

$$\tilde{\omega} = \begin{pmatrix} 0 & -\mathbb{1} \\ \mathbb{1} & 0 \end{pmatrix}, \quad \tilde{J} = \begin{pmatrix} 0 & U^{-1} \\ U & 0 \end{pmatrix} \quad (7.85)$$

where $U_{ij} = u_{ij}$. This defines a Kähler structure, with Kähler metric

$$\tilde{g} = \sum_{i,j} u_{ij} dx^i \otimes dx^j + u^{ij} d\tilde{y}^i \otimes d\tilde{y}^j \quad (7.86)$$

which agrees with the Buscher rules. This again defines a generalised Calabi-Yau structure, though it is not immediately clear that $\tilde{g}$ is classically Ricci flat. To see this, we may use a different approach, where we rewrite α_{mix} .

To this end, write

$$\alpha_{\text{mix}} = b^1 \wedge dy^1 + \dots + b^k \wedge dy^k \quad (7.87)$$

where $b^i = \sum_j u_{ji} dx^j$. Then the T-dual structures are given by

$$\tilde{\phi}_J = \exp(id x^1 \wedge d\tilde{y}^1 + \dots + id x^k \wedge d\tilde{y}^k) \quad (7.88)$$

$$\tilde{\phi}_\omega = \left(\sum_j u_{j1} dx^j + id\tilde{y}^1 \right) \wedge \dots \wedge \left(\sum_j u_{jk} dx^j + id\tilde{y}^k \right) \quad (7.89)$$

If we define new coordinates $\tilde{x}^i = u_i := \partial_{x_i} u$, so that $\frac{\partial \tilde{x}^i}{\partial x^j} = u_{ij}$, then

$$d\tilde{x}^i = \sum_j u_{ij} dx^j, \quad \text{so that} \quad dx^i = \sum_j u^{ij} d\tilde{x}^j \quad (7.90)$$

and hence

$$\tilde{\phi}_J = \exp\left(i \sum_j u^{ij} d\tilde{x}^j \wedge d\tilde{y}^i\right) \quad (7.91)$$

$$\tilde{\phi}_\omega = (d\tilde{x}^i + id\tilde{y}^1) \wedge \dots \wedge (d\tilde{x}^i + id\tilde{y}^k). \quad (7.92)$$

The associated Kähler metric is therefore

$$\tilde{g} = \sum_{i,j} u^{ij} (d\tilde{x}^i \otimes d\tilde{x}^j + d\tilde{y}^i \otimes d\tilde{y}^j), \quad (7.93)$$

whose Ricci form is $-i\bar{\partial}\partial \log \det(u^{ij}) = 0$, so $\tilde{g}$ is Ricci flat.

Example 7.94 (Harmonic Functions and Doubly Symplectic Manifolds).

Consider $\mathcal{Q}_1 = \mathcal{B} \times \mathbb{T}^2$ for some 2-manifold $\mathcal{B}$. Consider local coordinates $\{x^1, x^2\}$ on $\mathcal{B}$ and $\{y^1, y^2\}$ on $\mathbb{T}^2$. We can endow $\mathcal{Q}_1$ with the following complex structures

$$J_+ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} \eta & 0 \\ 0 & \eta \end{pmatrix} \quad (7.95)$$

where $\eta = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Consider the metric $g = uE$, where E is the Euclidean metric, and $u \in C^\infty(\mathcal{B})$. Then $J_\pm$ preserve the metric, and so define a bi-Hermitian structure. The associated two-forms $\omega_\pm = gJ_\pm$ are

$$\omega_+ = u(dx^1 \wedge dx^2 + dy^1 \wedge dy^2), \quad \omega_- = u(dx^1 \wedge dy^1 + dx^2 \wedge dy^2). \quad (7.96)$$

We can then calculate

$$d_-^c \omega_- = -d_+^c \omega_+ = (u_2 dx^1 - u_1 dx^2) \wedge dy^1 \wedge dy^2, \quad (7.97)$$

where $u_i = \frac{\partial u}{\partial x^i}$. We thus take $H \in \Omega^3(\mathcal{Q}_2)$ as

$$H = (u_2 dx^1 - u_1 dx^2) \wedge dy^1 \wedge dy^2. \quad (7.98)$$

Then H is closed if and only if $\Delta u = 0$. It follows that $\mathcal{K} = (\mathcal{J}_+, \mathcal{J}_-)$ is a H -twisted generalised Kähler structure, where

$$\mathcal{J}_\pm = \frac{1}{2} \begin{pmatrix} J_+ + J_- & -(\omega_+^{-1} \mp \omega_-^{-1}) \\ \omega_+ \mp \omega_- & -(J_+^* \pm J_-^*) \end{pmatrix}. \quad (7.99)$$

Then

$$\begin{aligned} L_\pm = \{ & \mp \partial_{x^1} + i\partial_{x^2} \pm i\partial_{y^1} + \partial_{y^2}, \mp i\partial_{x^1} - \partial_{x^2} \pm iudx^1 + udx^2, \\ & i\partial_{x^1} + \partial_{y^1} + iudx^1 + udy^1, +i\partial_{x^2} \mp i\partial_{y^1} \pm udx^1 + udy^2 \}, \end{aligned} \quad (7.100)$$

which can be written as $\text{gr}(\exp(\alpha_\pm)|_{E_\pm}) \oplus \text{Ann}(E_\pm)$, where $E_\pm = \rho(L_\pm)$, and $\alpha_\pm = u(idx^1 + dy^1) \wedge (idx^1 \pm dx^2)$. Hence generating spinors are given by

$$\phi_\pm = \exp(\mp udx^2 \wedge dy^1 - iudx_1 \wedge dy^1 \pm uidx^1 \wedge dx^2) \wedge (\pm idx^1 + dx^2 \pm dy^1 - idy^2). \quad (7.101)$$

Furthermore, letting $\Theta_{\pm} = -dx^1 \pm idx^2 + idy^1 \pm dy^2$, which is closed, we have

$$\begin{aligned}
 d_H \phi_{\pm} &= d(\alpha_{\pm}) \wedge \exp(\alpha_{\pm}) \wedge \Theta_{\pm} + H \wedge \exp(\alpha_{\pm}) \wedge \Theta_{\pm} \\
 &= d(\alpha_{\pm}) \wedge \Theta_{\pm} + H \wedge \Theta_{\pm} \\
 &= (\pm iu_1 + u_2)dx^1 \wedge dx^2 \wedge dy^1 \wedge dy^2 + (\mp iu_1 - u_2)dx^1 \wedge dx^2 \wedge dy^1 \wedge dy^2 \\
 &= 0
 \end{aligned} \tag{7.102}$$

and hence $\mathcal{K}$ is generalised Calabi-Yau.

We now apply T-duality in the y_2 direction. Thus let $\mathcal{Q}_2$ be a principal circle bundle over $\mathcal{B} \times S^1$, with coordinate $\tilde{y}^2$ on the fibers. On the correspondence space $M = \mathcal{Q}_1 \times_{\mathcal{B} \times S^1} \mathcal{Q}_2$, take

$$B = dy^2 \wedge \tilde{\theta} \tag{7.103}$$

where $\tilde{\theta} = d\tilde{y}^2 - y^1 u_2 dx^1 + y^1 u_1 dx^2$. Note that $\mathcal{Q}_1$ therefore has Chern class

$$[u_2 dx^1 \wedge dy^1 - u_1 dx^2 \wedge dy^1] \in H^2(\mathcal{B}, \mathbb{R}). \tag{7.104}$$

With the chosen B , we have that $dB = H$, and there is a T-duality relation $R: (\mathbb{T}\mathcal{Q}_1, H) \dashrightarrow (\mathbb{T}\mathcal{Q}_2, 0)$, with generators given by

$$\{ (\partial_{x^i}, \partial_{x^i} - (-1)^i y^1 u_{3-i} \partial_{\tilde{y}^2}), (dx^i, dx^i), (\partial_{y^1}, \partial_{y^1}), (dy^1, dy^1), (\partial_{y^2}, \Theta), (dy^2, \partial_{\tilde{y}^2}) \}_{i=1,2}. \tag{7.105}$$

By Proposition 6.241, we may write the T-dual spinors as $\tilde{\phi}_{\pm} = \exp(\tilde{\alpha}_{\pm})$, where

$$\tilde{\alpha}_{\pm} = \mp u dx^2 \wedge dy^1 - iu dx_1 \wedge dy^1 \pm u i dx^1 \wedge dx^2 + (\pm i dx^1 + dx^2 \pm dy^1) \wedge (-i\tilde{\Theta}). \tag{7.106}$$

By T-duality, this defines a generalised Calabi-Yau structure, i.e. $d\tilde{\phi}_{\pm} = 0$. Again,

we may decompose $\tilde{\alpha}_{\pm} = \pm A + i\tilde{\omega}_{\pm}$, where

$$A = y^1 u_1 dx^1 \wedge dx^2 - u dx^2 \wedge dy^1 + dx^1 \wedge d\tilde{y}^2. \quad (7.107)$$

It follows that $dA_{\pm} = 0$ and hence $d\tilde{\omega}_{\pm} = 0$, so that $\tilde{\phi}_{\pm}$ both define twists of genuine symplectic structures on $\mathcal{Q}_2$.

Chapter 8

Conclusions

In this thesis, we argue that the language of Courant algebroid relations gives T-duality a robust geometric meaning. Specifically, T-duality is described by a Courant algebroid relation given by the composition of two reduction relations, fitting the diagram (4.13), i.e.

$$\begin{array}{ccc}
 & E & \\
 Q(K_1)^\top \nearrow & & \nwarrow Q(K_2) \\
 \underline{E}_1 & \xrightarrow{\quad R \quad} & \underline{E}_2
 \end{array} \tag{8.1}$$

The T-duality relation R is required to be a generalised isometry between generalised metrics V_1^+, V_2^+ on $\underline{E}_1, \underline{E}_2$ respectively. Upon imposing this restriction, the T-duality relation recovers the classical transformation laws of T-duality given by the Buscher-Rules. If the base manifolds are restricted to the case that the diagram

$$\begin{array}{ccc}
 & M & \\
 \swarrow \varpi_1 & & \searrow \varpi_2 \\
 (\mathcal{Q}_1, \underline{\mathcal{F}}_1) & & (\mathcal{Q}_2, \underline{\mathcal{F}}_2)
 \end{array} \tag{8.2}$$

defines a Hausdorff-Morita equivalence, Theorem 4.45 states that a generalised metric on $\mathcal{Q}_1$ with appropriate symmetry with respect to $\underline{\mathcal{F}}_1$ (be it local or global symmetry) can be transported to a T-dual generalised metric on $\mathcal{Q}_2$.

Once a generalised isometry is established, the duality transformation of other structures important to T-duality and the wider mathematical field can be described, such as generalised Riemannian geometry and twisted differential cohomology. This is the content of Chapters 5 and 6.

The starting point of exploring the T-duality of generalised Riemannian geometry is a divergence operator on the Courant algebroid $\underline{E}_1$. The appropriate invariance conditions are given and the question of transporting the divergence to $\underline{E}_2$ is answered in Theorem 5.57. This again agrees with the usual transformation laws of T-duality, where the transformation for the divergence is the dilaton shift. The divergence, along with the generalised metric, allows one to define the generalised Ricci curvature and generalised scalar curvature. Generalised isometries, if moreover they are constrained to relate the divergences, are such that the associated generalised Riemannian structures must also be related. Thus the question of transporting the generalised Riemannian structures to $\mathcal{Q}_2$ is succinctly answered using the T-duality relation. When this is the case, the Ricci-Buscher rules are provided, allowing efficient computation of the classical Ricci tensor of the T-dual Riemannian metric. Since the generalised Ricci tensor and generalised scalar curvature are necessarily preserved by the T-duality relation, the question of whether the generalised Ricci flow is compatible with T-duality can be answered directly: if the initial isometries of the flow preserve the metric and divergence for all times up to $T > 0$, then the T-dual generalised Ricci flow exists for up to time T . In the case that $\mathcal{Q}_1$ is compact or complete, this condition is automatically satisfied, and hence the generalised Ricci flow on the dual $\mathcal{Q}_2$ always exists for some time. This includes the possibility that $\mathcal{Q}_2$ is non-compact/non-complete, see Section 7.2.2.

In Chapter 6, we discuss how Courant algebroid relations induce relations for spinors. When $\text{Cl}(\underline{E}_i)$ are equipped with spinor modules $\mathbf{S}_i$, we consider a spinor module $\mathbf{S}_R \subset \mathbf{S}_1 \times \mathbf{S}_2$ for the Clifford algebra $\text{Cl}(R)$ generated by the relations. Since $\mathbf{S}_R$ is necessarily preserved by $\text{Cl}(R)$, this explains how $\mathbf{S}_R$ relates the action of $\text{Cl}(\underline{E}_1)$ on $\mathbf{S}_1$ to that of $\text{Cl}(\underline{E}_2)$ on $\mathbf{S}_2$. Similarly, $\mathbf{S}_R$ can be constrained to preserve differential graded operators, which we call DGO relations. We can in the correspondence case explicitly construct a DGO relation for the T-duality relation R , through use of [30, 31]. If the Fourier-Mukai transform exists, the DGO relation $\mathbf{S}_R$ is shown to reproduce it. The language of DGO relations allows this procedure to be extended to the case with non-compact fibers, where the Fourier-Mukai transform does not exist, answering some of the questions raised in [86].

In parallel with the treatment of spinors, we discuss how Lagrangian and Dirac

structures are related. Dirac structures and pure spinors are in one-to-one correspondence, and we explain how this correspondence extends to the language of Courant algebroid relations. Dirac structures and their relations have important applications in the study of quasi-Hamiltonian reduction. In [78], the reduction is described in the spinor language by the fiber integration forms, and hence restricts to compact groups. As mentioned in [80], there are many examples of quasi-Hamiltonian actions of non-compact groups, which the approach of [80] subsumes. The language of Clifford relations introduced in Chapter 6 resolves this discrepancy.

Naturally leading from Dirac structures, generalised complex structures are discussed. The language of Courant algebroid relations and their DGO relations is shown to perfectly encapsulate the notion of related generalised complex structures, generalised Kähler structures and generalised Calabi-Yau structures. In the T-duality setting, we again show how, under an invariance assumption on a generalised complex/Kähler/Calabi-Yau structure on $\mathcal{Q}_1$, this structure can be transported to a structure on $\mathcal{Q}_2$ of the same kind. An explicit type change formula is given, which is in agreement with [46] for torus bundles, but can easily be extended to the non-compact case.

The interplay between these structures and Courant algebroid relations — particularly the T-duality relation — examined in this thesis provides a coherent geometric formulation of non-linear sigma models and supergravity. This framework encompasses the bosonic string, its low energy effective action and renormalization group flow, and the Ramond-Ramond fields, the constituent components for such theories [12, 87], culminating in Proposition 6.202. Though not explored in this thesis, D-branes are encapsulated by the language of generalised complex geometry [88], and so could be incorporated into this picture.

Recently, in [89], the integration of generalised Kähler structures has been examined, where by integration we mean the Lie-algebra-Lie-group-like correspondence extended to algebroids. This problem is closely related to the discussion of the generalised Kähler potential, which itself is related to the associated bi-Hermitian structure of the Kähler manifold. It would be interesting to see how the discussion of Remark 6.275 could be used to related generalised Kähler potentials. The full discussion of relating the integrated generalised Kähler structures would in all like-

lihood necessitate the integration of Courant algebroid relations, a process not yet fully understood.

Appendix A

Change of Splitting for Compositions of Transverse Generalised Isometries

The composition of transverse generalised isometries $R_1: (E_1, W_1) \dashrightarrow (E_2, W_2)$ and $R_2: (E_2, W_2) \dashrightarrow (E_3, W_3)$ may fail due to the fact that the lifts of $W_2^\pm$ for the decomposition (3.80) may differ for R_1 and R_2 . To see the solution, let us first investigate the case of relations R_i which are graphs of classical Courant algebroid morphisms given by isomorphisms of exact Courant algebroids. Suppose that $\Phi_1: E_1 \rightarrow E_2$ and $\Phi_2: E_2 \rightarrow E_3$ are Courant algebroid isomorphisms, where E_i are exact Courant algebroids over M_i . Let K_i be isotropic involutive subbundles of E_i , let W_i be pre- K_i -transverse generalised metrics, and assume that Φ_i is a regular transverse generalised isometry between W_i and W_{i+1} .

Then there exist the bundles $\widetilde{W}_1^\pm$, $\widetilde{W}_2^\pm$, $\widetilde{W}_2'^\pm$ and $\widetilde{W}_3'^\pm$ such that

$$\Phi_1(\widetilde{W}_1^\pm) = \widetilde{W}_2^\pm \quad \text{and} \quad \Phi_2(\widetilde{W}_2'^\pm) = \widetilde{W}_3'^\pm . \quad (\text{A.1})$$

We can define the subbundles $\widetilde{W}_1'^\pm := \Phi_1^{-1}(\widetilde{W}_2'^\pm)$. If $\widetilde{W}_1'^\pm$ are also lifts of $W_1^\pm$ to W_1 , then

$$\Phi_2 \circ \Phi_1(\widetilde{W}_1'^\pm) = \widetilde{W}_3'^\pm , \quad (\text{A.2})$$

and $\Phi_2 \circ \Phi_1$ is a regular transverse generalised isometry between W_1 and W_3 . If

$s_2^\pm: W_2^\pm \rightarrow W_2$ and $s_2'^\pm: W_2^\pm \rightarrow W_2$ are the splittings with images $\widetilde{W}_2^\pm$ and $\widetilde{W}_2'^\pm$ respectively, the requirement that $\widetilde{W}_1'^\pm$ are also lifts of $W_1^\pm$ to W_1 is that the differences $s_2^\pm - s_2'^\pm$ map to $\Phi_1(K_1)$. Similarly one could define $\widetilde{W}_3^\pm := \Phi_2(\widetilde{W}_2^\pm)$ which, provided that $s_2^\pm - s_2'^\pm$ map to $\Phi_2^{-1}(K_3)$, define new splittings such that $\Phi_2 \circ \Phi_1(\widetilde{W}_1^\pm) = \widetilde{W}_3^\pm$.

When R_i are general Courant algebroid relations, the condition on the splittings $s_2^\pm$ and $s_2'^\pm$ may be written as

$$\text{pr}_1(R_1 \cap (E_1 \times \text{im}(s_2^\pm - s_2'^\pm))) \subset K_1 \quad \text{or} \quad \text{pr}_2(R_2 \cap (\text{im}(s_2^\pm - s_2'^\pm) \times E_3)) \subset K_3, \quad (\text{A.3})$$

which should be viewed as pointwise inclusions. These are similar to the conditions given in Theorem 4.45, though here the splittings $s_2^\pm$ and $s_2'^\pm$ are given *a priori* whereas in the proof of Theorem 4.45 we had to construct these splittings. That is, a portion of the proof is showing that Φ is a transverse generalised isometry. The pointwise nature of the present condition is also in contrast with the construction in Section 4.1.2, which provides a smooth splitting and hence shows that Φ is a regular transverse generalised isometry.

Appendix B

The σ -Adjoint Action

In this appendix we prove some properties of the σ -adjoint action that we use throughout the main text. Recall that it is given by Equation (4.34) as

$$\mathbf{ad}_{e_1}^\sigma e_2 := \llbracket e_1, e_2 \rrbracket - \rho^* \sigma^* \llbracket e_1, \sigma \rho(e_2) \rrbracket , \quad (\text{B.1})$$

for $e_1, e_2 \in \Gamma(E)$ and an isotropic splitting $\sigma : TM \rightarrow E$ of an exact Courant algebroid E over M . This operation was introduced (but not thusly named) in [5, Definition 5.21].

Lemma B.2. $\rho(\mathbf{ad}_{e_1}^\sigma e_2) = \mathcal{L}_{\rho(e_1)} \rho(e_2)$ for all $e_1, e_2 \in \Gamma(E)$.

Proof. This follows immediately from the fact that the anchor map $\rho : E \rightarrow TM$ is a bracket homomorphism and $\rho \circ \rho^* = 0$. $\square$

Lemma B.3. The σ -adjoint action satisfies the anchored Leibniz rule

$$\mathbf{ad}_{e_1}^\sigma (f e_2) = f \mathbf{ad}_{e_1}^\sigma e_2 + (\mathcal{L}_{\rho(e_1)} f) e_2 , \quad (\text{B.4})$$

for all $e_1, e_2 \in \Gamma(E)$ and $f \in C^\infty(M)$.

Proof. For any $e_1, e_2 \in \Gamma(E)$ and $f \in C^\infty(M)$ we compute

$$\begin{aligned}
 \mathbf{ad}_{e_1}^\sigma(f e_2) &= \llbracket e_1, f e_2 \rrbracket - \rho^* \sigma^* \llbracket e_1, f \sigma \rho(e_2) \rrbracket \\
 &= f \llbracket e_1, e_2 \rrbracket + (\mathcal{L}_{\rho(e_1)} f) e_2 - \rho^* \sigma^* (f \llbracket e_1, \sigma \rho(e_2) \rrbracket + (\mathcal{L}_{\rho(e_1)} f) \sigma \rho(e_2)) \\
 &= f \mathbf{ad}_{e_1}^\sigma e_2 + (\mathcal{L}_{\rho(e_1)} f) (e_2 - \rho^* \sigma^* \sigma \rho(e_2)) ,
 \end{aligned} \tag{B.5}$$

and the result now follows from $\sigma^* \circ \sigma = 0$. $\square$

Lemma B.6. The pairing $\langle \cdot, \cdot \rangle$ is preserved by $\mathbf{ad}_e^\sigma$, for all $e \in \Gamma(E)$.

Proof. Let $e, e_1, e_2 \in \Gamma(E)$. Then

$$\begin{aligned}
 \mathcal{L}_{\rho(e)} \langle e_1, e_2 \rangle &= \langle \llbracket e, e_1 \rrbracket, e_2 \rangle + \langle e_1, \llbracket e, e_2 \rrbracket \rangle \\
 &= \langle \mathbf{ad}_e^\sigma e_1, e_2 \rangle + \langle \rho^* \sigma^* \llbracket e, \sigma \rho(e_1) \rrbracket, e_2 \rangle + \langle e_1, \mathbf{ad}_e^\sigma e_2 \rangle + \langle e_1, \rho^* \sigma^* \llbracket e, \sigma \rho(e_2) \rrbracket \rangle \\
 &= \langle \mathbf{ad}_e^\sigma e_1, e_2 \rangle + \langle \llbracket e, \sigma \rho(e_1) \rrbracket, \sigma \rho(e_2) \rangle + \langle e_1, \mathbf{ad}_e^\sigma e_2 \rangle + \langle \sigma \rho(e_1), \llbracket e, \sigma \rho(e_2) \rrbracket \rangle \\
 &= \langle \mathbf{ad}_e^\sigma e_1, e_2 \rangle + \langle \llbracket e, \sigma \rho(e_1) \rrbracket, \sigma \rho(e_2) \rangle + \langle e_1, \mathbf{ad}_e^\sigma e_2 \rangle \\
 &\quad + \mathcal{L}_{\rho(e)} \langle \sigma \rho(e_1), \sigma \rho(e_2) \rangle - \langle \llbracket e, \sigma \rho(e_1) \rrbracket, \sigma \rho(e_2) \rangle \\
 &= \langle \mathbf{ad}_e^\sigma e_1, e_2 \rangle + \langle e_1, \mathbf{ad}_e^\sigma e_2 \rangle ,
 \end{aligned} \tag{B.7}$$

as required. $\square$

Lemma B.8. Under the Courant algebroid isomorphism $E \cong (\mathbb{T}M, H)$ induced by σ , the Dorfman bracket $\llbracket \cdot, \cdot \rrbracket_H$ is preserved by $\mathbf{ad}_{X+\alpha}^\sigma$ if and only if $\iota_X H$ is a closed two-form, for all $X + \alpha \in \Gamma(\mathbb{T}M)$.

Proof. Under the isomorphism $E \cong (\mathbb{T}M, H)$ we can identify $\sigma(X) = X$ for all

$X \in \Gamma(TM)$. Then

$$\begin{aligned}
 & \text{ad}_{X+\alpha}^\sigma \llbracket Y + \eta, Z + \xi \rrbracket_H - \llbracket \text{ad}_{X+\alpha}^\sigma(Y + \eta), Z + \xi \rrbracket_H - \llbracket Y + \eta, \text{ad}_{X+\alpha}^\sigma(Z + \xi) \rrbracket_H \\
 &= -\iota_{[Y,Z]} \iota_X H + \llbracket \iota_Y \iota_X H, Z + \xi \rrbracket_H + \llbracket Y + \eta, \iota_Z \iota_X H \rrbracket_H \\
 &= -\iota_{[Y,Z]} \iota_X H - \iota_Z \text{d} \iota_Y \iota_X H + \mathcal{L}_Y \iota_Z \iota_X H \\
 &= \iota_Z \iota_Y \text{d} \iota_X H,
 \end{aligned} \tag{B.9}$$

where in the final step we make use of various Cartan calculus identities. Thus the Dorfman bracket is invariant under this action if and only if $\iota_X H$ is closed. $\square$

Let $Z \in \Gamma(TE)$ be a vector field on E covering a vector field $\bar{Z} \in \Gamma(TM)$ on M . The *generalised Lie derivative* along Z at $e \in E$ is defined by

$$(\mathcal{L}_Z v)_e := \frac{\text{d}}{\text{d}s} (\Phi_s^* v)_e \Big|_{s=0}, \quad \mathcal{L}_Z f := \mathcal{L}_{\bar{Z}} f \tag{B.10}$$

for all $v \in \Gamma(E)$ and $f \in C^\infty(M)$, where $\Phi_s \in \text{Aut}(E)$ is the flow of Z through e . This definition easily extends to tensor powers of E via the Leibniz rule.

Lemma B.11. On the space of vector fields $X \in \Gamma(TM)$ for which ad_X^σ preserves the Dorfman bracket, there are infinitesimal Courant algebroid automorphisms $Z_X \in \Gamma(TE)$ such that

- $\overline{Z_X} = X$,
- $\mathcal{L}_{Z_X} v = \text{ad}_X^\sigma v$, for all $v \in \Gamma(E)$,
- $[Z_X, Z_Y] = Z_{[X,Y]}$ and the map $X \mapsto Z_X$ is $\mathbb{R}$ -linear.

Proof. For a section e of E and an integral curve $\gamma(t) \subset M$ of X , the vector field Z_X is given by the tangent to the curve $(e - t \text{ad}_X^\sigma e)_{\gamma(t)} \subset E$ at $t = 0$; a standard argument (cf. [27]) shows that Z_X does not depend on the choice of auxiliary section or curve. Then the first two properties follow immediately, while the final property follows from $[\text{ad}_X^\sigma, \text{ad}_Y^\sigma] = \text{ad}_{[X,Y]}^\sigma$, which can be proven using various Cartan identities and the fact that $\iota_{[X,Y]} H$ is closed (and hence $\llbracket [X, Y]_H, \cdot \rrbracket_H = \llbracket \sigma([X, Y]), \cdot \rrbracket_H$.) $\square$

Remark B.12. If $\sigma_0(X) = X$ and $\text{ad}_X^{\sigma_0}$ preserves the Dorfman bracket as above, then $\mathcal{L}_X H = \text{d} \iota_X H = 0$, and hence X integrates to a Courant algebroid automorphism induced by a diffeomorphism $\varphi \in \text{Diff}(M)$ which preserves H , i.e. $\varphi^* H = H$.

Remark B.13. If the infinitesimal Courant algebroid automorphism corresponding to $e \in \Gamma(E)$ (in the sense of [61, Proposition 2.7]) preserves a generalised metric $V^+ \subset E$, i.e. $\llbracket e, v^+ \rrbracket \in \Gamma(V^+)$ for all $v^+ \in \Gamma(V^+)$, then V^+ is also invariant under the σ -adjoint action ad_e^σ , since $\rho^* \sigma^* \llbracket e, \sigma \rho(v^+) \rrbracket = 0$ and hence ad_e^σ acts through the usual Dorfman bracket on $\Gamma(V^+)$. For instance, if $B \in \Omega^2(M)$ and $\sigma_B(X) = X + \iota_X B$ is an infinitesimal Courant algebroid isometry for (g, b) , then $\mathcal{L}_X(g+b) = 0$ and $\iota_X \text{d}B = \text{d} \iota_X b$, so $\mathcal{L}_X H = 0$. In this sense, it is a generalisation of inner automorphisms allowing for $\iota_X H$ to be closed.

Lemma B.14. In the setting of Definition 4.102, the space of D -invariant sections $\Gamma_D(E)$ is involutive.

Proof. This is a consequence of Lemma B.8. □

Lemma B.15. If $V^+ \subset E$ is a D -invariant generalised metric and v is a D -invariant section of E , then the projections $v^\pm \in \Gamma(V^\pm)$ are also D -invariant.

Proof. Since $\text{ad}_X^\sigma v = 0$ and $\text{ad}_X^\sigma v^+ \in \Gamma(V^+)$ for all $X \in \mathfrak{k}_\tau$, it follows that $\text{ad}_X^\sigma v^- \in \Gamma(V^+)$. Thus for any $w^+ \in \Gamma(V^+)$, we have

$$\langle \text{ad}_X^\sigma v^-, w^+ \rangle = \mathcal{L}_X \langle v^-, w^+ \rangle - \langle v^-, \text{ad}_X^\sigma w^+ \rangle = 0. \quad (\text{B.16})$$

Hence $\text{ad}_X^\sigma v^-$ must vanish, and so then must $\text{ad}_X^\sigma v^+$. □

Appendix C

A Note on Fiber Integration

The fibered integral over a co-dimension $n - i$ fibre F of a k -form ρ is defined as

$$\left(\int_F \rho \right) (X_1, \dots, X_{k-i}) = \int_F \beta \quad (\text{C.1})$$

where β is fiber volume form such that

$$\rho(Y_1, \dots, Y_i, \tilde{X}_1, \dots, \tilde{X}_{k-i}) = \beta(Y_1, \dots, Y_i) \quad (\text{C.2})$$

for $Y_1, \dots, Y_i$ vertical, and X_k on M/F . Thus, if $X \in T(M/F)$, then

$$\left(\iota_X \int_F \rho \right) (X_2, \dots, X_{k-i}) = \left(\int_F \rho \right) (X, X_2, \dots, X_{k-i}). \quad (\text{C.3})$$

On the other hand

$$\left(\int_F \iota_{\tilde{X}} \rho \right) (X_2, \dots, X_{k-i}) = \int_F \beta' \quad (\text{C.4})$$

where β' is such that

$$\beta'(Y_1, \dots, Y_i) = (\iota_{\tilde{X}} \rho)(Y_1, \dots, Y_i, \tilde{X}_2, \dots, \tilde{X}_{k-i}) = (-1)^i \rho(Y_1, \dots, Y_i, \tilde{X}, \tilde{X}_2, \dots, \tilde{X}_{k-i}). \quad (\text{C.5})$$

Thus

$$\iota_X \int_F \rho = (-1)^i \int_F \iota_{\tilde{X}} \rho. \quad (\text{C.6})$$

Similarly,

$$\xi \wedge \int_F \rho = (-1)^i \int_F \varpi^* \xi \wedge \rho, \quad \text{and} \quad d \int_F \rho = (-1)^i \int_F d\rho \quad (\text{C.7})$$

This has the consequence of introducing a grading on the BEM-isomorphism τ , i.e.

$$\tau(e \cdot \rho) = (-1)^i \psi(e) \cdot \tau(e). \quad (\text{C.8})$$

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